

UNDERGROUND CBRN-HARDENED BLAST-RESISTANT PROTECTIVE STRUCTURE, WITH SENTRY POST

Pune, Maharashtra · Master project report

Item	State
Report revision	PR2 · 13 September 2026
Architectural revision	Rev F
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Design report revision (historical)	Rev D
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Governing authority	master/MASTER_PROJECT_STATE.md — Parts A, B, F and L
Status	FOR REVIEW — NOT FOR CONSTRUCTION

This report states the project once, in the order an engineer would need it to reproduce the design: the science first, then the site, then the loads, then the calculations, then the openings, then what is drawn, then the services, then how it is built — and then what is still open.

It is a statement of the design as it stands, not a history of how it got there. Every value carries the evidence class the master gives it. Where the project does not hold a number, this report says so in the same sentence rather than filling it in.

Twenty-five parts, four appendices and forty-four drawn figures. The figures are generated from the confirmed geometry in project coordinates. They are **not** the issued drawings and must never be used for setting out or fabrication.

Where this report and the master disagree, the master governs.

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PART 1 — THE PROJECT AND THIS DOCUMENT

1.1 What this report is

This is the **master project report** for a buried reinforced-concrete protective structure at Pune, Maharashtra, and for the separate sentry post that watches its approach. It states the project once, in the order an engineer would need it to reproduce the design: the physics first, then the ground, then the loads, then the calculations, then what is drawn, then what is still open.

It is written to stand on its own. Everything needed to follow an element from a first-principles load to a bar mark on a drawing is in here, with the clause of the Indian Standard that permits each step named beside it.

It is a statement of the design as it stands, not a history of how it got there. The design went through revisions, and every one of them is recorded in the master. None of them is narrated here. Where a decision is load-bearing — where an engineer reading the drawing would otherwise ask *why is this wall 400 and not 200* — the reason is given as a reason, in the present tense, as part of the design. Where it is only project history, it is left in the master, and **Appendix D** carries it in full — the conflicts, the rulings and the revision list — so that nothing is lost by keeping it out of the design.

The figures are drawn for this report. Forty-four of them, generated from the confirmed geometry rather than traced, at reading scale. They are **not** the issued drawings: they carry no title block, no revision box and no bar mark, and they must never be used for setting out or fabrication. The issued package is the 80 DXF sheets of Part 14.

1.2 Authority, and what governs what

Document	Role
master/MASTER_PROJECT_STATE.md	The single source of truth. Parts A, B, F and L are the authoritative current design. Where this report and the master disagree, the master governs
master/QUICK_STATE.md	Orientation digest only. Not an authority
This report	The project stated once, in full, for reading — design basis, science, calculations, services, works management, open items
Discipline packages	Structural CAD/·Drainage/·HVAC/·Electrical/·EMP Protection/·Fire and Life Safety/·Schedule of Finishes/·Site Selection and Geotechnical/·Site and Concealment/·WORKS MANAGEMENT/·DRAWING QAQC/
Drawings	80 DXF. The ten Rev F architectural files are source , not build artefacts; the discipline sheets are generated and must be regenerated, never hand-edited

This report changes no design value. No dimension, level, load, thickness, bar, quantity, rate, date or float is altered by it, no .std file is touched, and **no evidence tag is converted, downgraded or deleted.** Where the project does not hold a number, this report says so in the same sentence rather than filling it in.

1.3 Evidence classes — and why they are the point of the document

Every value in this project carries a class. They are the master's own and they are used here unchanged. A reader who ignores them will read an assumption as a measurement, which is the one mistake this document set exists to prevent.

Tag	Meaning	How to treat it
[C] CONFIRMED	Read directly from a project file, or computed in the project from confirmed inputs	Use as fact
[R] RECONSTRUCTED	Derived by back-calculation from partial evidence	Use, but state the derivation when relying on it
[A] ASSUMED	An engineering assumption. No test, no client confirmation	Must be re-confirmed before construction
[U] UNRESOLVED	Two or more conflicting values and insufficient evidence to choose	Do not guess
[N] NOT AVAILABLE	The information does not exist in any project material	Do not invent

The register is not decoration. Seven assumptions remain open (Part 24.2) and **every one of them is a physical test on this plot** — a borehole, a piezometer, a plate load test, a percolation test, a packer test. Eighteen open items remain (Part 24.3), and **four of them are the ones a reviewer should look at first**: an investigation that reached about 1.5 m against a formation at (–)6.800; two 1 400 mm holes in the protective boundary whose bonding is still undesigned; a CBRN shelter that cannot state which way a release drifts; and a sealed box with a 4 kW heat surplus and an unmodelled rejection path.

1.4 Project identity

Field	Value	Class
Project title	Underground CBRN-hardened, blast-resistant protective structure with associated sentry post	[C]
Location	Pune, Maharashtra, India — CTW / College of Military Engineering campus	[C]
Project type	Buried reinforced-concrete protective structure, military-operational, for actual construction	[C]
Academic context	B.E. Civil Engineering, final semester, three-presentation capstone	[C]
Objective	Protect 9 occupants for 96 h against a nuclear air-blast design basis threat, with CBRN, EMP and fallout hardening	[C]
Architectural revision	Rev F — the ten uploaded DXF files	[C]
Design report revision	Rev D — older than the drawings, and where the two disagree the drawings govern (Part 3.4)	[C]
Structural revision	Phase 2 Rev A + M1 — the master's own designation	[C]
Presentation status	P1 delivered. P2 (AutoCAD + STAAD + manual calculations) is the active deliverable. P3 not yet scoped	[C]
Software	STAAD.Pro (Bentley), AutoCAD/DXF, plus a purpose-built dependency-free Python DXF toolchain	[C]

1.5 The three structures, and they are deliberately not equal

#	Structure	Status
1	Main underground box — 8 bays, 22.0 × 6.2 m external, roof 2.0 m below grade	Blast designed. This is the protected volume
2	Entry headhouse and covered approach stairwell — above-ground bermed RC, sitting on the shelter roof and on adjacent fill	Headhouse blast designed ; the stairwell is NOT , and is declared expendable
3	Sentry post — two-storey RC framed building, ≥ 10 m clear of the shelter excavation, on its own footings on in-situ rock	NOT blast designed. A recorded decision, not an omission — Part 11.1

1.5.1 The protective boundary — the single most important idea in the design

The protective boundary is **Blast Doors 1 and 2 at level (–)6.100, together with walls W6 and W7, the perimeter walls, the mat and the pressure slab.**

Everything above the blast doors — the stair shaft, the headhouse, the entry stairwell — is OUTSIDE it. That is deliberate. The stair shaft is open to atmosphere through the roof void, and the design accepts that it fills with blast pressure rather than pretending it does not. That single admission is the reason walls W6 and W7 are **400 mm thick** and the reason both blast doors are rated for the full incident overpressure (Part 9.2). A design that assumed the shaft stayed at atmospheric pressure would have made them 200 mm, and would have been wrong on the face that matters most.

Gas-tight envelope = Bays 1 to 6 only — 67.80 m² floor, 216.96 m³ (stated on the issued sheet as 67.8 / 217.0) [C]. Bay 7 (stair shaft) and Bay 8 (generator) are **outside** it; Bay 8 is the "grey zone".

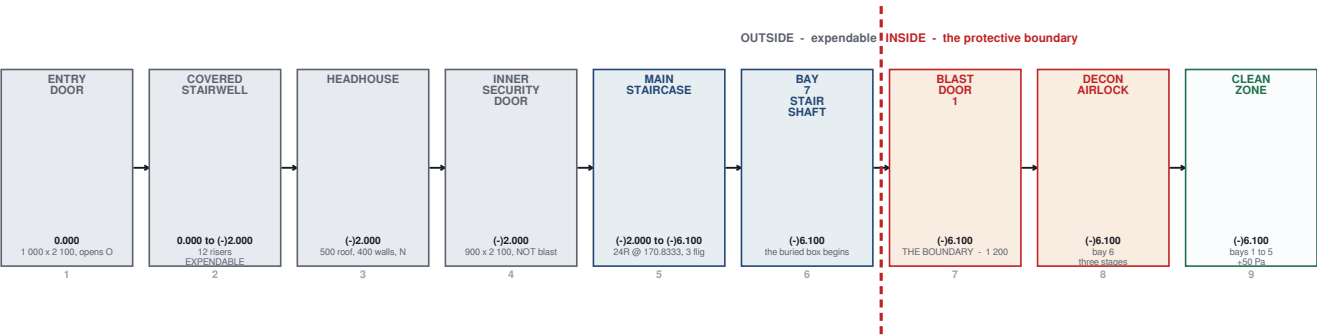
The design holds TWO protective boundaries and they do not coincide. The gas-tight boundary is bays 1–6. The **EMP boundary is bays 1–8**, because otherwise the 15 kVA generator, its control panel and every cable in bay 8 would stand unprotected against the pulse. **Only the gas-tight one has ever been drawn.** Part 18.

1.5.2 Access and egress

Route	Path
Primary	Grade → covered entry stairwell, 12R @ 166.667 / 300 → platform (–)2.000 → inner security door → headhouse → stair void → main dog-leg stair, 24R @ 170.8333 / 280, 3 flights × 8 → arrival landing (–)6.100 → Blast Door 1 , 1200 × 2100, ≥ 7 bar → Bay 6
Secondary	Blast Door 2 , 1200 × 2100, from the stair shaft into Bay 8 (generator / grey zone)
Emergency	Two 1 400 mm dia vertical escape shafts, ESC 1 (Bay 1) and ESC 2 (Bay 8), 250 mm RC collars, heads at +0.150 and +0.700

R1, the main stair, is the only route that does not require climbing a shaft, and the only one usable by an injured or unconscious person. ESC 1 and ESC 2 are **6.250 m** and **6.800 m** vertical climbs (Part 19).

THE WAY IN - NINE STEPS FROM GRADE TO THE CLEAN ZONE, AND WHERE THE PROTECTIVE BOUNDARY ACTUALLY STARTS



EVERYTHING WEST OF THAT LINE IS DESIGNED TO BE LOST.

The covered stairwell is explicitly expendable and is not blast rated. The headhouse has no earth cover. The entry door opens outward so that blast cannot drive it into the stairwell, and it is not rated either. None of that is an oversight: the boundary is Blast Door 1 and the 400 mm wall it sits in, and the approach is a weather and security enclosure in front of it.

WHICH IS WHY THE STAIR VOID IS THE EMP FINDING.

The same route that makes the shelter usable puts a 2 800 x 3 160 hole through a 900 mm pressure slab, and the reinforcement cage stops at its edge. A bonded conducting hatch answers the escape shafts. Nothing equivalent is specified here, and nothing is invented for it.

AND IT IS ALSO THE ONLY ROUTE AN INJURED PERSON CAN USE.

Routes R2 and R3 are 1 400 mm shafts with vertical ladders and a six-metre climb. They are escape shafts, not exits.

Figure 1 The way in, from grade to the clean zone, and where the protective boundary actually starts.

1.6 The mission, in numbers

	Value	Class
Occupancy	9 persons	[C]
Endurance	96 hours	[C]
Gas-tight floor area / volume	67.80 m ² / 216.96 m ³	[C]
Gross internal volume, bays 1–8	20.800 × 5.000 × 3.200 = 332.8 m ³	[C]
Design ventilation flow	300 m ³ /h, 2 × 300 trains, true N+1	[C]
Air changes on the gross volume	≈ 0.9 per hour	[R]
Closed-mode CO ₂ limit without scrubbing	9.9 h to 1.0 %	[C]
Closed-mode limit on the soda-lime store	48 h	[C]
Oxygen store	2 × 50 L at 150 bar = 15 m ³ = 80 h at 4.5 m ³ /day	[C]
Standby power	15 kVA generator, bay 8; connected load 7.360 kVA = 49 %	[C] / [R]
Battery autonomy, essential loads	4 h, 149 Ah at 48 V	[C] by ruling
Potable water	1 000 L tank, bay 1 = 27.8 L/person/day. Not firefighting water	[C]

1.7 Working rules this project is held to

These are not general good practice — they are the specific disciplines this project adopted, and several of them exist because something once went wrong.

1. **The master governs.** A digest is never an authority.
2. **Inspect native files before modifying them.** Parse the DXF; read the .std. Never edit from memory or from a summary.
3. **Never guess or silently resolve** an [UNRESOLVED], [ASSUMED] or [NOT AVAILABLE] item. Do not convert one into a confirmed fact, and do not delete or downgrade a tag.
4. **Do not fabricate missing generator scripts or unavailable output drawings.**
5. **Distinguish file reconciliation from analysis.** STAAD.Pro is not available in this environment. **Editing and verifying a .std file is not running an analysis** and must never be reported as one.
6. **No drawing shows an arrangement that differs from a calculation.** If the steel changes, the drawing changes in the same turn.
7. **Every significant change is recorded, and no record is overwritten without preserving it.** This report is the current statement of the design; the master is the ledger behind it.

8. **A new drawing package registers itself with the QA tool.** A package that does not is invisible to the drawing index, and stays invisible for as long as nobody looks (Part 14.3).
9. **A change that stops at "reinforcement" and never reaches "drawings" is not finished.**

1.7.1 Frozen geometry

The main staircase is frozen: 24 risers, 170.8333 mm riser, 280 mm tread, 3 flights × 8, total rise 4 100 mm. The landing levels, the 1 200 flight widths, the 200 mm well and the 2 533 mm headroom go with it. **Nothing in this report changes any of it,** and no service, duct, pipe, drain or diffuser is placed in the flights, the well or the landings.

1.8 How this report is arranged

Part	Contents
1	The project, the authority, the evidence classes, the rules
2	The engineering science behind the design — blast physics, structural dynamics, radiation attenuation, buoyancy, earth pressure, filtration, EMP coupling, seismic response, fire and the thermal problem
3	Codes and standards, clause by clause
4	Site selection, the soil report and geotechnics — the plot, the investigation, the ground parameters, the water
5	Geometry and configuration — the complete dimensional register
6	Materials, concrete mix design and detailing rules — M35 and M30 proportioned, cover, development and lap lengths
7	Loading — every load derived and every combination stated
8	Analysis models — what they are, what they prove, what they cannot
9–11	Structural design calculations , element by element: the box; the stairs, headhouse and appurtenances; the sentry post
12	Openings, blast doors, escape hatches and closures — every hole through the protective boundary
13	Reinforcement register and quantities
14	Drawings and drawing quality
15	HVAC, CBRN filtration and the sealed atmosphere
16	Water, sewage and drainage
17	Electrical and power
18	EMP protection
19	Fire and life safety
20	Site layout, external works, finishes and concealment
21	Works management — the bill, the cost estimate, the programme and the critical path, procurement, resources, QA/QC and risk
22	The environmental management plan
23	What this design does not demonstrate
24	Assumptions, open items and rulings
25	The verification performed for this report
A–D	Notation · IS clause index · reproducing the project · traceability: decisions, conflicts and revision history

Why the services get five parts and not one. A protective structure is not a building with services added to it. The filter train sets the overpressure, the overpressure sets the leak direction, the leak direction sets which openings matter, and the openings are what the EMP and the blast design are both fighting. Each of Parts 15 to 19 is a discipline in its own right with its own findings, and collapsing them into one section hides exactly the couplings that make this building difficult.

PART 2 — THE ENGINEERING SCIENCE BEHIND THE DESIGN

What this part is. The design decisions in Parts 6 to 11 are only defensible if the physics underneath them is understood. This part sets out that physics: where the loads come from, why the code rules take the form they do, and which physical effect each dimension was bought for.

It introduces no project value. Every number quoted here is carried from the project's own register and is cross-referenced to the Part where it is used. Where the physics would allow a quantity the project does not hold — a weapon yield, an ambient design temperature, a soil modulus — **the gap is named, not filled.**

2.1 The blast wave

2.1.1 What a shock front is

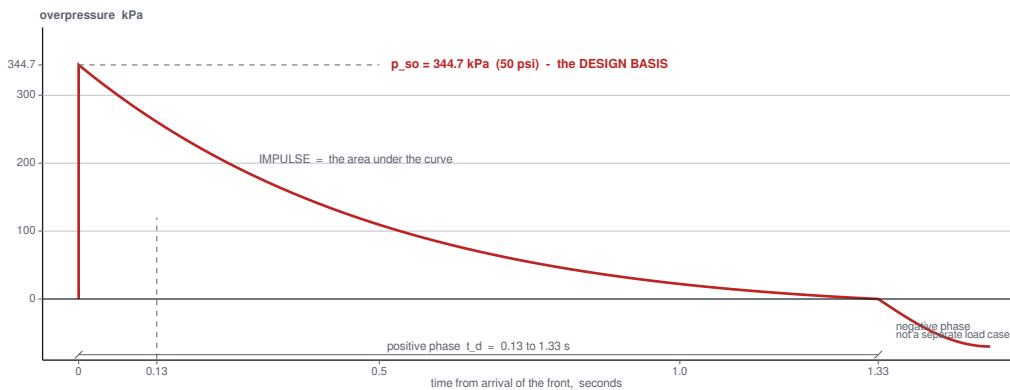
An explosion releases energy in a time far shorter than the time the surrounding air needs to move out of the way. The air is compressed faster than an acoustic wave can carry the information away, so the disturbance steepens into a **shock front**: a near-discontinuity across which pressure, density, temperature and particle velocity all jump.

Across that front, conservation of mass, momentum and energy give the **Rankine–Hugoniot** relations. For a strong shock in air the two consequences that matter to a structural engineer are:

STATIC OVERPRESSURE	p_{so}	the rise in ambient pressure at the front. It acts EQUALLY IN ALL DIRECTIONS, like any pressure. A buried roof, a buried wall and a free-standing wall all feel it.
DYNAMIC PRESSURE	q	the kinetic pressure of the air actually moving behind the front, $q = 0.5 \rho u^2$. It acts ONLY in the direction of flow, and it is what produces DRAG on an obstacle standing in it.

They are different loads with different physics, and the design uses both. The roof and walls of the buried box are loaded by p_{so} . The drag term q is what a bermed wall standing in the flow would take, and it is the reason the headhouse walls have to be argued about rather than assumed — Part 7.5.

THE DESIGN BLAST WAVE



Shape indicative; the DESIGN INPUTS are the peak and the duration, both confirmed. The yield behind them is deliberately not stated [N].

Figure 2 The design pressure-time history at the structure, and the idealised triangular pulse taken from it.

2.1.2 Reflection, and why the reflected pressure is not simply doubled

When a shock meets a rigid surface head-on, the air behind the front is brought to rest and its momentum is converted into further compression. For a weak (acoustic) wave the reflected pressure would be exactly twice the incident. For a strong shock in a gas that can be heated and partially dissociated, the reflection coefficient rises well above 2 — for the design basis threat of this project it is close to 4:

$p_{so} = 344.7 \text{ kPa (50 psi)}$	incident static overpressure	[C]
$p_r = 1366 \text{ kPa}$	reflected pressure	[C]
$q = 282 \text{ kPa}$	dynamic pressure	[C]

reflection coefficient $p_r / p_{so} = 1366 / 344.7 = 3.96$

That reflected value is not used on the buried box — a buried surface is not a free-standing reflecting face — but it **is** the number that decides the sentry post. Part 11.1 computes the reflected force on the sentry post's 4.0×6.7 m face as **36 600 kN, about 3 730 tonnes**, and that single figure is why hardening it was rejected rather than attempted.

2.1.3 Positive phase duration and impulse

Behind the front, pressure decays back to ambient over the **positive phase duration** t_d , then falls below ambient into a **negative phase** in which the flow reverses. The area under the pressure–time curve is the **impulse**, and for short loading it is impulse rather than peak pressure that damages a structure.

$t_d = 0.13$ to 1.33 s the design basis positive phase duration [C]

The yield behind those numbers is deliberately not stated. $p_{so} = 344.7$ kPa and $t_d = 0.13$ – 1.33 s **are** the design inputs, and every load case in this project uses them. The weapon yield that would produce them is [N], and it **stays unstated rather than invented**. Nothing in the design requires it. The one thing it would change is the prompt-radiation cover depth, and that is said openly in 2.4.

The negative phase is not designed for as a separate load case, and that is recorded in Part 23. Where it matters it is allowed for physically rather than numerically: the escape-shaft head hatches take **quarter-turn dogs so the leaf resists uplift as well as downward pressure**, because the negative phase and the rebound both lift it (Part 12).

2.1.4 Why a buried roof takes the full overpressure

A metre of soil is not a shield against overpressure. The wave in the air couples into the ground and propagates as a stress wave; the soil is a medium that **transmits** pressure, it does not absorb it in the way intuition suggests. Codified practice handles this with a **soil transmission factor** K_a applied to the free-field overpressure:

design surface pressure on a buried element = $K_a \times p_{so}$
IS 4991 Cl. 7.2 and Table 3

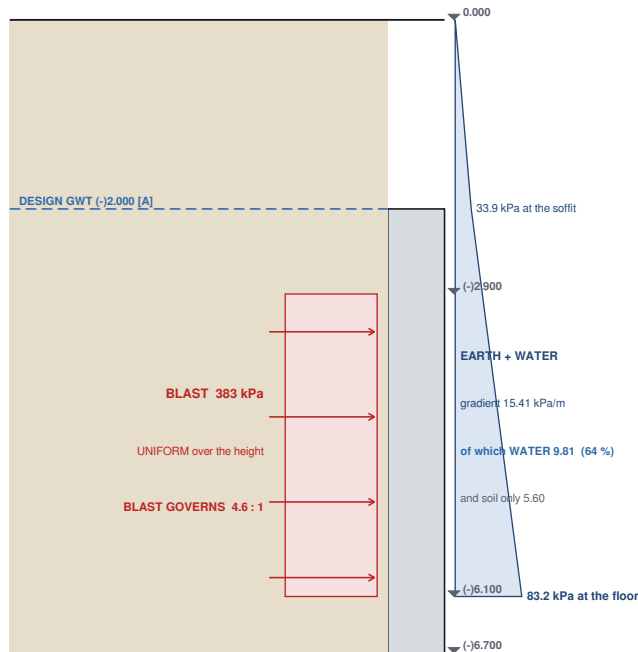
For **saturated** soil, $K_a = 1.0$ — water is nearly incompressible and transmits the pressure essentially undiminished. This project's founding horizon is permanently below the design water table, and the governing case is therefore the saturated one:

	Value	Class
K_a , saturated	1.0	[C] IS 4991 Cl. 7.2 / Table 3
K_a , dry compacted berm fill	≈ 0.5	[A] — deliberately not relied on

The single most consequential choice in the loading. Because $K_a = 1.0$, **2.0 m of engineered cover buys no reduction in blast pressure at all**. The roof of the buried box is designed for the same 383 kPa a bare roof would take. Every metre of that cover was bought for something else — radiation attenuation — and 2.5 names the price.

The same logic governs the headhouse walls, which stand behind about 2.5 m of berm. Crediting the berm and using the drag pressure would give 113 kPa. **Since K_a for a dry berm cannot be verified on this site, those walls are designed for the full 383 kPa envelope** (Part 7.5). The cost of that decision is one link cage; the utilisation goes from 17 % to 59 %.

LATERAL PRESSURE ON A PERIMETER WALL - THE TWO CASES COMPARED



The static case is a triangle that starts at the water table; the blast case is a rectangle over the whole height, because $K_a = 1.0$ transmits it undiminished.

Figure 3 Lateral pressure on a perimeter wall - the static case and the blast case compared.

2.2 From a pressure history to a design load: structural dynamics

2.2.1 The single-degree-of-freedom idealisation

A blast-loaded slab is a distributed system, but its response is dominated by its first mode. The standard idealisation replaces it with a single-degree-of-freedom (SDOF) oscillator of equivalent mass and stiffness carrying a time-varying force. The response then depends on **one ratio**: the load duration to the natural period, t_d / T .

$t_d / T < \sim 0.1$	IMPULSIVE	the load is over before the structure has moved. Only the IMPULSE matters.
~ 0.1 to ~ 10	DYNAMIC	genuine transient interaction; the full SDOF solution is needed.
$t_d / T > \sim 10$	QUASI-STATIC	the structure completes many cycles while the load is still applied. The peak response approaches that of a step load.

2.2.2 Which regime this structure is in — and the calculation that settles it

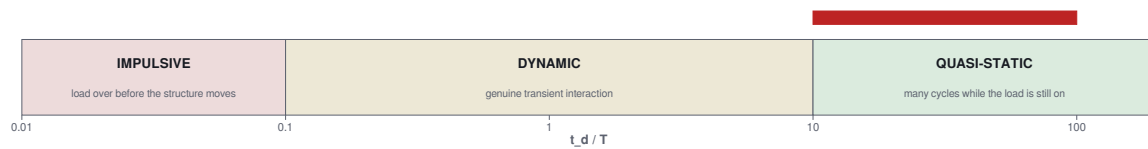
The governing element is the 900 mm pressure slab spanning 5.0 m clear. Its fundamental frequency is computed in Part 9.4 from the cracked flexural stiffness and the participating mass, including a share of the soil cover:

$m = 0.900 \times 2500 + 0.25 \times 2.0 \times 2000$	$= 3250 \text{ kg/m}^2$
$0.5 EI_g = 0.5 \times 2.958 \times 10^4 \times 0.900^3 / 12$	$= 8.985 \times 10^8 \text{ N.m}^2/\text{m}$
$f_1 = (22.373 / 2\pi) \cdot \sqrt{EI / (m \cdot L^4)}$	$= 74.9 \text{ Hz}$
$T = 1 / f_1$	$= 13.4 \text{ ms}$
$t_d / T = 130 \text{ ms} / 13.4 \text{ ms} \text{ to } 1330 \text{ ms} / 13.4 \text{ ms}$	$= 10 \text{ to } 100$

$t_d / T = 10$ to 100 places this structure firmly in the **QUASI-STATIC regime**. That is the single most important dynamic result in the project, and it is what makes a static analysis with a dynamic load factor an honest way to obtain the demand. It is not an approximation of convenience — it follows from a thick slab on a short span being extremely stiff, and a nuclear air blast being a comparatively long-duration load.

The coefficient **22.373** is the first eigenvalue constant for a uniform beam **fixed at both ends** ($\beta_L = 4.730$, $\beta_L^2 = 22.373$). Using the fixed-fixed value is consistent with the plastic mechanism adopted for strength in 2.2.4.

WHICH DYNAMIC REGIME THE STRUCTURE IS IN



Roof $T = 13.4$ ms against t_d 0.13-1.33 s. A thick slab on a short span is very stiff, and a nuclear air blast is a long-duration load. THAT is what makes a static analysis with a load factor an honest way to get the demand.

Figure 4 Impulsive, dynamic and quasi-static response - where this structure sits, and why.

2.2.3 The dynamic load factor, and the ductility it is bought with

For a perfectly elastic SDOF system under a **step load**, the classical result is a dynamic amplification of exactly **2.0**: the mass overshoots the static deflection by as much again before the restoring force brings it back.

A structure that is allowed to yield does not have to store that energy elastically. Equating the work done by the load to the energy absorbed by an elastic-plastic resistance function, and defining the **ductility ratio** μ as the ratio of maximum to yield displacement, gives the form codified in IS 4991 Cl. 10.3.3:

$$DLF = \mu / (\mu - 0.5)$$

VALIDATION $\mu = 1$ (no ductility at all) $\rightarrow DLF = 1/0.5 = 2.00$
which is exactly the elastic step-load factor. The expression reduces correctly at its own limit.

ADOPTED $\mu = 5$ "moderate, repairable damage" IS 4991 Cl. 10.3.3
 $DLF = 5 / 4.5 = 1.111$

DESIGN BLAST PRESSURE = $344.7 \times 1.111 = 383$ kPa

A ductility ratio is a promise about detailing, not a discount. Taking $\mu = 5$ asserts that the section can reach five times its yield rotation without losing capacity. That promise is kept in two places in this project and both are checkable:

First, the neutral axis depth. For the roof slab the design gives $x_u/d = 0.139$ against the Fe500 limit of 0.46 (IS 456 Cl. 38.1(f)).

A section with a neutral axis at 14 % of the effective depth is very lightly reinforced in flexural terms, the steel strain at failure is large, and the section is unambiguously under-reinforced and ductile. **That number is the proof that $\mu = 5$ is defensible**, and it is why it is printed next to the result in Part 9.4.

Second, the shear detailing. Shear failure is brittle, and a brittle failure mode makes a ductility claim meaningless. Every blast-governed element in this project carries designed shear links even where nominal steel would satisfy IS 456, and the mat's link grid was added for exactly this reason, and not because IS 456 asked for it (Part 9.3).

What is NOT demonstrated is the support rotation that $\mu = 5$ implies — $\theta \leq 2^\circ$ corresponds to $\delta \leq 87$ mm = span/57 and requires a **non-linear SDOF check** (IS 4991 Fig. 6, or Biggs). **That is Phase 3 work and it is not claimed here.** See Part 23.

THE DYNAMIC LOAD FACTOR

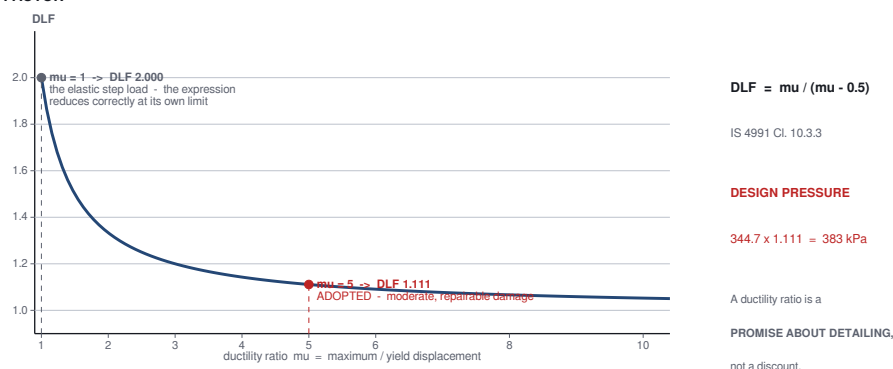


Figure 5 The dynamic load factor against ductility ratio, and the value adopted.

2.2.4 Why the plastic hinge mechanism, and why $wL^2/16$

Flexural design of the blast-governed elements uses the plastic mechanism moment for a fixed-fixed one-way strip:

$$M_p = w \cdot L_n^2 / 16$$

This is the collapse mechanism value, not the elastic value. An elastic fixed-fixed strip carries $wL^2/12$ at the supports and $wL^2/24$ at midspan. Once the support sections yield, they rotate at constant moment and the span redistributes until the midspan section also reaches its capacity; setting external work equal to internal work for the three-hinge mechanism gives a required capacity of $wL^2/16$ at every hinge, which is **less than the elastic peak of $wL^2/12$** .

This is only legitimate because the section is ductile enough to redistribute — the same promise $\mu = 5$ makes. The two are one assumption, checked once, and used consistently: the mechanism supplies the demand and $x_u/d = 0.139$ supplies the licence.

2.2.5 Strain-rate effects: why 25 % is added, and where it is not

Concrete and reinforcing steel are both **rate-sensitive**. Loaded in milliseconds rather than minutes, both show higher apparent strength — concrete because micro-crack growth is a time-dependent process that cannot keep pace, steel because dislocation motion at the yield plateau is similarly rate-limited. IS 4991 Cl. 10.3.1 codifies this as a flat 25 % increase for the blast case only:

```
BLAST CASE ONLY          IS 4991 Cl. 10.3.1
fck,dyn = 1.25 x 35 = 43.75 N/mm2
fy,dyn  = 1.25 x 500 = 625 N/mm2

NO DYNAMIC INCREASE ON SHEAR  IS 4991 Cl. 10.3.1.1
every tau_c in this project is read at the STATIC M35 value
BOND the clause permits +25 % (Ld 40phi -> 32phi). NOT TAKEN.
all detailing uses the static 40phi
```

Two deliberate refusals, and both are conservative.

Shear gets no bonus, by the code's own rule. Diagonal tension failure is brittle and aggregate-interlock dependent, and the rate sensitivity of that mechanism is not the same as that of the compression block. The consequence is real and visible: the mat's one-way shear check, run at the static τ_c for the first time in this project, is what produced the requirement for a **T12 link grid at 250 × 250 throughout the mat**, at twice the shear requirement, where nominal steel would otherwise have been accepted.

The 25 % bond bonus is available and is refused. Every development and lap length in this project is the static value. A 25 % shorter anchorage is not worth the argument at a construction joint that also has to be watertight and electrically continuous.

2.2.6 Why $\gamma = 1.0$ on the blast combination

Blast appears in combination 103 with a load factor of **1.0**, not 1.5. That is not a relaxation:

The blast case is an **extreme event** checked against **ultimate** capacity computed with **dynamic** material strengths. Applying a 1.5 partial factor to the load *while also* taking the 25 % material bonus would double-count the same conservatism from both ends. The project takes the material bonus and drops the load factor, states so, and does not mix the two.

IS 4991 Cl. 11.1 forbids combining wind or earthquake with blast, and Cl. 11.2 forbids live load on the roof at the time of blast. Both exclusions are applied (Part 7.9).

2.3 Ground shock, soil-structure interaction and arching

A buried structure and the soil around it are one system. Three effects matter, and this project's position on each is stated rather than assumed away:

Transmitted overpressure. Handled by K_a (2.1.4), taken at the saturated upper bound of 1.0.

Arching. A buried structure stiffer than the surrounding soil attracts load; one more flexible sheds it to the soil either side. Real buried boxes usually benefit from **positive arching**, which would reduce the roof pressure below the free-field value. **This project takes no arching credit at all.** That is conservative and it is declared, because the credit depends on the relative stiffness and on the compaction actually achieved, neither of which is measurable in advance.

Transient soil-structure interaction — participating mass, interface pressure ratios, the phasing between the ground shock arriving at the roof and at the walls — is **not demonstrated** and is listed in Part 23 as such. A static analysis with an SDOF-derived load factor gives the correct *demand*; it is not a transient interaction analysis and is never presented as one.

2.4 Nuclear radiation, and what 2.0 m of cover actually buys

2.4.1 Two different radiation problems

	Prompt radiation	Fallout
What it is	Gamma rays and neutrons emitted during the detonation itself	Gamma emission from activated debris deposited afterwards
When	Seconds	Hours to days
Spectrum	Hard; neutrons are the hard part to stop	Softer gamma
What stops it	Mass , and for neutrons hydrogen	Mass

2.4.2 Attenuation is exponential in mass, not in thickness

For a narrow beam through a uniform absorber, intensity falls as

```
I = I0 . exp( -mu . x )      mu = linear attenuation coefficient
                             x  = thickness

More useful in shielding practice, the same law written with MASS per unit
area, because it is areal density that does the work:

I = I0 . exp( -(mu/rho) . rho.x )

TENTH VALUE LAYER  TVL = ln(10) / mu  -- the thickness that cuts the dose
                                     by a factor of 10.
```

The practical consequence for a buried shelter is that **what protects the occupants is the mass above them, and it has to be above the slab**. The same tonnage placed against a wall does nothing for the roof.

For neutrons the mechanism is different and it matters: fast neutrons are slowed most efficiently by **elastic collision with nuclei of similar mass**, which means hydrogen. Moisture in soil and the chemically bound water in concrete are genuinely useful; dense dry rock is less effective per unit mass against neutrons than against gamma.

2.4.3 Why 2.0 m, and what each metre is for

Requirement	Does it need cover?	Verdict
Blast	No . A buried roof takes the full p_{so} regardless of depth — IS 4991 Cl. 7.2, $K_a = 1.0$	Cover buys nothing here
Fallout	1.0 m already gives a protection factor of $\approx 2\,200$ against a requirement of $\sim 1\,000$ [C]	The first metre covers it, twice over
Prompt neutron and gamma	Needs mass, above the slab	This is what the second metre is for

Stated plainly: the second metre of cover is bought entirely for prompt neutron and gamma attenuation. Reducing the cover from 4.0 m to 2.0 m in an earlier revision saved 2 m of rock excavation, 2 m of shaft, one stair flight and 2 m of headroom, and cost nothing in blast capacity because there was nothing to lose.

And the honest caveat. The prompt-radiation requirement scales with the yield, which is [N] and deliberately so (2.1.3). **If the design basis threat is ever stated and is larger, the cover depth is the dimension that changes** — along with the burster slab. The project records this rather than pretending the depth is threat-independent.

2.4.4 The burster slab

The 200 mm M30 slab inside the cover, reinforced T12 @ 150 both ways, is not there to stop anything. It is there to **break up a penetrating item** so that its energy is dispersed in the 500 mm of crushed basalt rubble below it rather than delivered to the roof as a point load.

It breaks up a penetrating item; it does not defeat one. No direct hit is designed for, and that is a stated position rather than an oversight. If a direct hit ever becomes a requirement, both the cover depth and the burster design change.

2.5 Groundwater, buoyancy and the compensated foundation

2.5.1 Archimedes, applied to a buried box

A structure below the water table displaces water and is pushed up by the weight of the water it displaces. The uplift is the water pressure at the underside integrated over the plan area:

```
u  = gamma_w x h_w = 9.81 x 4.700      = 46.11 kPa
A  = 22.000 x 6.200      = 136.40 m2
```

U	$= 46.11 \times 136.40$	$= 6289 \text{ kN}$
-----	-------------------------	---------------------

Two points are easy to get wrong and are got right here:

1. **The uplift acts on the real underside, not on a model's mid-surface area.** A plate model of the mat has its elements at mid-depth; using that area under-states the force.
2. **Spring supports in a linear analysis take tension.** In the elastic-foundation model the mat therefore **never lifts, whatever the load case**. Flotation cannot be demonstrated — or refuted — by that model. It is a hand check on real dimensions and real weights (Part 9.3).

Anyone who says "the analysis shows no uplift" has misunderstood their own model.

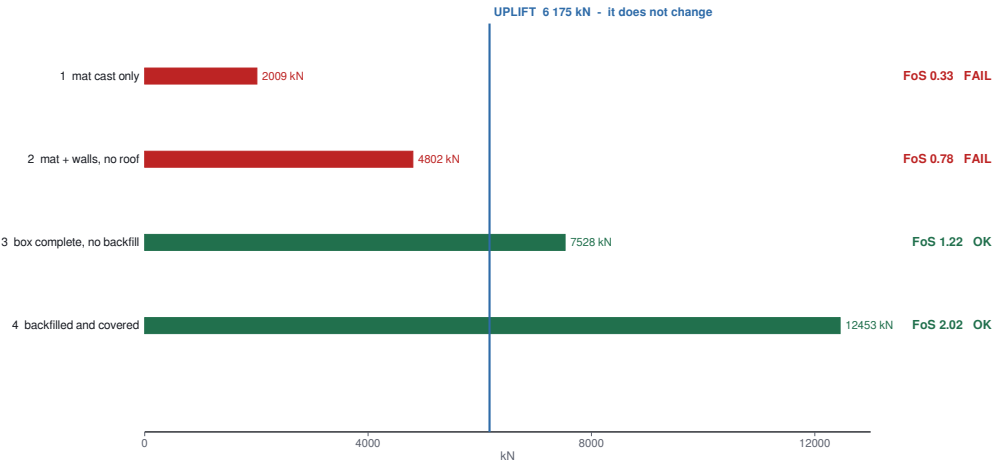
2.5.2 Why the construction stage governs

Buoyancy competes with weight, and the weight arrives late. Before the roof is on and the cover is placed, the box is a light, closed vessel in a wet hole:

Stage	Weight	Uplift	FoS at GWT (-)2.000
1 — mat cast only	2 009 kN	6 175 kN	0.33 FAIL
2 — mat + walls, no roof	4 802 kN	6 175 kN	0.78 FAIL
3 — box complete, no backfill	7 528 kN	6 175 kN	1.22 MARGINAL
4 — backfilled and covered	12 453 kN	6 175 kN	2.02 OK

(The stage figures above are tabulated on a 21.600 m box length. Both uplift and resistance scale linearly with length, so every factor of safety in the table is unchanged at the built 22.000 m; the governing hand check in Part 9.3 uses the built 136.40 m² and 6 289 kN.)

FLOTATION - THE CONSTRUCTION STAGE GOVERNS



Buoyancy competes with weight, and the weight arrives late. Require FoS >= 1.2. Figures at the pre-lengthening box; both sides scale with length, so every factor is unchanged.

Figure 6 Flotation at each construction stage. The bare box is the stage that governs.

Flotation, not bearing and not settlement, is the governing foundation problem — and it governs during construction, not in service. That is why the mitigation in Part 9.3 is a **design output** with four mandatory items, not a note passed to the contractor.

2.5.3 The compensated foundation, and why "settlement negligible" is true

The excavation removes more ground than the structure puts back:

overburden removed, 1.13 m soil at 19.5 kN/m ³	= 22.10 kPa
overburden removed, 5.67 m basalt at 25.0 kN/m ³	= 141.67 kPa
TOTAL GROSS PRESSURE REMOVED AT FORMATION	= 163.77 kPa
replaced by, in service	= 58.40 kPa
NET CHANGE IN PRESSURE AT FORMATION	= -105.37 kPa

In service the structure is lighter than the ground it replaces, by about 105 kPa. It is a fully compensated foundation and then some. **This is the physical reason "settlement negligible" is true rather than merely hoped for** — and it is the same fact, seen from the other side, that makes flotation the governing case. One physical observation explains both conclusions.

2.5.4 Effective stress, and why water is two-thirds of the wall load

Below the water table, soil and water carry the lateral load separately. The soil skeleton transmits its share through grain contacts at the **submerged** unit weight; the water transmits its own hydrostatic pressure, in full, in every direction:

```
gamma' = gamma_sat - gamma_w = 21 - 9.81           = 11.19 kN/m3
K0      = 1 - sin(phi)      (Jaky), phi ~ 30 deg    = 0.50
lateral gradient = K0.gamma' + gamma_w
              = 0.50 x 11.19 + 9.81                = 15.41 kPa/m
of which SOIL = 5.60 kPa/m      WATER = 9.81 kPa/m
```

Water is 64 % of the lateral gradient and soil is 36 %. That single split is why the design groundwater table is called the most important number in the project, and why the master tags it [ASSUMED] in bold. K_0 uses **Jaky's** relation for a normally consolidated soil at rest — at-rest, not active, because a stiff buried box does not move enough to mobilise active pressure.

2.6 The modulus of subgrade reaction, and why both bounds must be run

A mat on ground is modelled as a plate on a **Winkler** foundation: a bed of independent springs of stiffness k_s (kN/m³ — pressure per unit settlement). It is a crude idealisation — real soil couples adjacent points, which springs do not — but it is the standard one, and its weakness is specific and known:

k_s does not change the total load; it changes how the load is distributed. A soft bed lets the mat dish and drives moment into the span. A stiff bed concentrates reaction under the stiff lines — walls, corners — and drives moment there instead. **The two bounds do not bracket a small error; they can put the peak moment in different places.**

	Value	Class
k_s	100 000 to 500 000 kN/m ³	[A] — run the mat model at BOTH bounds

The project has built the upper-bound model (Underground_Shelter_ks500000.std, differing from the reference model **only** in the subgrade line and the four corner $K_F Y$ values, verified by diff). **Neither bound has been run, because STAAD.Pro is not available in this environment.** k_s remains [ASSUMED] at both bounds, and a plate load test is the only thing that closes it.

2.7 CBRN: keeping the outside outside

2.7.1 Overpressure is the primary barrier

A gas-tight envelope leaks. The defence is not perfect sealing but **positive pressure**: hold the inside above ambient, and any leak flows outward.

	Value	Class
Operating overpressure	+50 to +100 Pa	[C]
Leak-test pressure	+300 Pa	[C]
Permitted leakage	≤ 0.15 vol/h at +300 Pa = 32.5 m³/h = 11 % of one train	[C] / [R]
Cascade, clean to dirty	0 → +10 → +20 → +35 → +50 Pa	[C]

The cascade maps exactly onto the three decon airlock stages, which is what makes a three-stage airlock work: at every door, air moves from cleaner to dirtier and never the reverse.

A consequence that is easy to miss: every water trap inside the envelope is a pressure boundary. A seal shallower than the overpressure blows through and the shelter vents to the drain. Part 16.3 shows the sizing — 75 mm deep seals hold **736 Pa**, which is **7.4 : 1** against the operating pressure and **2.5 : 1** against the test. **And an unused gully evaporates dry**, after which it leaks air in both directions; priming is therefore a written maintenance task, not a nicety.

2.7.2 Filtration: three different physical problems

Threat	Physical form	What removes it
Radioactive particulate, biological agents	Aerosol, ~0.1–10 μm	HEPA, EN 1822 class H14 , ≥ 99.995 % at MPPS

Threat	Physical form	What removes it
Chemical warfare agents	Vapour	Impregnated activated carbon (ASZM-TEDA)
Coarse dust, the blast pulse	—	Pre-filter; blast valves

A HEPA filter does not remove a vapour and a carbon bed does not remove a particle. They are in series because they solve different problems. HEPA performance is quoted at the **most penetrating particle size**, which is *not* the smallest: below about 0.3 μm , diffusion capture improves as particles get smaller, while above it interception and impaction take over. The minimum in between is the number the standard quotes, which is why an H14 rating is a stronger statement than "99.995 % of dust".

Carbon beds self-heat. Adsorption is exothermic and a loaded bed is a genuine fire hazard in a protected ventilation system — which is why the filter bay appears in Part 19.2's fire-load schedule rather than being treated as inert plant.

2.7.3 Blast valves: protecting the filters from the shock

The filter train cannot take 383 kPa. **Blast valves** are fast-acting closures on each envelope penetration that shut on the pressure rise itself and reopen afterwards. This project carries **five**, all confirmed on the issued services sheet, recessed to IS 4991 Cl. 6.2.1, each with a manual quarter-turn gas-tight damper inboard of it, and with a **hand crank on each filter fan** so that ventilation does not depend on electrical power at all.

2.7.4 Closed mode: the arithmetic of a sealed box

With every valve shut there is no fresh air. Two consumables then run the clock:

```

CO2 ACCUMULATION
allowable rise 0.04 % -> 1.0 %           = 0.0096 volume fraction
occupied volume, airlock shut             = 184.96 m3
9 occupants x 0.020 m3/h/person           = 0.18 m3/h of CO2
time to 1.0 % = 0.0096 x 184.96 / 0.18    = 9.9 h [C]
over 96 h closed                          = 17.28 m3 of CO2

OXYGEN
2 x 50 L at 150 bar = 15 m3 at 4.5 m3/day = 80 h [C]

SODA LIME (CO2 scrubbing)  40 kg store      = 48 h [C]
```

HV-F1 — the soda lime, not the oxygen, limits closed mode. Unscrubbed CO_2 gives 9.9 h; the soda-lime store gives 48 h; the oxygen store gives 80 h. **The binding consumable is the one to count and the one to write on the drill card.** Closed mode covers 48 of the 96 h endurance — **the other 48 h require filtration.**

And the scrubber is therefore not optional. Without it the envelope reaches 1.0 % CO_2 in 9.9 hours against a 96-hour design occupancy. That ratio is the whole argument. The recirculation loop is sized at **75 m^3/h** — the worst cell of the efficiency-against-target matrix, taken deliberately because designing to the worst cell costs almost nothing here (Part 15).

2.7.5 The decon airlock, and the manning constraint hiding in it

The three-stage airlock has to be purged between users:

```

volume      5 x (2.0 x 2.0 x 3.2)         = 64 m3
purge at the design flow 300 m3/h         = 12.8 min [R]
-> 4-5 persons per hour
```

That is an operational constraint, not a ventilation figure. Four to five people per hour is the maximum rate at which anyone can enter the shelter through a working decon procedure, and it belongs on the drill card. The master carries it as a confirmed item *requiring attention* for exactly that reason.

2.8 EMP: why concrete and rebar are not a shield

2.8.1 Where the pulse comes from

A high-altitude nuclear detonation produces no blast, no thermal pulse and no fallout at the ground — the burst is far above the sensible atmosphere. What reaches the ground is an electromagnetic field, and the mechanism that makes it is worth stating because it explains the shape of the requirement:

```

prompt gamma rays leave the burst and reach the upper atmosphere

-> COMPTON SCATTERING off air molecules ejects electrons forward
    at relativistic speed, leaving a positive space charge behind
```

-> the geomagnetic field DEFLECTS those electrons into helical paths, which turns a radial current into a TRANSVERSE one

-> a transverse current sheet radiates. The source region is a shell tens of kilometres thick and hundreds of kilometres across, so the ground sees a nearly PLANE WAVE arriving over an enormous footprint at essentially the same instant

Two consequences follow, and both are structural. The gamma flux is deposited in a few nanoseconds, so the radiated field rises in a few nanoseconds — which puts energy at frequencies up to the order of a gigahertz. And because the source is a sheet hundreds of kilometres across, **there is no "far side of the building": every face is illuminated, every cable is illuminated, and the field does not fall off with distance in any way a designer can use.**

A high-altitude detonation therefore produces three distinct components, separated in time by orders of magnitude:

Component	Time scale	What it couples to
E1	nanoseconds, very fast rise	Short conductors — cables, boxes, PCBs. The component that destroys electronics
E2	microseconds to milliseconds	Similar to lightning; conventional surge protection is relevant
E3	seconds	Long conductors — power lines, pipelines. Quasi-DC induced currents

HEMP is not a personnel hazard. The occupants are protected from blast, CBRN and fallout by the box. **EMP protection exists so that the shelter can still function afterwards.** It protects equipment. That is why the answer is an enclosure and not a room, and why nobody should expect to shelter inside it.

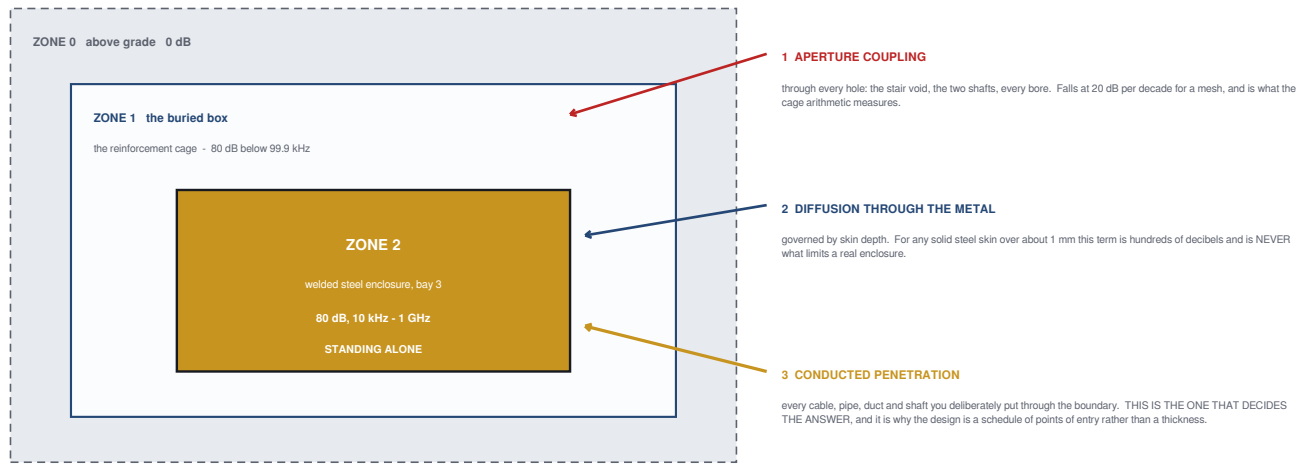
2.8.2 The three ways in, and the one that decides the answer

A shielded volume can be entered in exactly three ways, and no others:

Path	Physics	How it is beaten
Aperture coupling	Field leaks through every hole in the shield. A regular array of holes behaves as a high-pass filter	Make the holes small relative to a wavelength, or terminate them
Diffusion through the metal	Field penetrates the skin itself, attenuated by skin depth	Trivially beaten: about 1 mm of steel is 80 dB at the hardest frequency in the band
Conducted penetration	Every cable, pipe, duct and shaft you deliberately put through the boundary carries current straight across it	A treatment at the boundary for every single one — and there is no other answer

HOW A PULSE GETS IN - THREE PATHS, AND THE ONE THAT ACTUALLY MATTERS

A shield fails in three ways and only three. Two of them are properties of the metal and are easy; the third is a property of everything you deliberately run through it, and it is the one every real installation loses on.



THE ZONE 2 ENCLOSURE IS DESIGNED TO THE FULL 80 dB STANDING ALONE. NO ATTENUATION FROM THE CONCRETE BOX IS CREDITED AT ANY FREQUENCY.
That is the whole package in one line, and it is the only defensible way to use a shield you can never survey. The cage is then MARGIN, not design: it earns its keep at the bottom of the band, where E3 lives and where 99.99 dB at 10 kHz is not nothing.

AND EMP PROTECTION IS NOT A PERSONNEL MEASURE.
The occupants are protected from blast, CBRN and fallout by the box. The enclosure exists so that the shelter can still FUNCTION afterwards - it protects equipment. That is why the answer is a cabinet-sized room inside a room, and why nobody should expect to shelter in it.

Figure 7 How a pulse gets in - the three coupling paths, and the zone architecture that answers them.

Diffusion is never the limit and conducted penetration almost always is. That single sentence is why an EMP design is a **schedule of points of entry** rather than a statement of a thickness, why the penetration register of Part 18.4 is the longest table in that Part, and why the fibre-optic signal route is described there as the bargain of the whole package: a dielectric that crosses the boundary is not a penetration at all.

2.8.3 Why the requirement has the shape it has

MIL-STD-188-125-1: 80 dB, 10 kHz to 1 GHz [C]

The lower end is set by E3 and the upper end by E1. E3 is a seconds-long, quasi-DC disturbance that couples into anything long — a buried cable run, a pipeline, a power feeder — so the standard has to reach down to where those behave as antennas. E1 rises in nanoseconds, which puts useful spectral content up to roughly a gigahertz, and above that there is little left to shield against. **The 80 dB is not a physical constant; it is an engineering judgement about how much margin a piece of equipment needs between the incident field and its own damage threshold, made once, in a standard, so that every installation does not have to make it again.**

This project designs to the 80 dB performance requirement and to nothing else. It holds no incident field strength, no waveform, and no E1/E2/E3 decomposition [N] — and **it needs none.** A performance requirement stated in decibels over a band can be met, and verified, from the geometry alone. That is a considerable practical advantage and it is the reason the shielding work in this project could be done at all.

2.8.4 A solid shield, and why its panel is never the problem

For a **solid** metallic shield the classical treatment is Schelkunoff's, which separates the loss into three physically distinct terms:

SE	=	A	+	R	+	B	decibels
A	ABSORPTION	the field decays exponentially inside the metal					
		$A = 8.686 t / \delta$					$\delta = \text{skin depth}$
		$\delta = 1 / \sqrt{\pi f \mu \sigma}$					
R	REFLECTION	the impedance mismatch at each metal face					

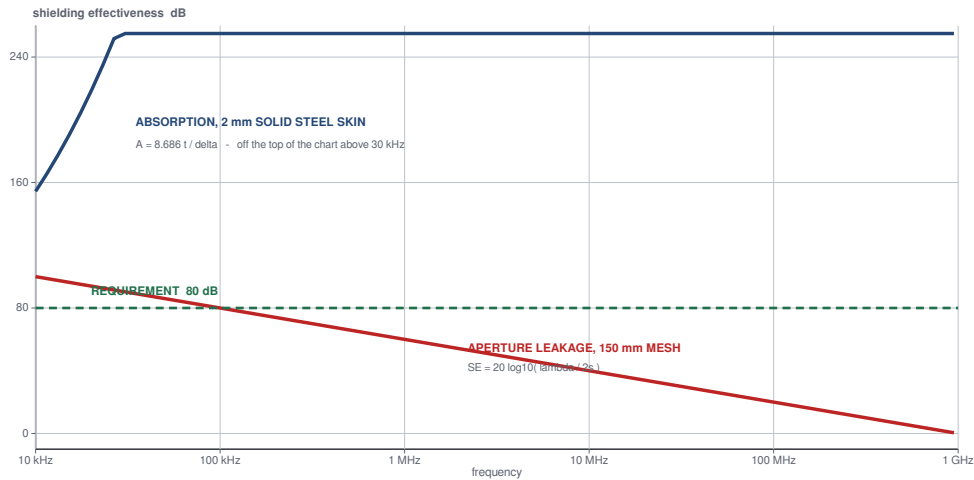
B MULTIPLE REFLECTION		$R = 168 - 10 \log_{10}(f \mu_r / \sigma_r)$ plane wave a correction, negligible once A > about 15 dB
--------------------------	--	--

For low-carbon steel — $\sigma = 1.0 \times 10^7$ S/m and $\mu_r = 200$, both [A]:

Frequency	Skin depth δ	Steel thickness giving 80 dB by absorption alone	A for a 2 mm skin
10 kHz	0.1125 mm	1.04 mm	154 dB
1 MHz	0.0113 mm	0.104 mm	1 544 dB
1 GHz	0.00036 mm	0.0033 mm	48 814 dB

Add the plane-wave reflection term — **97 dB at 10 kHz** for the same steel — and a 2 mm skin is a quarter of a *thousand* decibels at the worst frequency in the band.

ABSORPTION IN STEEL AGAINST APERTURE LEAKAGE IN A MESH - THE TWO CURVES THAT DECIDE THE ARCHITECTURE



SKIN DEPTH IN LOW-CARBON STEEL, AND THE THICKNESS THAT ALONE GIVES 80 dB

frequency	skin depth delta	80 dB needs	absorption of a 2 mm skin
10 kHz	0.1125 mm	1.0365 mm of steel	154 dB
1 MHz	0.0113 mm	0.1037 mm of steel	1544 dB
1 GHz	0.0004 mm	0.0033 mm of steel	48814 dB

$\sigma = 1.0e7$ S/m and $\mu_r = 200$ are assumed [A]. ONE MILLIMETRE OF STEEL GIVES 80 dB BY ABSORPTION ALONE AT 10 kHz, the hardest frequency in the band - so the measured performance of a welded enclosure is decided ENTIRELY by its seams, its door and its penetrations, and never by its panel.

Figure 8 Absorption in a solid steel skin against aperture leakage in a 150 mm mesh - the two curves that decide the architecture.

The numbers are absurd, and that is the finding. No real enclosure measures 250 dB. **The panel is never what limits a shielded room** — what limits it is the door, the seams, and every conductor that crosses. This is exactly why Part 18.5 treats the **24.8 m of seam** as the risk and the 25.28 m² of panel as an irrelevance, and why the acceptance test is a survey of joints rather than a measurement of steel.

And it is why a mesh is a completely different animal from a skin. A mesh has no absorption term worth the name; its performance is *entirely* aperture leakage, and aperture leakage gets worse with frequency at a fixed 20 dB per decade. **The two curves cross, and the architecture of this design is a consequence of where they cross.**

2.8.5 The aperture arithmetic that defeats the concrete

A conducting screen pierced by a regular array of apertures behaves as a high-pass filter. For apertures of pitch s the classical single-aperture estimate is

$$SE = 20 \cdot \log_{10}(\lambda / 2s)$$

$$\lambda = c / f$$

Applied to this project's reinforcement cage, whose bar spacing is **150 mm both curtains both ways** — itself an EMP requirement, stricter than IS 456 Cl. 26.3.3 needs:

Frequency	SE of the cage
10 kHz	99.99 dB

Frequency	SE of the cage
100 kHz	79.99 dB
1 MHz	59.99 dB
100 MHz	19.99 dB
1 GHz	0.00 dB
Mesh cutoff $c / 2s$	999.31 MHz
Highest frequency meeting 80 dB	99.93 kHz
Fraction of the required band met	one decade of five — 20 %

And a coincidence worth saying out loud: $2s = 300$ mm, and the wavelength at 1 GHz is 299.79 mm. The cage stops shielding at almost exactly the frequency at which the standard stops asking. That is arithmetic, not design.

So the cage is a genuine low-frequency measure and must never be described as a MIL-STD-188-125-1 boundary. It earns its keep where it can: E3, the slow component that drives long conductors, lives at the bottom of the band where the cage is strongest, and 99.99 dB at 10 kHz is not nothing.

Three further reasons the cage figure is an upper bound, and every one of them is real:

- 1. The array correction is not applied.** The classical $-10 \log_{10}(n)$ term for n identical apertures in one screen is omitted throughout. For a screen with thousands of openings that is not a small omission — Part 18.4 works it through for the honeycomb panel, where it turns a +53 dB margin into **+8.3 dB**.
- 2. The crossings are tied, not welded — and whether they are tied or welded is itself $[N]$ in this project.** A shield made of bars in electrical contact only through a wire tie has a contact resistance at every node that no aperture formula knows about, and the contact degrades with corrosion over the life of the structure.
- 3. No absorption is credited to the concrete.** That is conservative and is the right way round, but it means the quoted figures are the mesh alone.

A measured cage will therefore be worse than the calculation — possibly much worse. Which is the third independent reason the Zone 2 enclosure is designed to the full requirement standing alone.

The second, independent reason is simply the holes: a 900 mm slab with a **2 800 × 3 160 stair void** in it — 8.85 m² — is not a shield at any frequency, and neither are two 1 400 mm escape shafts.

2.8.6 Waveguide below cutoff — the one place geometry works for you

A conducting tube of diameter d will not propagate below its TE₁₁ cutoff:

```
f_c(TE11) = 1.8412 . c / (pi . d)
and BELOW cutoff the attenuation is roughly
A = 32 . L / d   dB      (L = tube length, d = diameter)
```

A penetration bounded by metal and long relative to its bore is therefore a very effective attenuator. This is why a small pipe through a thick wall is not a problem and a large one is:

Penetration	Bore	Wall	Bounded by	Cutoff	WBC	Verdict
BV-1 / BV-2	100 Ø	600	steel	1 757 MHz	192 dB	PASS
BV-3	100 Ø	400	steel	1 757 MHz	128 dB	PASS
BV-4 / BV-5	350 Ø	600	steel	502 MHz	54.8 dB	FAIL, both criteria
PD-05	50 Ø	600	steel	3 514 MHz	384 dB	PASS
ESC 1 / ESC 2	1 400 Ø	3 050 / 3 600	concrete	126 MHz	none	FAIL
Stair void	3 160	900	concrete	47.4 MHz	none	FAIL

Read the three-figure numbers correctly. No practical penetration achieves 192 or 384 dB. Real assemblies flatten out around 100–120 dB and what limits them is **workmanship** — the bond at the frame, the gasket, the one sleeve nobody welded. A large number means *"the bore is not the problem here."*

A depth credit is taken only where the bore is bounded by metal. Concrete is a lossy dielectric, not a waveguide wall. The escape shafts' 250 mm collars carry five T25 bars each side — five bars is not a conducting tube, so as built they earn nothing.

Lining a 1 400 mm shaft does not fix it: above 125.5 MHz it propagates however it is lined. The treatment that works is a **bonded conducting hatch at the head**, which *terminates* the shaft instead of trying to attenuate down it.

2.8.7 Bonding inductance — where EMP design actually lives

A bond strap is not a resistor; at EMP frequencies it is an inductor, and its impedance rises linearly with frequency:

$$L = 2e-7 \cdot l \cdot [\ln(2l / (w+t)) + 0.5 + 0.2235(w+t)/l] \quad \text{henries}$$

$$X = 2 \cdot \pi \cdot f \cdot L$$

Strap	L	X at 1 MHz	X at 10 MHz	X at 100 MHz
600 mm × 25 × 3	512.2 nH	3.22 Ω	32.2 Ω	321.8 Ω
100 mm × 50 × 3	38.9 nH	0.24 Ω	2.45 Ω	24.5 Ω

A 600 mm strap is 322 Ω at 100 MHz. That is not a bond; it is a resistor with a nice green sleeve on it. Shortening it to 100 mm and widening it to 50 mm takes the inductance from 512 nH to 39 nH — a factor of **13.2** — and it costs nothing.

The five rules that follow are the ones that get built wrong: every bond ≤ 100 mm long; width-to-length at least 5 : 1; **flat strap only** — never a round wire, never a pigtail, never "loop it round to the nearest stud"; clean bare metal both ends, protected after making off; and **the shield bonds to the structure at ONE place**, because a second bond is a loop and a loop is an antenna.

2.8.8 Earthing — and why 5 Ω is the wrong target to chase

Electrode	ρ = 1 000 Ω·m	ρ = 10 000 Ω·m
One 3 m × 16 mm rod	335 Ω	3 349 Ω
Rods needed for 5 Ω, <i>ignoring interaction</i>	67	670
The structure itself — 136.4 m ² , r _{eq} 6.589 m, p/4r	38 Ω	379 Ω

Deccan basalt is **1 000 – 10 000 Ω·m** [C].

≤ 5 Ω is not achievable with rods here, and it is not an EMP number in any case. It is a power-safety and lightning requirement from IS 3043 / IEEE 142; it is real and it still applies. **But an EMP shield works by being equipotential, and equipotential is decided by bonding inductance, not by earth resistance.** The mat and its cage are already a large concrete-encased electrode, an order of magnitude better than a rod and costing nothing because it is already built. **Bond to the structure; do not chase rods.**

2.9 Seismic response, and why a buried box is the easy case

Seismic design force follows from a design spectrum reduced by a response reduction factor:

$$A_h = (Z/2) \cdot (I/R) \cdot (S_a/g)$$

$$V_b = A_h \cdot W$$

IS 1893 (Pt 1) Cl. 6.4.2
Cl. 7.6.3

	Underground box	Sentry post
Zone factor <i>z</i> (Zone III)	0.16	0.16
Importance <i>I</i>	1.5	1.5
Response reduction <i>R</i>	4.0	3.0
<i>S_a/g</i>	2.5	2.5
<i>A_h</i>	0.075	0.100
Seismic weight <i>w</i>	14 002 kN	731.80 kN (model)
Base shear <i>V_b</i>	1 050 kN	73.18 kN

For the buried box, seismic is negligible and the calculation says so numerically, not by assertion: 525 kN per long wall gives $\tau = 0.063$ N/mm², and IS 13920 Cl. 10.4 boundary elements are checked and **not triggered**. A buried box moves with the ground; there is no tall, flexible mass to excite.

For the sentry post seismic governs over wind by 2.4 : 1 (73.18 kN against 29.9 kN), and the choice of *R* = 3.0 is the conservative one:

R = 3.0 is taken **because the infill panels are not positively separated from the frame.** *R* = 5.0 would require a special moment-resisting frame with separated infill, which this project does not claim. The consequence is quantified: *R* = 5.0 would give *A_h* = 0.060 against 0.100, so the design base shear of **73.18 kN is about 1.67 × what a separated SMRF would need.** *R* = 3.0 is kept on exactly that basis: it is paid for once, in steel, and it removes an argument about infill separation that this frame could not win.

2.10 Fire in a volume that cannot be ventilated

Every conventional fire strategy assumes smoke can be got out. **Here it cannot.**

Internal volume	332.8 m ³ gross
Design air supply	300 m ³ /h
Air changes	≈ 0.9 per hour
Smoke extract	NONE — and there cannot be one. Every opening in the envelope is a blast valve or a blast door; a smoke vent would be a hole in the protective boundary

In **Mode 3 CLOSED** the shelter has **zero** air exchange, and opening a valve to clear smoke breaks the protection the closed mode exists to provide.

So the strategy is not the usual one. It is: prevent, detect early, extinguish while it is small — and if it cannot be extinguished, LEAVE, because you cannot wait it out. Part 19 sets out the routes and the decision rule, and records the findings a fire plan in this situation is obliged to raise.

2.11 The thermal problem of a sealed box

Nine people, lighting, plant and a dehumidifier in a sealed 332.8 m³ volume put heat in and have nowhere to put it. The 96-hour balance is bounded three ways, because no single bound is honest on its own:

```

electrical load dissipated INSIDE the envelope      = 5.733 kW
9 occupants at 70 W sensible                       = 0.630 kW
TOTAL SENSIBLE GAIN                                = 6.363 kW
over 96 h = 6.363 x 345600 s                        = 2.199 GJ

BOUND 1  adiabatic, AIR ONLY: 216.96 m3 x 1.2 x 1.005 = 261.7 kJ/K
          dT = 8404 K -- PHYSICALLY MEANINGLESS.
          THE AIR HOLDS NOTHING; THE STRUCTURE IS THE ENTIRE STORY.
BOUND 2  air + the concrete the heat can REACH in 96 h
          thermal penetration sqrt(alpha.t) = sqrt(5.5e-7 x 345600) = 0.436 m
          414.38 m2 of wetted concrete surface, responding depth 0.30 m [A]
          124.3 m3 x 2500 x 0.88                = 273 493 kJ/K
          dT                                     = 8.03 K
BOUND 3  the air-to-surface film, h = 3.0 W/m2K [A]
          dT_film = 6363 / (3.0 x 414.38)        = 5.12 K

AIR TEMPERATURE RISE OVER 96 h = 8.03 + 5.12      = 13.2 K
from a ground temperature of 26 degC at 6 m depth [A]
-> about 39 degC by hour 96, AND STILL RISING, because nothing has
    reached equilibrium

COOLING DUTY TO HOLD 30 degC
the structure can absorb 273 493 x 4 K             = 1.094 GJ
heat to be rejected 2.199 - 1.094                 = 1.105 GJ
mean sensible duty 1 105 079 / 345 600            = 3.20 kW
plus latent                                         = 0.405 kW
TOTAL MEAN DUTY 3.60 kW = 1.02 TR
ADOPT 4 kW (1.14 TR) as the closed-mode cooling duty [R]

```

The rejection path adopted is **the ventilation air in open mode only, adding no new penetration** — and the arithmetic of that path is itself the finding:

```

ventilation 300 m3/h -> 0.1005 kW/K -> 63.3 K needed to reject 6.363 kW
at a realistic 5 K difference, 300 m3/h -> 0.50 kW = 7.9 % of the gain

```

300 m³/h is a CONTAMINANT rate, not a heat-rejection rate, and it is about an order of magnitude short. The second train is standby, not simultaneous, so 600 m³/h is not a case the design contemplates. **And in Pune it can be worse than nothing:** whenever ambient exceeds the internal temperature, ventilating *adds* sensible heat. **No ambient design temperature exists anywhere in this project** [N], so the number of hours cannot be stated — only the direction.

The design therefore accepts the ≈ 39 °C condition rather than claiming to solve it. The structure and the surrounding rock carry essentially the whole load in both modes, and the open item is stated in exactly those terms: *the rejection path is the ground through the structure, and it has not been modelled.* A transient soil–structure **thermal** model is the missing piece — the sibling of the transient soil–structure **interaction** already deferred to Phase 3 (RC4–V1, Part 24).

2.12 What the science section does not supply

Not supplied	Why, and where it is recorded
Weapon yield	[N], and it stays unstated — the overpressure and duration are the design inputs
Negative-phase and rebound load case	Not designed as a separate case; handled physically at the hatches. Part 23
Support rotation / non-linear SDOF	Phase 3. Part 23
Shock propagation down the entry shaft, and the resulting door loading	Needs CFD or a shock tube. Vendor / specialist. Part 23
Transient soil–structure interaction	Not demonstrated. Part 23
Transient soil–structure thermal model	RC4–V1, open (Part 24)
Ambient design temperature	[N] — nowhere in the project
Incident EMP field, waveform, E1/E2/E3 decomposition	[N]. The EMP work is designed to the 80 dB performance requirement , which needs none of them
Concrete permittivity / conductivity; whether rebar crossings are tied or welded	[N] — and the second materially changes 2.8.2

PART 3 — CODES, STANDARDS AND REFERENCES

3.1 The governing caveat, stated before anything else

IS 4991:1968 Cl. 1.1 expressly EXCLUDES nuclear explosions from its scope.

It is used in this project **for its blast loading rules and its dynamic material strengths only**, as a documented and conservative extrapolation. **Every structural capacity in this project is computed to IS 456:2000**, with IS 13920:2016 for the ductile detailing of the sentry post frame.

A conventional static STAAD.Pro analysis gives the correct DEMAND. It is not, and must never be presented as, proof of blast resistance.

This caveat appears on every deliverable in the project, and it appears here first because a reader who does not have it cannot correctly weigh anything in Parts 7 to 11.

3.2 Code register — with the clauses actually relied upon

No clause number in this register was invented. Where a clause could not be confirmed from the material available it is not listed. A code named **by title only** is one the project uses for its existence, not for a quotation; **no clause is ever quoted from a document that is not in the workspace.**

Code / reference	Used for	Clauses relied upon
IS 456:2000	All reinforced concrete design	Cl. 6.2.3.1 (E _c); Cl. 13.4 (construction joints); Cl. 22.2(a) (effective span); Cl. 23.2.1 + Fig. 4 (deflection); Cl. 24.4, 24.5 + Fig. 7 (two-way slabs, load to beams); Cl. 25.1.2 (short column); Cl. 25.4 (minimum eccentricity); Cl. 26.2.1, 26.2.1.1, 26.2.5.1 (L _{cr} , laps); Cl. 26.3.3 (bar spacing); Cl. 26.4.2, 26.4.2.1, 26.4.2.2 + Table 16 (cover); Cl. 26.5.1.1, 26.5.1.2, 26.5.1.5, 26.5.1.6 (beam steel, links); Cl. 26.5.2.1, 26.5.2.2 (slab steel); Cl. 26.5.3.1, 26.5.3.2 (column steel, ties); Cl. 31.6, 31.6.1, 31.6.3.1 (punching); Cl. 32.2, 32.5(a)(b)(c) (walls); Cl. 33.1(b), 33.2 (stairs); Cl. 34.2.4.1 (footing shear); Cl. 36.4.2 (γ _m); Cl. 38.1, 38.1(c), 38.1(f) (limit state, stress block, x _{u,max}); Cl. 39.1, 39.3, 39.6 (columns, biaxial); Cl. 40.1, 40.2.1.1, 40.2.3, 40.4(a) (shear); Tables 3, 5, 16, 18, 19, 20, 26, 27, 28; Annex D (D-1.1 to D-1.8, D-2); Annex G-1.1(b), G-1.1(c)
IS 875 (Part 1):1987	Dead loads, unit weights	Table 1
IS 875 (Part 2):1987	Imposed loads	Plant / storage 5.0 kPa; office / observation post 3.0; accessible roof 1.5
IS 875 (Part 3):2015	Wind — sentry post only	Cl. 6.3, 6.3.4, 7.2, 7.2.1, 7.3.3.13, 7.4; Tables 1, 2, 4, 26; Annex A (V _b = 39 m/s)
IS 875 (Part 5):1987	Load combinations	with IS 456 Table 18
IS 1893 (Part 1):2016	Seismic	Cl. 6.3.1.2, 6.3.2.2 (combinations, directional); Cl. 6.4.2 + Fig. 2 (A _h , S _a /g); Cl. 7.2.3 (I), 7.2.6 (R), 7.3.1, 7.3.2 (seismic weight); Cl. 7.6.1, 7.6.2, 7.6.2(c), 7.6.3 (period, base shear, distribution); Cl. 7.11.1 (drift); Table 3 (Z)
IS 13920:2016	Ductile detailing — sentry post	Cl. 6.1.1, 6.1.2, 6.1.3 (beam geometry); Cl. 6.2.1(b), 6.2.2, 6.2.3, 6.2.4 (beam steel); Cl. 6.3.3, 6.3.4 , 6.3.5.1, 6.3.5.2 (capacity-design shear, hoops); Cl. 7.1, 7.2.1 (strong column-weak beam), 7.3, 7.4 (columns); Cl. 8.1, 8.2 (special confining reinforcement); Cl. 10.4 (boundary elements — checked, not triggered)
IS 3370 (Parts 1, 2):2021	Crack width, surface-zone steel	0.2 mm limit; 0.35 % surface-zone steel over a 250 mm zone
IS 1786:2008	Fe500D reinforcement	D grade — guaranteed minimum elongation
IS 1904:1986	Presumptive bearing capacity	Table 1 (hard rock, 3 240 kPa)
IS 2950 (Part 1):1981	Raft foundations	General provisions
IS 12070	Rock foundations	Cl. 6 (settlement on sound rock); Table 2 (broken bedrock, 10 kg/cm ² — the SEMT report's own basis)
IS 1498:1970	Soil classification	CH, GM, GP, SC; free-swell-index bands — the SEMT report's classification basis
IS 2720 Pts IV, VIII, X, XIII, XL	Soil testing	Classification (Pt IV); compaction, OMC-MDD (Pt VIII); UCS (Pt X); direct shear (Pt XIII); free swell (Pt XL) — the SEMT report's own test methods, cited as it cites them
IS 2720 (Part 28)	In-situ density	Backfill compaction testing, WBS A7015
IS 1121 (Part I)	Rock strength	Compressive strength of rock cores — the SEMT report's Appendix C basis. <i>(The report prints the year as "1874"; that is a typographical error, transcribed as printed, not corrected)</i>
IS 2470 (Part 1):1985	Septic tank	Cl. 6.2 (24 h detention), 6.3 (30 L/person/yr sludge), 6.5 (L:B 2–4), 6.6 (min width 750, min depth 1.0 m), 6.9 (vent); Table 1 (≤ 10 users: 1.5 × 0.75 × 1.0)

Code / reference	Used for	Clauses relied upon
IS 2470 (Part 2):1985	Soak pit / dispersion	Cl. 4 (percolation test — mandatory), Cl. 5 (dispersion trench)
IS 4991:1968	Blast loading rules and dynamic material strengths ONLY	Cl. 1.1 (EXCLUDES NUCLEAR — quoted in every deliverable) ; Cl. 6.2.1 (reflected pressure, recessing); Cl. 7.2 + Table 3 (buried roof / walls, K_a); Cl. 7.4 (drag on bermed faces); Cl. 10.3.1 (dynamic f_{ck} , f_y); Cl. 10.3.1.1 (no dynamic increase on shear; +25 % bond) ; Cl. 10.3.3 (DLF = $\mu/(\mu-0.5)$); Cl. 11.1 (no wind / EQ with blast); Cl. 11.2 (no live load on the roof at blast); Fig. 6 (SDOF — Phase 3)
IS 1077	Common burnt clay bricks	Sentry post masonry infill, 190 × 90 × 90 modular
IS 10262:2019	Concrete mix proportioning	Cl. 5.2 + Tables 1, 2 (target mean strength); Table 3 (entrapped air); Table 4 (water content, nominal maximum aggregate size); Table 5 (coarse aggregate volume, grading zone); Cl. 5.5.1 (congested / pumpable reduction); Cl. 5.6 (absolute-volume proportioning) — Part 6.5
IS 383:2016	Aggregates for concrete	Grading zone II fine aggregate, 20 mm nominal maximum coarse aggregate
IS 269:2015	Ordinary Portland cement	43 grade, the procurement basis of Part 21
IS 2645:2003	Integral waterproofing admixtures	The integral crystalline admixture, and the quality-plan hold point on it
IS 732:2019	Electrical wiring installations	Part 17
IS 3043:2018 / IEEE 142	Earthing	$\leq 5 \Omega$ target — a power-safety and lightning number, not an EMP one
IS 694:2010 · IS 1554 (Pt 1):1988	Cables	Final circuits; sub-mains and generator cabling
IS 13416	Construction-phase fire safety	In the register for exactly that
SP 16:1980	Design aids	Column interaction charts — cross-checked only ; the P-M values in Part 11.5 were computed from first principles, not read off
SP 34:1987	Detailing	Shape codes; Cl. 5.5 (opening-corner detailing)
NBC 2016 Part 4	Fire and life safety	Stair geometry, 1 100 guarding, means of escape
BS 8666	Bar shape codes	Used alongside SP 34 in every bar bending schedule
UFC 3-340-02	US criteria, cited as US criteria	§4-27 (opening trimmers); §4-30 (direct shear)
UFC 3-350-04AN	US criteria	Referenced in the Phase 1 report
MIL-STD-188-125-1	EMP	§5.4 (access), §5.5 (waveguide below cutoff), §5.7.2.1 (power PCI), §5.7.4.1 (fibre), §5.7.6 (RF); 80 dB, 10 kHz – 1 GHz
IEEE Std 299	EMP verification	Shielding effectiveness survey
FEMA 453	Shelter ventilation	0.25 cfm/ft ² ; carbon adsorber performance
EN 1822	HEPA classification	H14, $\geq 99.995\%$ at MPPS
Glasstone & Dolan	Weapons effects	Unclassified, public — blast and radiation data only
Biggs, <i>Structural Dynamics</i>	SDOF method	Referenced for the Phase 3 support-rotation check

THE CODE FRAMEWORK - WHAT EACH STANDARD IS USED FOR, AND WHAT HAPPENS WHEN TWO DISAGREE

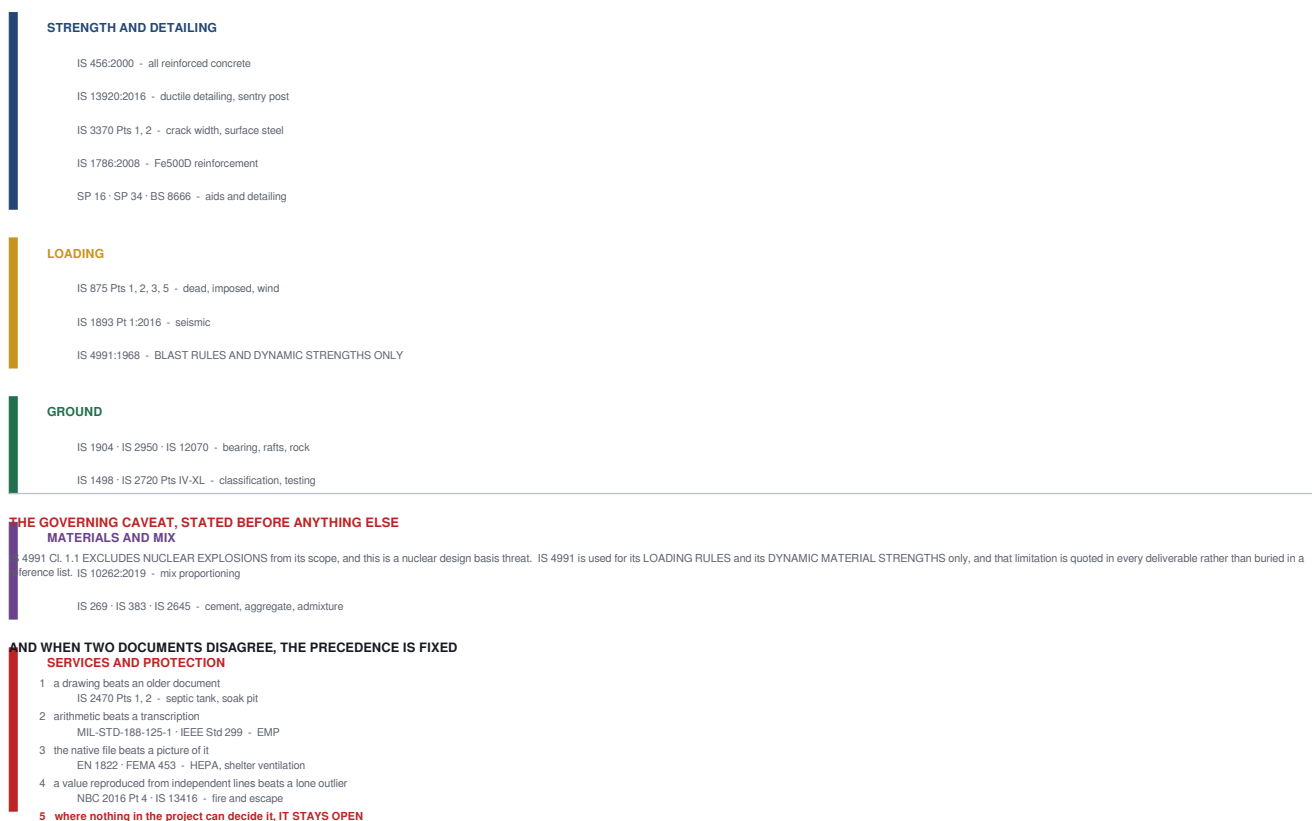


Figure 9 The code framework - what each standard is used for, and the precedence applied when two disagree.

3.3 Codes named by title only

These are named because the work touches their subject, and **no clause is quoted from any of them**, because no copy is in the workspace:

IS 1742 · IS 5329 · NBC 2016 Part 9 (drainage) · IS 3103 and the ISHRAE / ASHRAE handbooks (ventilation) · IS 3696 (Part 2) (ladders) · IS 3764 (excavation safety and slopes) · IS 2502 (bar bending — **explicitly NOT held and NOT cited**, which is why the schedules take no bend deduction and say so).

3.4 The hierarchy used to resolve a conflict

Where two project documents disagree, the design does not average them and does not pick the more convenient one. It applies a fixed precedence, and that precedence is part of the design record because it is what makes the resolutions reviewable:

1. **A drawing beats an older document.** The drawings are Rev F and the design report is Rev D; where they disagree, the drawing governs. This is the rule that settles the largest class of disagreement in the project, and it is applied without exception.
2. **Arithmetic beats a transcription.** $(-)8.000 - (-)7.600 = 0.400$, so the sump base is 400 thick and a sheet note reading "300" is a transcription error, not a second opinion.
3. **The native file beats a picture of it.** A .std file is primary; a screen capture of the same model is secondary and may be stale.
4. **A value reproduced from independent lines beats a lone outlier.** Nine statements of 300 m³/h against one of 250 — with 250 failing the sheet's own FEMA criterion — is not a genuine conflict.
5. **Where nothing in the project can decide it, it stays open.** No amount of choosing between recorded values supplies a datum, a test result or a client decision. This rule is why the register in Part 24 is as long as it is, and the register being long is the point.

PART 4 — SITE SELECTION, THE SOIL REPORT AND GEOTECHNICS

Read 4.2.2 before anything else in this Part. Everything that follows — the parameters, the bearing checks, the water, the rockhead quantity — rests on a sub-soil investigation that reached about **1.5 m** below a structure that founds at **(–)6.800**. That gap is not a criticism of the investigation, which was competent and is internally consistent. It is a statement of what this design is standing on, and it is why the confirmatory investigation sits on the critical path of the construction programme rather than being a preliminary.

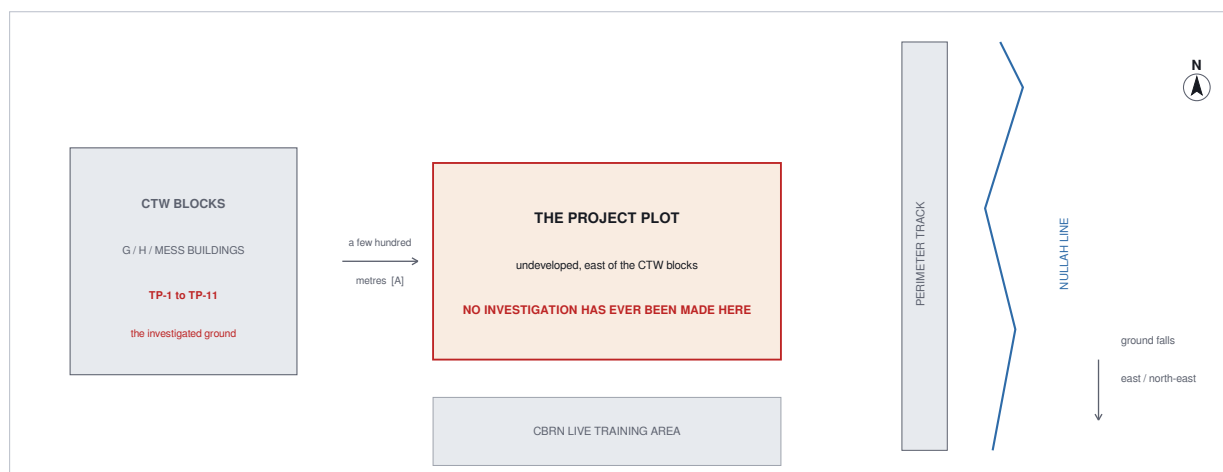
4.1 The site, and what is and is not known about it

Item	Value	Class
Installation	CTW / College of Military Engineering campus, Pune, Maharashtra	[C]
Plot	Undeveloped rectangle east of the two CTW blocks , west of the perimeter track and nullah line, CBRN Live Training Area to the south	[C]
Latitude / longitude	NOT ESTABLISHED	[N]
Plot boundary and area	NOT DIMENSIONED ANYWHERE. No scale bar, grid, boundary dimension or coordinate appears on any supplied site image	[N] — master gap D3
Site benchmark	NONE. Nothing ties any supplied level to the project datum	[N]
Project datum	finished grade = 0.000 , local	[C]
Site orientation	+X = EAST, +Y = NORTH	[A] — a project convention, not a survey

"No site plan" cannot be closed by imagery. Until a boundary, a dimension, a benchmark and a coordinate exist, the berm, the hardstanding, the access, the five external drainage runs, the concealment layout and the cut-and-fill cannot be drawn. The **setting** is recorded; the **survey** is not, and no drawing in this report or in the issued package pretends otherwise.

THERE IS NO PHOTOGRAPH, SATELLITE IMAGE, CONTOUR SHEET OR SURVEY DRAWING OF THIS PLOT IN THE PROJECT. The site figures in this Part are therefore **schematic**: they are drawn from the project's own coordinate system and from the relationships the record does state — which block is west, which training area is south, which fence is east — and they carry no dimension that the project does not hold. **A figure that looks like a site plan is not a site plan.** The one that would be is drawn after the topographic survey, and that survey is activity **A1070** in the programme.

SITE SETTING - SCHEMATIC, NOT TO SCALE



Arrangement drawn from the recorded written description. No dimensioned site plan, boundary, benchmark or coordinate exists [N].

Figure 10 The site in its setting, and the three concealment decisions that follow from it.

4.1.1 The selection, and what it got right

The site was chosen at the P1 stage. Its recorded SWOT, verbatim: **Strengths** — proximity to FCBRNP · availability of electric lines · good road connectivity · good water supply · **no pipelines.** **Weakness** — *located near perimeter fence.* **Opportunities** — training value. **Threat** — interference with the CTW schedule.

Three of those strengths are load-bearing for the design that followed, and none had been written down as such:

1. **"No pipelines."** For a 22 × 6.2 m box excavated to (–)6.800 with a continuous gas-tight, EMP-bonded envelope, a buried main crossing the footprint would have been a first-order problem: **every envelope penetration is a blast, gas and EMP discontinuity**. A clear plot is worth more here than on an ordinary building.
2. **"Availability of electric lines."** The only other record of an incoming mains supply anywhere in the project is a line in the owner's own construction schedule. This slide is earlier. It gives no capacity — EL-V4 — but it confirms the supply exists.
3. **"Good road connectivity."** The works programme carries **994 m³ of rock excavation by hydraulic breaker with no blasting** and tipper haulage throughout. That is a haul-route requirement and the site meets it.

4.1.2 What the selection did not consider

Not considered	What it costs, quantified
Ground conditions on the plot itself	The selection rests on a soil report for three <i>other</i> buildings. The confirmatory investigation A1075 (12 d, critical path) must be located on this plot — SG-V1
Depth of investigation	Nothing is known below about 1.5 m; the structure founds at (–)6.800 — SG-V2, the governing item
Ground slope	The plot is on the edge of a slope falling east. No cut-and-fill item exists in the bill because no site plan exists
"Near perimeter fence"	It is also a concealment weakness, and the distance has never been dimensioned — SG2-V2, CAM-V2

A buried box, 8 bays, roof 2.0 m below grade
 B entry headhouse, +0.900, no earth cover
 C covered entry stairwell, head +2.450
 D sentry post 4 000 x 5 000, parapet +7.000 [A]
 E septic tank ST-01, soak pits SK-01 to SK-04
 F SH-1 fresh-air intake, centred (-2 400, 3 100)
 G SH-2 generator air, X 22 598 - 23 196; Y not recorded [N]

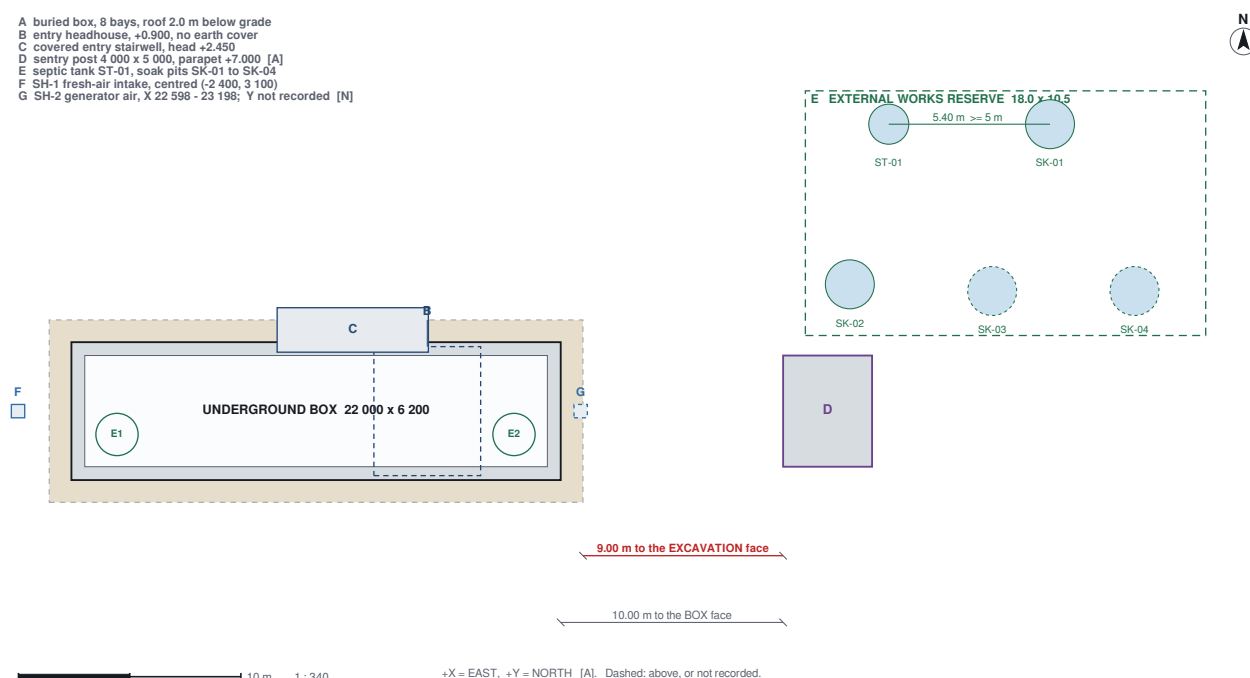


Figure 11 Key plan. Everything on the plot at true project coordinates - the buried box, the headhouse, the covered entry stairwell, the two escape shafts, the sentry post and the reserve.

4.2 The sub-soil investigation, and the one fact that governs it

#	Document	What it is	Class
1	SEMT/67/15 — <i>Report on Sub-Soil Investigation for CTW Ph-III ACCN Project at CME Pune</i> , Soil Engineering & Material Testing Wing, College of Military Engineering. Raised on CTW letter 8722/Ph-III/CTW/116/Q dated 20 May 2015	A real investigation: 11 trial pits at 3 locations , Appendices A (classification), B (soil test results), C (rock cores)	[C]
2	P1 presentation deck — <i>Construction of CBRN Hardened Underground Ops Room</i> , 32 slides	The project's own first-phase presentation: location, contour, watershed, pipelines, elevation profile, SWOT, wind, precipitation, temperature, seismic, soil	[C] as a record of what was presented

4.2.1 It is not an investigation of this plot

The SEMT report investigates **G Building (TP-1 to TP-4)**, **H Building (TP-5 to TP-8)** and the **Mess Building (TP-9 to TP-11)**. The project plot is a separate undeveloped area east of the CTW blocks. The report may be carried across only on the strength of its own paragraph 5:

"The area under investigation is underlain by the Deccan Trap formation. These are volcanic lava flows, which were poured out between the late cretaceous and early Eocene times. These basaltic flows are horizontally bedded and more or less uniform in character over a wide area."

That carry-across is an assumption [A], not a finding of the report. And the report itself shows how much variation "uniform over a wide area" permits: across three locations a few hundred metres apart, rockhead moved from **0.9 m to 1.5 m** and the surface stratum changed from a 1.0 m high-plasticity clay to a 0.2 m clayey-sand murrum.

4.2.2 The governing fact: the investigation reached 1.5 m, the structure founds at 6.8 m

	Level
Deepest stratum boundary recorded anywhere in the report	(-)1.500
Sentry footing F1, on in-situ basalt	(-)2.000
Underground box, internal floor	(-)6.100
Mat soffit	(-)6.700
FORMATION — what the mat actually bears on	(-)6.800
Sump SU-01 base	(-)8.000

The investigation reached about 1.5 m. The structure founds at 6.8 m.

There are **5.3 m of completely unlogged ground** between the bottom of the deepest trial pit and the formation the 600 mm mat bears on, and **6.5 m** to the sump base. **Only the sentry post footing F1 at (-)2.000 lies inside the investigated horizon, and only just.**

Nothing the report says about rockhead continuity, groundwater, jointing, red-bole seams or modulus of subgrade reaction can be read as covering the founding horizon of the shelter. **This is open item SG-V2, and it is the one a reviewer should look at first.**

4.2.3 The report is internally consistent, and that was checked before it was used

Chain	Check	Result
Soil	$SBC = q_{ult} / 2.5$, the report's own stated factor of safety	6 of 6 rows reproduce
Rock, step 1	$UCS = \text{max load} / \text{specimen area}$	6 of 6 rows reproduce
Rock, step 2	$SBC = UCS / 25$, rounded — the factor the report applies but never prints	6 of 6 rows reproduce

The report is internally consistent and is usable. That is worth establishing before anything is built on it, and it is not an assumption: every one of the eighteen rows above was recomputed from the report's own printed inputs, and every one reproduces.

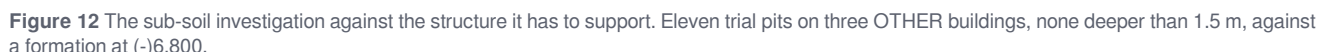
4.2.4 The trial pits and the strata, in full

Eleven pits at three locations, excavated in open trench by JCB. The report's own words for the method are *"Excavation in open trench / trial pit by JCB was carried out and 11 trial pits were made."* [C]

Location	Pits	Depth	Stratum	IS 1498	SBC kg/cm ²	SBC kPa
G Building	TP-1 to TP-4	GL – 1.0 m	High plasticity CLAY	CH	0.27	26.5
		1.0 – 1.2 m	Murrum	GM	4.27	418.7
		1.2 – 1.5 m	Broken rock	—	10.00	980.7
		below 1.5 m	Basalt	—	20.00	1 961.3
H Building	TP-5 to TP-8	GL – 0.18 m	High plasticity CLAY	CH	0.25	24.5
		0.18 – 0.7 m	Murrum	GM	4.66	457.0
		0.7 – 0.9 m	Broken rock	—	10.00	980.7
		below 0.9 m	Basalt	—	21.00	2 059.4
Mess Building	TP-9 to TP-11	GL – 0.2 m	Murrum	SC	2.07	203.0
		0.2 – 0.5 m	Murrum	GP	5.18	508.0
		0.5 – 1.0 m	Broken rock	—	10.00	980.7
		below 1.0 m	Basalt	—	21.00	2 059.4

All rows [C], read off the report. The broken-rock value is the report's own citation of **IS 12070:1987 Table 2** for broken bedrock, and its footnote is worth quoting because it describes the horizon the box is cut through for five metres:

THE SUB-SOIL INVESTIGATION AGAINST THE STRUCTURE IT HAS TO SUPPORT - 1 : 100



Six samples, classified and tested to **IS 2720** Parts IV, VIII, X, XIII and XL. The $1 - \sin \psi$ column is not in the report; it is computed here to test the project's own K_0 against measurement.

The report's own three notes on Appendix B are load-bearing and are reproduced verbatim:

Note (3) matters more than it looks. Every SBC in the table is for a **1.2 m wide footing at 1.5 m depth**. The mat of this structure is **6.2 m wide at 6.7 m depth**. The table's numbers are not transferable to it as they stand, and the design does not transfer them — it uses the **rock** values, at the horizon the mat actually bears on, and it declares the presumptive value it uses as presumptive.

4.2.6 Appendix C — the rock cores

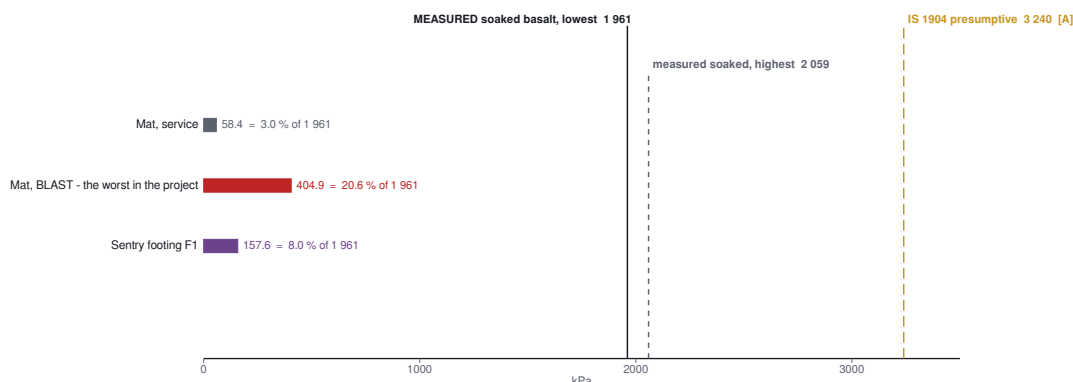
Tested to **IS 1121 (Part I)**; bearing to **IS 12070 Cl. 6**. The P/A and $UCS / 2.5$ columns are computed here; all six rows reproduce, which establishes that the bearing factor the report applies but never prints is **25**.

Condition	Location	Max load kg	Area cm ²	P/A	Printed UCS	$UCS / 2.5$	Printed SBC kg/cm ²	SBC kPa
UNSOAKED	G Building, TP-1	42 356	52.06	813.6	814	32.56	33	3 236.2
UNSOAKED	H Building, TP-8	21 924	29.15	752.1	752	30.08	30	2 942.0
UNSOAKED	Mess, TP-9	57 573	63.96	900.1	900	36.00	36	3 530.4
SOAKED	G Building, TP-1	24 563	48.30	508.6	509	20.36	20	1 961.3
SOAKED	H Building, TP-8	14 133	27.56	512.8	513	20.52	21	2 059.4
SOAKED	Mess, TP-9	27 033	51.12	528.8	529	21.16	21	2 059.4

Which column applies to this structure	the SOAKED one
Why	the design water table is (–)2.000 and the formation is (–)6.800, so the founding horizon is 4.8 m below the water table, permanently submerged
Soaked band	1 961 – 2 059 kPa
Unsoaked band	2 942 – 3 530 kPa
The presumptive value this design uses, IS 1904:1986 Table 1	3 240 kPa [A] — inside the <i>unsoaked</i> band, 65 % above the <i>soaked</i> one

Soaking costs this rock about 43 per cent of its unconfined strength, and that is exactly the sort of fact a presumptive table cannot tell you. It changes nothing in this design, because nothing in this design is close to governing on bearing — but it is the reason the soaked column is the one quoted throughout Part 9, and the reason the presumptive 3 240 is labelled [A] every time it appears rather than being quietly promoted.

BEARING - DEMAND AGAINST CAPACITY



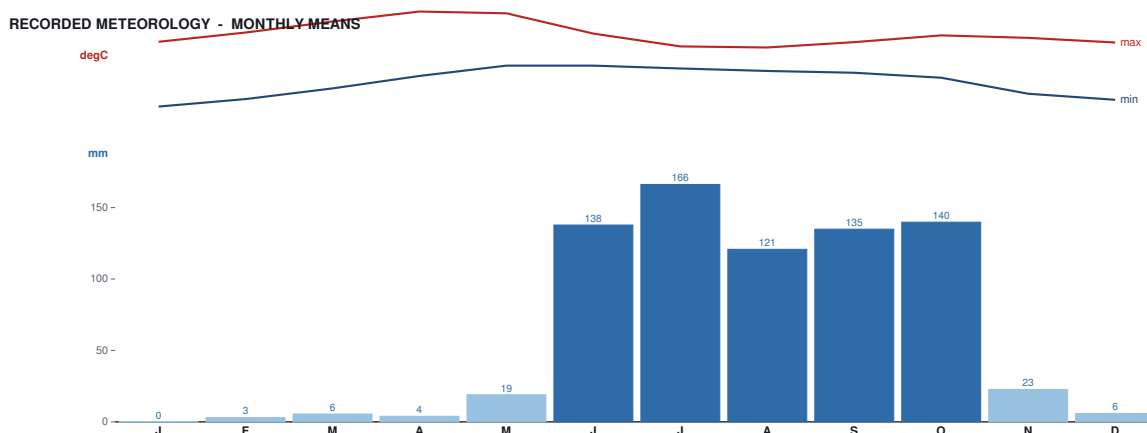
The founding horizon is 4.8 m below the design water table, so the SOAKED value is the one that applies. Worst utilisation 20.6 %, factor of 4.8 in hand.

Figure 13 Bearing demand against capacity. The worst case in the project uses 20.6 per cent of the lowest measured soaked strength.

4.2.7 The meteorological record

Monthly rows are [C], read off the first-phase presentation, each slide captioned "*last 10 years avg*". **No station, period of record or source is named on any of them.** The year row is summed here.

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	YEAR
Wind km/h	1.4	2.1	2.5	3.6	6.2	6.6	5.7	5.2	3.3	1.6	1.4	1.2	3.4
Rain mm	0.1	3.0	5.5	3.9	19.0	137.8	166.2	120.8	134.9	139.8	22.7	5.9	759.6
Max °C	29.8	32.4	35.5	38.3	37.8	32.1	28.5	28.2	29.7	31.6	30.9	29.6	38.3
Avg °C	20.7	23.2	26.0	29.3	30.6	27.6	25.4	24.8	25.4	25.6	23.1	21.7	25.3
Min °C	11.6	13.7	16.7	20.2	23.1	23.1	22.3	21.6	21.1	19.7	15.2	13.5	11.6



Rainfall total 759.6 mm against the soil report's 500-600 mm - an unresolved conflict in the supplied documents. October, at 139.8 mm, is the figure to go back and check: the monsoon has withdrawn by then.

Figure 14 The meteorological record used by the design, month by month.

Quantity	Presentation record	Soil report	Status
Annual rainfall	759.6 mm	500–600 mm, para 9	[U] CONFLICT, +27 % to +52 % . Open, and not resolved by preferring one source over the other
Temperature	monthly means; max 38.3, min 11.6	absolute extremes 40/12 summer, 26/4 winter	Not a conflict — different statistics, and mutually consistent
Wind	monthly means 1.2–6.6 km/h	—	Not a conflict . The wind design in Part 7.8 uses $v_b = 39$ m/s, a 3-second gust at a 50-year return. A monthly mean must never be used to reduce it
Seismic	Zone III, IS 1893:2016	Zone III, IS 1893 of 1984	Corroborated . $z = 0.16$ reproduces exactly

Almost nothing in this design comes from those tables, and that is worth saying plainly. The only rainfall figure the design uses is the **50 mm/h** intensity on the drainage sheet, and a monthly total cannot produce a short-duration intensity. The soak-pit sizing needs a percolation test, not a rainfall record. Where the record *does* bite is the **programme**: the excavation stands open across a monsoon, and 700-odd millimetres of it falls in four months.

And October is the figure to go back and check. At **139.8 mm** it is 1.04 times September and above August, in a month when the south-west monsoon has withdrawn from interior Maharashtra. It is **not corrected here** — correcting a datum whose source is unknown would be inventing evidence, not removing an error.

4.3 Ground parameters — the design set

Parameter	Value	Class
Ground	Deccan basalt (trap), with red-bole / vesicular seams at flow contacts	[A]
Rockhead	(–)1.500 to (–)2.000	[A]
Presumptive safe bearing capacity	3 240 kPa (IS 1904:1986 Table 1, hard rock)	[A]
Bulk unit weight γ	20 kN/m ³	[A]
Saturated unit weight γ_{sat}	21 kN/m ³	[A]
Submerged $\gamma' = 21 - 9.81$	11.19 kN/m³	[C] derived
At-rest coefficient $K_0 = 1 - \sin \phi$	0.50 ($\phi \approx 30^\circ$)	[A]
Design groundwater table	(–)2.000 (monsoon)	[A] — the single most important number to confirm
Modulus of subgrade reaction k_s	100 000 to 500 000 kN/m³	[A] — run the mat model at BOTH bounds
Settlement	Negligible on sound basalt (IS 12070)	[A]
Soil transmission factor K_a , saturated	1.0 (IS 4991 Cl. 7.2 + Table 3)	[C]
K_a , dry compacted berm fill	≈ 0.5	[A] — deliberately not relied on

The hazard is not the basalt — it is the flow contacts. A single red-bole seam under the mat produces the differential-support case that **SIZES the mat** (Part 9.3, Case 2, 84 % utilised). Horizontally bedded flows are precisely the geometry that produces flow contacts at intervals, and **the report neither finds nor excludes one, because it never reached the founding horizon.** Over-excavate any red-bole or vesicular seam at founding level and replace with M15 lean concrete.

4.3.1 What the evidence changed, and what it did not

Not one parameter above is changed by the supplied documents. What each gained is a stated provenance and a quantified margin:

Row	What the evidence says	Effect
Ground	Deccan Trap confirmed — <i>"horizontally bedded and more or less uniform"</i>	Corroborated. The flow-contact hazard is neither found nor excluded
Rockhead	0.9–1.5 m at three locations	The report's band is entirely at or above the assumed 1.5–2.0 m. Conservative for founding depth, unconservative for rock excavation quantity: +118 m³, ≈ 2 days (SG-F5)
SBC 3 240	Measured basalt 1 961–2 059 kPa soaked , 2 942–3 530 unsoaked	The founding horizon is 4.8 m below the design GWT , so soaked governs . Worst utilisation rises 12.5 % → 20.6 % ; every element still passes with a factor of 4.8 in hand. 3 240 stands — it is an IS 1904 presumptive value and is declared as one (SG-F3)
$\gamma_{20} / \gamma_{\text{sat } 21}$	95 % MDD at OMC gives 19.12–19.61 ; γ_{sat} back-figures to 21.26	Bulk 20 is 2–5 % conservative for lateral load; γ_{sat} agrees within 1.2 %
K_0 0.50	Measured ϕ 27–35° → K_0 0.426–0.546	Conservative against all three granular murrum samples. Worst case moves the wall design load under 1 % — the walls are blast-governed (SG-F9)
GWT (–)2.000	<i>"Water table was not encountered in any trial pit" — in pits ~1.5 m deep</i>	The provenance, recorded for the first time: no water was found, so 2 m was CHOSEN. <i>Not encountered here means not reached</i> (SG-F6)
k_s	Nothing — no plate load test	Untouched. Both bounds still outstanding
Settlement	Net pressure at formation ≈ –105 kPa	This is <i>why</i> "negligible" is true — and the same fact that makes flotation the governing problem (SG-F4)

4.3.2 Bearing: it does not govern anything

Check	Demand	/ 3 240 [A]	/ 1 961 soaked	Verdict
Mat, service	58.4 kPa	1.80 %	2.98 %	PASS
Mat, blast	404.9 kPa	12.50 %	20.64 %	PASS
Sentry footing F1	157.6 kPa	4.86 %	8.04 %	PASS

Even at the **broken** basalt value of 981 kPa — the horizon 5 m *above* the formation — the mat under full blast would be at **41 %**. **The bearing case is nowhere near governing anything.**

4.4 Groundwater — the central question

"Water table was not encountered in any trial pit. However, water table may raise during/after rainy season." — SEMT/67/15 para 13 [C]

"Proposed structure safe for water table at depth of 2 m below GL" — P1 deck slide 29 [C]

The provenance of (–)2.000, recorded for the first time: no water was found, so a depth of 2 m below ground level was CHOSEN, and the structure was then designed to be safe for it. That is an assumption adopted in the absence of data. It is a defensible one — it puts the table near the surface, which is the conservative direction for uplift and for lateral load. **It is not a measurement, and nothing in this project turns it into one.**

Three reasons the observation does not reach the design question:

1. **Depth.** The pits reached about 1.5 m. The design GWT is at (–)2.000 — **already below the deepest pit** — and the founding horizon is at (–)6.800. **Here, not encountered means not reached.**
2. **Season.** The report itself says the table *"may raise during/after rainy season"*. The request is dated 20 May 2015; if the field work followed promptly it was the **pre-monsoon minimum**. The report nowhere states the field-work date, so this is inference and is tagged [U], not asserted.
3. **Ground type.** Deccan Trap groundwater is not a simple water table in a porous medium. It sits in the vesicular and jointed zones **at the flow contacts** — the same feature that sizes the mat. Such water is commonly perched and strongly seasonal, and **an open trial pit in the dry season is close to the worst available instrument for finding it.**

And it is the number that matters most. Of the 15.41 kPa/m lateral gradient, **9.81 is water**. The hydrostatic uplift on the mat is **46.11 kPa = 6 289 kN over 136.4 m²**. What closes it is a **standpipe piezometer read through a full monsoon** — and the programme's own monitoring window, 12 November to 4 December, **is not in the monsoon**.

4.5 Black cotton soil — confirmed, and it reaches two elements

The surface soil is **BLACK COTTON, CH, free swell index 60–65 % — the top band of the IS 1498 scale**. [C], by measurement, from SEMT/67/15 Appendix A. It is a measured property of the ground, not an inference, and it reaches two elements of this design.

Nothing structural founds in it. The mat is at (–)6.700 and footing F1 at (–)2.000, both in basalt — and Part 11.6's insistence that F1 bears *"on in-situ rock, never on backfill"* is now backed by a measurement rather than by instinct. **Two elements are exposed anyway:**

Element	Exposure	Status
The covered entry stairwell's stepped raft	Its top founds at about (–)0.300, inside the 0.18–1.0 m CH horizon . This is heave , not bearing — and the stairwell is expendable against <i>blast</i> , which heave is not	STRIP AND REPLACE . Strip the CH horizon to 1.000 m below existing ground over 4.410 × 4.000 m and replace with granular fill at FSI ≤ 20 % (IS 2720 Pt XL) in 200 mm layers to ≥ 95 % MDD , extending 1.000 m beyond every raft edge. 8 m³ in the bill; rate [A]
The 300 mm turf concealment layer	Re-laid <i>from the site's own stockpile</i> over the 150 mm granular filter. If the site's own topsoil is this CH clay it cracks in the dry season — a concealment defect — and the wet/dry cycles pump fines into the filter, which is the one thing the filter exists to stop	OPEN — SG–V7 . The load is unaffected: 300 at 18 kN/m ³ = 5.40 kPa is the right figure for a black cotton soil either way

The two collide, and the collision is recorded rather than designed away. Strip-and-replace produces a stockpile of **exactly the material SG–V7 warns against re-laying** — and the excavation face below produces 76 to 118 m³ more of it. Somebody has to decide what happens to that spoil, and Part 22 is where the decision belongs.

4.6 The excavation face

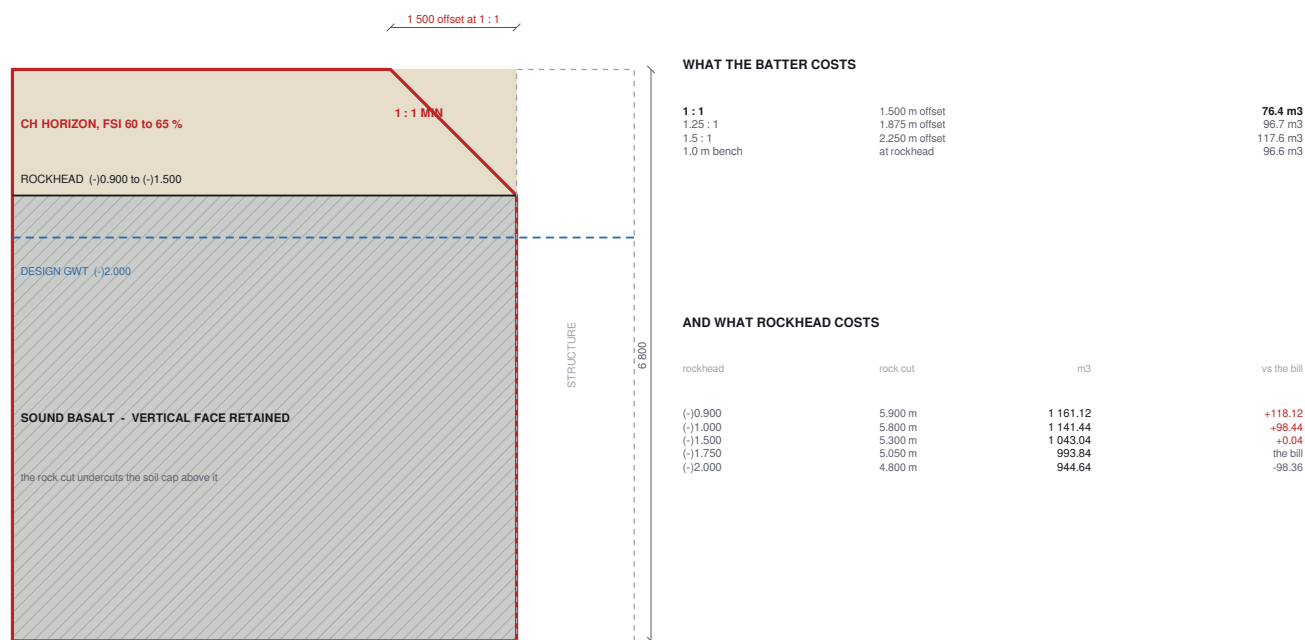
The measured basis is **1.000 m working space and vertical unbenched faces for the full 6.800 m**. In basalt that is reasonable. But rockhead is at 0.9–1.5 m with a CH horizon at FSI 60–65 % on top of it:

So the face is about **1.0–1.5 m of soil — including very-high-swelling clay — standing vertically on 5.3–5.9 m of rock, with the rock excavation undercutting it**. It also stands open **across a monsoon**, because the flotation sequence in Part 9.3 requires the box to be complete before the hole can be backfilled. **That soil cap is the one part of this excavation that has no business being vertical.**

ADOPTED	batter 1 : 1 MINIMUM over the soil cap, grade to rockhead, all four sides; VERTICAL FACE RETAINED IN ROCK below rockhead
extra excavation	1 : 1 1.500 m offset 76.4 m3
	1.25 : 1 1.875 m offset 96.7 m3
	1.5 : 1 2.250 m offset 117.6 m3
	(a 1.000 m bench at rockhead instead would be 96.6 m3)
WORKING SPACE UNCHANGED --	the 1.000 m is at FORMATION, 5.3 m below the battered zone, so no BOQ working-space quantity moves

1 : 1 is a MINIMUM, not the answer. A batter in FSI 60–65 % clay, undercut and wetted across a monsoon, may need to be considerably flatter. IS 3764 is named **by title only**. **The angle and its seasonal variation are the geotechnical engineer's, and the quantity moves with the angle** [A]. The design states a face treatment rather than an unexamined vertical, which is what makes the excavation measurable — **but the slope-stability assessment behind the angle is still outstanding, and no quantity in the bill is safe from it.**

THE EXCAVATION FACE - 1.0 TO 1.5 m OF VERY-HIGH-SWELLING CLAY STANDING ON 5.3 TO 5.9 m OF BASALT 1 : 90



1 : 1 IS A MINIMUM, NOT THE ANSWER.

A batter in clay at free-swell index 60 to 65 per cent, undercut by the rock cut below it and wetted across a monsoon, may need to be considerably flatter. The angle and its seasonal variation belong to the geotechnical engineer, and the quantity moves with the angle. The working space is unchanged - the 1.000 m is at formation, 5.3 m below the battered zone. Total excavation does not change with rockhead either: soil falls by the same volume as rock rises, so what is at stake is the RATE difference over about 118 m³, not a rock rate over 118 m³.

Figure 15 The excavation face: a very-high-swelling clay cap standing on basalt, and what the batter and the rockhead band cost.

4.7 Rockhead, and what it costs

Location	Sound basalt below	On the project datum
G Building (TP-1 to TP-4)	1.500 m	(-)1.500
H Building (TP-5 to TP-8)	0.900 m	(-)0.900
Mess Building (TP-9 to TP-11)	1.000 m	(-)1.000
Report band		(-)0.900 to (-)1.500 [C]
Assumed band		(-)1.500 to (-)2.000 [A]

The two bands touch at exactly one point, (-)1.500.

Rockhead	Depth of rock cut	Rock volume m³	vs BOQ
(-)0.900	5.900 m	1 161.12	+118.12
(-)1.000	5.800 m	1 141.44	+98.44
(-)1.500	5.300 m	1 043.04	+0.04
(-)1.750 ← BOQ E-02b	5.050 m	993.84	-49.16
(-)2.000	4.800 m	944.64	-98.36

Overrun at the report's shallow rockhead: +118.1 m³ (+11.3 %), about 2 extra days at the programme's 60 m³/day output. **Total excavation does not change** — soil falls by the same volume as rock rises — so what is at stake is the **rate difference** over about 118 m³, not a rock rate over 118 m³. **The bill is not re-quantified on this:** the bill measures to (-)1.750, the risk is declared, and the risk register carries it. What this Part supplies is the number, and the number is about two days rather than the one the register assumes.

One consequence is favourable and is recorded because it is free: the sliding argument rests on the box being "socketed ~4.8 m into basalt". On the report's band the socket is 5.3 m to 5.9 m. **The argument gets stronger.**

PART 5 — GEOMETRY AND CONFIGURATION

5.1 Coordinate system

ORIGIN	= south-west EXTERNAL corner of the underground box, at ground level
X	= EAST (along the length, 0 to 22000)
Y (plan)	= NORTH (across the width, 0 to 6200)
LEVELS	= metres relative to finished site grade 0.000, negative downwards
UNITS	= millimetres in all DXF geometry; metres in all level annotation

Site orientation +X = EAST, +Y = NORTH is a project convention, [A], not a survey. No benchmark, bearing or coordinate ties it to anything on the ground. Every drawing and every calculation in this project uses it. **The STAAD underground model does not:** it is built on mid-surface geometry with its origin at the **underside of the mat**, so project level = model Y - 6.700. Part 8.2.

5.2 Level schedule

Level	Value	Description
Grade	0.000	Finished site level, crowned, falls 1:50 away
Top of pressure slab	(-)2.000	= headhouse floor level
Roof soffit	(-)2.900	900 mm slab
Internal floor / top of mat	(-)6.100	3 200 mm clear height
Underside of mat	(-)6.700	600 mm mat
Formation / underside of PCC	(-)6.800	100 mm blinding
Design groundwater table	(-)2.000	monsoon [A]
Rockhead	(-)1.500 to (-)2.000	[A]
Stair landing L1	(-)4.7333	8 risers up from the floor
Stair landing L2	(-)3.3667	16 risers up from the floor
Headhouse roof soffit	+0.400	2 400 mm clear internally
Headhouse roof top	+0.900	no earth cover; berm graded to this level
Entry stairwell roof at head	+2.450	soffit +2.200
Sentry post GF FFL	+0.450	= base of the STAAD model
Sentry post first floor	+3.650	storey height 3 200
Sentry post roof	+6.700	storey height 3 050
Sentry post parapet top	+7.000	
Sump pit invert	(-)7.600	pit base slab (-)8.000

5.3 The main box

Item	Value	Class
External length	22 000 mm	[C]
External width	6 200 mm	[C]
Internal length	20 800 mm	[C]
Internal width — the clear span of the roof	5 000 mm — do not widen	[C]
Perimeter wall thickness	600 mm	[C]
Roof (pressure) slab	900 mm	[C]
Mat foundation	600 mm	[C]
Blinding / PCC	100 mm M15	[C]
Internal clear height	3 200 mm	[C]
Engineered cover over roof	2 000 mm , six layers	[C]

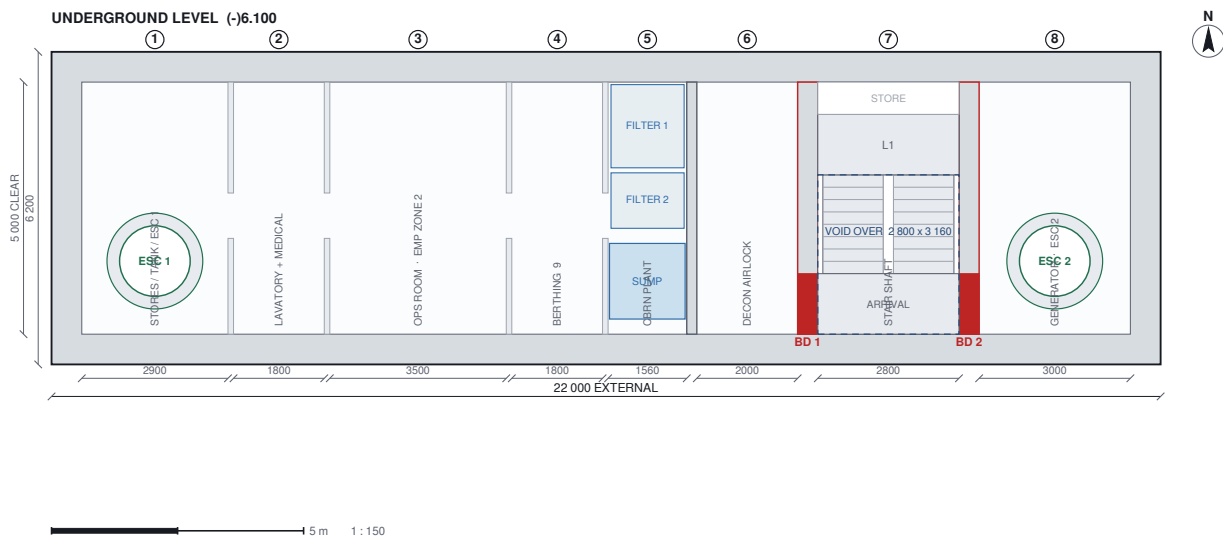


Figure 16 The underground box in plan - eight bays, the two escape shafts, the stair shaft and every internal wall.

Why length is structurally free here and width is not. The roof aspect ratio is $20.8 / 5.0 = 4.2$, and IS 456 Table 26 is tabulated only to $l_x / l_y = 2.0$, so **two-way action cannot be claimed**. The roof therefore spans one-way across the 5 m width and its moment depends on that width alone. **Adding length adds no roof moment at all; widening 5.0 → 6.0 m would add 44 %**. Every accommodation problem in this box is therefore solved along its length, and the 5 000 mm clear span is the one internal dimension that must not be touched.

LONGITUDINAL SECTION ON THE BOX CENTRELINE

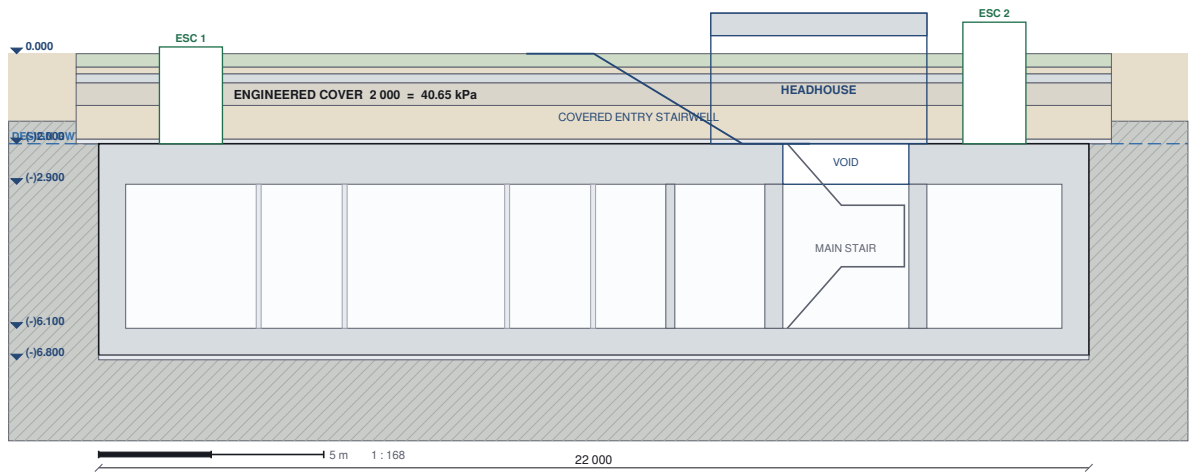


Figure 17 Longitudinal section on the spine, from the west end to the generator bay.

5.4 Bay schedule, and internal walls

Bay	X range (mm)	Clear width	Use
1	600 – 3500	2900	Emergency stores, 1 000 L potable tank, ESC 1
2	3610 – 5410	1800	Lavatory (1800×2000) + medical (1800×3000)
3	5520 – 9020	3500	Ops room and hazard plotting; EMP Zone 2 enclosure
4	9130 – 10930	1800	Berthing, 3 × 3-tier bunks, 9 berths

Bay	X range (mm)	Clear width	Use
5	11040 – 12600	1560	CBRN plant: 2 × 300 m³/h filter trains, CO ₂ /O ₂ , dehumidifier, clean sump
6	12800 – 14800	2000	Decon airlock, 3 stages (2000×2000, 2000×1500, 2000×1500)
7	15200 – 18000	2800	Stair shaft
8	18400 – 21400	3000	Generator 15 kVA (grey zone), ESC 2 at X 19 900

Mark	X range (mm)	Thickness	Notes
W8 partitions ×4	3500–3610, 5410–5520, 9020–9130, 10930–11040	110	Each with a permanent 900 mm gap at Y 2500–3400
W5	12600 – 12800	200	Bay 5/6, fire and gas-tight only , no pressure differential
W6	14800 – 15200	400	Bay 6/7 — the protective boundary . Blast Door 1 opening Y 600–1800
W7	18000 – 18400	400	Bay 7/8 — the protective boundary . Blast Door 2 opening Y 600–1800

W6 and W7 are 400 mm because the stair shaft equalises to the full incident overpressure. They are not internal partitions with a door in them; they are the last two pieces of the protective envelope, and each carries a 1 200 × 2 100 opening in it. Every other internal wall in this box is 110 mm or 200 mm, and the difference between 200 and 400 here is the difference between a wall designed for a pressure difference and a wall designed for a blast. Part 9.2.

TRANSVERSE SECTION THROUGH THE BOX

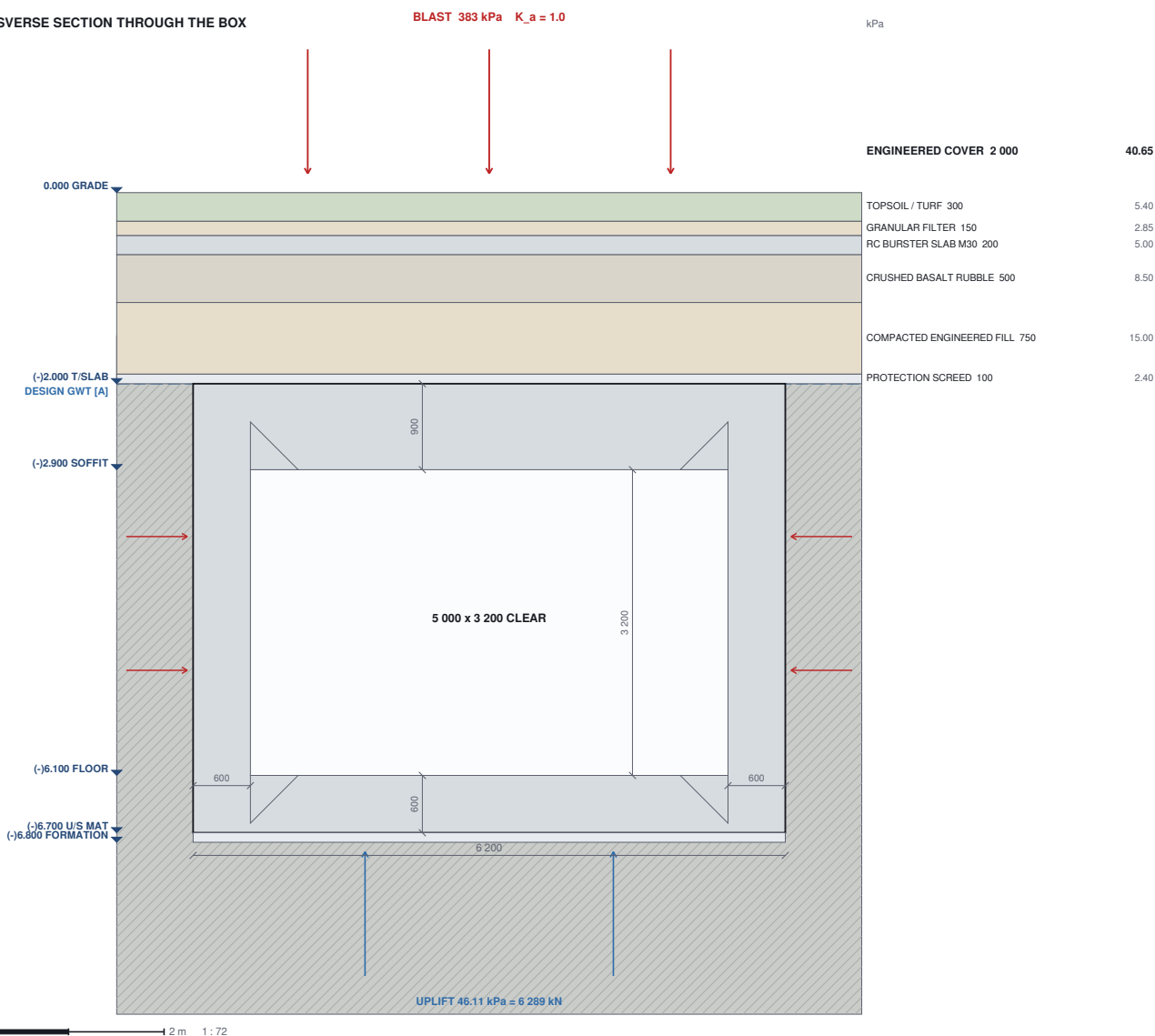


Figure 18 Transverse section through the box, with the full engineered cover build-up and the load that arrives from each direction.

5.5 Stair shaft, roof void and cantilever pad

Stair shaft (Bay 7) clear X 15200 - 18000 Y 600 - 5600 = 2800 x 5000
 Void in the pressure slab X 15200 - 18000 Y 600 - 3760 = 2800 x 3160
 Cantilever pad (remaining slab) X 15200 - 18000 Y 3760 - 5600 = 2800 x 1840
 Flight A (flights 1 and 3, stacked) X 15300 - 16500 (1200 wide)
 Well X 16500 - 16700 (200)
 Flight B (flight 2) X 16700 - 17900 (1200 wide)
 Landing L1 (-)4.7333 Y 3760 - 4960, full 2800 width
 Landing L2 (-)3.3667 Y 600 - 1800, full 2800 width (over the arrival landing)
 Arrival landing (-)6.100 = the mat surface, Y 600 - 1800
 Store under landing L1 Y 4960 - 5600
 MAIN STAIR 24R @ 170.8333, tread 280, 3 flights x 8R, rise 4100, headroom 2533

5.6 Escape shafts, and the 750 mm rule

	ESC 1	ESC 2
Centre	(2050, 2050)	(19900, 2050)
Clear opening	1 400 mm dia	1 400 mm dia
Collar	250 mm RC, OD 1 900	250 mm RC, OD 1 900
Clearance, opening edge to wall face	750 mm all round	800 / 800 mm
Head level	+0.150	+0.700
Climb from (-)6.100	6.250 m	6.800 m
Escape route	R2	R3 — shares Bay 8 with the generator

THE 750 mm CLEARANCE RULE. Opening edge to structural wall face ≥ 750 mm = 250 collar + ~400 trimmer band + 100 tolerance. **A bay must therefore be at least $1\,400 + 2 \times 750 = 2\,900$ mm wide to hold an escape shaft.** It is the reason bay 1 is 2 900 and bay 8 is 3 000, and it is the reason neither of them can be shortened to buy space anywhere else in the box.

5.7 Headhouse

External X 13600 - 18400 Y 200 - 6000 = 4800 x 5800
 Internal X 14000 - 18000 Y 600 - 5600 = 4000 x 5000
 Walls 400 thk (HW1 south, HW2 north, HW3 west, HW4 east)
 Roof 500 thk, soffit +0.400, top +0.900, NO earth cover
 Floor = the top of the 900 pressure slab, (-)2.000
 Clear internal height 2400
 Inner security door 900 x 2100 in HW2 (north), X 14450 - 15350. NOT blast rated.
 Berm graded against all four walls to +0.900 at 1.5 : 1

Wall	Runs	Length	Sits over
HW1 south	in X	4 000	Box south perimeter wall — direct
HW2 north	in X	4 000	Box north perimeter wall — direct
HW3 west	in Y	5 000	Bay 6, mid-slab. NO wall below — a line load on the pressure slab, checked at 26 % two-way / 41 % on the one-way bound (Part 10.4)
HW4 east	in Y	5 000	Wall W7 (18000-18400) — they align exactly

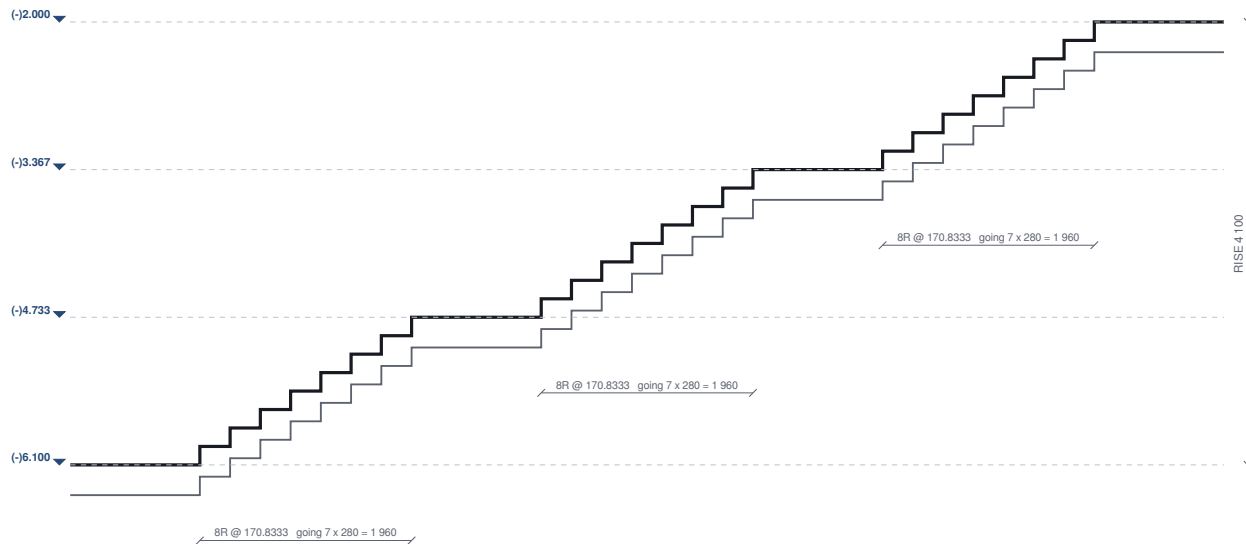
5.8 Covered entry stairwell

External X 9250 - 16050 Y 5750 - 7750 = 6800 x 2000
 Internal X 9500 - 15800 Y 6000 - 7500 = 6300 x 1500
 Walls 250 RC (both sides, headwall, east wall)
 Top landing X 9500 - 11000 at 0.000, 250 thk
 Flight X 11000 - 14300, 12R @ 166.6667, going 300, 11 x 300 = 3300, waist 250
 Platform X 14300 - 15800 at (-)2.000, 1500 x 1500, 250 thk
 Roof 250 RC raking, soffit 2200 above the flight; IT STAYS 250 OVER THE PLATFORM
 Entry door 1000 x 2100 at grade in the headwall (X 9250 - 9500), opens OUTWARD
 300 mm channel + grating, full 1500 width, at X 8950 - 9250
 Stepped RC raft 300 thk on compacted fill; MOVEMENT JOINT where it meets the headhouse
 1.0 m3 external sump at the platform, own soakaway

Berm 1.5:1 against the walls to +0.900, toe at grade over 1350

The raking roof stays 250 mm over the platform. It does not thicken to the 500 mm headhouse roof there, and the reason is geometric rather than structural: **the headhouse footprint Y 200–6000 does not reach the platform at Y 6000–7500 at all.** The structural design designs a 250 roof, the load register loads one and the reinforcement register registers one, which is the whole of the position.

MAIN STAIRCASE, BAY 7 - DEVELOPED SECTION. TWENTY-FOUR RISERS, THREE FLIGHTS OF EIGHT 1 : 70



THIS GEOMETRY IS FROZEN. 24 risers at 170.8333 mm, 280 mm tread, three flights of eight, total rise 4 100 mm, 1 200 mm wide, 200 mm well, 200 mm waist, 2 533 mm headroom.

24 x 170.8333 = 4 100.0 exactly, which is (-)6.100 to (-)2.000. 2R + T = 621.7, inside the 550 to 700 band. R + T = 450.8. R x T = 47 833 - all three of the classical checks pass.

It is also the only escape route usable by an injured person, which is why the width and the landings matter as much as the rise.

Figure 19 The main staircase, developed. Twenty-four risers, three flights of eight - frozen geometry.

5.9 Sentry post

External plan	4000 x 5000	Internal 3600 x 4600
Column grid	A-B 3650 c/c (X)	1-2 4650 c/c (Y)
Grid coordinates	A: x = 175	B: x = 3825 1: z = 175 2: z = 4825
Columns C1	350 x 350, 4 No., both storeys	
Beams B1	250 x 450, spanning A-B (3650 c/c), on grids 1 and 2	
Beams B2	250 x 450, spanning 1-2 (4650 c/c), on grids A and B	
Slab S1	150 thk two-way, 3650 x 4650 c/c, both floors	
Plinth beam PB	250 x 400 at +0.450	
Footings F1	1500 x 1500 x 600, 4 No., on in-situ basalt at (-)2.000	
Ground storey infill	190 BRICK MASONRY in the 200 structural zone	
First storey infill	armoured vision panels, 1200 wide	
Spiral stair	external, 1000 R, 250 dia central pole	
Door D1	900	
Storey heights	ground 3200 (+0.450 -> +3.650), first 3050 (+3.650 -> +6.700)	
Siting	>= 10 m clear of the shelter excavation	
Site position	X 32000 - 36000, Y 600 - 5600 [A] -- see 5.9.2	

5.9.1 The ground-storey infill is brick masonry, and what that costs

The ground-storey infill is **190 mm one-brick modular brickwork to IS 1077, laid in CM 1:6, inside a 200 mm structural zone**, with the residual 10 mm taken up at the internal face in the plaster. **190 and not 230** because 230 conventional brickwork would project 30 mm past the column faces and change the 4 000 x 5 000 external envelope. Quantity: **12.20 m³ over 64.19 m² of face**, about **6 400 bricks**.

Four consequences follow, and each is designed for or recorded rather than left implicit:

1. **Masonry over an opening needs a lintel.** A reinforced panel would not have. Designed — Part 11.7.
2. **Masonry infill in a frame needs a tie detail.** Designed — Part 11.7.
3. **The seismic weight falls.** Brick at about 20 kN/m³ over 0.190 x 2.600 gives roughly **9.88 kN/m** against the **13.000 kN/m** carried in the analysis model, so w falls, v_b falls, and **every member designed to 73.18 kN base shear is over-designed.** That is a direction, not a verification: the model is not re-run to claim the margin, and the conservative 13.000 kN/m is what the

design is held to. Open item WM-V6.

4. **The infill is not ballistic protection, and it is not claimed to be.** Masonry does not give what a reinforced panel would. **Ballistic protection is not a requirement of this post**, and that is a stated position rather than an oversight — Part 11.1 explains why the whole post is deliberately not hardened.

5.9.2 The sentry post's position — one assumed position instead of two, and it still fails the rule as written

The post sits EAST of the box, at X 32 000 – 36 000. The elevation's own note sets that range — 4 000 wide, which is the post's actual external dimension. **An elevation cannot give a Y**, so **Y 600 – 5600, centred on the box longitudinal centreline at Y 3100**, is a project convention.

And it fails its own clearance rule as that rule is written. "≥ 10 m clear of the shelter EXCAVATION": X 32 000 is **10.00 m** clear of the **box face** at X 22 000, but the **excavation** face with its 1 000 mm working space is at X 23 000 — giving **9.00 m**. The note cites the excavation while applying the box face. Satisfying it as written moves the post to X 33 000 – 37 000, which passes every check. **Both readings are recorded and neither is adopted over the other**, because the east position is a drawing convention rather than a survey coordinate: U4 stays [ASSUMED] and is closed by the topographic survey, not by argument.

And it costs the post its stated function. One of the site package's five reasons for +X = east was *"the sentry post covers the approach from the north"*. With the post 10 m beyond the **far** end of the box it covers nothing — the approach is at the **west** end. **No stated rule fails, but a reviewer will ask.**

5.10 Every opening through the protective boundary

Mark	Where	Leaf	Level	Rating	Class
Blast Door 1	W6 , X 14800–15200 (400 thk), opening Y 600–1800 — stair shaft → Bay 6	1200 × 2100	(–)6.100	≥ 7 bar, gas-tight, rebound-rated	[C] geometry · the door itself is vendor data
Blast Door 2	W7 , X 18000–18400 (400 thk), opening Y 600–1800 — stair shaft → Bay 8	1200 × 2100	(–)6.100	≥ 7 bar	[C] · vendor
Inner security door	HW2 , headhouse north wall, X 14450–15350	900 × 2100	(–)2.000	NOT blast rated	[C]
Entry door	Entry stairwell headwall, X 9250–9500, opens outward	1000 × 2100	0.000	NOT blast rated — outside the boundary, expendable	[C]
D-05, the W5 gas-tight door	W5 , X 12600–12800	—	(–)6.100	gas-tight, not blast rated	W5 is a <i>fire and gas-tight</i> separation with no pressure differential across it , and escape route R1 crosses it. The door is required and is to be designed — Part 12.4
W8 partition gaps ×4	3500–3610, 5410–5520, 9020–9130, 10930–11040	permanent 900 gap, Y 2500–3400	(–)6.100	no doors, by design	[C]
ESC 1 / ESC 2 heads	Bays 1 and 8, through the pressure slab	1 400 dia bore	+0.150 / +0.700	1.54 m² hole in the protective boundary each	See Part 10.5

Both blast doors ARE the protective boundary. They sit in the same plane as W6 and W7. The stair shaft beyond them equalises to the full incident overpressure, so the doors and the walls beside them take the same **383 kPa** — which is why those walls are 400 mm. Each frame is a **cast-in steel frame anchored into and welded to the reinforcement cage**, which is both the blast fixing and the only EMP continuity the opening has.

This table is the dimensional register only. Part 12 is where every one of these openings is designed — the loads on the leaves, the hatch that closes a 1.54 m² hole in the pressure slab, the trimmers around each void, and the four things a closure in a protective boundary has to do at the same time.

PART 6 — MATERIALS, CONCRETE MIX DESIGN AND DETAILING RULES

6.1 Concrete and reinforcement

Item	Shelter / stairs / headhouse	Sentry post	Source
Concrete	M35	M30	IS 456 Table 5 — M35 is the minimum for very severe exposure , Table 3
Free water / cement ratio	≤ 0.45	—	IS 456 Table 5
Minimum cement content	≥ 340 kg/m ³	—	IS 456 Table 5
Admixture	Integral crystalline waterproofing	—	[R]
Blinding	M15, 100 thk	—	
Burster slab (in the cover)	M30, 200 thk	—	
Reinforcement	Fe500D to IS 1786:2008	Fe500	
$E_c = 5000\sqrt{f_{ck}}$	29 580 N/mm²	27 386 N/mm²	IS 456 Cl. 6.2.3.1
Poisson's ratio	0.20	0.20	[A] — the standard value for concrete, adopted as such
Unit weight, reinforced concrete	25 kN/m ³	25 kN/m ³	IS 875 (Pt 1) Table 1
γ_m concrete / steel	1.5 / 1.15	same	IS 456 Cl. 36.4.2

Why Fe500D and not plain Fe500. The **D** designation carries a guaranteed minimum elongation. A ductility ratio of 5 is claimed in the blast case (Part 2.2.3); claiming it on steel with no guaranteed elongation would be claiming it on nothing. The sentry post, which is not blast designed, uses Fe500.

6.2 Cover — IS 456 Cl. 26.4.2 / Table 16

Face	Cover
Cast against blinding / trimmed rock	75 mm
Formed earth face	50 mm
Internal faces	40 mm
Stair shaft faces (wet / dirty zone)	30 mm
Sentry post beams and slabs	30 mm
Sentry post columns	40 mm — Cl. 26.4.2.2, never less than 40 nor the bar diameter
Footings cast against earth	50 mm — Cl. 26.4.2.1
Maximum bar spacing, both curtains	150 mm — an EMP requirement , stricter than IS 456 Cl. 26.3.3

The 5 mm cover reduction IS 456 Table 16 permits for M35 and above is deliberately not taken. This structure is permanently below the design water table on a very severe exposure; 5 mm is not worth the argument.

And the 150 mm bar spacing is an EMP decision, recorded as one. It is stricter than IS 456 needs and it is carried in the master, in the structural CAD package, in eight bar bending schedules and on the drawings. Part 2.8.5 is the arithmetic behind it, and it is the reason the cage gives 99.99 dB at 10 kHz.

6.3 Development and lap lengths — IS 456 Cl. 26.2.1

```

ld = phi . sigma_s / (4 . tau_bd),      sigma_s = 0.87 fy = 435 N/mm2
tau_bd (Cl. 26.2.1.1):  M30 -> 1.5      M35 -> 1.7
      x 1.6 for deformed bars in tension
      / 0.8 (i.e. x 1.25) for bars in compression

```

Grade	$\tau_{bd} \times 1.6$	L_d tension	L_d compression
M35	2.72	40 ϕ	32 ϕ
M30	2.40	46 ϕ	37 ϕ

Bar	L_d M35	Lap M35	L_d M30	$L_{d,c}$ M30
T8	320	400	368	296
T10	400	500	460	370
T12	480	600	552	444
T16	640	800	736	592
T20	800	1000	920	740
T25	1000	1250	1150	925

Lap policy. IS 456 Cl. 26.2.5.1(c) gives lap = L_d or 30 ϕ , whichever is greater = 40 ϕ . Cl. 26.2.5.1 requires $\times 1.4$ if more than 50 % of bars are lapped at one section. **All laps are staggered so that ≤ 50 % are spliced at any section, and the lap is specified as 50 ϕ — 25 % above requirement.**

Blast bond bonus: refused. IS 4991 Cl. 10.3.1.1 permits +25 % on bond, which would take L_d to 32 ϕ . **Not taken. All detailing uses the static 40 ϕ** (Part 2.2.5).

6.4 Waterproofing and the tank

The structure is a **tanked box, not a drained one**. That single decision sets the entire water strategy, and four rules follow from it:

1. **Nothing penetrates the roof.** No rainwater outlet, no vent stack, no soil stack through the 2 000 mm cover — that would breach both the radiation mass and the roof membrane. **There is no downpipe on the buried roof and none is added.**
2. **One services envelope crossing.** The service entry plate. The sump rising main uses it; no second drainage penetration is created.
3. **Three streams, never combined.** Clean (seepage and condensate) $\rightarrow$ storm soakaway. Foul (peacetime) $\rightarrow$ septic tank $\rightarrow$ soak pit. Decon effluent $\rightarrow$ 1 000 L tank $\rightarrow$ **tanker only**.
4. **Drainage is not in the safety path.** No drainage failure may block egress, breach the envelope or flood an entrance.

Element	Specification
Membrane	Continuous tank: on the blinding, turned up the external face, lapped to the roof membrane
Admixture	Integral crystalline waterproofing throughout
Construction joints (~6 m)	Reinforcement fully continuous, two waterstops, and a welded Cu / galvanised EMP strap
Movement joints	NONE inside the protective envelope. A movement joint is a guaranteed blast, gas and EMP discontinuity
Floor falls	Formed in the screed, never cut into the 600 mm mat — cutting them would reduce the section and the 75 mm cover to the bottom curtain
Pipe crossings	Puddle flange welded to the sleeve, membrane dressed and clamped to it

The box is never drained to relieve groundwater. A relief drain under the mat would defeat the tank and would not reduce uplift by design intent — the structure is held down by mass and by the staged construction sequence, not by pressure relief. **The only temporary relief permitted is the six construction-stage knock-out plugs, grouted after backfill** (Part 9.3).

6.5 Concrete mix design — M35 and M30 to IS 10262:2019

6.5.1 What this section is, and what it is not

The grades are confirmed: **M35** for the shelter, the stairs and the headhouse; **M30** for the sentry post and the burster slab; **M15** for the blinding. The durability limits that go with the M35 are confirmed too — free water/cement ratio **not greater than 0.45**, minimum cement content **340 kg/m³**, from IS 456 Table 5 at very severe exposure.

The mix proportions are not confirmed anywhere in this project, and nothing below turns them into confirmed values. What follows is worked through **the IS 10262:2019 Annex A procedure, in its own step order**, so that a reader can check each step against the code rather than against a result.

THE IS 10262 PROCEDURE, STEP BY STEP - AND WHERE EACH GRADE COMES OUT

step	IS 10262:2019	M35 shelter, stairs, headhouse	M30 sentry post, burster
A-1	Stipulations for proportioning	M35 · 20 mm · 100 mm slump	M30 · 20 mm · 75 mm slump
A-2	Test data for materials	SG 3.15 / 2.84 / 2.65 [A]	SG 3.15 / 2.84 / 2.65 [A]
A-3	Target mean strength	43.25 N/mm²	38.25 N/mm²
A-4	Approximate air content, Table 3	1.0 %	1.0 %
A-5	Selection of w/c, against the IS 456 cap	0.40 cap 0.45	0.44 cap 0.45
A-6	Water content, Table 4 + slump + admixture	158 L/m ³	157 L/m ³
A-7	Cement content, and the minimum check	400 kg/m ³ min 340	360 kg/m ³ min 320
A-8	Coarse aggregate volume, Table 5	0.577 after congestion	0.633
A-9	Mix calculations by absolute volume	CA 1 149.6 · FA 786.4	CA 1 287.5 · FA 696.5
A-10	MIX PROPORTIONS FOR TRIAL 1	1 : 1.966 : 2.874 w/c 0.395	1 : 1.935 : 3.576 w/c 0.436
A-11	Trial mixes, moisture correction, cubes	REQUIRED - hold point Q-06	REQUIRED - hold point Q-06

EVERY INPUT MARKED [A] IS ASSUMED BY THIS REPORT, AND NO MIX IS CONFIRMED ANYWHERE IN THE PROJECT.

Every mix design is a design against materials you have in your hand. This one is against materials nobody has yet seen: there is no aggregate source, no grading curve, no specific gravity, no water absorption, no admixture product and no supplier standard deviation in the project. Each is [N]. What A-1 to A-10 produce is a STARTING POINT FOR TRIAL BATCHING, which is exactly what the code calls it.

It lands independently on the same 400 and 360 kg/m³ the bill's procurement take-off already assumes. That is a CONSISTENCY CHECK, NOT A CONFIRMATION: both remain [A], and both are replaced the day a laboratory trial is run.

Figure 20 The IS 10262 procedure step by step, and where each grade comes out.

Every mix design is a design against materials you have in your hand. This one is against materials nobody has yet seen: there is no aggregate source, no grading curve, no specific gravity, no water absorption, no admixture product and no supplier standard deviation anywhere in the project. **Each of those is [N].** Steps A-1 to A-10 therefore produce a **starting point for trial batching**, which is exactly what the code calls it, and **A-11 is where a laboratory replaces it.** QA/QC hold point Q-06 is that moment.

6.5.2 A-1 · Stipulations for proportioning

	M35	M30
(a) Grade designation	M35 [C]	M30 [C]
(b) Type of cement	OPC 43 grade, IS 269 [C]	OPC 43 grade, IS 269 [C]
(c) Maximum nominal size of aggregate	20 mm [A]	20 mm [A]
(d) Minimum cement content	340 kg/m³ — IS 456 Table 5, very severe [C]	320 kg/m³ — IS 456 Table 5, severe [C]
(e) Maximum free water/cement ratio	0.45 — IS 456 Table 5 [C]	0.45 [C]
(f) Workability	100 mm slump [A]	75 mm slump [A]
(g) Exposure condition	Very severe — permanently below the design water table [C]	Severe — the burster slab sits inside the engineered cover [C]
(h) Method of placing	Pumped into a congested cage; two single continuous pours [C]	Ordinary RC members [C]
(i) Degree of supervision	Good — 16 hold points and 42 ITP items (Part 21.6) [C]	Good [C]
(j) Type of aggregate	Crushed angular (basalt) [A]	Crushed angular [A]
(k) Maximum cement content	450 kg/m³ — IS 456 Cl. 8.2.4.2, shrinkage [C]	450 kg/m ³ [C]
(l) Chemical admixture	Superplasticiser + integral crystalline waterproofing [C]/[N]	Superplasticiser [A]

Stipulation (h) is the one that earns its place. The pressure slab carries **T25 at 150 centres in both directions in both curtains**, plus T12 four-leg links at 250 over the end 1 500, and it is cast in **one continuous pour of 112 m³**. That is why the M35 slump is 100 mm and not 75: a stiffer mix in that cage is a honeycombing risk on the one element in the project that is 51 % utilised in flexure and **can never be inspected or repaired from the outside**.

6.5.3 A-2 · Test data for materials

	Value	Class
Cement used	OPC 43 grade to IS 269	[C]
Specific gravity of cement	3.15	[A]
Specific gravity of coarse aggregate	2.84 — basalt	[A]
Specific gravity of fine aggregate	2.65	[A]
Specific gravity of chemical admixture	1.145	[A]
Water absorption, coarse and fine aggregate	—	[N]
Free surface moisture, coarse and fine aggregate	—	[N]
Sieve analysis, coarse aggregate	Conforming to IS 383 for 20 mm graded	[A]
Sieve analysis, fine aggregate	Grading zone II, IS 383	[A]
Standard deviation s	5.0 N/mm² — IS 10262 Table 2	[A]

$s = 5.0$ is a table value standing in for a measurement. IS 10262 Table 2 is explicit that the tabulated standard deviation applies **only until enough results exist from the actual plant**, at which point the real value replaces it. No plant has been appointed on this project, so the table value is all there is. **If the appointed supplier's s is worse than 5.0, the target strength rises and the mix changes.** That is a procurement condition, not a footnote.

6.5.4 A-3 · Target strength for mix proportioning

IS 10262:2019 Cl. 5.2 takes the **greater** of two expressions:

$f'_{ck} = f_{ck} + 1.65 s$	s	from IS 10262 Table 2
$f'_{ck} = f_{ck} + X$	X	from IS 10262 Table 1
M35 $s = 5.0$ $X = 6.5$		
$35 + 1.65 \times 5.0 = 43.25$	$35 + 6.5 = 41.50$	
TARGET MEAN STRENGTH = 43.25 N/mm ²	<- GOVERNS	
M30 $s = 5.0$ $X = 6.5$		
$30 + 1.65 \times 5.0 = 38.25$	$30 + 6.5 = 36.50$	
TARGET MEAN STRENGTH = 38.25 N/mm ²	<- GOVERNS	

6.5.5 A-4 · Approximate air content

IS 10262:2019 Table 3, entrapped air for 20 mm nominal maximum size aggregate: **1.0 per cent**, for both grades. No air-entraining agent is used and none is required — this is not a freeze-thaw exposure.

6.5.6 A-5 · Selection of water/cement ratio

M35	maximum permitted, IS 456 Table 5, very severe	= 0.45	
	ADOPTED	= 0.40	[A]
M30	maximum permitted, IS 456 Table 5, severe	= 0.45	
	ADOPTED	= 0.44	[A]

Why the M35 goes to 0.40 when the code cap is 0.45. Three reasons, and none of them is conservatism for its own sake:

- The structure is permanently below the design water table.** Very severe exposure with a permanent head on the outside face is the case IS 456 Table 5 sets 0.45 for, not a case with margin to spare.
- The waterproofing is an integral crystalline admixture, and crystalline systems work by growing in a dense, low-permeability matrix.** A wetter mix is a worse host for the chemistry the tanking depends on.
- The 5 mm cover reduction IS 456 Table 16 permits for M35 and above is deliberately refused** (Part 6.2). Refusing the cover bonus and taking the w/c margin is **one consistent position** about a structure whose external face can never be inspected again.

6.5.7 A-6 · Selection of water content

IS 10262:2019 Table 4 gives the maximum water content for a **50 mm slump** with angular coarse aggregate: **186 L/m³ at 20 mm nominal maximum size**. The code's correction is **+3 per cent for each 25 mm of slump above 50**.

M35	target slump 100 mm	->	+3 % x (100-50)/25	= +6 %
	186 x 1.06			= 197.16 L/m ³
	superplasticiser, water reduction 20 %		[A]	
	197.16 x 0.80			= 157.73
	ADOPTED			= 158 L/m ³
M30	target slump 75 mm	->	+3 % x (75-50)/25	= +3 %
	186 x 1.03			= 191.58 L/m ³
	superplasticiser, water reduction 18 %		[A]	
	191.58 x 0.82			= 157.10
	ADOPTED			= 157 L/m ³

A 20 per cent water reduction is inside the ordinary range for a polycarbonate-ether superplasticiser and it is still an assumption. No product is specified anywhere in the project, so the reduction is [A] and the trial mix is what confirms it. **If the admixture delivers less, the water rises and the cement rises with it to hold the water/cement ratio — the ratio is the constraint, not the cement content.**

6.5.8 A-7 · Calculation of cement content

M35	cement = water / (w/c) = 158 / 0.40		= 395.0 kg/m ³
	ADOPTED		= 400 kg/m ³
	resulting free w/c = 158 / 400		= 0.395
	CHECK 400 >= 340 minimum	IS 456 Table 5	PASS
	400 <= 450 maximum	IS 456 Cl 8.2.4.2	PASS
	0.395 <= 0.45 cap	IS 456 Table 5	PASS, 12 % inside
M30	cement = 157 / 0.44		= 356.8 kg/m ³
	ADOPTED		= 360 kg/m ³
	resulting free w/c = 157 / 360		= 0.436
	CHECK 360 >= 320 minimum		PASS
	360 <= 450 maximum		PASS
	0.436 <= 0.45 cap		PASS

6.5.9 A-8 · Proportion of volume of coarse aggregate and fine aggregate content

IS 10262:2019 Table 5 gives the volume of coarse aggregate per unit volume of total aggregate for **20 mm nominal maximum size with grading zone II fine aggregate: 0.62 at a water/cement ratio of 0.50**, corrected **+0.01 for each 0.05 the water/cement ratio falls below 0.50**.

M35	w/c 0.395 -> 0.62 + 0.01 x (0.50 - 0.395)/0.05	= 0.641
	IS 10262 Cl. 5.5.1 -- reduce by 10 % for pumped or congested concrete, AND THIS CASE IS CONGESTED	
	0.641 x 0.90	= 0.577
	fine aggregate volume fraction 1 - 0.577	= 0.423
M30	w/c 0.436 -> 0.62 + 0.01 x (0.50 - 0.436)/0.05	= 0.633
	no congestion reduction -- ordinary RC members	= 0.633
	fine aggregate volume fraction	= 0.367

6.5.10 A-9 · Mix calculations, by absolute volume

M35, one cubic metre:

a)	volume of concrete		= 1.00000 m ³
b)	volume of cement	400 / (3.15 x 1000)	= 0.12698
c)	volume of water	158 / (1.00 x 1000)	= 0.15800
d)	volume of admixture	4.00 / (1.145 x 1000)	
	dosed at 1.0 % by mass of cement	[A]	= 0.00349
e)	volume of entrapped air, 1.0 %		= 0.01000

f)	volume of all-in aggregate	= a - (b+c+d+e)	
		= 1 - 0.29847	= 0.70153 m ³
g)	mass of coarse aggregate	= f x 0.577 x 2.84 x 1000	= 1149.6 kg
h)	mass of fine aggregate	= f x 0.423 x 2.65 x 1000	= 786.4 kg
	FRESH DENSITY	400 + 158 + 1149.6 + 786.4 + 4.00	= 2498.0 kg/m ³

M30, one cubic metre:

a) volume of concrete		= 1.00000 m ³
b) volume of cement	360 / (3.15 x 1000)	= 0.11429
c) volume of water	157 / (1.00 x 1000)	= 0.15700
d) volume of admixture	2.88 / (1.145 x 1000)	
	dosed at 0.8 % by mass of cement [A]	= 0.00252
e) volume of entrapped air, 1.0 %		= 0.01000

f) volume of all-in aggregate	= 1 - 0.28380	= 0.71620 m ³
g) mass of coarse aggregate	= f x 0.633 x 2.84 x 1000	= 1287.5 kg
h) mass of fine aggregate	= f x 0.367 x 2.65 x 1000	= 696.5 kg
FRESH DENSITY	360 + 157 + 1287.5 + 696.5 + 2.88	= 2503.9 kg/m ³

6.5.11 A-10 · Mix proportions for trial number 1

Per cubic metre	M35	M30
Cement, OPC 43	400 kg	360 kg
Water	158 L	157 L
Fine aggregate, zone II	786.4 kg	696.5 kg
Coarse aggregate, 20 mm graded	1 149.6 kg	1 287.5 kg
Chemical admixture	4.00 kg (1.0 %)	2.88 kg (0.8 %)
Free water/cement ratio	0.395	0.436
MIX BY MASS	1 : 1.966 : 2.874	1 : 1.935 : 3.576
Fresh density	2 498 kg/m ³	2 504 kg/m ³

CONCRETE MIX DESIGN - M35 AND M30 TO THE IS 10262 METHOD, AS TRIAL MIXES

Every input marked [A] is assumed by this report. NO MIX IS CONFIRMED ANYWHERE IN THE PROJECT, and none of this replaces a laboratory trial.

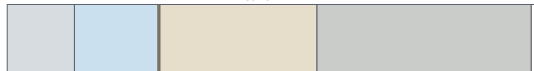
M35

shelter, stairs, headhouse - very severe, permanently below the design water table

characteristic strength f_{ck}	35 N/mm ²	[C]
standard deviation s , Table 2	5.0 N/mm ²	[C]
$f_{ck} + 1.65 s$	43.25 N/mm ²	
$f_{ck} + X$, Table 1	41.50 N/mm ²	
TARGET MEAN STRENGTH	43.25 N/mm²	
free w/c cap, IS 456 Table 5	0.45	[C]
minimum cement, IS 456 Table 5	340 kg/m ³	[C]
target slump	100 mm	[A]
water, Table 4 at 50 mm, 20 mm agg	186 L/m ³	
slump correction +3 % / 25 mm	197.2 L/m ³	
superplasticiser reduction	20 %	[A]
WATER ADOPTED	158 L/m³	
w/c adopted	0.40	[A]
cement = water / (w/c)	395.0 kg/m ³	
CEMENT ADOPTED	400 kg/m³	
resulting free w/c	0.395	

ONE CUBIC METRE BY ABSOLUTE VOLUME - total 1.0000

cement 0.1270 · water 0.158 · admixture 0.0035 · aggregate 0.7015 · air 0.010



BY MASS 1 : 1.966 : 2.874 at w/c 0.395

cement 400 · fine 786.4 · coarse 1149.6 · water 158 · admixture 4.00 kg/m³ = 2498 kg/m³

M30

sentry post and burster slab

characteristic strength f_{ck}	30 N/mm ²	[C]
standard deviation s , Table 2	5.0 N/mm ²	[C]
$f_{ck} + 1.65 s$	38.25 N/mm ²	
$f_{ck} + X$, Table 1	36.50 N/mm ²	
TARGET MEAN STRENGTH	38.25 N/mm²	
free w/c cap, IS 456 Table 5	0.45	[C]
minimum cement, IS 456 Table 5	320 kg/m ³	[C]
target slump	75 mm	[A]
water, Table 4 at 50 mm, 20 mm agg	186 L/m ³	
slump correction +3 % / 25 mm	191.6 L/m ³	
superplasticiser reduction	18 %	[A]
WATER ADOPTED	157 L/m³	
w/c adopted	0.44	[A]
cement = water / (w/c)	356.8 kg/m ³	
CEMENT ADOPTED	360 kg/m³	
resulting free w/c	0.436	

ONE CUBIC METRE BY ABSOLUTE VOLUME - total 1.0000

cement 0.1143 · water 0.157 · admixture 0.0025 · aggregate 0.7162 · air 0.010



BY MASS 1 : 1.935 : 3.576 at w/c 0.436

cement 360 · fine 696.5 · coarse 1287.5 · water 157 · admixture 2.88 kg/m³ = 2504 kg/m³

Figure 21 Concrete mix design - M35 and M30 to the IS 10262 method, as trial mixes.

The coarse aggregate as two fractions. IS 383 graded 20 mm is normally supplied and batched as 20 mm and 10 mm single sizes. At an assumed **60 : 40** split [A]:

	M35	M30
20 mm fraction	689.8 kg	772.5 kg
10 mm fraction	459.8 kg	515.0 kg

Batching by the bag, which is how it is actually issued on site:

	M35 — 8.0 bags/m ³	M30 — 7.2 bags/m ³
Cement	1 bag = 50 kg	1 bag = 50 kg
Fine aggregate	98.3 kg	96.7 kg
Coarse aggregate	143.7 kg	178.8 kg
Water	19.75 L	21.81 L
Admixture	0.500 kg	0.400 kg

6.5.12 A-11 · Trial mixes, corrections and the checks that close it

All the aggregate masses above are on a saturated surface-dry basis. On site they never are, and the correction is the step most often skipped:

WATER ACTUALLY ADDED = free water
 - free surface moisture carried by the aggregate
 + water the aggregate will absorb to reach SSD

AGGREGATE ACTUALLY BATCHED = SSD mass, adjusted for the same terms

Both correction terms are [N] in this project — no water absorption and no free-moisture figure exists for an aggregate nobody has yet sourced. **The correction is therefore stated as a procedure, not performed**, and it is performed on the day by the batching plant against the day's moisture test. **A mix batched on nominal aggregate moisture is not the mix designed above.**

The checks the design has to pass, and does:

Check	Requirement	M35	M30
Free water/cement ratio	≤ 0.45, IS 456 Table 5	0.395 PASS	0.436 PASS
Minimum cement content	≥ 340 M35 · ≥ 320 M30	400 PASS	360 PASS
Maximum cement content	≤ 450, IS 456 Cl. 8.2.4.2	400 PASS	360 PASS
Grade against exposure	M35 minimum, very severe	M35 PASS	M30, above ground
Absolute volumes close on 1.000 m ³	exact	PASS	PASS
Fresh density	plausible 2 400 – 2 550	2 498	2 504
Agreement with the procurement take-off	400 / 360 kg/m ³ in Part 21.2	exact	exact

The last row is a consistency check, not a confirmation. The bill's cement quantity is built on an assumed 400 kg/m³ for the M35 and 360 for the M30. This mix design, worked independently from the code tables, lands on the same two figures — which means **the 194.2 t of cement and 828.7 t of aggregate in Part 21.2 are internally consistent with a mix that satisfies IS 456**. Both are still assumptions, and both are replaced on the day a laboratory trial is run.

What the trial itself must establish, before the first structural pour:

1. **Workability as batched and after the transport time**, not workability at the mixer.
2. **Compressive strength at 7 and 28 days**, against the target mean strengths of A-3 — not against 35 and 30.
3. **Compatibility of the superplasticiser with the integral crystalline waterproofing admixture.** These are two chemical systems in the same mix, and their interaction affects both the water reduction claimed in A-6 and the permeability the whole tanking strategy depends on. **Both are [N], so the interaction cannot even be guessed at here.**
4. **The trial at the real placing temperature.** Part 21.4 puts the mat pour in March and the pressure-slab pour in June, in Pune. **A mix trialled at 27 °C and placed at 38 °C is a different mix**, and the one that matters is the one in the shutter.

6.5.13 What must be supplied before any of this is used

THE MIX DESIGN IS A TRIAL MIX, AND IT IS NOT A CONFIRMED PROJECT VALUE

CONFIRMED [C]

- M35 for the shelter, stairs and headhouse; M30 for the sentry post and the burster slab; M15 blinding
- free water / cement ratio not greater than 0.45
- minimum cement content 340 kg/m³ for the M35
- Fe500D to IS 1786 in the shelter, Fe500 in the sentry post
- integral crystalline waterproofing admixture throughout

ASSUMED BY THIS REPORT [A]

- target slump, and therefore the water content
- the superplasticiser and the water reduction credited to it
- specific gravities - cement 3.15, coarse 2.84, fine 2.65
- fine aggregate grading zone II, 20 mm nominal coarse aggregate
- the 10 % reduction in coarse aggregate volume for congestion

NOT AVAILABLE [N]

- the site's actual aggregate - no source, no grading, no specific gravity, no water absorption
- the admixture - no product, no dosage, no compatibility data
- trial mix cube results at 7 and 28 days
- the standard deviation of the actual supplier, which is what Table 2 is a stand-in for

THE PROJECT'S OWN PROCUREMENT TAKE-OFF ASSUMES 400 kg/m³ FOR THE M35 AND 360 kg/m³ FOR THE M30. This design lands on the same two figures, which is a consistency check and not a confirmation:

both are assumptions, and both are replaced the day a laboratory trial to IS 10262 is run on the aggregate the site will actually receive. QA/QC item Q-06 is the hold point.

Figure 22 What the mix design rests on, and what it cannot settle.

Input	Status	Who supplies it
Coarse and fine aggregate source, grading, specific gravity, water absorption	[N]	The contractor, at award
Whether the fine aggregate is genuinely grading zone II	[A]	Sieve analysis to IS 383
Superplasticiser product, dosage and water reduction	[N]	Vendor, with a compatibility trial
Integral crystalline waterproofing admixture, product and dosage	[N] — the admixture itself is [R]	Vendor, to IS 2645
Supplier's standard deviation s	[A] — table value	The plant, after 30 results
Trial mix cubes at 7 and 28 days	[N]	The laboratory, before the first structural pour

PART 7 — LOADING

7.1 Blast

DESIGN BASIS THREAT	nuclear air blast, $p_{so} = 344.7$ kPa (50 psi)	[C]
Positive phase duration	$t_d = 0.13$ to 1.33 s	[C]
Reflected pressure	$p_r = 1366$ kPa	[C]
Dynamic pressure	$q = 282$ kPa	[C]
Ductility ratio	$\mu = 5$ (moderate, repairable damage)	IS 4991 Cl. 10.3.3
DLF = $\mu / (\mu - 0.5)$	$5 / 4.5 = 1.111$	
validation: $\mu = 1 \rightarrow DLF = 2.00 =$ the elastic step-load factor		
*** DESIGN BLAST PRESSURE	$344.7 \times 1.111 = 383$ kPa ***	
applied to the ROOF and to the WALLS ($K_a = 1.0$, IS 4991 Cl. 7.2)		
Roof natural period T	13.4 ms $\rightarrow t_d/T = 10$ to $100 \rightarrow$ QUASI-STATIC	
Dynamic material strengths, BLAST CASE ONLY	IS 4991 Cl. 10.3.1	
$f_{ck,dyn} = 1.25 \times 35 = 43.75$ N/mm ²	$f_{y,dyn} = 1.25 \times 500 = 625$ N/mm ²	
NO dynamic increase on shear	IS 4991 Cl. 10.3.1.1	

7.2 Gravity, soil and water

Load	Value	Applied to	Source
Self weight, reinforced concrete	25 kN/m ³	all	IS 875 (Pt 1) Table 1
Roof self, 0.900×25	22.50 kPa	roof	
Mat self, 0.600×25	15.00 kPa	mat	
Engineered cover, 2 000 layered	40.65 kPa	roof	[C] — Part 7.3
Superimposed dead (services, finishes)	2.0 kPa roof / 1.0 kPa mat		[R]
Live load, internal floor	5.0 kPa — plant and storage, not 2.0 residential	floor	IS 875 (Pt 2)
Earth + water lateral gradient	$K_0 \gamma' + \gamma_w = 0.50 \times 11.19 + 9.81 = \mathbf{15.41}$ kPa/m	walls	[C] derived
— at roof soffit (–)2.900	33.9 kPa	walls	
— at floor (–)6.100	83.2 kPa	walls	
Hydrostatic uplift on the mat	$4.700 \times 9.81 = \mathbf{46.11}$ kPa; total 6 289 kN over 136.4 m ²	mat	[C]
Construction surcharge	20 kPa vertical / 10 kPa lateral		[A]
Staircase, smeared on the two shaft walls	2.947 kPa	W6 / W7	[C]

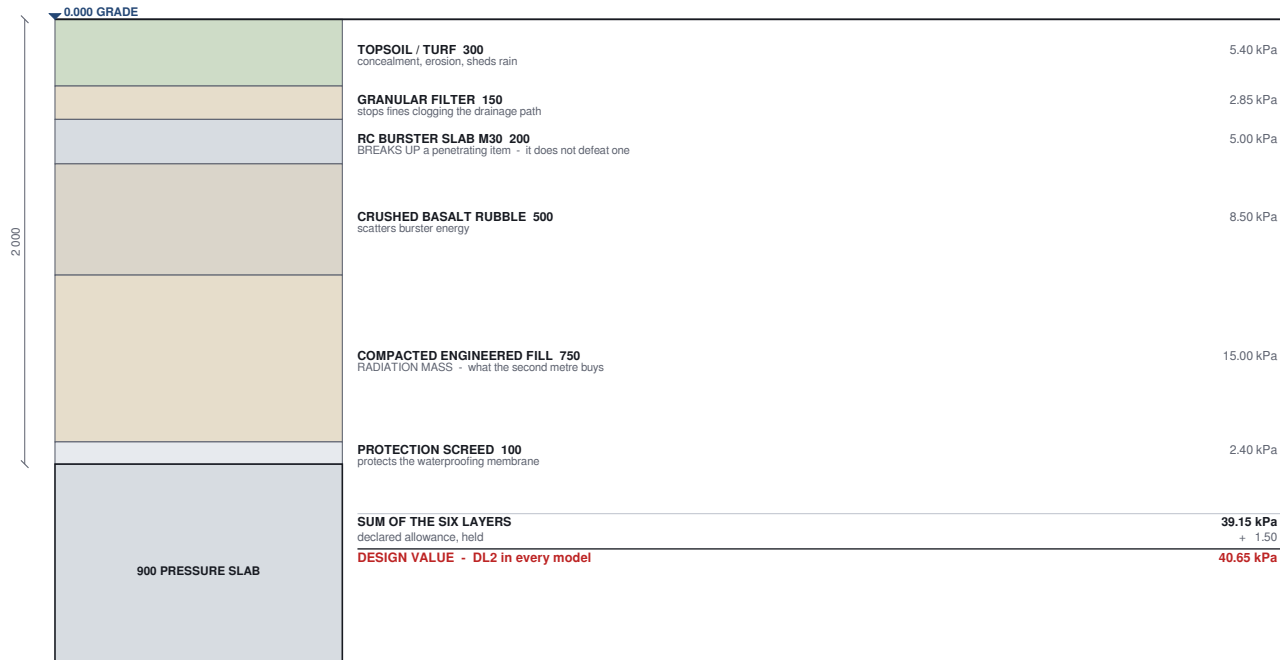
Of the 15.41 kPa/m lateral gradient, **9.81 is water and only 5.60 is soil**. Water is nearly two-thirds of the lateral load. **This is why the groundwater level is the most important number the site investigation must confirm.**

7.3 The engineered cover — 2 000 mm, 40.65 kPa

From the top	Thickness	Unit weight	Load kPa	Function
Topsoil / turf	300	18	5.40	Concealment , erosion, sheds rain
Granular filter	150	19	2.85	Stops fines clogging
RC burster slab M30, T12 @ 150 B/W	200	25	5.00	Breaks up a penetrating item
Crushed basalt rubble 25–75 mm	500	17	8.50	Scatters burster energy
Compacted engineered fill @ 95 % MDD	750	20	15.00	Radiation mass
Protection screed over the membrane	100	24	2.40	Protects the waterproofing
Sum of the six layers	2 000		39.15	
Declared allowance, held	—		+1.50	
TOTAL — DESIGN VALUE	2 000		40.65	

The design value is 40.65 kPa, and the table shows exactly where it comes from. The six layers sum to 39.15 kPa. The design carries 40.65, which is the layer sum plus a declared allowance of 1.50 kPa for the tolerance on six compacted layers laid over a 136 m² roof. The allowance is stated rather than absorbed, because a 1.5 kPa difference between a layer table and a load case is exactly the sort of thing that later reads as an error. 40.65 is the number in the roof total, in the quantity register and in DL2 in every underground .std file, and it is the larger of the two.

THE ENGINEERED COVER - 2 000 mm IN SIX LAYERS



The cover buys NO reduction in blast pressure: K_a = 1.0 saturated, so a buried roof takes the full p_so.
1.0 m already covers fallout at PF 2 200. THE SECOND METRE IS BOUGHT ENTIRELY FOR PROMPT NEUTRON AND GAMMA ATTENUATION.

Figure 23 The engineered cover, layer by layer: 2 000 mm and 40.65 kPa.

7.3.1 The burster slab is laid to falls

The 200 mm burster slab is **not laid flat**. It is laid to a **1:50 crossfall, crowned on the box longitudinal centreline (Y = 3100) and falling each way to the box edges** — that is, **parallel to the finished grade**, which is itself crowned and falls 1:50 away. Over the 3 100 mm half-width the slab drops **62 mm** from crown to box edge.

Why. The granular filter sits directly on the burster slab, and the drainage basis already states that water infiltrating the topsoil is *"intercepted by the granular filter and dispersed at the berm toe"*. **On a flat slab it cannot be** — it ponds on the slab and finds the construction joints. The crossfall gives the filter layer the gradient that claim depends on. **No pipe is introduced anywhere in the cover.**

Thicknesses and load. The fall is taken up entirely in the **compacted engineered fill** below the rubble, which is **750 nominal at the crown thinning to 688 at the box edge**. Every other layer keeps its nominal thickness, so the cover weighs 40.65 kPa at the crown and less toward the edges. **The fall costs nothing in load, and nothing in the bill.**

7.4 Total roof load — combination 103

Component	kPa
Blast, 344.7 × 1.111	383.00
Engineered cover	40.65
Superimposed dead	2.00
Self weight	22.50
TOTAL w	448.15

Live load is excluded — IS 4991 Cl. 11.2: "No live load shall be considered on roof at the time of blast."

Static ULS on the roof (COMB 101)
1.5 × (40.65 + 2.0 + 22.5 + 20) = 127.7 kPa
BLAST GOVERNS 448.15 / 127.7 = 3.51 : 1

7.5 Headhouse loads

Element	Load	Basis
Roof (flush horizontal at the berm crest)	396.5 kPa = 383 blast + 12.5 self + 1.0 SIDL	IS 4991 Cl. 7.2
Walls	383 kPa, acting EITHER FACE	IS 4991 Cl. 7.2 — the buried-element rule, applied to a bermed one
<i>The alternative that is not taken</i>	$113 \text{ kPa} = C_d \cdot q = 0.4 \times 282$	IS 4991 Cl. 7.4, drag on a bermed face

Why the upper bound and not the drag value. The walls sit behind about 2.5 m of berm, and **earth transmits pressure**. Taking the drag value credits the berm with a transmission factor that cannot be verified on this site, and 2.5 m of dry compacted fill is not the same material as the saturated ground the buried box sits in. So the walls are designed for **the full 383 kPa**. **The cost of that decision is one link cage, T12 4-leg at 250; the utilisation rises from 17 % to 59 %.** Removing an unverifiable assumption from the design of a protective structure for the price of a link cage is a trade worth making every time.

And the pressure acts from inside too. The inner security door is not blast rated and the entry stairwell is expected to be lost, so **the headhouse fills and the walls are pushed outwards**. They are reinforced symmetrically for 383 kPa either way — *a stated requirement, not an accident of detailing*.

7.6 Entry stairwell loads — static only, NOT blast rated

Element	Load
Flight, waist 250	waist 7.150 + steps 2.083 + finishes 1.000 + LL 5.000 = 15.233 → $w_u = 22.85 \text{ kPa}$
Side walls 250	$K_0 \cdot 0.50 \times 20 \times 2.9 + 0.5 \times 10$ surcharge = 34 kPa at base
Raking roof 250	self 6.25 + waterproofing 2.0 + earth lap 5.4 + imposed 20 = 33.65 → $w_u = 50.5 \text{ kPa}$
Top landing 250	18.4 + flight reaction 36.5 = 54.9 kPa

The 20 kPa imposed load on the raking roof is deliberate: it is there for a stray vehicle on the berm. It is not a code minimum and it is not an accident.

7.7 Sentry post loads

Item	Value	Verification
Slab dead, first floor	4.750 kPa	$0.150 \times 25 + 1.0$ finish
Slab dead, roof	5.250 kPa	$3.75 + 1.5$ screed / waterproofing
Imposed, first floor	3.000 kPa	IS 875 (Pt 2), observation post
Imposed, roof	1.500 kPa	IS 875 (Pt 2), accessible
Peak two-way intensity, DL floor	8.669 kN/m	4.75×1.825
Peak two-way intensity, DL roof	9.581 kN/m	5.25×1.825
Peak two-way intensity, LL floor	5.475 kN/m	3.00×1.825
Peak two-way intensity, LL roof	2.737 kN/m	1.50×1.825
Infill on the first-floor beams	13.000 kN/m	$0.200 \times 2.600 \times 25$ — held, because the brick infill actually built is lighter ($\approx 9.88 \text{ kN/m}$) and the modelled value is therefore conservative
Roof projection + parapet	4.162 kN/m	[C] as drawn — half re-derived, see below
— parapet term, $0.300 \times 0.150 \times 25$	1.125 kN/m	[C], reproduces exactly
— roof projection term	3.037 kN/m	[N] — the projection dimension is recorded nowhere

Item	Value	Verification
$w_{u \text{ floor}} = 1.5(4.75 + 3.00)$	11.625 kPa	governs the slab
$w_{u \text{ roof}} = 1.5(5.25 + 1.50)$	10.125 kPa	

Half of that line can be reproduced and half cannot. The parapet is 300×150 , and $0.300 \times 0.150 \times 25 = 1.125 \text{ kN/m}$ reproduces exactly. The residual **3.037 kN/m** is the roof projection, and **the projection dimension is recorded nowhere in this project.** Back-solving gives 810 mm (slab alone) or 578 mm (slab plus finish); **neither is round, neither is drawn and neither is adopted.** **4.162 kN/m is used as given, and no member force changes** — U8-F1.

Load path to the beams — IS 456 Cl. 24.5, 45° yield lines:

```
Peak intensity on any beam = w x short span / 2 = w x 3.650/2 = w x 1.825 m
SHORT beams B1 (3650 long) -> TRIANGLE
LONG beams B2 (4650 long) -> TRAPEZOID
w_eq = w_peak x [ 1 - 1/(3 r^2) ],    r = 4650/3650 = 1.2740
1 - 1/(3 x 1.2740^2) = 0.79463 -> 0.7946 adopted
```

The trapezoid factor **is exact**; using the same factor for both the bending moment and the fixed-end moment is the simplification, and it is worth about **2 %** on B2's support moment, inside its 89 % utilisation. **Accepted on that margin, and declared as assumption A12 rather than absorbed.**

7.8 Seismic and wind

7.8.1 Underground box — IS 1893 (Part 1):2016

```
Z = 0.16 (Zone III)      I = 1.5      R = 4.0      Sa/g = 2.5
Ah = (Z/2)(I/R)(Sa/g) = 0.08 x 0.375 x 2.5          = 0.075
W = 14 002 kN -> Vb = 0.075 x 14 002                  = 1050 kN
Per long wall 525 kN
tau = 525e3 / (600 x 0.8 x 21600)                    = 0.063 N/mm2
NEGLIGIBLE
IS 13920 Cl. 10.4 boundary elements CHECKED and NOT TRIGGERED
```

7.8.2 Sentry post — IS 1893 (Part 1):2016

```
Z = 0.16    I = 1.5    R = 3.0    Sa/g = 2.5 (rock, T on the plateau)
R = 3.0 because the infill panels are NOT separated from the frame.
R = 5.0 would require a special moment-resisting frame with positively
separated infill, which this project does not claim.
Ah = 0.08 x 0.50 x 2.5          = 0.100

Fundamental period -- all three estimates land inside 0.10-0.40 s,
so Sa/g = 2.5 regardless of which is used:
bare RC MRF, Cl 7.6.2          Ta = 0.075 h^0.75
                                = 0.075 x 6.250^0.75          = 0.297 s
with infill, Cl 7.6.2(c)      Ta = 0.09 h / sqrt(d)
                                X 0.281 s          Z 0.252 s

HAND-CALCULATED seismic weight
Roof +6.700 : slab 89.1 + parapet/projection 69.1 + beams 31.1
              + half-columns 18.7 + half first-storey infill 107.9 = 315.9 kN
Floor +3.650 : slab 80.6 + beams 31.1 + columns 38.3
              + half ground-storey infill 114.1 + 25 % LL 12.7    = 276.8 kN
TOTAL W = 592.7 kN -> Vb,hand = 0.100 x 592.7          = 59.3 kN

*** STAAD MODEL VALUE, WHICH GOVERNS DESIGN: Vb = 73.18 kN ***
stated in the load-case titles and CONFIRMED by the reactions:
Fx = 18.295 kN per column x 4          = 73.18 kN
```

The hand check and the model differ by 139.1 kN, and the difference is fully explained. The model prints its own seismic-weight summary: roof 331.46 + floor 400.34 = **W = 731.80 kN**, and $V_b = 0.100 \times 731.80 = 73.18$ kN returns $A_h = 0.1000$ **exactly**. Rebuilding both storey weights from the model's own load blocks reproduces 331.46 and 400.34 to the kilonewton, **and the 139.1 kN gap to the hand check resolves completely**:

- **107.9 kN** is half the first-storey infill, which the hand check's floor line does not allocate; - **31.1 kN** is the model taking the full 450 mm beam depth in `SELFWEIGHT` while also applying the full slab pressure, where the hand check nets the beam to 300 mm.

Nothing is unexplained, and every member is already designed to the higher value.

7.8.3 Wind on the sentry post — IS 875 (Part 3):2015

```
Vb = 39 m/s (Pune)    k1 = 1.08 (100-yr, Table 1)    k2 = 1.00 (Cat 2, 10 m)
k3 = k4 = 1.00
Vz = 39 x 1.08
pz = 0.6 Vz^2
pd = pz x Kd 0.90 x Ka 0.897 x Kc 1.0
F = Cf 1.3 x Ae (4.0 x 6.7 = 26.8 m2) x pd
SEISMIC 73.18 kN    vs    WIND 29.9 kN    ->    SEISMIC GOVERNS 2.4 : 1
```

$k_1 = 1.08$ assumes a **100-year design life** [A]. That is a deliberate choice rather than a default: a 100-year life on a load path that seismic already beats 2.4 : 1 costs nothing at all, and a protective installation is not a 50-year building.

7.9 Load combinations

7.9.1 Underground box — five combinations, matching the STAAD model

No.	Title	Factors
101	ULS static	1.5 (DL + SIDL + LL + SOIL + UPLIFT)
102	ULS uplift	0.9 DL + 1.5 UPLIFT
103	BLAST	1.0 (DL + SIDL + SOIL + UPLIFT + BLAST) ← governs every element
104	SLS crack width	1.0 (DL + SIDL + LL + SOIL + UPLIFT), IS 3370 Pt 2
105	Construction	1.5 (DL + SOIL + UPLIFT + SURCHARGE)

$\gamma = 1.0$ on blast because it is an extreme event checked against **ultimate** capacity with **dynamic** material strengths (IS 4991 Cl. 10.3.1). Applying 1.5 while also taking the 25 % material bonus would be inconsistent (Part 2.2.6).

Wind and earthquake are absent from 103 — IS 4991 Cl. 11.1 forbids combining them with blast.

7.9.2 Sentry post — sixteen combinations, read from the .std file

No.	Combination
101	1.5 DL + 1.5 LL
102-105	1.2 DL + 1.2 LL + 1.2 EQ ($\pm X$, $\pm Z$)
106-109	1.5 DL + 1.5 EQ ($\pm X$, $\pm Z$)
110-113	0.9 DL + 1.5 EQ ($\pm X$, $\pm Z$)
201	1.0 DL + 1.0 LL — SERVICE
202 · 203	1.0 DL + 1.0 EQ+X · 1.0 DL + 1.0 EQ+Z — DRIFT CHECK, IS 1893 Cl. 7.11

A correction worth recording. This table used to read "*15 combinations*" and stopped at 202, on the authority of the screen captures. **The .std file itself carries sixteen LOAD COMB cases — 203, the second drift check, was missing from the master.** No combination factor, member force or bar changes: 203 is a serviceability drift check that was already in the model being designed to. **The file is primary; the screenshots were not.**

PART 8 — ANALYSIS MODELS

8.1 What the models are, and what they are not

STAAD.Pro is not available in this environment. No analysis has been run in this project and no result — moment, displacement, shear or reaction — exists for the underground box or for the three mesh models. Editing, reconciling and validating a .std file is **not** running an analysis and is never reported as one.

Every underground design value in Parts 9 and 10 was produced by hand calculation, to IS 456 with IS 4991 loading rules. The model supplies the demand distribution and the visual check that loads are applied to the correct faces. **For the sentry post, only the forces were taken from the model** — the base shear and the seismic axial couples — and no STAAD concrete design output was relied upon anywhere.

8.2 Model inventory

Model	File in current/staad/	Role	Status
Underground box — reference	Underground_Structure_WITH_LOADS_worked_example (4).STD	The Phase 2 Rev A plate model of the box as designed. Every box value in Part 9 is checked against it	[C]
Sentry post	Sentry_Post_Framed_Seismic.std	Two-storey RC frame, seismic	[C]
Entry stairwell	Entry_Stairwell.std	Covered approach stairwell, static only, not blast rated	[C]
Underground box — COARSE	Underground_Shelter_Mesh_Coarse.std	Mesh study — 324 joints, 336 plates	[C]
Underground box — MEDIUM	Underground_Shelter_Mesh_Medium.std	Reference mesh — 1 113 joints, 1 138 plates	[C]
Underground box — FINE	Underground_Shelter_Mesh_Fine.std	h/2 refinement — 4 500 joints, 4 552 plates	[C]
Underground box — k_s upper bound	Underground_Shelter_ks500000.std	The reference model at $k_s = 500\,000$; ELASTIC MAT SUBGRADE and the four corner KFY ($\times 5$) are the only lines that differ , verified by diff	[C]

Units: metres and kilonewtons throughout. **Coordinate system:** STAAD global — **Y is vertical**, X and Z horizontal.

The underground model's datum is not the project datum. It uses **mid-surface** geometry and its origin $Y = 0$ is at the **underside of the mat**, real level (-6.700) .

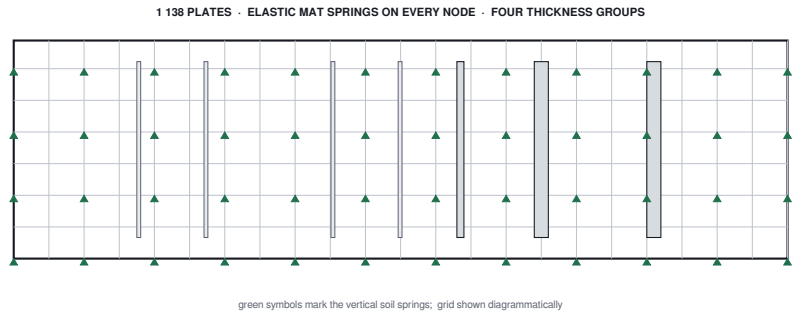
project level = model $Y - 6.700$. Apply the offset before comparing any model output to a drawing. $Y = 0.300$ is the mid-plane of the 600 mat; $Z = 0.300$ and $Z = 5.900$ are the mid-planes of the 600 perimeter walls; $X = 0.300$ is the mid-plane of the west end wall.

8.3 The underground box model

Form. A **plate + beam mid-surface model** of the whole box — mat, four perimeter walls, internal walls and the pressure slab — meshed on a regular grid over the full 22×6.2 m footprint, with support symbols at every mat node.

Shell elements	1 138 , in four thickness groups: mat 600 / roof 900 / perimeter 600 / internal 200–400
Supports	ELASTIC MAT DIRECT Y SUBGRADE 100000 — vertical springs on every mat node, generated from k_s
Corner joints	Explicit KFY per corner, from the corner tributary areas: 6 891 / 8 269 / 7 219 / 8 663 kN/m
Analysis command	PERFORM ANALYSIS PRINT STATICS CHECK — no P-Delta

THE UNDERGROUND BOX MODEL - WHAT IT IS, WHAT IT CARRIES, AND THE ONE QUESTION IT CANNOT ANSWER



ELEVEN PRIMARY LOAD CASES, AND THE FIVE COMBINATIONS BUILT FROM THEM

1	DL1 self weight	auto	7	HY1 hydrostatic uplift on the mat	46.11 kPa
2	DL2 earth cover	40.65 kPa	8	BL1 blast on the roof	383 kPa
3	DL3 services and finishes	2.0 kPa	9	BL2 blast on the external walls	383 kPa
4	LL1 internal floor	5.0 kPa	10	LL2 construction surcharge	20.0 kPa
5	DL4 staircase on the shaft walls	2.947 kPa	11	BL3 BLAST IN THE STAIR SHAFT ON W6 AND W7	383 kPa
6	EP1 earth + water on the external walls	33.9 to 83.2 kPa			

COMB 103 = 1.0 (DL + SIDL + SOIL + UPLIFT + BLAST) - GOVERNS EVERY ELEMENT

SPRING SUPPORTS TAKE TENSION, SO THE MAT NEVER LIFTS IN THIS MODEL, IN ANY LOAD CASE.
Flotation cannot be demonstrated OR refuted by it. The check in Part 9.3.5 is a hand calculation on real dimensions and real weights. Anyone who says the analysis shows no uplift has misunderstood their own model.

Figure 24 The underground box model - what it is, what it carries, and the one question it cannot answer.

Load cases, read verbatim from the file:

#	Title	Value
1	DL1 SELF WEIGHT (SELFWEIGHT Y -1)	auto
2	DL2 EARTH COVER ON ROOF 40.65 KN/M2	40.65 kPa
3	DL3 SIDL SERVICES AND FINISHES	2.0 kPa
4	LL1 LIVE LOAD ON INTERNAL FLOOR	5.0 kPa
5	DL4 STAIRCASE ON THE TWO SHAFT WALLS	2.947 kPa
6	EP1 EARTH + WATER ON EXTERNAL WALLS	33.9 → 83.2 kPa
7	HY1 HYDROSTATIC UPLIFT ON MAT	46.11 kPa
8	BL1 BLAST ON ROOF 383 KN/M2	383 kPa
9	BL2 BLAST ON EXTERNAL WALLS 383 KN/M2	383 kPa
10	LL2 CONSTRUCTION SURCHARGE	20.0 kPa
11	BL3 BLAST IN THE STAIR SHAFT ON W6 AND W7 383 KN/M2	383 kPa, acting out of the shaft on both faces

Load case 11 is the whole argument of Part 9.2, expressed as a load. A model that applies blast only to the external walls has an internally consistent but wrong picture of this structure: the stair shaft is open to atmosphere through the roof void, so it equalises to the full incident overpressure and loads W6 and W7 from inside. BL3 is that demand, it is in combination 103, and the 400 mm thickness of those two walls exists to carry it.

8.3.1 The modelling warning that must not be lost

Spring supports in STAAD take tension unless explicitly declared compression-only. In this model the mat therefore never lifts, in any load case. The flotation check of Part 9.3 cannot be performed in STAAD and is not. It is a hand calculation on real dimensions and real weights. Anyone who says "the analysis shows no uplift" has misunderstood their own model.

8.4 The sentry post model

Joints	Twelve — base 1, 2, 11, 12 at Y = +0.450; first floor +100; roof +200
Members	16 — 1–8 columns PRIS YD 0.35 ZD 0.35; 9–16 beams PRIS YD 0.45 ZD 0.25
Supports	1 2 11 12 FIXED — all four column bases fully fixed
Member releases	NONE. The command appears nowhere in the file — every beam–column joint is fully continuous, which is what a moment frame requires and what the portal-method forces in Part 11.1 assume
Infill	Modelled as MEMBER LOAD only , never as a strut
Slab S1	Not modelled as plates — the slab load is applied to the beams as triangular / trapezoidal loading
Analysis command	PERFORM ANALYSIS PRINT STATICS CHECK — no P-Delta

Fixed bases are justified, not assumed: the columns sit on isolated footings **on in-situ rock**. Rotational restraint is real. **On backfill it would not be defensible** — which is precisely why the post is sited ≥ 10 m clear of the shelter excavation.

The statics check passes in every case — difference between loads and reactions = 0.000.

L/C	ΣF_x	ΣF_y	ΣF_z	ΣM_x	ΣM_y
1 (DL)	0.000	-624.548	0.000	1561.369	0.000
2 (LL)	0.000	-76.376	0.000	190.941	0.000
3 (EQ+X)	73.180	0.000	0.000	0.000	182.950
5 (EQ+Z)	0.000	0.000	73.180	436.631	-146.360

Two confirmations taken from the reaction table and used directly in design:

```

1  Fx = 18.295 kN per column x 4 = 73.18 kN
   -- the portal-method distribution used in design is EXACTLY what the
   -- model produces.
2  Fy = +/- 36.127 kN (EQ+X) and +/- 27.891 kN (EQ+Z)
   -- these ARE the seismic axial couples used in the column design.
   -- They were READ from the model, not calculated by hand.
```

And one difference, stated rather than smoothed over. Back-calculating the model's own vertical distribution from its load resultants gives $Q_{\text{roof}} = 55.57$ kN and $Q_{\text{floor}} = 17.61$ kN, against the hand wh^2 distribution's 59.51 and 13.67. **The design used the hand distribution at the model's total of 73.18 kN, which gives the LARGER ground-storey column moment. Conservative, and recorded.**

8.5 Mesh sensitivity

A plate model's answer depends on its mesh, and a design that quotes plate moments without demonstrating mesh independence has not finished its argument. Three mesh-density variants of the box model exist: **COARSE** (about $1.9 \times$ the reference element size), **MEDIUM** (an unmodified copy of the reference mesh) and **FINE** (an exact $h/2$ refinement).

Geometry, thicknesses, materials, supports and soil springs, loads, load cases and combinations are **identical in all three**; only the plate mesh density differs. The opening boundaries and the footprint are pixel-identical across all three: **net roof area 104.048 m² and hole area 15.792 m² match exactly**. Corner spring K_{FY} values and the $EP1$ earth-plus-water row pressures were **re-derived per mesh from the same physical formulas the reference model itself encodes** —

```

KFY = k_s x A_trib
EP1 p = 15.4071 x depth - 10.8198 kN/m2, depth = 6.700 - Y
      fitted from the reference model's own six data points, residual <= 0.01
```

— **not re-guessed.**

The convergence conclusion is OUTSTANDING. The three models are built and validated; **STAAD.Pro is not available in this environment, so no result exists for any of them.** Saying the mesh converges would be inventing the study's own conclusion, and the study exists precisely so that the conclusion does not have to be assumed.

8.6 The input files, and the class of defect that only appears when a model is opened

The first time any of these models was opened in STAAD.Pro it reported errors. All four box models were then re-read line by line, and what came out is worth stating in full, because none of it is visible in a drawing, a summary or a screen capture.

#	Defect	Fix
1	Three <code>LOAD 6</code> south-wall lines were 80 columns against the file's own <code>INPUT WIDTH 79</code> . STAAD reads 79 columns and discards the rest, so it read <code>-77.6 / -57.3 / -37.0</code> where the file says <code>-77.64 / -57.36 / -37.07</code> . The north wall (79 columns) was read correctly, so <code>LOAD 6</code> no longer balanced — 4 848.4 kN in +Z against 4 843.6 kN in -Z, an out-of-balance of 4.8 kN on a case that is self-equilibrating by construction, and the file asks for exactly the statics check that reports it	Split on the STAAD – continuation character. Every data line in all four box models is now ≤ 79 columns
2	<code>LOAD 10</code> was defined after <code>LOAD 11</code> in all four models, so the primary case numbers ran 1...9, 11, 10	Reordered. Physically neutral — combinations reference cases by number and 101–105 are byte-identical before and after
3	Corner spring <code>KFY</code> at joint 1 was 27 562 where the model's own formula gives 27 562.5 and the study documentation gives 27 563, while the sibling half-value was rounded up	27 563. A 0.5 kN/m difference on one spring; the point is that the file now agrees with its own documentation
4	A load title and one <code>JOB CLIENT</code> line also exceeded 79 columns (titles only, no numeric consequence)	Shortened

Verified after the edits, on all four box models: the 336 / 1 138 / 4 552-plate topology is unchanged and clean — every element resolves to defined joints, no repeated node, no degenerate or non-planar quadrilateral, no duplicate coordinate or element, no unused joint, no gap in the numbering; every joint still satisfies the model's own numbering rule; and, **re-read as STAAD reads them at 79 columns, COARSE, MEDIUM and FINE now produce identical panel-by-panel load resultants for all eleven load cases** — mat, roof, each of the four external walls, W6 and W7 — **which they did not before.**

Not one of those corrections changed an analysis model. No geometry, thickness, material, support stiffness, load magnitude, load case number or combination factor moved. **What they changed is whether the file says what its author believed it said** — and a 79-column input truncation that silently turns `-77.64` into `-77.6` on three lines out of several hundred is exactly the class of defect no amount of reading a summary will ever find.

8.7 What must be obtained before the STAAD work can be extended

1. **A STAAD.Pro run.** Everything below waits on it.
2. The underground model's **post-processing output** — plate M_x/M_y , support reactions, deflections.
3. The **mesh convergence table**, which is the conclusion the mesh study exists to produce.
4. **Both k_s bounds run** — the upper-bound model exists and has never been executed.
5. The **sentry post re-run at the built brick infill** (w_{M-V6}) — confirmation, not risk: brick is lighter than the modelled infill, so every member is already over-designed.

PART 9 — STRUCTURAL DESIGN CALCULATIONS: THE UNDERGROUND BOX

Every value in Parts 9 to 11 has been computed numerically in this project, and the drawings carry exactly this reinforcement. No drawing shows an arrangement that differs from these calculations.

The flexure equations used throughout — IS 456 Annex G-1.1(b) and Cl. 38.1:

$$\begin{aligned} \mu &= 0.87 f_y \cdot A_{st} \cdot d \cdot [1 - (A_{st} \cdot f_y) / (b \cdot d \cdot f_{ck})] \\ x_u &= 0.87 f_y A_{st} / (0.36 f_{ck} b) && \text{IS 456 Cl. 38.1} \\ x_{u,max} / d &= 0.46 \text{ for Fe500} && \text{IS 456 Cl. 38.1(f)} \\ \mu_{lim} &= 0.133 f_{ck} b d^2 && \text{IS 456 Annex G-1.1(c)} \end{aligned}$$

Notation: T16 @ 150 EF EW = 16 mm Fe500D deformed bars at 150 mm centres, Each Face, Each Way. 4L = four-legged links.

9.1 Perimeter walls W1-W4 — 600 thk — 68 %

INPUTS	t 600, clear height 3200, cover 75 earth / 40 internal, T16 d = 600 - 75 - 8 = 517 mm Blast 383 kPa (Ka 1.0); static earth+water 83.2 kPa at base -> BLAST GOVERNS 4.6 : 1
FORMULA	$M_p = w \cdot L_n^2 / 16$ (fixed-fixed plastic mechanism)
SUBSTITUTE	$M_p = 383 \times 3.200^2 / 16$
RESULT	$M_p = 245.1 \text{ kNm/m}$
CHECK	$\mu_{lim} = 0.133 \times 43.75 \times 1000 \times 517^2 = 1556 \text{ kNm/m}$ -> singly reinforced OK $A_{st,req} (f_y, dyn 625, f_{ck}, dyn 43.75) = 894 \text{ mm}^2/\text{m}$ IS 456 Cl. 32.5(a) min vertical $0.0012 \times 600 = 720 \text{ mm}^2/\text{m}$ IS 456 Cl. 32.5(b) min horizontal $0.0020 \times 600 = 1200 \text{ mm}^2/\text{m}$ IS 3370 Pt 2 surface zone 0.35 % x 250 each face = 875 mm ² /m/face
ADOPTED	T16 @ 150 c/c EACH FACE EACH WAY = 1340 mm ² /m $\mu = 362.8 \text{ kNm/m}$ -> UTILISATION 68 % $x_u = 46.3$ -> $x/d = 0.089 < 0.46$
SHEAR	V at d = 383 (1.600 - 0.517) = 414.8 kN/m $\tau_{av} = 414.8e3 / (1000 \times 517) \text{ Cl. 40.1} = 0.802 \text{ N/mm}^2$ $\rho_t = 0.259 \% \rightarrow \tau_{uc} \text{ (Table 19, M35)} = 0.375 \text{ N/mm}^2$ $\tau_{uc,max} \text{ (Table 20, M35)} 3.70 > 0.802 \text{ OK Cl. 40.2.3}$ $V_{us} = (0.802 - 0.375) \times 517 = 220.8 \text{ kN/m}$ $A_{sv}/s_v = 220800 / (0.87 \times 500 \times 517) \text{ Cl. 40.4(a)} = 0.982 \text{ mm}^2/\text{mm}$ -> T12 2-leg at 230 Cl. 26.5.1.5 limit min(0.75d = 388, 300) = 300
ADOPTED	T12 CLOSED LINKS @ 200 c/c (Asv/sv = 1.131 OK)
AXIAL	$N = 448.15 \times 2.50 + 3.2 \times 0.6 \times 25 = 1168 \text{ kN/m}$ $f = 1.95 \text{ N/mm}^2 = 11 \% \text{ of } 0.4 f_{ck, dyn} (17.5)$ -> P-M not critical
CRACK	IS 3370 Pt 2, limit 0.2 mm; surface steel 875 < 1340 OK

Why 600 mm, stated honestly. The wall is **not strength-governed** — it is 68 % utilised. 600 mm is set by five things at once, and none of them is bending:

1. **75 mm cover** against blinding and trimmed rock; 2. **congestion** — two curtains, closed links, waterstops and cast-in frames all in one section; 3. **IS 3370 crack control** under sustained hydrostatic load; 4. **the EMP double curtain at 150 mm** (Part 2.8.5); 5. **the 1 168 kN/m axial path** from the roof.

Saying so matters: an optimiser who sees 68 % and thins the wall loses four of the five reasons it is that thick.

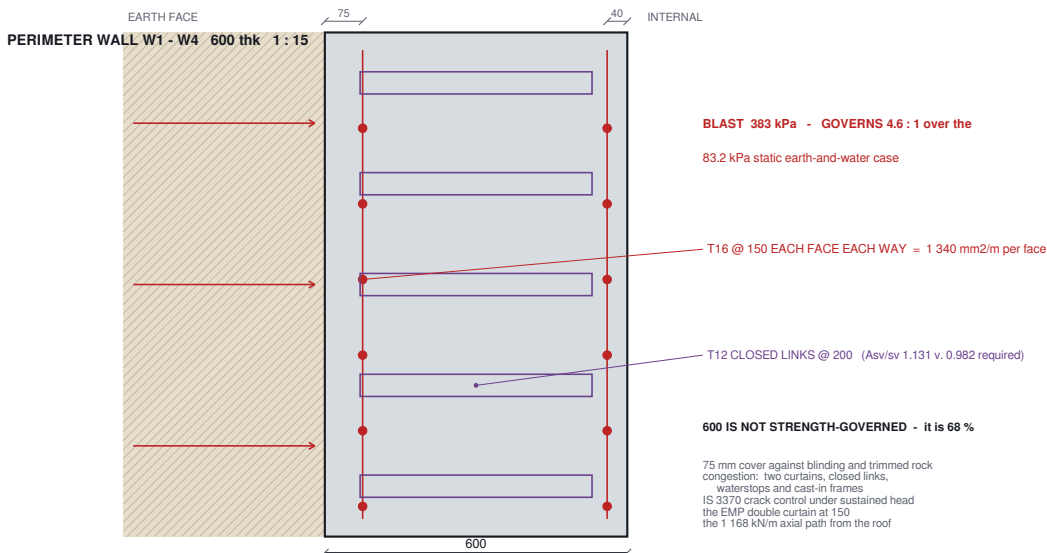


Figure 25 Perimeter wall W1 to W4, 600 thick - section and reinforcement.

9.2 Walls W6 and W7 — 400 thk — 69 %

9.2.1 Why these two walls are not like the other internal walls

FINDING F1 The stair shaft is open to atmosphere through the 2800 x 3160 roof void, the headhouse and the entry stairwell.

Fill time $V / (A \cdot c) = 105 / (3.3 \times 340) = 0.094 \text{ s}$
 against the positive phase duration $t_d \text{ } 0.13 - 1.33 \text{ s}$

-> THE SHAFT EQUALISES TO FULL p_{so} , and it does so in well under one t_d .

W6 and W7 therefore sit in the SAME PLANE as the 7-bar blast doors, separating a PRESSURISED shaft from Bay 6 / Bay 8 at AMBIENT.

That paragraph is why W6 and W7 are 400 mm and every other internal wall is 110 or 200. They are not partitions: they are the last two panels of the protective boundary, standing in the same plane as two 7-bar doors and separating a **pressurised shaft from bays at ambient. The design does not pretend the shaft stays at ambient. It accepts that the shaft fills, and carries the consequence in the thickness.**

9.2.2 Why 200 mm is impossible — not a detailing problem

$d = 200 - 40 - 6 = 154 \text{ mm}$
 $M_{u,lim} = 0.133 \times 43.75 \times 1000 \times 154^2 = 138.0 \text{ kNm/m}$
 Demand $M_p = 383 \times 3.200^2 / 16 = 245.1 \text{ kNm/m}$

$138 < 245 \rightarrow \text{NO STEEL RATIO MAKES IT WORK}$

Two-way action was CHECKED BEFORE REJECTION:
 panel 3200 x 3800, fixed on three edges, yield line
 $\sim 0.7 \times \text{one-way} = 172 \text{ kNm/m}$ -- still above 138.

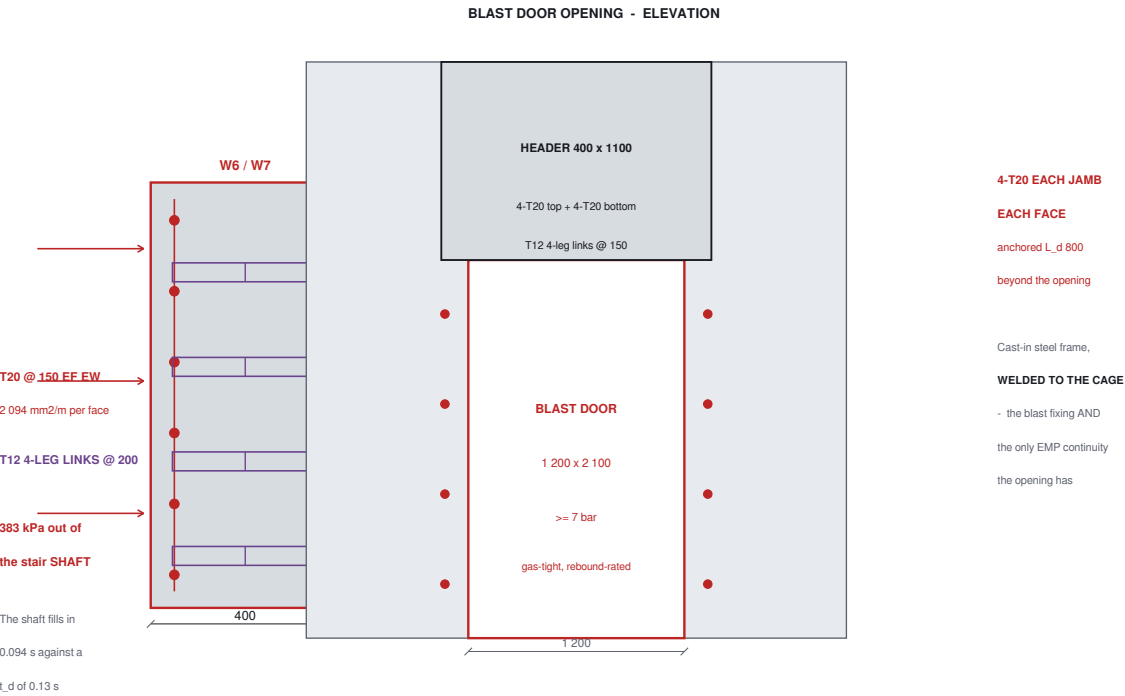
The section is over its singly-reinforced capacity before any steel is chosen. This is the one place in this structure where the answer is not "add bars" but "change the geometry", and it is worth being explicit about that: no detailing, no grade change and no two-way idealisation rescues a 200 mm wall here.

And the 400 mm cannot be found anywhere convenient. It cannot come out of Bay 8, because the 750 mm escape-shaft clearance rule (Part 5.6) fixes Bay 8's minimum width at 2 900 mm. It cannot come out of the shaft, because $2\,800 = 2 \times 1\,200 \text{ flights} + 200 \text{ well}$, exactly. **So the box is 400 mm longer than the accommodation alone would need** — which the one-way roof makes structurally free, and which is the reason Part 5.3 spends a paragraph on the aspect ratio.

9.2.3 Design of the 400 mm wall

	$d = 400 - 50 \text{ (shaft face)} - 8$	$= 342 \text{ mm}$
	$M_p = 383 \times 3.200^2 / 16$	$= 245.1 \text{ kNm/m}$
	$M_{u,lim} = 0.133 \times 43.75 \times 1000 \times 342^2$	$= 680.6 \text{ kNm/m} \quad \text{OK}$
	$A_{st,req}$	$= 1400 \text{ mm}^2/\text{m}$
ADOPTED	T20 @ 150 EF EW	$= 2094 \text{ mm}^2/\text{m}$
	$M_u = 355.3 \text{ kNm/m} \rightarrow \text{UTILISATION } 69 \%, \quad x/d = 0.211$	
SHEAR	$V \text{ at } d = 383 \text{ (1.600 - 0.342)}$	$= 481.8 \text{ kN/m}$
	$\tau_{u,v} 1.409 \mid \tau_{u,c} 0.540 \text{ (pt } 0.612 \%) \mid \tau_{u,c,max} 3.70 \quad \text{OK}$	
	$V_{us} 297.2 \text{ kN/m}; \quad A_{sv}/s_v 1.998 \rightarrow \text{T12 4-leg at } 226$	
	$Cl. 26.5.1.5 \text{ limit } \min(0.75d = 257, 300)$	$= 257$
ADOPTED	T12 4-LEGGED LINKS @ 200 c/c	
BLAST DOOR OPENING	1200 x 2100	
	Interrupted steel 2094 x 1.2 = 2513 mm ² /face	
	$\rightarrow 1256 \text{ each jamb}$	
ADOPTED	4 No. T20 EACH JAMB EACH FACE (1257 mm ²), anchored L_d 800 beyond	
HEADER	400 x 1100 over 1200 clear	
	$w = 383 \times 2.1/2$	$= 402 \text{ kN/m}$
	$M = w L^2 / 12$	$= 48.2 \text{ kNm}$
	$V = 241 \text{ kN}; \quad \tau_u = 0.578 \text{ N/mm}^2$	
ADOPTED	4-T20 TOP + 4-T20 BOTTOM, T12 4-leg links @ 150	
FRAME	Cast-in steel frame anchored into and WELDED to the cage (EMP).	
	Door: proprietary, $\geq 7 \text{ bar}$, rebound-rated, gas-tight [vendor]	

W6 / W7 400 thk - THE PROTECTIVE BOUNDARY



200 thk is IMPOSSIBLE here, not merely awkward: $M_{u,lim} = 138.0 \text{ kNm/m}$ against a demand of 245.1 kNm/m . No steel ratio closes that gap, so the GEOMETRY changed.

Figure 26 Walls W6 and W7 at 400, and the blast door openings they carry.

9.3 Mat foundation — 600 thk — 84 %, the most heavily worked element

9.3.1 Bearing governs nothing

BEARING	Service 58.4 kPa	$= 1.8 \%$ of 3240 kPa
	Blast 404.9 kPa	$= 12.5 \%$ (20.6 % on the measured soaked value -- Part 4.3.2)
	BEARING GOVERNS NOTHING.	

9.3.2 Two flexural cases, and the second one sizes the mat

CASE 1 -- NET UPLIFT (COMB 102)	
$u = 4.700 \times 9.81$	$= 46.11 \text{ kPa}$

	self 0.600 x 25	= 15.00 kPa
	net 31.1 kPa UP over the 5000 clear span	
	Mp = 31.1 x 5.0^2 / 16	= 48.6 kNm/m
		NOMINAL
CASE 2 -- SOFT / RED-BOLE ZONE	*** GOVERNS ***	
	A 3.0 m band of red-bole or vesicular material removed, at the WORST POSITION.	
	q = 404.9 kPa (blast bearing) over a 3.0 m unsupported span	
	M = q . L^2 / 12 = 404.9 x 9 / 12	= 303.7 kNm/m
	d = 600 - 75 - 8	= 517 mm
	Ast,req	= 1115 mm2/m
ADOPTED	T16 @ 150 EF EW = 1340 mm2/m	
	Mu = 362.8 kNm/m -> UTILISATION 84 %	
THICKNESS STUDY		
	500 -> Ast,req 1407 = 105 % of provided	FAIL
	600 -> 1115 = 84 %	ADOPTED
	700 -> 926 = 70 %	
	800 (as once drawn) ~ 57 % = 33 % unused reserve	

Case 2 is the geology, expressed as a load case. The flow contacts in a Deccan Trap sequence are exactly where a soft seam sits, and a single red-bole band under the mat removes support over its width. **That case, not bearing and not uplift, is what sets the 600 mm mat and its T16 @ 150 EF EW.** It is also why Part 4.2.2's unlogged 5.3 m matters so much: **the case that sizes the mat is a case nobody has looked for at the founding horizon.**

9.3.3 One-way shear, and the link grid it produces

ONE-WAY SHEAR	
	V at d from the soft-band edge
	= 404.9 (1.500 - 0.517)
	= 398.0 kN/m
	tau_v 0.770 tau_c 0.375 (pt 0.259 %) tau_c,max 3.70 OK
	*** tau_c read at the STATIC M35 value --
	IS 4991 Cl. 10.3.1.1 gives NO dynamic increase on shear ***
	Vus = 204.2 kN/m ; Asv/sv
	= 0.908 mm2/mm
ADOPTED	T12 CLOSED LINKS ON A 250 x 250 GRID THROUGHOUT
	supplied Asv/sv = (4 x 113.1) / 250 = 1.810 = 2.0 x required
	-- and the grid DOUBLES AS THE SPACER SYSTEM between the two curtains

MAT FOUNDATION 600 thk - THE CASE THAT SIZES IT 1 : 70

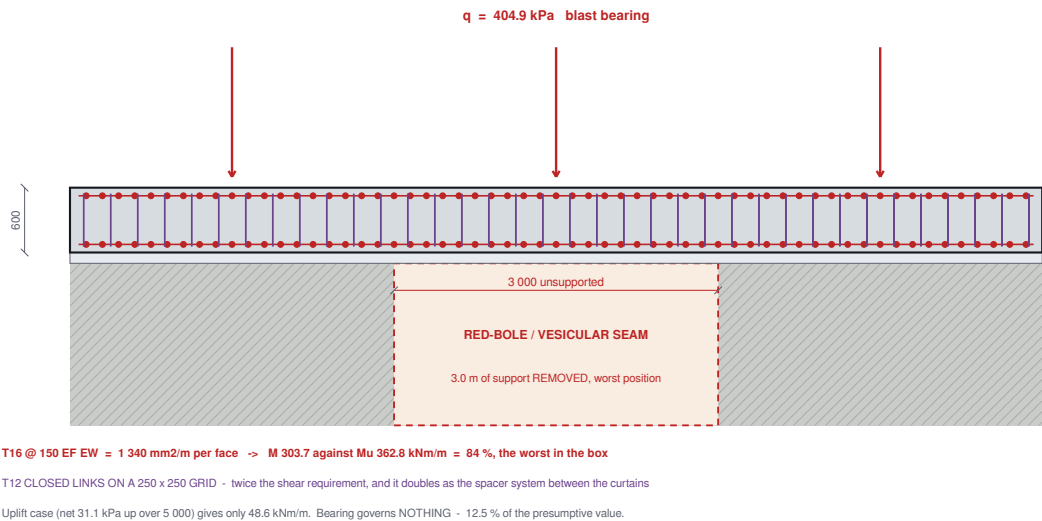


Figure 27 The mat foundation, 600 thick, and the soft-band case that sizes it.

9.3.4 The checks that do not apply, declared rather than ignored

Check	Verdict
Punching, IS 456 Cl. 31.6	NOT APPLICABLE. No column or pedestal bears on the mat; the headhouse walls bear on the roof . <i>Declared, not ignored</i>
Sliding	The box is socketed ~4.8 m into basalt (5.3–5.9 m on the measured rockhead) with the 300 mm annulus backfilled in M15 lean concrete. Not a credible mechanism — the reasoning is stated and no spurious factor of safety is computed

Check	Verdict
Overturning	Buried box 22.0 × 6.2 × 4.7 m under a symmetrical lateral load. NOT APPLICABLE
Settlement	Negligible on sound basalt, IS 12070 Cl. 6 — and Part 2.5.3 gives the physical reason. The hazard is the flow contacts, not the settlement

9.3.5 Flotation — the construction stage governs

Stage	Weight	Uplift	FoS @ (-)2.000	FoS flooded
1 mat cast only	2 009 kN	6 175 kN	0.33 FAIL	0.23 FAIL
2 mat + walls, no roof	4 802 kN	6 175 kN	0.78 FAIL	0.55 FAIL
3 box complete, no backfill	7 528 kN	6 175 kN	1.22 MARGINAL	0.86 FLOATS
4 backfilled + cover	12 453 kN	6 175 kN	2.02 OK	1.41 OK

Require FoS >= 1.2.

$$\text{ULS COMB 102} = 0.9 \times 12453 / (1.5 \times 6175) = 1.21$$

Uplift acts on the REAL UNDERSIDE, not on the plate model's mid-surface area.

*** THE STAGE TABLE ABOVE IS TABULATED ON A 21.600 m BOX LENGTH ***
 21.600 × 6.200 = 133.92 m² → 46.11 × 133.92 = 6175 kN (table)
 22.000 × 6.200 = 136.40 m² → 46.11 × 136.40 = 6289 kN (BUILT)
 Both uplift and resistance scale with length (× 22.0/21.6),
 so EVERY FACTOR OF SAFETY IS UNCHANGED:
 0.325 / 0.778 / 1.219 / 2.017,
 and COMB 102 stays 0.9 × 12684 / (1.5 × 6289) = 1.21
 THE GOVERNING HAND CHECK USES 136.40 m² / 6289 kN.

Mandatory mitigation — a design output, not a contractor's problem:

1. **Continuous dewatering** from the start of excavation until backfill and cover are complete.
2. **Temporary pressure-relief valves / knock-out plugs in the mat**, 6 No., grouted up **only after backfill**.
3. **Programme the sub-structure to complete before the monsoon**, or bund and positively drain with standby pumping and generator back-up.
4. **Backfill SYMMETRICALLY**, in layers.

And mitigation 1 is NOT met by the current programme — the departure is recorded, in writing. The owner's schedule ends dewatering on 11-05-27; side backfill runs 20-07-27 to 30-07-27 and the burster slab is not cast until 21-08-27. **The gap spans the whole 2027 monsoon with the box at stage 3 — FoS 1.22 at the design water table and 0.86 flooded to grade.** **The residual flotation risk is ACCEPTED IN WRITING, and no date moves.** Mitigations 2, 3 and 4 stay mandatory. This is a **declared departure from mitigation 1**, carried as a departure rather than quietly dropped — which is the only honest way to hold it, and it is why the monsoon groundwater monitoring sits on the critical path of the programme (Part 21.4).

9.4 Pressure (roof) slab — 900 thk — THE GOVERNING ELEMENT — 51 %

9.4.1 One-way, and what that buys

ONE-WAY ARGUMENT Aspect ratio 20.8 / 5.0 = 4.2
 IS 456 Table 26 is tabulated only to $l_y/l_x = 2.0$,
 so TWO-WAY ACTION CANNOT BE CLAIMED.
 → M_p depends on L_n of the 5 m WIDTH only.
 → LENGTH IS STRUCTURALLY FREE; widening 5.0 → 6.0 m
 would raise the roof moment 44 %.

9.4.2 Natural period — the calculation that puts this in the quasi-static regime

$m = 0.900 \times 2500 + 0.25 \times 2.0 \times 2000 = 3250 \text{ kg/m}^2$
 $0.5 \text{ EIG} = 0.5 \times 2.958 \times 10^{10} \times 0.900^3 / 12 = 8.985 \times 10^8 \text{ N.m}^2/\text{m}$
 $f_1 = (22.373 / 2 \pi) \cdot \sqrt{EI / (m \cdot L_n^4)} = 74.9 \text{ Hz}$
 $T = 13.4 \text{ ms}$
 $t_d / T = 10 \text{ to } 100 \rightarrow \text{QUASI-STATIC}$

9.4.3 Flexure

FLEXURE $d = 900 - 75 - 12.5 = 812.5 \text{ mm}$
 $M_p = w \cdot L_n^2 / 16 = 448.15 \times 25 / 16 = 700.2 \text{ kNm/m}$
 $M_{u, \text{lim}} = 0.133 \times 43.75 \times 1000 \times 812.5^2 = 3841 \text{ kNm/m OK}$
 $A_{st, \text{req}} (\text{pt } 0.20 \%) = 1632 \text{ mm}^2/\text{m}$

```

Cl. 26.5.2.1 minimum 0.12 % x 900 = 1080 mm2/m
Cl. 26.5.2.2 max bar dia D/8 = 112 OK for T25

ADOPTED T25 @ 150 EF EW = 3272 mm2/m
Mu = 1362.4 kNm/m -> UTILISATION 51 %
xu = 113.0 -> x/d = 0.139 << 0.46

*** x/d = 0.139 IS THE PROOF THAT mu = 5 IS DEFENSIBLE ***

```

9.4.4 Shear, including the direct-shear check

```

SHEAR V at the support face = 448.15 x 2.50 = 1120 kN/m
V at d from the face = 448.15 x (2.50 - 0.8125) = 756.3 kN/m
tau_v 0.931 | tau_c 0.450 (pt 0.403 %) | tau_c,max 3.70 OK
Vus 390.8 kN/m ; Asv/sv 1.106 -> T12 4-leg at 409
Cl. 26.5.1.5 limit 300 GOVERNS

ADOPTED T12 4-LEG @ 250 IN THE END 1500 EACH SIDE
T12 2-LEG @ 300 ELSEWHERE

DIRECT SHEAR Vd,max = 0.16 fck,dyn b d = 5688 kN/m
against 1120 applied = 19.7 %
[UFC 3-340-02 Sec. 4-30, cited as US criteria, NOT IS 456]

```

Direct shear is a blast-specific failure mode that IS 456 does not address. At a support, a very short, very high shear pulse can shear the section off cleanly before flexural response develops. It is checked here against a US criterion, **and the criterion is named as a US criterion** rather than blended into the Indian Standard work.

9.4.5 Why 900 mm and not 800

t	d	Mp	Ast,req	tau_v	verdict
600	512.5	689	2673	1.71	links at 164, congested
800	712.5	696	1868	1.12	the STRUCTURAL OPTIMUM
900	812.5	700	1633	0.93	ADOPTED
1000	912.5	704	1453	0.78	11 % over

900 is adopted over the structural optimum of 800 for three stated reasons:

1. **Reserve at the two 1 400 mm collars**, where the slab is locally interrupted;
2. **Headroom against a raised design basis threat while the yield is unconfirmed** — the honest consequence of Part 2.1.3's [N];
3. **Support rotation kept inside 2° without resorting to UFC lacing reinforcement.**

9.5 Roof openings

9.5.1 The two 1 400 dia escape-shaft openings

```

Interrupted steel = 3272 x 1.400 = 4581 mm2/face/direction
Trimmers each side = 4581 / 2 = 2291 mm2

ADOPTED 5 No. T25 EACH SIDE, EACH FACE, EACH DIRECTION (2454 mm2)
anchored Ld = 1000 beyond the opening
Collar 250 RC; T16 @ 150 hoops (3 layers) + T16 @ 150 radials
Slab thickened 900 -> 1200 over a 600 mm annulus

```

Circular is the right shape: hoop action, and no re-entrant stress concentration. This detail is what SETS the 750 mm clearance rule (Part 5.6) — 250 collar + about 400 of trimmer band + 100 tolerance — and that rule propagates all the way back into the bay widths, because a bay narrower than 2 900 mm cannot hold an escape shaft at all.

9.5.2 The 2 800 x 3 160 stair void, and the cantilever it leaves

```

The void leaves an 1840 mm CANTILEVER of 900 slab, 2800 wide, at 448.15 kPa.

M(root) = 448.15 x 1.840^2 / 2 = 758.6 kNm/m
*** THIS EXCEEDS THE 700.2 kNm/m MIDSPAN Mp ***

Ast,req TOP = 1772 mm2/m < 3272 provided -> UTILISATION 56 % OK
V(root) = 448.15 x 1.840 = 824.6 kN/m
tau_v 1.015 > tau_c 0.450 ; Vus 459.4 kN/m ; Asv/sv 1.300

```

-> T12 4-leg at 348 ; Cl. 26.5.1.5 limit 300

ADOPTED T12 4-LEG LINKS @ 250 THROUGHOUT THE PAD (NEW requirement)
 plus Free edge thickened 900 -> 1200 over 600, with 6-T25 top +
 6-T25 bottom and T12 closed links @ 150
 plus 6 No. T25 each face, top and bottom, in a 900 band over W6 and W7
 plus Diagonal trimmers 4 No. T25 each face at 45 deg at BOTH re-entrant
 corners, 2000 long each way, anchored L_d beyond
 plus 1100 mm guarding to the free edge (NBC 2016 Pt 4)

The cantilever root moment exceeds the midspan moment of the element it hangs off. 758.6 against 700.2 kNm/m. The flexural steel already in the slab covers it at 56 %, so the bars do not change — but the shear does not pass on concrete alone, and the four-leg links through the pad are a requirement that comes from this check and from nowhere else. A 1.84 m cantilever of 900 mm slab at 448 kPa is not a detail; it is a structural element, and it is designed as one.

ROOF PLAN - OPENINGS AND TRIMMERS 1 : 150

5-T25 EACH SIDE / FACE / DIRECTION at the collars, L_d 1 000 beyond

collar 250 RC · slab thickened 900 -> 1 200 over a 600 annulus



4-T25 EACH FACE at 45 deg, 2 000 each way, at BOTH re-entrant corners

6-T25 top and bottom in the thickened free edge · 6-T25 band over W6 and W7

T12 4-LEG LINKS @ 250 THROUGHOUT THE PAD - a requirement in its own right

Figure 28 The roof in plan: the stair void, the two escape shaft collars and their trimmers.

PART 10 — STAIRS, HEADHOUSE AND APPURTENANCES

10.1 Main staircase, Bay 7 — FROZEN GEOMETRY

STATUS: INSIDE the protective envelope. NOT a blast element. The shaft pressurises, but that pressure acts on the shaft walls and on the blast doors — **not as a net load on a slab that is open on both faces.** The stair is therefore designed to IS 456 with normal partial factors, **not** with IS 4991 dynamic strengths. That is stated explicitly on the drawing.

Geometry check — NBC 2016 Part 4:

Item	Provided	Limit	Verdict
Riser	170.833 mm	≤ 190	OK
Tread / going	280 mm	≥ 250	OK
Flight width	1 200 mm	$\geq 1\ 000$	OK — stretcher-capable
Risers per flight	8	≤ 12 preferred	OK
Headroom	2 533 mm	$\geq 2\ 200$	OK
Total rise	4 100 = 24 × 170.833	(-)6.100 → (-)2.000	closes exactly

```

FLIGHT -- waist 200
  theta = arctan(170.833 / 280) = 31.4 deg,  cos theta = 0.8535
  waist self    0.200 x 25 / 0.8535          = 5.858 kPa
  steps         0.5 x 0.170833 x 25   IS 456 Cl. 33.2 = 2.135 kPa
  finishes      = 1.000 kPa
  live          IS 875 (Pt 2), plant + escape route = 5.000 kPa
  total 13.99  ->  wu = 1.5 x 13.99          = 21.0 kPa

  Leff = going + min(landing/2, 1000) each end  IS 456 Cl. 33.1(b)
        = 1960 + 600 + 600                      = 3160 mm
  d = 200 - 30 - 6                                = 164 mm
  M = wu . Leff^2 / 8                              = 26.2 kNm/m
  Ast,req 380 ;  Cl. 26.5.2.1 min 0.12 % x 200 = 240

ADOPTED  T12 @ 150 MAIN (754)  ->  Mu 50.3,  UTILISATION 52 %
         T10 @ 200 DISTRIBUTION (393)

DEFLECTION  Leff/d = 19.3 ; basic 20 ; fs = 0.58 x 500 x 380/754 = 146
             pt 0.46 % ; MF ~ 1.9 -> permissible 38 > 19.3      OK
SHEAR  V = 33.2 kN/m ; tau_v 0.202 ;
        tau_c 0.479 x k 1.20 (Cl. 40.2.1.1)          = 0.575  OK, no links
TOP STEEL  T12 @ 150 for 0.25 Leff = 800 into the flight, anchored Ld 480

LANDINGS L1 / L2 / ARRIVAL -- 200 thk, spanning ACROSS the shaft
  self 7.50 + finishes 1.50 + live 7.50 + flight reaction 27.70 = 44.20 kPa
  Leff = 2800 clear + 200 bearing      Cl. 22.2(a)              = 3000 mm
  M = 44.2 x 3.000^2 / 8                  = 49.7 kNm/m
  Ast,req 745 ;  T12 @ 150 gives 754 = 100 %  -- TOO TIGHT

ADOPTED  T12 @ 125 BOTTOM (905)  ->  Mu 59.5,  UTILISATION 84 %
         T12 @ 125 TOP for 900 from each support
         T10 @ 200 distribution

DEFLECTION  3000/164 = 18.3 ; fs 239 ; pt 0.552 % ; MF ~ 1.35
             -> 27 > 18.3                                     OK
SHEAR  V 66.3 kN/m ; tau_v 0.404 < tau_c 0.519 x 1.20 = 0.622  OK

```

T12 @ 150 on the landings gives exactly 100 % utilisation, and 100 % is not a design. It was tightened to @ 125. That is the kind of decision that never appears in a summary and always appears in a site query.

Compatibility notes. The arrival landing at (-)6.100 is the mat surface — there is no separate slab. Flight 3 lands directly on the 900 mm pressure-slab pad at (-)2.000. The landings prop W6 and W7 at (-)4.733 and (-)3.367, **but only over their 1 200 mm depth — and the 400 mm wall design in Part 9.2 does NOT rely on that prop.**

10.2 Entry (approach) stairwell

STATUS: OUTSIDE the protective boundary, NOT blast rated, and expected to be LOST in the design event. Designed to IS 456 with normal partial factors.

```

FLIGHT -- 12R @ 166.6667 / 300, 1500 wide, waist 250
theta = 29.05 deg, cos theta = 0.8742
waist 7.150 + steps 2.083 + finishes 1.000 + live 5.000 = 15.233
                                -> wu      = 22.85 kPa
Leff = 3300 + 750 + 750          IS 456 Cl. 33.1(b) = 4800 mm
d = 250 - 30 - 8                 = 214 mm
M = 22.85 x 4.800^2 / 8           = 65.8 kNm/m
Ast,req 744 ; min 0.12 % x 250 = 300

ADOPTED T16 @ 200 MAIN (1005) -> Mu 87.3, UTIL 75 %, x/d 0.162
        T10 @ 200 DISTRIBUTION (393)
        T16 @ 200 TOP for 1200 into the flight, anchored Ld 640

DEFLECTION 4800/214 = 22.4 ; basic 20 ; fs 215 ; pt 0.470 %
MF ~ 1.5 -> 30                                OK
SHEAR V 54.8 kN/m ; tau_v 0.256 ; tau_c 0.484 x k 1.10 = 0.532 OK

SIDE WALLS 250 -- retained 2.9 m, propped by the roof and the raft
sigma_h at base = 0.5 x 20 x 2.900 + 0.5 x 10      = 34 kPa
M ~ w L^2 / 12 on an equivalent 20 kPa UDL         = 14.0 kNm/m
d = 250 - 50 - 6 = 194 ; Ast,req 168
Cl. 32.5(a) min 0.0012 x 250 = 300 ; Cl. 32.5(b) 0.0020 x 250 = 500
Cl. 32.5(c) two curtains required (t > 200)        OK
ADOPTED T12 @ 200 EF EW (565/face)
-- MINIMUM STEEL GOVERNS, 31 % utilised
SHEAR V ~ 49 kN/m ; tau_v 0.253 < tau_c 0.484      OK

RAKING ROOF 250 -- 1500 clear
self 6.25 + waterproofing/screed 2.0 + earth lap 5.4
+ IMPOSED 20                                     = 33.65
                                -> wu      = 50.5 kPa
(20 kPa imposed taken DELIBERATELY, for a stray vehicle on the berm)
M = w L^2 / 12                                    = 9.5 kNm/m
d = 250 - 30 - 6 = 204 ; Ast,req 108 ; min 300 GOVERNS
ADOPTED T12 @ 200 EF EW -> Mu 48.2, 20 % utilised
SHEAR V 38 kN/m ; tau_v 0.186                    OK

TOP LANDING 250 (0.000)
wu 54.9 kPa ; Leff 1750 ; M 21.0 kNm/m ; Ast,req 229
ADOPTED T12 @ 200 EF EW (565) -> Mu 50.6, 41 % utilised

PLATFORM 250 ((-)2.000, on fill) ADOPTED T12 @ 200 EF EW, nominal
HEADWALL 250 ADOPTED T12 @ 200 EF EW
Door lintel over 1000 clear: 250 x 350, 3-T12 top + 3-T12 bottom, T8 @ 150
STEPPED RAFT 300 on compacted fill ADOPTED T12 @ 200 EF EW

```

10.2.1 The opening corner — the detail most often got wrong

At the TOP of the flight the tension face turns through 209 degrees.

A bar bent round that corner has its bend resultant directed OUT of the concrete: it spalls the cover and the bar loses its anchorage.

MAIN BARS SHALL NOT BE BENT ROUND IT.

Each layer is continued STRAIGHT, CROSSED, and anchored Ld = 640 into the OPPOSITE face, plus a U-bar T16 @ 200 across the corner.
[SP 34:1987 Cl. 5.5 and standard detailing practice]

At the FOOT of the flight the corner CLOSES (151 degrees) -- bars may turn normally.

This is a geometry problem, not a preference. An obtuse re-entrant corner in a tension face puts the resultant of the bend into the cover, and the cover has nothing to push against. Two corners, two different answers, and the drawing shows both.

10.3 Entry headhouse — roof, 500 thk — 90 %, the highest utilisation in the project

```

w = 383 (blast) + 12.5 (self) + 1.0 (SIDL)          = 396.5 kPa
Panel 4000 x 5000 clear, monolithic with 400 walls on all four
sides, ly/lx = 1.25
d = 500 - 75 - 10                                    = 415 mm

```

```

Mu,lim = 0.133 x 43.75 x 1000 x 415^2 = 1002 kNm/m

THREE ANALYSES, AND THE MOST CONSERVATIVE IS ADOPTED
| # | Method | M | |
| 1 | One-way fixed-fixed strip, Mp = w . Ln^2 / 16 | 396.5 | <- ADOPTED
| 2 | IS 456 Table 26 Case 1, alpha_x- = 0.045 at 1.25 | 285.5 |
| 3 | Yield line, FIXED on 4 edges, isotropic | 162.2 |

Method 3 formula: wu = 48 m / [ a^2 ( sqrt(3 + (a/b)^2) - a/b )^2 ]
VALIDATION: as b -> infinity, 48/3 = 16 = the fixed-fixed strip w L^2/16
              (the 24-coefficient version gives 8 = SIMPLY SUPPORTED w L^2/8)

Ast,req (one-way basis) = 1879 mm2/m
Cl. 26.5.2.1 minimum 0.12 % x 500 = 600 mm2/m

ADOPTED T20 @ 150 EF EW = 2094 mm2/m -> Mu = 438.5 kNm/m
UTILISATION 90 % one-way | 65 % Table 26 | 37 % yield line
xu 72.3 -> x/d = 0.174 << 0.46

SHEAR *** NEW AND MANDATORY -- NOT IN THE PHASE 1 REPORT ***
V at d = 396.5 (2.000 - 0.415) = 628.4 kN/m
tau_v 1.514 | tau_c 0.502 (pt 0.505 %) | tau_c,max 3.70 OK
(tau_c read at the STATIC M35 value -- IS 4991 Cl. 10.3.1.1)
Vus 420.0 kN/m ; Asv/sv 2.327 -> T12 4-leg at 194
Cl. 26.5.1.5 limit min(0.75d = 311, 300) = 300
ADOPTED T12 4-LEG @ 175 IN THE END 1200 EACH SIDE; @ 250 ELSEWHERE

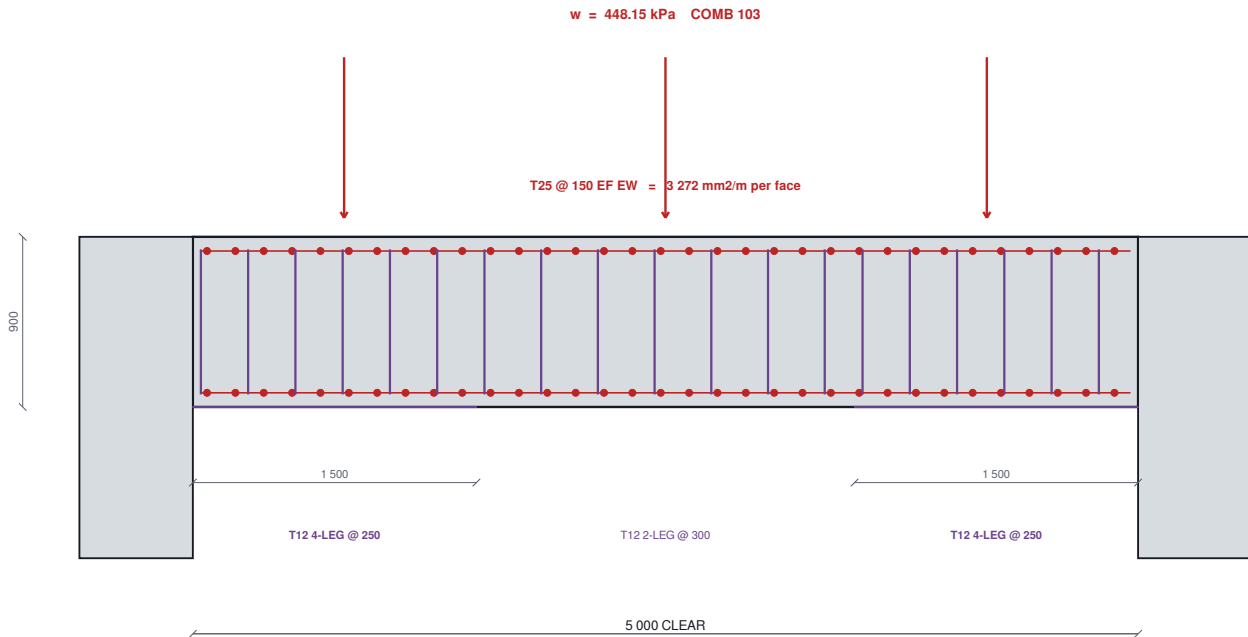
DETAILING
No opening in the headhouse roof (the stair void is in the FLOOR).
Corner torsion steel (Cl. D-1.8) NOT required -- all four edges carry
full T20 @ 150 hogging steel continuous into the walls.
Haunch 300 x 300 at all four wall-roof junctions, diagonal T16 @ 150.
Top steel continuous over every support, anchored Ld = 800 into the wall.

```

Three analyses, and the reason the crudest one is adopted. The one-way strip is the most conservative of the three by a factor of 2.4 against the yield-line value. A yield-line solution is only as good as the assumption that all four edges are genuinely fixed and that the panel can develop a full mechanism; under a blast load on a structure whose neighbour is expendable, that is not a bet worth taking for a saving of one bar size. **The project takes the strip, prints all three, and says which one it used.**

The corrected yield-line coefficient, recorded as an error. The yield-line value was once computed as 324.4 kNm/m using the **simply-supported** coefficient 24 instead of the **fixed-four-edges** coefficient 48. It is **162.2**. The validation that catches it is in the formula itself: as $b \rightarrow \infty$, $48/3 = 16 = wL^2/16$, which is the fixed-fixed strip. **No reinforcement changed, because the one-way value was the one adopted and nothing depended on the yield-line number** — but the error is recorded rather than quietly fixed.

And the shear governs the detailing. $\tau_v = 1.514$ against $\tau_c = 0.502$ is a factor of three, so the four-leg links through the end bands are **mandatory**, not nominal. A blast-loaded flat slab that passes in flexure and fails in shear has no ductility to claim, and the ductility is what the whole dynamic load factor rests on (Part 2.2.3).

PRESSURE SLAB 900 thk - THE GOVERNING ELEMENT 1:40

Aspect 20.8 / 5.0 = 4.2, so the slab spans ONE-WAY across the 5 m width and lengthening the box adds no roof moment. Mp 700.2 against Mu 1 362.4 kNm/m - 51 % utilised, $x_u/d = 0.139$.

Figure 29 The pressure slab, 900 thick - the governing element.

10.4 Headhouse walls HW1-HW4 — 400 thk — 59 %

	CASE A (report)	CASE B (ADOPTED)
Lateral pressure	113 kPa drag	383 kPa full envelope
$M_p = w \times 2.400^2 / 16$	40.7 kNm/m	137.9 kNm/m
$A_{st, req}$	221 mm ² /m	766 mm ² /m
Cl. 32.5(a) min vertical	480 mm ² /m -- governs	480
Cl. 32.5(b) min horizontal	800 mm ² /m -- governs	800
Utilisation on T16@150 EF EW	17 %	59 %
Shear V at d	97.0 kN/m	328.6 kN/m
$\tau_{u,v}$	0.283	0.961
$\tau_{u,c}$ (pt 0.392 %)	0.444	0.444
Verdict	NO LINKS	LINKS REQUIRED

$d = 400 - 50 - 8 = 342$; $\mu_{u, lim} = 681 \text{ kNm/m}$
 ADOPTED T16 @ 150 EF EW = 1340 mm²/m -> $\mu_u 235.2$, UTIL 59 %, $x_u/d 0.135$
 $V_{us} 176.9 \text{ kN/m}$; $A_{sv}/s_v 1.189$ -> T12 4-leg at 381
 Cl. 26.5.1.5 limit min(0.75d = 256, 300) = 256 GOVERNS
 ADOPTED T12 4-LEGGED LINKS @ 250 c/c THROUGHOUT ALL FOUR WALLS

Two-way check (information only, fixed on 4 edges):

HW1/HW2 4000 x 2400 -> $m = 69.9 \text{ kNm/m}$ (one-way strip 137.9)
 HW3/HW4 5000 x 2400 -> $m = 79.8 \text{ kNm/m}$ (one-way strip 137.9)

AXIAL Roof reaction, two-way (Cl. 24.5):

HW3/HW4 trapezoid 2379 kN / 5.0 m = 476 kN/m
 HW1/HW2 triangle 1586 kN / 4.0 m = 396 kN/m
 One-way bound: $396.5 \times 2.0 = 793 \text{ kN/m}$; self 29 kN/m
 Worst N = 822 kN/m -> $f = 2.06 \text{ N/mm}^2$
 = 12 % of 0.4 $f_{ck, dyn}$ OK

*** PRESSURE ACTS FROM INSIDE TOO. The inner security door is NOT blast rated and the entry stairwell is expected to be lost, so the headhouse FILLS and the walls are pushed OUTWARDS. THE WALLS ARE REINFORCED SYMMETRICALLY FOR 383 kPa EITHER WAY -- a stated requirement, not an accident of detailing. ***

DOOR OPENING 900 x 2100 IN HW2 -- and a geometric trap

Interrupted steel 1340 x 0.900 = 1206 mm²/face -> 603 each jamb
 ADOPTED 2 No. T20 EACH JAMB EACH FACE (628), anchored $L_d 800$ beyond

Door head at (-)2.000 + 2.100 = +0.100 ; roof soffit +0.400
 -> ONLY 300 mm OF WALL ABOVE THE DOOR -- too shallow for a lintel.

The 300 wall and the 500 roof act TOGETHER as an 800 mm deep edge band:
 $w = 383 \times 2.100/2 = 402 \text{ kN/m}$; $M = w L^2/12 = 27.1 \text{ kNm}$; $V = 181 \text{ kN}$

$d = 800 - 40 - 8 - 10 = 742$; $\tau_{uv} = 0.610 \text{ N/mm}^2$
 ADOPTED 400 x 800 EDGE BAND: 4-T20 TOP + 4-T20 BOTTOM,
 T12 4-LEG @ 150, over the opening and 600 each side

Door frame cast in and WELDED to the cage (EMP), even though the leaf is a security door and is not blast rated.

10.4.1 HW3 — the wall that has nothing under it

The pressure slab spans one-way in Y over 5000 clear. HW3 runs IN the Y direction, so its load is carried by a slab strip acting as a beam.

$$b_{\text{eff}} = 0.400 + 2 \times 1.05 \quad (45 \text{ deg spread through the 900 slab}) = 2.5 \text{ m} \quad [A]$$

CASE 1 -- two-way roof distribution (IS 456 Cl. 24.5, correct)

Line load 476 (trapezoid) + 29 (self)	= 505 kN/m
Equivalent UDL = $505/2.5 + 22.5 + 2.0$	= 226 kPa
$M = w \cdot l_n^2 / 16 = 226 \times 25 / 16$	= 354 kNm/m
Against the slab's 1362 kNm/m capacity	= 26 % OK

CASE 2 -- one-way roof bound

Line load 793 + 29 = 822 kN/m $\rightarrow$ 353 kPa $\rightarrow$ M = 552 kNm/m	= 41 % OK
---	-----------

ADOPTED No slab change needed. DETAILED AS A LINE LOAD, NOT AS A SUPPORT.
 Robustness: 4 No. T25 ADDITIONAL TOP AND BOTTOM in a 1200 mm band
 beneath HW3, lapped 1250 (50 phi) into the main mesh.

$b_{\text{eff}} = 2.5 \text{ m}$ is an assumption, and it is accepted on the margin. The strip is at 26 % on the correct two-way distribution and 41 % on the conservative one-way bound, so more than half the capacity is spare and b_{eff} would have to be badly wrong to matter. Declared as assumption A11 rather than absorbed into the calculation.

10.5 Escape shaft head hatches

The protective boundary is defined as the blast doors plus the perimeter walls, the mat and THE PRESSURE SLAB. ESC 1 and ESC 2 are 1 400 mm dia bores straight through the pressure slab. They begin in bay 1 — inside the gas-tight envelope — and in bay 8, and they finish at grade.

Each shaft head is therefore a 1.54 m² hole in the protective boundary, and the only thing that can close it is a hatch.

The EMP penetration register requires a bonded conducting hatch at each head — but that is an electromagnetic requirement, and it says nothing about what the head has to resist structurally.

The head is part of the protective boundary and takes the full 383 kPa design blast. There is no other defensible reading: a closure in the same plane as the pressure slab, over a bore that runs straight into the gas-tight envelope, cannot be designed to anything less than the slab around it.

total force on the leaf	$383 \times \pi \times 0.700^2$	= 589.6 kN
$M = w a^2 (3 + \nu) / 16$		= 38.71 kNm/m
V at the seating = $w a / 2$		= 134.1 kN/m
flat Fe250 plate	$t = \sqrt{6M / \sigma} = 30.5 \rightarrow$ say 32 mm	
mass of a 1600 dia x 32 leaf		= 505 kg

505 kg is the point, not the thickness. Half a tonne cannot be lifted by a person escaping up a 6.8 m ladder in the dark. That is why the answer is not a flat plate.

ADOPTED: a ribbed steel weldment — 1 600 dia, 12 mm face, 8 No. radial ribs 150 x 10 and a 150 x 12 perimeter ring — on a steel seating ring cast into the 250 mm collar, bearing 150 mm all round, with four quarter-turn dogs so the leaf resists UPLIFT as well as downward pressure (the negative phase and the rebound both lift it), and counterbalanced or spring-assisted, openable from inside by one person without a key or a tool. Indicative leaf mass $\approx$ 322 kg — still a mechanically assisted item.

Not designed and not invented [N]: the rib proportioning (an orthotropic plate problem — the flat-plate demand above is exact and is the brief), the counterbalance mechanism, and every EMP figure.

The structural hatch and its EMP bonding are one item, and they must be procured as one. Bonding is not a finish applied afterwards — it is electrical continuity between the leaf, the seating ring and the collar reinforcement, and it has to be designed into the weldment and the seat. An opening whose structure and whose bonding sit in different packages is an opening that ends up with neither. EM-V6 stays open for the EMP half, and Part 12 is where both halves are stated together.

10.6 The escape shaft ladders

A **6.250 m** climb out of Bay 1 and a **6.800 m** climb out of Bay 8 need a designed way up, and the shafts get one.

One ladder type serves both shafts:

```
20 mm dia galvanised MS rungs, 400 clear width, EQUAL PITCH within each shaft
    ESC 1    297.6 mm over 21 spaces
    ESC 2    295.7 mm over 23 spaces
>= 200 behind the rung    .    >= 750 clear climbing space in front
2 No. 50 x 10 galvanised flat stringers
cast-in lugs to the 250 collar and the roof-slab bore,
expansion-anchored brackets at 1.5 m over the lower 3.200 m
grab rails 1100 above the head
```

IS 3696 (Part 2) and NBC 2016 Part 4 are named **by title only** — neither is in the workspace and no unconfirmed clause is cited.

The ladder is designed; the climb is not solved. Three things are still missing, and they are stated rather than detailed away:

- **no fall-arrest** — deferred, and it is a client decision rather than a detailing one;
- **no rest platform** — a 1 400 mm bore cannot take one without blocking the escape;
- **the injured-person question** — **a vertical ladder cannot pass a stretcher**, and whether a casualty is expected to use a shaft at all is a client question.

PART 11 — THE SENTRY POST

11.1 Why it is not blast designed, and what that decision cost

The sentry post is NOT designed for the blast, and that is a RECORDED decision, not an omission.

Reflected pressure on the 4.0 x 6.7 m face
 $p_r \times A_e = 1366 \text{ kPa} \times 26.8 \text{ m}^2 = 36\,600 \text{ kN}$ (about 3730 t)
 Hardening would need roughly 700 mm of RC on all four above-ground faces.

Not justified for a peacetime observation post. The post is declared expendable, and the shelter's protection does not depend on it in any way.

All member design uses the STAAD base shear $V_b = 73.18 \text{ kN}$, not the hand-calculated 59.3 kN (Part 7.8.2).

THE SENTRY POST - A TWO-STOREY RC FRAME ON ITS OWN FOOTINGS, DELIBERATELY NOT BLAST DESIGNED 1 : 90

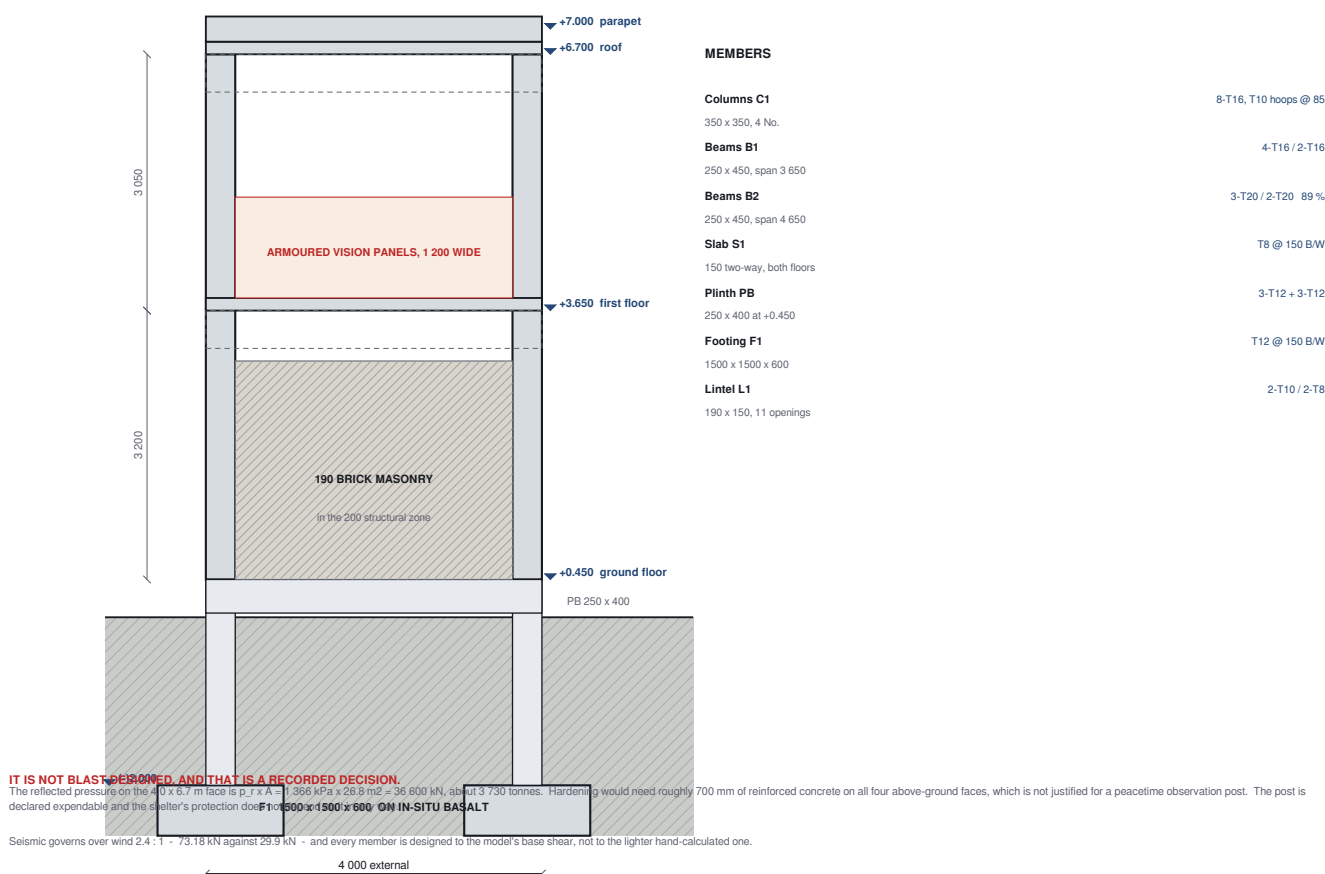


Figure 30 The sentry post - a two-storey RC frame on its own footings, deliberately not blast designed.

11.1.1 Seismic member forces — portal method at $V_b = 73.18 \text{ kN}$

```

Q_roof = 73.18 x (315.9 x 6.25^2) / Sigma = 59.51 kN
Q_floor = 73.18 x (276.8 x 3.20^2) / Sigma = 13.67 kN

Column shear, ground storey = 73.18 / 4 (= STAAD Fx 18.295) = 18.30 kN
Column shear, first storey = 59.51 / 4 = 14.88 kN
Column moment, ground = 18.30 x 3.200 / 2 = 29.27 kNm
Column moment, first = 14.88 x 3.050 / 2 = 22.69 kNm

Beam moment, first-floor joint (corner joint, one beam) = 29.27 + 22.69 = 51.96 kNm
Beam moment, roof joint = 22.69 kNm

Axial from overturning -- READ DIRECTLY FROM THE STAAD REACTIONS:
EQ+X Fy = +/- 36.127 kN EQ+Z Fy = +/- 27.891 kN
  
```

-> 36.127 GOVERNS

11.1.2 The frame analysis assumption, stated rather than hidden

Single-bay portal in each direction. Beams are NOT fully fixed; the joint rotates. Slope-deflection with symmetric loading:

$$\begin{aligned} I_{\text{beam}} &= 250 \times 450^3 / 12 &= 1.8984\text{e}9 \text{ mm}^4 \\ I_{\text{col}} &= 350^4 / 12 &= 1.2505\text{e}9 \text{ mm}^4 \\ K_c \text{ (first-floor joint)} &= 4EI/3200 + 4EI/3050 &= 3.203\text{e}6 \text{ . E} \\ M_{\text{support}} &= -FEM + (2 E I_b / L_b) \cdot \theta, \\ \theta &= FEM / (2 E I_b / L_b + K_c) \end{aligned}$$

Beam	FEM	2EI/L	Reduction	M_support	M_midspan
B1, 1.5(DL+LL)	39.49	1.040e6 E	x 0.755	29.81	30.90
B1, DL only	22.53	--	x 0.755	17.01	--
B2, 1.5(DL+LL)	70.60	0.8165e6 E	x 0.797	56.26	49.64
B2, DL only	39.21	--	x 0.797	31.25	--

11.2 Slab S1 — 150 thk two-way

Cover 30 (Table 16, moderate) ; T8 bars
 $dx = 150 - 30 - 4 = 116 \text{ mm}$
 $dy = 150 - 30 - 8 - 4 = 108 \text{ mm}$
 Clear spans 3400 and 4400

Effective span, IS 456 Cl. 22.2(a) -- the LESSER of clear + d, or c/c:
 $lx = \min(3400 + 116, 3650) = 3516$
 $ly = \min(4400 + 108, 4650) = 4508$
 $ly / lx = 1.282 < 2 \rightarrow \text{TWO-WAY} \quad \text{Cl. 24.4 / Annex D}$

MOMENT COEFFICIENTS -- IS 456 Table 26 (Annex D-1.1) Case 9: four edges discontinuous but RESTRAINED, with corner torsion steel per Cl. D-1.8
 -> Table 26, NOT Table 27.
 $\alpha_x = 0.0777$ (interpolated at 1.282) $\alpha_y = 0.056$

$w_u = 1.5 (4.750 + 3.000) = 11.625 \text{ kPa}$
 (the FIRST FLOOR governs; the roof is 10.125)

$M_x = 0.0777 \times 11.625 \times 3.516^2 = 11.17 \text{ kNm/m}$
 $M_y = 0.0560 \times 11.625 \times 3.516^2 = 8.05 \text{ kNm/m}$
 $A_{st, req} \times 229 \mid y \ 176$
 Cl. 26.5.2.1 min 0.12 % x 150 = 180 mm²/m
 Cl. 26.5.2.2 max bar dia D/8 = 18.75 OK
 Cl. 26.3.3(b) max spacing 3d or 300 OK

ADOPTED T8 @ 150 c/c BOTH WAYS BOTTOM = 335 mm²/m
 $M_u = 16.09 \text{ (x)} / 14.92 \text{ (y)} \text{ kNm/m} \rightarrow \text{UTILISATION } 69 \% / 54 \%$
 $x_u = 13.5 \rightarrow x/d = 0.116 / 0.125$

DEFLECTION $lx/dx = 3516/116 = 30.3$; basic taken as SIMPLY SUPPORTED = 20
 $fs = 0.58 \times 500 \times 229/335 = 198$; $pt = 0.289 \%$; MF (Fig. 4) ~ 1.68
 Permissible $20 \times 1.68 = 33.6 > 30.3$ OK
 (as continuous, basic = 26 and the margin doubles -- it passes on the conservative assumption, so that assumption is the one used)

SHEAR $V_u \sim w_u \cdot lx / 3 = 13.6 \text{ kN/m}$; $\tau_{u,v} 0.117$;
 $\tau_{u,c} 0.390 \times k \ 1.30 \text{ (D} \leq 150) = 0.507 \text{ OK}$

Detailing — IS 456 Annex D:

Clause	Requirement	Adopted
D-1.2 / D-1.3	Middle strip = $\frac{3}{4}$ width, edge strips $\frac{1}{8}$ each side; Table 26 moments apply to MIDDLE STRIPS ONLY	—
D-1.4	50 % of midspan bottom steel to within 0.1 L of a discontinuous edge; remainder curtailed at 0.25 L	—
D-1.6	At every discontinuous edge, top steel = 50 % of the midspan bottom steel, extending 0.1 L = 400 mm into the span	T8 @ 300 TOP, 400 mm band, all four edges
D-1.7	Edge strips: minimum reinforcement only	—
D-1.8	Torsion at corners — both edges discontinuous at all four corners $\rightarrow$ steel = $\frac{3}{4}$ of the midspan area, in four layers (2 top + 2 bottom, each way), over $l_x/5 = 700 \text{ mm}$ each way	T8 @ 200, 4 LAYERS, 700 x 700 AT EACH CORNER

The torsion mats are a HOLD POINT for reinforcement inspection. Table 26 is valid **only because they are provided**. Omit them and the slab must be reassessed on Table 27, which gives larger coefficients. **This is why the drawing carries a panel headed "why Table 26 and not Table 27".**

11.3 Beam B1 — 250 × 450, span 3 650

```
IS 13920:2016 section limits
b >= 200                                250            OK
b/D = 0.556 >= 0.3                      OK
clear span / D = 3300/450 = 7.33 >= 4    OK

d = 450 - 30 - 8 - 8                    = 404 mm
Mu,lim = 0.133 x 30 x 250 x 404^2        = 162.8 kNm

| Combination                            | M_support hogging |
| 1.5(DL + LL)                          | 29.8 |
| 1.2(DL + LL + EL) = 1.2(19.87 + 51.96) | 86.2 |
| 1.5(DL + EL) = 1.5(17.01 + 51.96)      | 103.5 <- GOVERNS |
| 0.9 DL + 1.5 EL                       | 93.2 |

Mu = 103.5 < 162.8 -> singly reinforced ; Ast,req = 661 mm2
ADOPTED 4-T16 TOP AT SUPPORTS (804) -> Mu = 122.6 kNm, UTIL 84 %, x/d 0.321

BOTTOM
Reversal 1.5(17.01 - 51.96) = -52.4 kNm sagging -> Ast,req 313
IS 13920 Cl. 6.2.3: positive steel at the joint face >= 50 % of negative
= 0.5 x 804 = 402 mm2 <- GOVERNS
Midspan sagging 30.9 kNm -> Ast,req 181
IS 13920 Cl. 6.2.1(b) rho_min = 0.24 sqrt(fck)/fy
= 0.00263 x 250 x 404 = 266
IS 456 Cl. 26.5.1.1 As,min = 0.85 b d / fy = 172
IS 13920 Cl. 6.2.2 rho_max = 0.025 = 2525 OK
Cl. 6.2.4 top or bottom >= 25 % of max at the joint face = 201 OK
ADOPTED 2-T16 BOTTOM CONTINUOUS (402) -> Mu = 66.0 kNm, x/d 0.160

SHEAR -- CAPACITY DESIGN, IS 13920 Cl. 6.3.3
Vu = 1.2(D+L) . L/2 +/- 1.4 (Mu^As + Mu^Bh) / L
Unfactored gravity total = 54.29 (UDL) + 25.81 (triangle) = 80.1 kN
1.2 x 80.1 / 2 = 48.1 kN
1.4 x (122.6 + 66.0) / 3.650 = 72.3 kN
Vu = 120.4 kN
tau_v = 1.192 N/mm2 ; tau_c,max (Table 20, M30) 3.50 OK

*** IS 13920 Cl. 6.3.4: EQ share 72.3 / 120.4 = 60 % >= 50 %, no axial
-> tau_c TAKEN AS ZERO in the plastic hinge region ***

Asv/sv = 120.4e3 / (0.87 x 500 x 404) = 0.685 -> T8 2-leg at 147
Cl. 6.3.5.1 hinge region (2d = 810 from each face):
sv <= min(d/4 = 101, 8 x 16 = 128), need not be < 100 -> 100 mm
Cl. 6.3.5.2 elsewhere: sv <= d/2 = 202 mm

ADOPTED T8 2-LEGGED HOOPS @ 100 c/c OVER 2d = 810 FROM EACH FACE;
T8 @ 150 c/c ELSEWHERE. First hoop within 50 mm of the column face.

DEFLECTION l/d = 3650 / 404 = 9.0 << 20 OK
```

Capacity design, in one sentence: the beam is designed for the shear that accompanies its own flexural failure, not for the shear from the analysis. That is what $1.4 (Mu^As + Mu^Bh) / L$ means — the hinges form at over-strength and the shear follows. **And when the earthquake share exceeds half the total, IS 13920 Cl. 6.3.4 requires τ_c to be taken as zero:** a hinge that has cycled has lost its aggregate interlock, so the links carry everything.

11.4 Beam B2 — 250 × 450, span 4 650 — 89 %

```
| Combination                            | M_support hogging |
| 1.5(DL + LL)                          | 56.3 |
| 1.2(DL + LL + EL) = 1.2(37.51 + 51.96) | 107.4 |
| 1.5(DL + EL) = 1.5(31.25 + 51.96)      | 124.8 <- GOVERNS |
| 0.9 DL + 1.5 EL                       | 106.1 |

Mu = 124.8 < 162.8 OK ; Ast,req = 822 mm2
ADOPTED 3-T20 TOP AT SUPPORTS (942) -> Mu = 139.8 kNm, UTIL 89 %, x/d 0.376
*** 3-T20 CHOSEN OVER 5-T16: five T16 bars need 256 mm in a 250 mm
```

```

        wide beam. That is exactly the check that becomes a site RFI
        if it is skipped. ***

BOTTOM    Cl. 6.2.3 -> 0.5 x 942 = 471 ; midspan 49.6 kNm -> Ast,req 297
ADOPTED   2-T20 BOTTOM CONTINUOUS (628) -> Mu = 98.9 kNm, 50 %, x/d 0.250

SHEAR     Unfactored gravity (UDL-equivalent) = 26.115 x 4.650 = 121.4 kN
          1.2 x 121.4 / 2 = 72.8 kN
          1.4 x (139.8 + 98.9) / 4.650 = 71.9 kN
          Vu = 144.7 kN
          tau_v = 1.433 ; tau_c,max 3.50 OK

EQ share 71.9 / 144.7 = 49.7 % -- a hair under the Cl. 6.3.4 trigger.
*** TAKEN AS TRIGGERED (tau_c = 0). A design must not turn on the
    third decimal place of a ratio. ***

Asv/sv = 0.823 -> T8 2-leg at 122 ; hinge spacing 100 governs
Outside the hinge V falls to ~124 kN and tau_c = 0.641 (pt 0.933 %)
may be used -> Asv/sv 0.337 -> T8 at 298
ADOPTED   T8 @ 100 OVER 810 FROM EACH FACE ; T8 @ 150 ELSEWHERE
DEFLECTION l/d = 4650 / 404 = 11.5 << 20 OK

```

11.5 Column C1 — 350 × 350

```

SLENDERNESS
  clear height, ground = 3650 - 450 - 450 (beam depth) = 2750 mm
  Effective length factor 1.2 (Table 28, unbraced, both ends restrained)
  lex = 3300 ; lex/D = 9.43 < 12 -> SHORT COLUMN Cl. 25.1.2 OK

MINIMUM ECCENTRICITY    Cl. 25.4
  emin = 1/500 + D/30 = 5.5 + 11.67 = 17.2, but >= 20 -> 20 mm
  Mmin = 276.8 x 0.020 = 5.5 kNm << applied OK

AXIAL, per column, unfactored: DL 148.4 kN, LL 19.1 kN,
                               EQ axial +/- 36.127 kN
  1.5(DL + LL) = 251.3 kN
  1.5(DL + EL) = 276.8 kN <- max
  1.5(DL - EL) = 168.4 kN <- min

LONGITUDINAL STEEL    Cl. 26.5.3.1: min 0.8 % = 980 ; max 6 % = 7350
ADOPTED   8-T16 = 1608 mm2 -> p = 1.31 %, bars on all four faces
  Cover 40 (Cl. 26.4.2.2) ; d' = 40 + 8 + 8 = 56 ; d'/D = 0.16

UNIAXIAL CAPACITY -- computed from the IS 456 Cl. 38.1 stress block and the
Fe500 design curve (Fig. 23B), NOT read off a chart:
  | Pu (kN) | Mu1 (kNm) |
  | 168.4   | 105.8     |
  | 251.3   | 111.7     |
  | 276.8   | 112.9     |

BIAXIAL -- IS 456 Cl. 39.6
  Puz = 0.45 fck Ac + 0.75 fy Asc
        = 0.45 x 30 x 120892 + 0.75 x 500 x 1608 = 2235 kN
  Pu / Puz = 276.8 / 2235 = 0.124 < 0.2 -> alpha_n = 1.0
  Worst case, EQ in Z with 30 % in X IS 1893 Cl. 6.3.2.2:
    Mux = 25.6 kNm ; Muy = 66.8 kNm
    25.6/112.9 + 66.8/112.9 = 0.227 + 0.592 = 0.819 <= 1.0 OK

STRONG COLUMN - WEAK BEAM    IS 13920 Cl. 7.2.1, Sigma Mc >= 1.4 Sigma Mb
  | Joint | SigmaMb | 1.4 SigmaMb | SigmaMc | Verdict |
  | First floor, X (B1) | 122.6 | 171.6 | 112.9+101=213.9 | OK |
  | First floor, Z (B2) | 139.8 | 195.7 | 213.9 | OK |
  | Roof, Z (B2 roof, 2-T20) | 98.9 | 138.5 | 101 | OK, marginal |

TRANSVERSE REINFORCEMENT
  General, Cl. 26.5.3.2: tie dia >= max(6, phi/4) = 8 ;
  spacing <= min(350, 16 phi) = 256, 300 = 256
  IS 13920 Cl. 7.4: hoops over the full length at <= half the least
  dimension = 175
ADOPTED   T8 HOOPS @ 150 c/c outside the confining zone

SPECIAL CONFINING REINFORCEMENT    IS 13920 Cl. 8.1
  lo >= max(350, clear height/6 = 458, 450) = 500 mm
  spacing <= 0.25 x 350 = 87.5, but 75 <= s <= 100 = 85 mm
  Ash = 0.18 s h (fck/fy) [ (Ag/Ak) - 1 ]
    h = 350 - 2 x 40 = 270 ; Ag = 122500 ; Ak = 270^2 = 72900
    = 0.18 x 85 x 270 x (30/500) x (1.680 - 1) = 168.5 mm2
  minimum = 0.05 x 85 x 270 x 30/500 = 68.9 mm2

```

ADOPTED T10 HOOPS + ONE CROSS-TIE EACH WAY @ 85 c/c
 (3 legs x 78.5 = 235.5 mm² OK)
 over 500 mm from every joint face, top and bottom of every column,
 AND THROUGH THE JOINT Cl. 8.2

DRIFT -- IS 1893 Cl. 7.11.1
 limit 0.004 x storey height = 12.8 mm (ground), 12.2 mm (first)
 Calculated drift under 3 mm. Not critical, but CHECKED.

The roof joint at 101 against 138.5 is marginal and is printed as marginal. Strong column-weak beam is the mechanism check that keeps the hinges in the beams; at the roof, where only one column continues above, the margin is naturally smallest. It passes, and it is flagged so that nobody reduces the roof beam's bottom steel without re-running it.

11.6 Isolated footing F1 — 1 500 × 1 500 × 600 on in-situ basalt at (-)2.000

Founded on IN-SITU ROCK, never on backfill. This is the whole reason the sentry post was relocated ≥ 10 m clear of the shelter excavation — and Part 4.5's measured black cotton soil, CH at FSI 60–65 %, is why that insistence is now backed by a measurement rather than by instinct.

SERVICE P = 167.5 (column) + 36.1 (EQ axial) + 33.75 (footing self) = 237.4 kN
 M (base, EQ) = 29.3 kNm
 e = 29.3 / 237.4 = 0.123 m < L/6 = 0.250 -> NO TENSION OK
 q_{max} = P/A (1 + 6e/L) = 105.5 x 1.494 = 157.6 kPa
 vs presumptive SBC 3240 kPa (IS 1904 Table 1) [A] = 4.9 % OK
 vs measured SOAKED basalt 1961 kPa = 8.0 % OK

ULS Governing 1.5(DL + EL): Pu = 1.5(148.4 + 36.127) = 276.8 kN
 Mu = 1.5 x 29.27 = 43.9 kNm
 eu = 0.128 ; qu,max = 276.8/2.25 x (1 + 6 x 0.128/1.5) = 190.0 kPa

FLEXURE Cantilever from the column face = (1500 - 350)/2 = 575 mm
 Mu = 190.0 x 0.575² / 2 = 31.4 kNm/m
 d = 600 - 50 (Cl. 26.4.2.1) - 8 = 542 mm
 Ast,req 136 ; Cl. 26.5.2.1 min 0.12 % x 600 = 720 <- GOVERNS

ADOPTED T12 @ 150 c/c BOTH WAYS BOTTOM (754 mm²/m), x/d 0.056

ONE-WAY SHEAR IS 456 Cl. 34.2.4.1, critical at d from the column face:
 575 - 542 = 33 mm from the edge ; Vu = 6.3 kN/m ; tau_v = 0.012 OK

PUNCHING IS 456 Cl. 31.6.1, critical perimeter at d/2:
 b0 = 4 x (350 + 542) = 3568 ; Vu = 276.8 - 190.0 x 0.892² = 125.6 kN
 tau_v = 0.065 ; ks = (0.5 + beta_c) <= 1.0 = 1.0 ;
 tau_c = 0.25 sqrt(30) Cl. 31.6.3.1 = 1.369
 1.369 >> 0.065 OK

ANCHORAGE Column starters T16: Ld,compression = 37 x 16 = 592 mm
 Available 600 - 50 - 24 = 526 straight + 8 phi bend 128 = 654 mm OK

*** THE 600 mm DEPTH IS GOVERNED BY THE ANCHORAGE OF THE COLUMN STARTERS,
 NOT BY BENDING AND NOT BY SHEAR -- stated so that nobody "optimises"
 it to 350. ***

11.7 Lintel L1 and the wall ties

Masonry infill in a frame needs two things a reinforced panel would not: a lintel over every opening, and a tie detail to the columns. Both are designed here, and **neither adds anything to the frame nor changes any frame member.**

LINTEL L1 -- ONE TYPE OVER EVERY OPENING IN THE SENTRY POST

190 wide x 150 deep, M30 / Fe500, cover 30, bearing 200 each end
 2-T10 bottom . 2-T8 top (hangers) . T6 two-legged links @ 150

Openings served -- ELEVEN in all:
 ground D1 900 and W1 1200
 first D1 900 and the EIGHT 1200-wide vision-panel openings
 The 1200 opening governs; one type covers all.

Design basis — and the reason it does not depend on an unknown.

Effective span = min(clear + d, c/c bearings) = min(1315, 1400)
 IS 456 Cl. 22.2 = 1.315 m

```

*** THE OPENING HEIGHTS ARE NOT STATED ON ANY DRAWING -- WM-V3, open.
L1 is deliberately designed to the bound that does not depend on the
height at all: masonry standing JUST BELOW the 60 deg arching height,
0.866 x 1.315 = 1.139 m
which is the HEAVIEST case any opening height can produce.
Above it, arching relieves the lintel; below it, there is less
masonry. THE DESIGN THEREFORE HOLDS WHATEVER THE HEIGHTS PROVE TO BE. ***

w = 0.190 x 20 x 1.139 = 4.33 kN/m
M = 0.935 kNm ; V = 2.85 kN
Mu = 1.403 kNm against Mu,lim = 0.133 fck b d^2 = 10.03 kNm
                                -> 14 % UTILISED

Ast required 29.5 mm2
IS 456 Cl. 26.5.1.1 minimum 0.85 b d / fy = 37.1 mm2 <- GOVERNS
2-T10 = 157 mm2 provided
tau_v = 0.195 against tau_c ~ 0.56 (M30, pt 0.72 %)
                                -> NO SHEAR STEEL REQUIRED
the T6 @ 150 links are the Cl. 26.5.1.6 nominal minimum
(maximum spacing 324)

*** THE LINTEL CARRIES MASONRY ONLY. The floor and the roof go to beams
B1/B2 at each level, which is the whole point of the frame. ***

```

Wall ties. 6 mm dia MS ties at **every fifth course** (≈ 450 mm) up both column faces, projecting **200 mm** into the bed joint, anchored to the column by a cast-in or drilled-and-grouted 10 mm dowel. **The top course is built tight to the beam soffit and the last joint packed** — because the analysis takes **$R = 3.0$** , the infill is **not** separated from the frame **and must not be**. Separating it would be the $R = 5.0$ special moment frame case, which this project does not claim.

The infill load is not reduced to match the brick. It stays at **13.000 kN/m** on the first-floor beams, where brick gives about 9.88. The lighter value would need a re-run of the seismic model before it could be used, and until that re-run exists the heavier value is what the members are designed to. WM-V6 is open, and it is open as a **confirmation** rather than a risk: every member is over-designed in the direction the correction would move.

PART 12 — OPENINGS, BLAST DOORS, ESCAPE HATCHES AND CLOSURES

A protective envelope is only as good as its weakest closure, and every closure in this structure is a hole somebody had to put there on purpose. The doors are the way in, the shafts are the way out, the void is where the staircase has to go, and the valves are how the air arrives. None of them is a defect. What each one is, is a place where four separate design problems — blast, gas-tightness, electromagnetic continuity and human use — have to be solved by the same piece of steel at the same time.

12.1 The register

EVERY OPENING THROUGH THE PROTECTIVE BOUNDARY

A protective envelope is only as good as its weakest closure. There are nine, and four of them are not blast rated because they are outside the boundary by design.

MARK	WHERE	LEAF / BORE	LEVEL	WHAT IT HAS TO DO
BLAST DOOR 1	W6, 400 thk, opening Y 600-1800	1 200 x 2 100	(-)6.100	≥ 7 bar, gas-tight, rebound-rated
BLAST DOOR 2	W7, 400 thk, opening Y 600-1800	1 200 x 2 100	(-)6.100	≥ 7 bar
D-05	W5, the fire and gas-tight wall	to be designed	(-)6.100	gas-tight; escape route R1 crosses it
ESC 1 HEAD	bay 1, through the pressure slab	1 400 dia bore	+0.150	ribbed steel weldment, 4 quarter-turn dogs
ESC 2 HEAD	bay 8, through the pressure slab	1 400 dia bore	+0.700	as ESC 1; 6.800 m climb
STAIR VOID	pressure slab, bay 7	2 800 x 3 160	(-)2.000	NOT A CLOSURE - it is the route in
INNER SECURITY DOOR	HW2, headhouse north wall	900 x 2 100	(-)2.000	NOT blast rated
ENTRY DOOR	entry stairwell headwall, opens OUTWARD	1 000 x 2 100	0.000	NOT blast rated - expendable
W8 GAPS x4	internal partitions, 110 thk	900 x 2 100 gap	(-)6.100	PERMANENTLY OPEN, no doors, by design

BOTH BLAST DOORS ARE THE PROTECTIVE BOUNDARY, NOT SOMETHING FITTED INTO IT.

They sit in the same plane as W6 and W7, the shaft beyond them equalises to the full incident overpressure, and so the doors and the walls beside them take the same 383 kPa. Each frame is a cast-in steel frame anchored into and welded to the reinforcement cage - which is both the blast fixing and the only EMP continuity the opening has.

THE DOORS THEMSELVES ARE VENDOR ITEMS AND ARE NOT DESIGNED HERE.

The geometry, the level, the frame and the rating are the project's; the leaf, the hinge, the seal, the latching and the rebound capacity are the manufacturer's, and no vendor data exists in the project [N].

The four W8 gaps are deliberate: the partitions are 110 mm and non-structural, they were never fire or blast barriers, and closing them would obstruct the escape spine. The consequence - bays 1 to 6 are one smoke compartment 20.8 m long - is recorded, not designed away.

Figure 31 Every opening through the protective boundary, on one page.

Mark	Where	Leaf / bore	Level	What it has to do	Class
Blast Door 1	W6 , 400 thk, opening Y 600-1800 — stair shaft → bay 6	1 200 × 2 100	(-)6.100	≥ 7 bar, gas-tight, rebound-rated	[C] geometry · leaf is vendor data
Blast Door 2	W7 , 400 thk, opening Y 600-1800 — stair shaft → bay 8	1 200 × 2 100	(-)6.100	≥ 7 bar	[C] · vendor
D-05	W5 , the fire and gas-tight wall	to be designed	(-)6.100	Gas-tight, fire-rated; not blast rated — no pressure differential across W5	Required
ESC 1 head	Bay 1, through the pressure slab	1 400 dia bore	+0.150	383 kPa down AND up , gas-tight, bonded, operable by one person	See 12.5
ESC 2 head	Bay 8, through the pressure slab	1 400 dia bore	+0.700	as ESC 1; 6.800 m climb	See 12.5
Stair void	Pressure slab, bay 7	2 800 × 3 160	(-)2.000	Not a closure at all — it is the route in	[C]
Inner security door	HW2, headhouse north wall	900 × 2 100	(-)2.000	Security only. NOT blast rated	[C]
Entry door	Entry stairwell headwall, opens OUTWARD	1 000 × 2 100	0.000	Weather and security. NOT blast rated — expendable	[C]

Mark	Where	Leaf / bore	Level	What it has to do	Class
W8 gaps x4	Internal partitions, 110 thk	900 x 2 100 gap	(-)6.100	Permanently open, by design	[C]
BV-1 ... BV-5	W1 bay 1 x2 · W6 x1 · east wall bay 8 x2	DN100 x3, DN350 x2	various	Close in < 2 ms and hold 1.3 s, unattended	[C]
Service entry plate	North wall, X 11398-12198	800 wide	—	The shield itself — solid, welded, bonded 360°	[C]

Six of those are inside the protective boundary and five are outside it, and the line between them is not where a reader expects: it runs through **Blast Door 1**, not through the entry door at grade.

12.2 The four duties a closure in the boundary discharges at once

BLAST DOOR 1 - THE CLOSURE IN W6, AND THE FOUR THINGS IT HAS TO DO AT ONCE

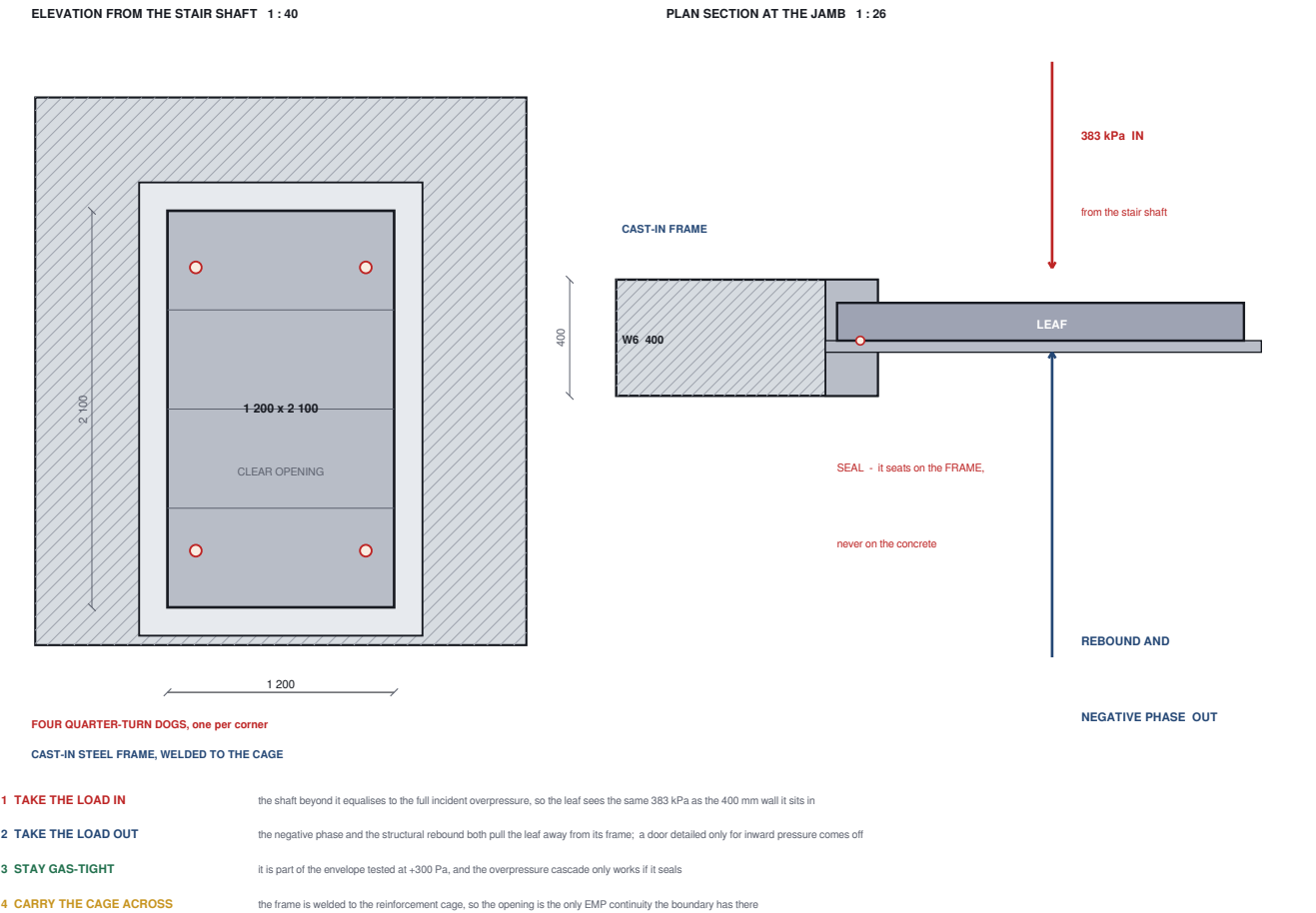


Figure 32 Blast door 1 in W6 - elevation, jamb section, and the four things the closure has to do at once.

- 1. Take the load IN.** The stair shaft beyond the doors is open to atmosphere through the roof void and **fills to the full incident overpressure in 0.094 s** against a positive phase of 0.13–1.33 s (Part 9.2.1). So the leaf sees the same **383 kPa** as the 400 mm wall it sits in. **That is why W6 and W7 are 400 and not 200**, and why both doors are rated to the full envelope rather than to a pressure difference.
- 2. Take the load OUT.** The **negative phase and the structural rebound both pull the leaf away from its frame**. A door detailed only for inward pressure comes off in the second half of the event, which is the half nobody photographs. Every closure in this boundary — both doors and both hatches — resists uplift as well as pressure, and the hatches do it with **four quarter-turn dogs** for exactly that reason.
- 3. Stay gas-tight.** Each closure is part of an envelope tested at **+300 Pa**, and the overpressure cascade of Part 15.4 only works if every one of them seals. **The seal seats on the cast-in frame, never on the concrete** — concrete is not a sealing face and does not become one.

4. **Carry the reinforcement cage across.** Each frame is a **cast-in steel frame anchored into and welded to the cage**, which makes the opening the only point of electromagnetic continuity the boundary has there. A bolted frame breaks it. **The rule is applied even to the non-blast-rated headhouse door**, because a shield that is continuous except in one place is not continuous.

12.3 Blast doors 1 and 2 — the opening, and what the structure does around it

The doors themselves are **proprietary vendor items and are not designed here** [N]. What the project fixes is the opening, the level, the frame, the rating — and the reinforcement that keeps the wall working with a 1 200 × 2 100 hole in it.

WALL W6 / W7	400 thk, T20 @ 150 EF EW
OPENING	1200 x 2100, Y 600 - 1800
JAMBS	interrupted steel 2094 x 1.2 = 2513 mm ² /face -> 1256 each jamb
ADOPTED	4 No. T20 EACH JAMB EACH FACE (1257 mm ²), anchored Ld = 800 beyond the opening
HEADER	400 x 1100 over 1200 clear
	w = 383 x 2.1/2 = 402 kN/m
	M = w L ² / 12 = 48.2 kNm
	V = 241 kN ; tau = 0.578 N/mm ²
ADOPTED	4-T20 TOP + 4-T20 BOTTOM, T12 4-leg links @ 150
FRAME	cast-in steel frame, anchored into and WELDED to the cage
LEAF	proprietary, >= 7 bar, rebound-rated, gas-tight [vendor]

The frames are on the critical path from day one. They must be welded into the W6 and W7 cages **before those walls are poured**, which puts a vendor lead time onto the programme's first procurement activity and leaves it with **19 days of float** — the tightest float on any long-lead item in the project (Part 21.5).

12.4 D-05 — the door in W5, and why a wall with no door in it is not a separation

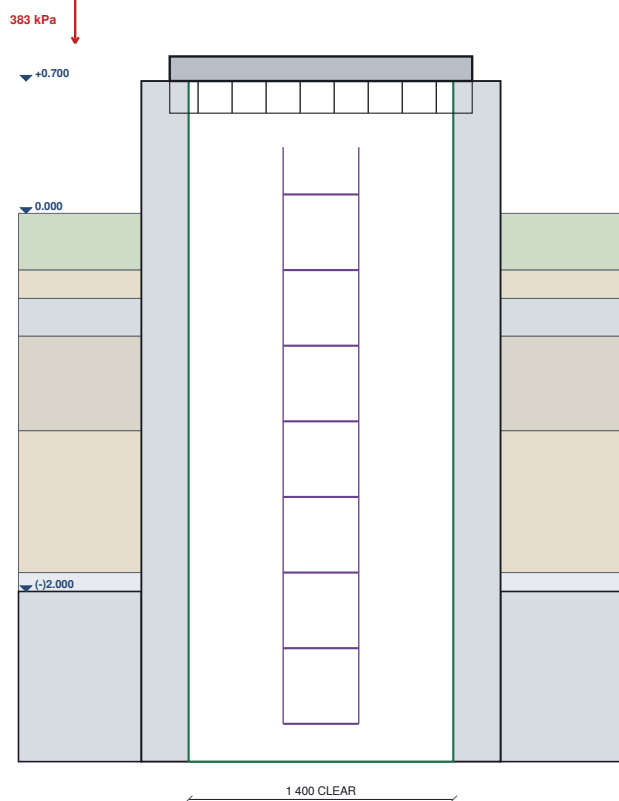
W5 is a 200 mm wall between bay 5 (the CBRN plant) and bay 6 (the decon airlock), designated **fire and gas-tight**, with **no pressure differential across it**. It is not a blast element.

It needs a door, and the door is a requirement of this design rather than an option, for two reasons that are independent of each other:

1. **Escape route R1 crosses it.** Every occupant leaving by the primary route passes from the clean zone through bay 6 to Blast Door 1. A separation the escape route cannot pass is not a separation; it is an obstruction.
2. **The compartment it protects contains the activated-carbon filter trains.** Carbon beds self-heat, and a filter fire is a known hazard in protected ventilation. **A fire separation with an unfilled opening in it separates nothing** — and what it fails to separate is the most likely fire in the shelter from the people it exists to protect.

What the door has to be: gas-tight to the +300 Pa envelope standard, fire-rated, and **not blast rated**, because there is no blast pressure difference across W5. **Its frame is still cast in and welded to the cage**, for the electromagnetic reason in 12.2.

12.5 The escape shaft heads — a 1.54 m² hole in the pressure slab, twice

ESCAPE SHAFT HEAD - A 1.54 m² HOLE IN THE PROTECTIVE BOUNDARY 1 : 40

THE HATCH - a structural closure

589.6 kN on the leaf at 383 kPa
 $M = w \cdot a^2 (3 + \nu) / 16 = 38.71 \text{ kNm/m}$
 a flat Fe250 plate would be 32 mm
 and 505 kg - UNLIFTABLE
ADOPTED ribbed steel weldment, 1 600 dia,
 12 mm face, 8 radial ribs 150 x 10,
 150 x 12 perimeter ring, seating ring
 CAST INTO the 250 collar, bearing 150
FOUR QUARTER-TURN DOGS - it must
 resist UPLIFT too: the negative phase
 and the rebound both lift it
 counterbalanced, openable from inside by
 one person without a key or a tool
 indicative leaf mass ~ 322 kg

THE LADDER

20 dia galvanised MS rungs, 400 clear,
 equal pitch ~ 297.6 mm in ESC 1 over 21
 spaces, 295.7 in ESC 2 over 23
 ~ 200 behind the rung, ~ 750 in front
 grab rails 1 100 above the head
**NO fall-arrest. NO rest platform. And a
 vertical ladder cannot pass a stretcher.**

250 RC COLLAR, OD 1 900

Figure 33 Escape shaft head - a 1.54 m² hole in the protective boundary, and the hatch that closes it.

Each shaft is a **1 400 mm clear bore straight through the 900 mm pressure slab**, in a 250 mm RC collar of 1 900 mm outside diameter, with its head at **+0.150 (ESC 1) or +0.700 (ESC 2)**. The pressure slab *is* the protective boundary, so **each head is a 1.54 m² hole in it**.

12.5.1 The structural brief, and why the obvious answer fails

total force on the leaf	$383 \times \pi \times 0.700^2$	= 589.6 kN
$M = w \cdot a^2 (3 + \nu) / 16$, $\nu = 0.30$ for a STEEL leaf		= 38.71 kNm/m
V at the seating = $w \cdot a / 2$		= 134.1 kN/m
a flat Fe250 plate, $t = \sqrt{6M / \sigma}$ at $\sigma = 250$		= 30.5 mm
	say	32 mm
mass of a 1600 dia x 32 leaf		= 505 kg

505 kilograms is the answer, and it is the wrong answer. Half a tonne cannot be lifted by a person escaping up a 6.8 m ladder in the dark, which is the only circumstance in which anyone will ever open it. **The flat-plate calculation is not the design — it is the brief**, and it is exact. What it proves is that the leaf has to be a ribbed weldment.

12.5.2 What is adopted

A ribbed steel weldment — 1 600 dia, 12 mm face, 8 radial ribs 150 x 10, a 150 x 12 perimeter ring — on a steel seating ring cast into the 250 mm collar, bearing 150 mm all round, with:

- **four quarter-turn dogs**, so the leaf resists **uplift** as well as downward pressure — the negative phase and the rebound both lift it;
- **counterbalance or spring assistance**, so it is **openable from inside by one person without a key and without a tool**;
- **bonded conducting construction**, with electrical continuity from the leaf through the seating ring to the collar reinforcement.

Indicative leaf mass ~ 322 kg — still a mechanically assisted item, which makes the counterbalance part of the escape route rather than an accessory to it.

Not designed and not invented [N]: the rib proportioning, which is an orthotropic plate problem; the counterbalance mechanism; and every electromagnetic figure for the assembly.

The structural hatch and its bonding are ONE item and must be procured as one. Bonding is not a finish applied afterwards — it is continuity designed into the weldment and the seat. **An opening whose structure sits in one package and whose bonding sits in another is an opening that ends up with neither.**

12.5.3 The trimming around the bore

Interrupted slab steel = 3272×1.400 = 4581 mm²/face/direction
 Trimmers each side = $4581 / 2$ = 2291 mm²

ADOPTED 5 No. T25 EACH SIDE, EACH FACE, EACH DIRECTION (2454 mm²),
 anchored $L_d = 1000$ beyond the opening
 Collar 250 RC; T16 @ 150 hoops in 3 layers + T16 @ 150 radials
 Slab thickened 900 → 1200 over a 600 mm annulus

Circular is the right shape — hoop action, and no re-entrant corner to concentrate stress. And this detail is what **sets the 750 mm clearance rule**: 250 collar + about 400 of trimmer band + 100 tolerance. A bay narrower than $1\,400 + 2 \times 750 = 2\,900$ mm cannot hold an escape shaft at all, which is why bay 1 is 2 900 and bay 8 is 3 000, and why neither can be shortened to buy space elsewhere in the box.

12.6 The stair void — the opening that is not a closure

2 800 × 3 160 through the 900 mm pressure slab. It is not a hole that a hatch closes; it is the route in, and it stays open for the life of the structure.

Structurally it is designed in Part 9.5.2: the **1 840 mm cantilever of 900 mm slab** it leaves has a root moment of **758.6 kNm/m**, which exceeds the midspan moment of the slab it hangs off, and it carries **T12 four-leg links at 250 throughout the pad**, a thickened free edge, diagonal trimmers at both re-entrant corners and 1 100 mm guarding.

Electromagnetically it is the finding of Part 18.

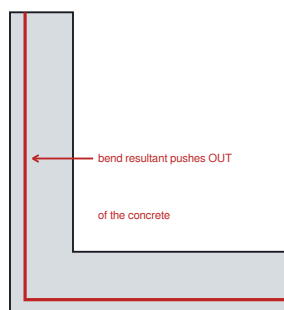
It never reaches 80 dB anywhere in the band, and above 47.4 MHz it is simply open. And it does not open into soil — **it opens into the headhouse, which is at +0.900 with no earth cover**, and from there to grade through a stairwell that is declared expendable. The entry route is an open electromagnetic path from grade to bay 7, and Blast Door 1 is the only thing across it — **whose radio-frequency performance is vendor data the project does not hold.**

This is not a criticism of the architecture. A shelter needs a staircase and the void is where a staircase has to go. It is a statement that **the blast boundary and the electromagnetic boundary are not the same surface, and only one of them has ever been drawn.**

12.7 The re-entrant corner, which is the detail most often got wrong

THE OPENING CORNER - THE DETAIL MOST OFTEN GOT WRONG SP 34 Cl. 5.5

TOP OF THE FLIGHT - THE TENSION FACE TURNS 209 deg



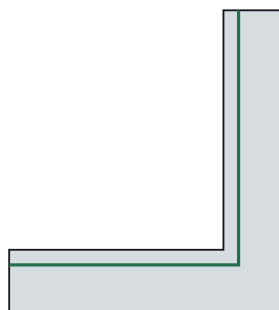
plus a U-bar T16 @ 200 across the corner.

MAIN BARS SHALL NOT BE BENT ROUND IT.

Each layer continued STRAIGHT, CROSSED,

anchored $L_d = 640$ into the OPPOSITE face,

FOOT OF THE FLIGHT - THE CORNER CLOSES, 151 deg



BARS MAY TURN NORMALLY.

The bend resultant is directed INTO the

concrete, which is what a closing corner does.

Figure 34 The opening corner - the detail most often got wrong.

Wherever the tension face of a member **turns outward** — at the top of the entry flight, at the corners of the stair void, at every haunch — the resultant of the bend in the main bars pushes **out of the concrete**, not into it. A bar bent round such a corner is a bar trying to

burst the cover off.

Main bars shall not be bent round an opening corner. Each layer is continued **straight, crossed**, and anchored L_d into the **opposite** face, with a U-bar across the corner. Where the corner **closes** — the foot of the flight — the bend resultant is directed **into** the concrete and bars may turn normally. **The two look identical on a drawing and are opposite in behaviour**, which is why SP 34 Cl. 5.5 exists and why it is cited here rather than assumed.

12.8 What the project does not hold about any of these

Item	Status
Blast door leaf, hinge, seal, latching, rebound capacity	[N] — vendor
Blast door radio-frequency performance	[N] — and Part 18 needs it
Blast valve make, closing mechanism, tested performance	[N] — vendor
D-05 leaf and frame	To be designed
Escape hatch rib proportioning and counterbalance	[N]
Escape hatch bonding detail	[N] — EM-V6
Any treatment for the stair void	[N] — none exists or is proposed anywhere
Service entry plate size	[N]
A penetration schedule for the ventilation and CBRN ducts	Does not exist

Every line in that table is a hole in the protective boundary whose closure is somebody else's drawing. The project fixes the opening, the level, the frame, the load and the duty — and stops. **Stopping there and saying so is the design position**; filling those rows in from a catalogue would be the failure.

PART 13 — REINFORCEMENT REGISTER AND QUANTITIES

13.1 Underground box

Element	Size	Cover	d (mm)	Main reinforcement	Links	A _{st} prov	x _u /d	Util.
W1 south perimeter	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W2 north perimeter	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W3 west end	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W4 east end	600	75/50/40	517	T16 @ 150 EF EW	T12 closed @ 200	1340/face	0.089	68 %
W5 Bay 5/6	200	40	154	T12 @ 150 EF EW	—	754/face	—	nominal
W6 Bay 6/7	400	50/40	342	T20 @ 150 EF EW	T12 4L @ 200	2094/face	0.211	69 %
W7 Bay 7/8	400	50/40	342	T20 @ 150 EF EW	T12 4L @ 200	2094/face	0.211	69 %
W8 partitions x4	110	25	—	A252 mesh both faces	—	252	—	non-structural
MAT	600	75/50	517	T16 @ 150 EF EW	T12 @ 250 × 250 grid	1340/face	0.089	84 %
ROOF midspan	900	75	812.5	T25 @ 150 EF EW	T12 4L @ 250 end 1500 / 2L @ 300 mid	3272/face	0.139	51 %
ROOF cantilever pad	900	75	812.5	T25 @ 150 top continuous	T12 4L @ 250 throughout	3272	0.139	56 %
Sump pit walls / base	300/400	50	244	T16 @ 150 EF EW	—	1340	—	nominal

Additional bars — underground box

Location	Requirement
Wall starters	T16 @ 150 EF, 900 mm horizontal leg into the mat , lap 800 (50 φ) above a 150 mm kicker
Haunches, all wall–roof and wall–mat junctions	500 × 500 , diagonal T20 @ 150
Mat edge, all free edges	T16 @ 150 U-bars closing both curtains
Sump pit opening	4-T20 trimmers each face, each side , L _d 800 beyond
Blast door jambs (W6, W7)	4-T20 each jamb, each face , L _d 800 beyond
Blast door header	400 × 1100: 4-T20 top + 4-T20 bottom, T12 4L @ 150
ESC 1 / ESC 2 collars	Collar 250 RC; 5-T25 each side, each face, each direction , L _d 1000 beyond; T16 @ 150 hoops × 3 layers + T16 @ 150 radials ; slab thickened 900 → 1200 over a 600 annulus
Stair void free edge	Thickened 900 → 1200 over 600; 6-T25 top + 6-T25 bottom, T12 closed @ 150
Stair void side bands	6-T25 each face, top and bottom, in a 900 band over W6 and W7
Stair void re-entrant corners (x2)	4-T25 each face at 45°, 2000 long each way , anchored L _d beyond
Band beneath headhouse wall HW3	4-T25 extra top and bottom in a 1200 band , lapped 1250
Construction joints (~6 m)	Reinforcement fully continuous ; two waterstops + welded Cu / galvanised EMP strap

13.2 Stairs

Element	Size	Cover	d	Main	Distribution	Top steel
Main stair flight waist	200	30	164	T12 @ 150 (754)	T10 @ 200 (393)	T12 @ 150 for 800 into the flight, L_d 480
Main stair landings L1/L2/arrival	200	30	164	T12 @ 125 bottom (905)	T10 @ 200	T12 @ 125 for 900 from each support
Entry stairwell flight waist	250	30	214	T16 @ 200 (1005)	T10 @ 200	T16 @ 200 for 1200 into the flight, L_d 640
Entry stairwell top landing	250	30	214	T12 @ 200 EF EW (565)	—	—
Entry stairwell platform	250	30	214	T12 @ 200 EF EW	—	on fill
Entry stairwell side walls	250	50	194	T12 @ 200 EF EW	—	minimum steel governs
Entry stairwell raking roof	250	30	204	T12 @ 200 EF EW	—	minimum steel governs
Entry stairwell headwall	250	30	214	T12 @ 200 EF EW	—	—
Entry door lintel (1000 clear)	250 × 350	30	306	3-T12 top + 3-T12 bottom	T8 @ 150	—
Stepped raft	300	50	244	T12 @ 200 EF EW	—	on compacted fill
Opening-corner U-bar (top of entry flight)	—	—	—	T16 @ 200 U-bar, bars CROSSED and anchored L_d 640 into the opposite face		SP 34 Cl. 5.5

13.3 Headhouse

Element	Size	Cover	d	Main	Links
Roof	500	75	415	T20 @ 150 EF EW (2094)	T12 4L @ 175 in the end 1200 each side; @ 250 elsewhere
Walls HW1–HW4	400	50/40	342	T16 @ 150 EF EW (1340)	T12 4L @ 250 throughout
Door jambs, HW2	—	—	—	2-T20 each jamb, each face, L_d 800 beyond	—
Door head edge band	400 × 800	40	742	4-T20 top + 4-T20 bottom	T12 4L @ 150, over the opening + 600 each side
Wall–roof haunch	300 × 300	—	—	diagonal T16 @ 150	—

13.4 Sentry post

Element	Size	Cover	d	Reinforcement
Slab S1 bottom	150	30	116/108	T8 @ 150 c/c BOTH WAYS (335 mm ² /m)
Slab S1 top, discontinuous edges	150	30	116	T8 @ 300, 400 mm (0.1 L) into the span, all four edges — Cl. D-1.6
Slab S1 corner torsion	—	—	—	T8 @ 200, FOUR LAYERS, 700 × 700 at all four corners — Cl. D-1.8
Beam B1 (3650)	250 × 450	30	404	4-T16 top at supports; 2-T16 bottom continuous
Beam B2 (4650)	250 × 450	30	404	3-T20 top at supports; 2-T20 bottom continuous
Beam hoops, both	—	—	—	T8 2-leg @ 100 over 2d = 810 from each face; @ 150 elsewhere; first hoop ≤ 50 from the face
Column C1	350 × 350	40	d' 56	8-T16 (1608 mm ² , p = 1.31 %)
Column confining hoops	—	—	—	T10 hoops + 1 cross-tie each way @ 85 c/c over 500 mm from every joint face, top and bottom, AND through the joint
Column general ties	—	—	—	T8 @ 150 c/c elsewhere
Plinth beam PB	250 × 400	30	354	3-T12 top + 3-T12 bottom
Footing F1	1500 × 1500 × 600	50	542	T12 @ 150 both ways bottom (754 mm ² /m)
Lintel L1 × 11	190 × 150	30	—	2-T10 bottom + 2-T8 top, T6 2-leg links @ 150, bearing 200 each end
Wall ties	—	—	—	6 mm MS at every 5th course (≈ 450), 200 into the bed joint, 10 mm dowel

13.5 The six that matter

ROOF	T25 @ 150 EF EW	.	T12 4L @ 250 end 1500 / 2L @ 300 mid	51 %
WALLS	T16 @ 150 EF EW	.	T12 closed @ 200	68 %

W6/W7	T20 @ 150 EF EW .	T12 4L @ 200	69 %
MAT	T16 @ 150 EF EW .	T12 @ 250 x 250 grid	84 %
HH ROOF	T20 @ 150 EF EW .	T12 4L @ 175 / 250	90 %
HH WALLS	T16 @ 150 EF EW .	T12 4L @ 250	59 %

13.6 Quantities — package SC1

Sentry post excluded — no sentry post bar, quantity or weight appears in the structural CAD schedules.

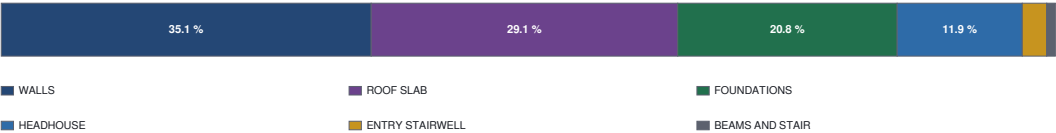
Bar	Unit mass kg/m	No. of bars	Total length m	Weight kg	% of total
T8	0.395	13	14.6	5.7	0.0 %
T10	0.617	85	102.6	63.3	0.1 %
T12	0.888	13 814	22 885.7	20 318.3	28.8 %
T16	1.578	3 698	15 020.2	23 707.1	33.6 %
T20	2.466	1 440	4 429.1	10 922.8	15.5 %
T25	3.853	654	4 006.7	15 439.4	21.9 %
		19 704	46 458.9	70 456.7	100 %

SEVENTY TONNES OF REINFORCEMENT - BY BAR SIZE, AND BY WHERE IT GOES

BY BAR SIZE



BY ELEMENT GROUP



AVERAGE STEEL DENSITY IS ABOUT 191 kg/m3. For a blast-hardened buried box whose bar spacing is set by a 150 mm EMP requirement rather than by strength, and whose sections are 900 roof, 600 walls and 600 mat, that is a plausible figure - and it is a sanity check, not a design check.

Sentry post excluded: no sentry post bar, quantity or weight appears in the structural CAD schedules. Its 1.97 t is carried separately in the bill.

Figure 35 Seventy tonnes of reinforcement, by bar size and by element group.

Element group	Weight kg	Weight t	%
FOUNDATIONS	14 665.2	14.665	20.8 %
WALLS	24 725.6	24.726	35.1 %
ROOF SLAB	20 490.5	20.490	29.1 %
BEAM-TYPE ELEMENTS	297.1	0.297	0.4 %
MAIN STAIRCASE	281.3	0.281	0.4 %
HEADHOUSE	8 364.2	8.364	11.9 %
ENTRANCE / ENTRY STAIRWELL	1 632.8	1.633	2.3 %
TOTAL	70 456.7	70.46	100 %

Concrete, in scope: mat 81.8 + pressure slab 112.0 (net of openings) + perimeter walls 103.7

- W6/W7 12.8 + W5 3.2 + headhouse 32.7 + main staircase 3.0 + entry stairwell 19.9 =

369.2 m³, excluding blinding and the sump pit.

Average steel density $\approx 191 \text{ kg/m}^3$. For a blast-hardened buried box whose bar spacing is set by the **150 mm EMP rule** rather than by strength, and whose sections are thick (900 roof, 600 walls and mat), that is a plausible figure. **It is a sanity check, not a design check.**

Quantity basis — declared, not assumed:

- **PBR-1** — cut length = Σ scheduled leg dimensions, **no bend deduction taken**; links add $2 \times 10 \phi$ for 135° hooks. Conservative by $\approx 2 \phi$ per 90° bend. Bend deductions to BS 8666 Table 3 shall be applied by the fabricator on the approved schedule. [A] — **no bar-bending standard is in the workspace; IS 2502 is NOT held and NOT cited.**
- **P-1** — stock bar 12 000 mm; longer runs split into equal pieces with 50ϕ laps added.
- **PBR-2** — a 4-legged link @ s is scheduled as **two** closed links per node on an $s \times s$ grid; a 2-legged link @ s as one.
- **PBR-3** — the stair void and the two escape openings are **deducted zone by zone**.
- Quantities are for **tender and estimating** and must be re-measured on the approved bar bending schedule.

Not quantified, and declared as such: the five blast valves with their sleeves and trimming (sizes are on the services sheet, not in the structural design); the service entry plate (size [N]); and the ventilation / CBRN duct penetrations (no penetration schedule exists). **All three are NOT DETERMINABLE from the information the project holds, and none is fabricated.**

PART 14 — DRAWINGS AND DRAWING QUALITY

14.1 Two distinct DXF sets — do not confuse them

Set	Origin	Revision	Purpose
Input set (10 files)	Supplied by the project owner	Rev F	Architectural / civil geometry — the primary geometry source . Directly editable
Output set (S-01 ... S-08)	Generated in this project	Phase 2 Rev A	Structural drawings with reinforcement. Generated — edit the script, never the DXF
Discipline sets	Generated in this project	per package	Structural CAD, drainage, HVAC, finishes, fire, concealment, EMP, electrical, site

14.2 The package as it stands: 80 DXF, 74 PASS

Discipline	Sheets
Architectural / general — Rev F	11 (A-101...A-301, plus the services sheet S-06)
Architectural — finishes	3 (A-601, A-611, A-612)
Structural — reinforcement	30 (R-001...R-805)
Drainage	11 (D-001...D-305) + 2 handout
HVAC	6 (M-001...M-203) + 2 handout
Fire and life safety	2 (F-101, F-102)
Site and concealment	1 (C-101)
EMP protection	6 (EM-001...EM-302)
Electrical	1 (E-001)
Site selection and geotechnical	5 (SG-001...SG-202)
TOTAL	80 DXF · 75 A1 · 4 A4 · 1 A0

THE DRAWING PACKAGE - EIGHTY DXF, AND THE TWO SETS THAT MUST NEVER BE CONFUSED

Architectural / general - Rev F		11	SOURCE
Architectural - finishes		3	generated
Structural - reinforcement		30	generated
Drainage		13	generated
HVAC		8	generated
EMP protection		6	generated
Site selection and geotechnical		5	generated
Fire and life safety		2	generated
Site and concealment		1	generated
Electrical		1	generated
TOTAL		80	75 A1 · 4 A4 · 1 A0

THE TEN Rev F ARCHITECTURAL FILES ARE SOURCE. EVERYTHING ELSE IS A BUILD ARTEFACT.
Source may be edited directly, a build artefact is regenerated from its script and must never be hand-edited, because the next regeneration would silently discard the edit. The index itself is generated from the DXF, so it cannot drift from the drawings - but a package that does not register its own prefix with the tool is invisible to it, and stays invisible for as long as nobody looks.

Figure 36 The drawing package - eighty DXF by discipline, and the two sets that must never be confused.

The drawing index is generated from the DXF files themselves, so it cannot drift from the drawings. Filenames are deliberately unchanged — every document in the project cites the current filenames — so the drawing number lives in the title block and the index carries both.

14.3 Drawing quality assurance

The drawing package is held to a stated drafting standard, and the whole package is scanned against it by a tool rather than by eye. The scan covers drafting, annotation, sheet composition and title blocks only — **it makes no engineering judgement and changes no engineering value.**

Measure	Before	After
DXF inspected / edited in place	—	65 / 65
Text-on-text overlaps, all 65 drawings	67	2
· of which, the 54 generated sheets	49	0
· of which, the 11 Rev F drawings	18	2
Annotation crossing hard line work (Rev F sheets)	50	10
A view drawn on top of its own notes panel	3 sheets	0
Entities outside the sheet border	0	0
Drawings with a border and title block	54 of 65	65 of 65
Panel / table body text below print size (1.4 mm on A1)	most of the package	0 — floor is now 2.0 mm
Generated packages rebuilding clean	54/54	54/54, 0 errors

No engineering value is changed anywhere by a quality pass. No dimension, level, bar mark, bar size, spacing, load, material grade, thickness or room size, and never the frozen staircase. A drafting tool that can change an engineering value is a drafting tool nobody should run.

The package currently scans at 80 DXF, 74 PASS.

And one rule of this project exists because it was once broken. A new drawing package **must register itself with the QA tool** — add its DXF/ prefix to DISCIPLINE in `qa_report_data.py` and its name to ORDER in `make_index.py`, then re-run both. **Twelve sheets can sit in the repository, correct and complete, and be invisible to the index for as long as nobody thinks to look.** A generated index is only as honest as its own inclusion list.

14.4 Sheet standard

SHEET SIZE	A1, 841 x 594 mm, DRAWN IN PAPER MILLIMETRES, PLOT 1:1 (one A0 -- the front elevation, which is 880 mm at 1:50 and cannot fit A1; four A4 handout sheets)
DXF FORMAT	AutoCAD R12 ASCII -- opens in AutoCAD, DraftSight, BricsCAD, LibreCAD and QCAD without translation
BORDER	outer 0,0-841,594 ; inner 10,10-831,584
TITLE BLOCK	180 x 62 mm, bottom right, at x0 = 651, y0 = 10
VIEW SCALES	each view carries its own scale note; views are mapped from model millimetres to paper millimetres by $\text{paper} = \text{origin} + \text{model} / \text{scale}$
TEXT HEIGHTS	3.4 view titles 2.4 panel headings 2.0 scale notes 1.6-1.7 general 1.55-1.62 panel body 1.4-1.5 small
VALIDATION	SECTION/ENDSEC balanced, EOF present, no undeclared layer, all geometry within the sheet extents

22 layers, identical in every generated file — CONC, CONC-HIDDEN, REINF-MAIN (red), REINF-DIST (green), REINF-LINK (magenta), REINF-SEC (cyan), DIM, TEXT, TEXT-TITLE, HATCH, GRID, SOIL, CENTRELINE, TITLEBLOCK, WATERPROOF (orange), STEELWORK (yellow), LEVELS, NOTES, TABLE, SERVICES, BLAST (red), and 0. **Dashed lines are drawn as explicit segments, not by linetype**, for maximum compatibility.

14.5 Two drafting decisions that are worth reading

The front elevation cannot be plotted at 1:50 on A1, and A0 is the only answer. The drawing is 880 mm wide at 1:50 against an A1 printable width of 821 mm. **The scale is a measurement statement and must not be changed to fit the paper,** and splitting the sentry post onto its own sheet would break a continuous 44 m elevation, which is the whole point of the drawing. **A0 keeps both the scale and the drawing intact.**

Six sheets do not fill their paper, and that is accepted as drawn. Six sheets share an empty strip. **Every one of them is correct, complete and legible; emptiness is not an error.** Re-scaling would make the stated scale a layout variable instead of a measurement statement — 1:20 is the right scale for a 400 mm wall section regardless of how much paper it leaves. Combining or renumbering would break cross-references, the index and the sheet numbering **for zero engineering benefit.** And reflowing the blocks **moves** the void, it does not remove it.

14.6 What cannot be regenerated, and is not invented

Seven of the eight S-series structural sheets are ABSENT and CANNOT be regenerated. Their generator toolchain — `proj.py`, `dxflib.py` and the eight sheet generators — **is not in the workspace.** S-01 ... S-05, S-07 and S-08 therefore cannot be rebuilt, and **they are not fabricated, and nothing in this report is a substitute for them.** S-06 **is present and is current.**

Also not in this workspace and not invented: the Phase 1 Rev D report, `SK02_Underground_Plan.png`, and the nineteen STAAD screen captures.

And the sentry post beams, columns and footings are not yet drawn. A future sheet s-09 would carry B1 4-T16 / 2-T16, B2 3-T20 / 2-T20, C1 8-T16 with T10 confining hoops @ 85, and F1 1500² × 600 with T12 @ 150 B/W. **Everything it needs is in Part 11; the sheet does not exist.**

14.7 The drainage items on the architectural sheets, and the question drawing them raised

The drainage items are carried on the Rev F architectural sheets: the sump pit on the side section, the septic tank and soak pit on the front elevation beyond a break, the sump pit's **cover** at (–)6.100, and a **1:200 key plan** carrying the sump, the septic tank and the soak pit at true project X and Y.

Drawing them asked a question the text had never had to answer. The mat is genuinely interrupted over the clean sump — the reinforcement register trims a **1 500 × 1 500 opening** with 4-T20 each face each side, the bar schedule calls it an opening, and the bill measures the mat gross by its 1.35 m³. **A 1 500 × 1 500 hole through the floor is what the project records, so that is what was drawn — and then Bay 5 is 1 560 mm clear and the pit is 1 500 of it.**

With the opening open there is no route past it to the two filter trains, the CO₂/O₂ plant or the dehumidifier; and the room finish schedule falls that room **1:80 direct to the sump**, so whatever closes it has to pass water. **A cover is not optional here — and the project contains no cover: no type, no depth, no duty, no fixing, no lifting arrangement, in any schedule, drawing or document.** The cover is drawn as one diagrammatic line, tagged [A], with its specification tagged [N], and it is carried as open item DR-A2-V1. **Nothing is invented to fill the gap, and the drawing does not pretend to specify what the project does not hold.**

This is what drawing something does that writing about it does not. Two [C] facts — a 1 500 opening and a 1 560 mm room — sat in two different schedules for four revisions and never met. They met on a sheet.

PART 15 — HVAC, CBRN FILTRATION AND THE SEALED ATMOSPHERE

This is the discipline that makes the structure habitable, and it is the one whose failures are silent. A wall that is too thin is visible in a calculation. A carbon bed that is spent is visible in nothing at all unless somebody fitted a gauge across it, and the first symptom is a casualty. Everything in this Part follows from that: the instrumentation is not an accessory, the hand crank is not a nicety, and the consumable that runs out first is the number that belongs on the drill card.

15.1 The basis, and the eleven figures that reproduce

The ventilation basis is stated on the issued services sheet. **Every figure on it has been re-derived from first principles rather than carried forward**, and all eleven reproduce:

Stated	Reproduced from	Result
Gas-tight envelope 67.8 m² / 217.0 m³	Bay widths 2900+1800+3500+1800+1560+2000 × 5.000 × 3.200	67.80 / 216.96
Clean zone 57.8 m²	Envelope less the 10.0 m ² decon airlock	57.80
Occupied volume, airlock shut 185.0 m³	57.80 × 3.200	184.96
FEMA 453 rate 264 m³/h	57.8 m ² = 622.2 ft ² × 0.25 cfm/ft ²	264.3 (0.10 %)
Leakage 32.5 m³/h	0.15 vol/h × 216.96	32.5 = 10.8 % of one train
Time to 1.0 % CO ₂ 9.9 h	(0.0096 × 184.96) / (9 × 0.02)	9.9
Airlock purge 13 min	5 × (2.0 × 2.0 × 3.2) = 64 m ³ at 300 m ³ /h	12.8 min
DN100 throat 10.6 m/s	300 / 3600 / (π × 0.05 ²)	10.61
DN350 throat 7.5 m/s	2600 / 3600 / (π × 0.175 ²)	7.51
Survival / working rates	5 × 9 = 45, 15 × 9 = 135 m ³ /h	reproduces
O ₂ store 15 m³	2 × 50 L at 150 bar	80 h at 4.5 m³/day

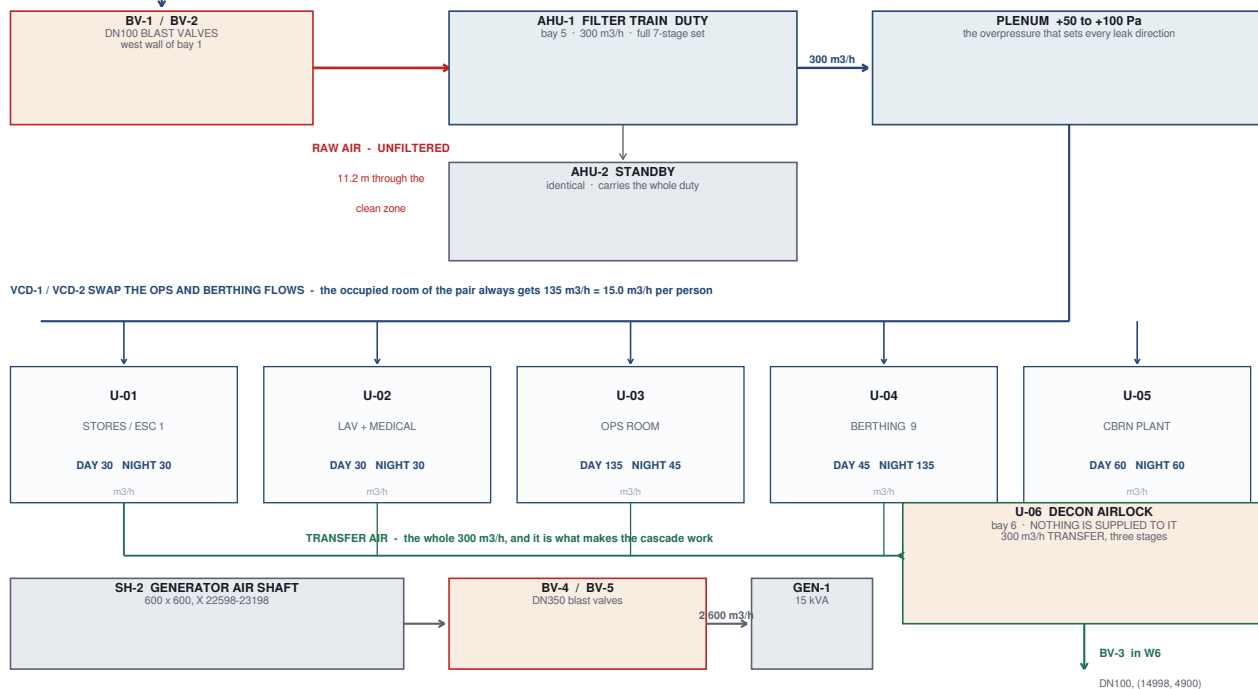
Every figure reproduces. The sheet is internally consistent, and that is worth establishing before anything is sized against it.

15.2 Design basis

Parameter	Value	Class
Occupancy / endurance	9 persons / 96 h	[C]
Design flow	300 m³/h, 2 × 300 trains, true N+1	[C]
— against the three criteria	6.7 × survival · 2.2 × working · 1.14 × FEMA	[R]
— per person	33.3 m³/h	[R]
Air changes, whole envelope / clean zone	1.38 / 1.62	[R]
Leakage	≤ 0.15 vol/h at +300 Pa = 32.5 m³/h , 11 % of one train	[C] / [R]
Operating overpressure	+50 to +100 Pa , tested at +300 Pa	[C]
Cascade	0 → +10 → +20 → +35 → +50 Pa	[C]
Closed mode	CO ₂ 0.18 m ³ /h; 9.9 h unscrubbed, 48 h on soda lime, 80 h on O ₂	[C] / [R]
Airlock purge	64 m ³ , 12.8 min, 4–5 persons/hour	[R]
Ductwork + plenum loss	161 Pa	[R]
Total fan duty	VENDOR DATA REQUIRED — 5 of 8 components	[N]
Cooling load	4 kW adopted; see Part 2.11 and RC4–V1	[R]

THE AIR PATH - ONE WAY IN, ONE WAY OUT, AND A SEPARATE LOOP FOR THE GENERATOR

SH-1 FRESH-AIR SHAFT 600 x 600

gooseneck head at +1.500, 12.3 m from the intake to the entry against a ≥ 10 m rule

THE GENERATOR LOOP DOES NOT TOUCH THE GAS-TIGHT ENVELOPE. Bay 8 is outside it, behind blast door 2 and W7, so running the set neither depressurises the clean zone nor spends carbon-bed life.

Figure 37 The whole air path - intake, the two trains, distribution, the cascade, the exhaust, and the separate generator loop.

Room-by-room distribution — developed in this project. The nine occupants are not in two places at once: they work in the ops room and sleep in the berthing bay. A single fixed split would starve whichever of the two is occupied, so the distribution is **balanced in two modes on two volume control dampers**:

Room	Use	Volume m³	Day m³/h	Night m³/h	Extract
U-01	Emergency stores / ESC 1	46.40	30	30	—
U-02	Lavatory + medical	28.80	30	30	45
U-03	Ops room and hazard plotting	56.00	135	45	—
U-04	Berthing, 9 berths	28.80	45	135	—
U-05	CBRN plant + sump	24.96	60	60	—
U-06	Decon airlock	32.00	300 transfer	300 transfer	to BV-3
TOTAL			300	300	

The occupied room of the pair gets 135 m³/h = 15.0 m³/h per person — exactly the working-shelter rate the sheet names as its second criterion. Holding that rate in *both* rooms at once would take 270 of the 300 m³/h and leave 30 for the stores, the lavatory *and* the plant room. If a reviewer requires that, the total must rise above 300 m³/h — which is a change to a confirmed value. Raised, not taken.

Nothing is supplied to Bay 6. The airlock is the exhaust path: the whole 300 m³/h transfers through the three decon stages and out at BV-3, which is what makes the confirmed cascade work. The cascade maps exactly onto the three airlock stages.

15.3 The filter train, stage by stage

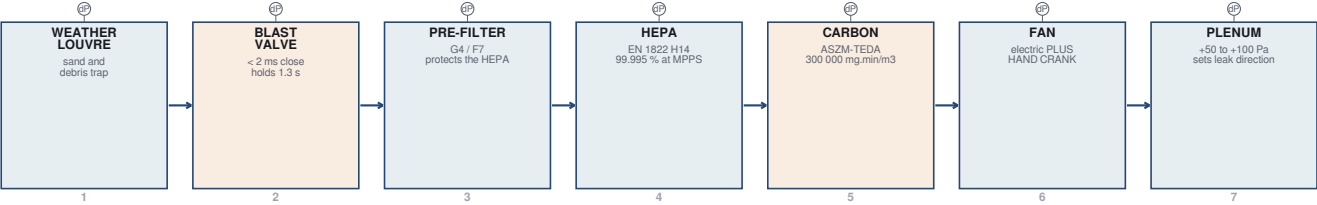
Seven stages, in the order the air meets them. Two identical trains, each able to carry the whole 300 m³/h duty on its own — true N+1, not 2 × 150.

#	Stage	Specification	What it is for
1	Weather louvre	sand and debris trap	Keeps rain, dust and debris out of the shaft, and out of every stage behind it

#	Stage	Specification	What it is for
2	Blast valve	< 2 ms close, holds 1.3 s	Not a damper, and not crew-operated: it has to work with nobody watching . The closing time is shorter than the rise time of the front
3	Pre-filter	G4 / F7	Protects the HEPA from coarse dust and multiplies its life
4	HEPA	EN 1822 H14, ≥ 99.995 % at MPPS	Particulate — biological agent, radioactive dust, fallout
5	Carbon	ASZM-TEDA, 300 000 mg·min/m³	Chemical agent vapour. The one stage with a finite, consumable life
6	Fan	electric drive plus hand crank	The hand crank is the reason ventilation survives a power failure
7	Plenum	+50 to +100 Pa	Sets the overpressure, and therefore the direction of every leak in the envelope

THE NBC FILTER TRAIN - SEVEN STAGES, IN THE ORDER THE AIR MEETS THEM

300 m³/h duty. TWO IDENTICAL TRAINS, EACH ABLE TO CARRY THE WHOLE DUTY ALONE - true N+1, not 2 x 150.



A DIFFERENTIAL-PRESSURE GAUGE ACROSS EVERY STAGE AND A FLOW METER ON THE TRAIN.

They are the only way to know a filter is spent. They turn a vendor's dirty-filter figure into a maintenance trigger.

STAGE 5 IS THE ONLY STAGE WITH A FINITE, CONSUMABLE LIFE.

Its change-out interval needs a challenge concentration and vendor breakthrough data. Neither exists - DATA REQUIRED [N].

FIVE OF THE EIGHT PRESSURE-LOSS COMPONENTS ARE VENDOR DATA.

Only 161 Pa - ductwork plus the 100 Pa plenum - is derivable. The fan must be selected on the DIRTY figures. A quoted total would be fabricated.

Figure 38 The NBC filter train - seven stages, in the order the air meets them.

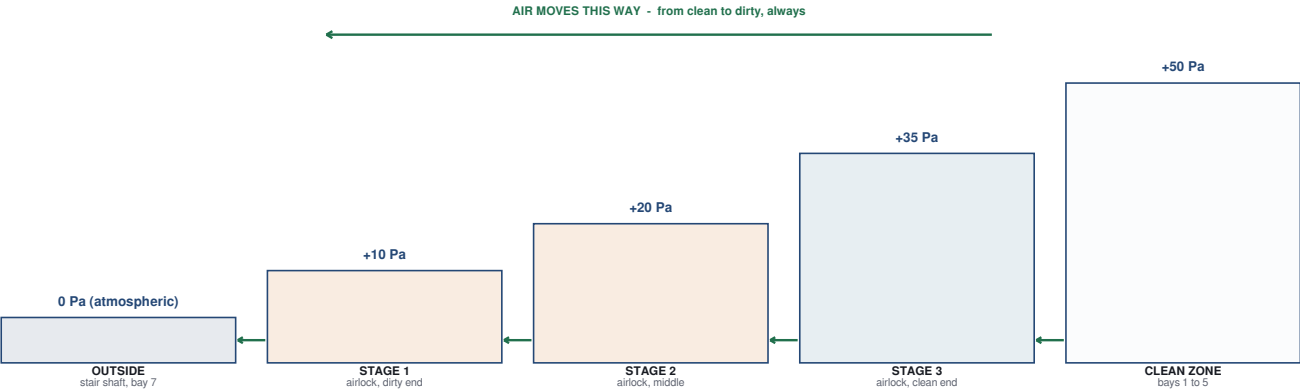
A differential-pressure gauge across every stage and a flow meter on the train are requirements, not options. They are the only way to know a filter is spent, and they are what turns a vendor's dirty-filter pressure drop from a data-sheet number into a maintenance trigger that someone acts on.

Stage 5 is the only stage that is consumed. Its change-out interval needs a challenge concentration and vendor breakthrough data, and **neither exists in this project** [N]. The interval is therefore not stated here, and nothing is invented for it — but the **consequence** is stated: every hour of peacetime running in mode 1 spends carbon life that can only be replaced in closed mode.

15.4 The overpressure cascade

The envelope is held above atmosphere so that every leak runs **outwards**. The cascade is confirmed at **0 → +10 → +20 → +35 → +50 Pa**, and it maps exactly onto the three stages of the decon airlock — a reconstruction [R], since the issued sheet states the cascade and the stages separately and never puts them side by side.

THE OVERPRESSURE CASCADE - 0 to +50 Pa IN FOUR STEPS, AND WHY THE AIRLOCK HAS THREE STAGES



THE CASCADE MAPS EXACTLY ONTO THE THREE AIRLOCK STAGES.

The cascade values are confirmed; the mapping onto the stages is a reconstruction [R] - the issued sheet does not state it. Purge: 5 air changes of stage 1 = 12.8 min, so 4 to 5 persons an hour.

Figure 39 The overpressure cascade, and the three airlock stages it maps onto.

Zone	Pressure	What holds it
Stair shaft, bay 7	0 Pa, atmospheric	Outside the gas-tight envelope
Airlock stage 1, dirty end	+10 Pa	Transfer from stage 2
Airlock stage 2	+20 Pa	Transfer from stage 3
Airlock stage 3, clean end	+35 Pa	Transfer from the clean zone
Clean zone, bays 1-5	+50 Pa	The plenum, at +50 to +100 Pa

The cascade is what makes the airlock an airlock rather than a lobby. Air moves from clean to dirty at every step, so a person carrying contamination into stage 1 is walking against the flow the whole way in. It works only if the whole 300 m³/h leaves through the airlock, which is why **nothing is supplied to bay 6** and why the exhaust is BV-3 in W6 rather than a duct back to the plant room.

And it is what sets the manning rate. Five air changes of stage 1 is 64 m³, which at 300 m³/h is **12.8 minutes** — so **four to five people an hour** is the maximum rate at which anyone enters this shelter through a working decon procedure. That is an operational number, not a ventilation one, and it belongs on the drill card next to the soda lime.

15.5 Findings

Ref	Finding
HV-F1	The soda lime, not the oxygen, limits closed mode — 9.9 h unscrubbed, 48 h on soda lime, 80 h on O ₂ . The consumable to count, and the one to write on the drill card. Closed mode covers 48 of the 96 h; the other 48 h require filtration
HV-F2	The raw-air duct is a protective element, not a duct. The fresh-air blast valves are in the west wall of Bay 1 and the filter trains are in Bay 5. Between them the air is unfiltered, and the duct carries it 11.2 m through the clean zone. Over that length the duct wall is the only barrier. Positions are not changed — they are confirmed on an issued sheet — but the duct is specified as protective: fully welded, no push-fit or slip joint, tested to the +300 Pa envelope standard, run visible and never boxed in, labelled RAW AIR — UNFILTERED at ≤ 2 m centres, re-tested after any work in bays 1-4. <i>Should the trains move to Bay 1? — raised, not taken</i>
HV-F3	Maintenance access is the tight dimension, not headroom. Each train is 1 450 wide in a 1 560 clear bay — 110 mm at the sides — so all access is along the bay, and a HEPA or carbon cassette must come in through Blast Door 1 (1200 × 2100) and along bays 6 and 5. Confirm the cassette dimensions against that route before the trains are ordered
HV-D1	No filter bypass for peacetime running. Without one, every peacetime hour spends carbon-bed life replaceable only in closed mode. A bypass is a deliberate leak path around the filters — a protective decision. Not added
HV-D2	BV-3 discharges into Bay 7, and the onward path to atmosphere — up the stair shaft, out through the headhouse — is not recorded anywhere. A path exists; its resistance is unknown and it is in series with the relief valve
HV-D4	No noise criterion exists anywhere in the project. Terminal selection cannot be closed, and a berthing space for nine is exactly where a noisy diffuser is felt
HV-D6	Fan static pressure — 5 of 8 loss components are vendor data. The derivable part is 161 Pa. Quoting a total without the vendor figures would be a fabricated number. The fan must be selected on the dirty figure, and the hand crank sized on the same duty

15.6 Operating modes

Mode	Condition
1 NORMAL	Peacetime, unfiltered or lightly filtered
2 FILTERED / PROTECTIVE	Full seven-stage train, overpressure held
3 CLOSED	Detonation to all-clear. All five blast valves shut. Soda lime and O ₂ only. 48 h limit
4 PURGE	Airlock purge cycle
5 GENERATOR RUNNING	BV-4 and BV-5 open. Bay 8 only — does not touch the gas-tight envelope. Independent of modes 1–4

Mode 3 and mode 5 have to be read together, and the reading is worth a factor of twelve. Mode 3 shuts all five blast valves at the shock. Mode 5 needs two of them — BV-4 and BV-5 — open for the generator. **The design position is: all five shut at the shock and hold 1.3 s, and BV-4 and BV-5 are then REOPENED for generator operation.** Bay 8 is outside the gas-tight envelope, behind blast door 2 and W7, so the clean zone is unaffected — which is what the mode 5 note says in its own terms.

The whole electrical battery sizing turns on it. If the generator cannot run in closed mode, the battery has to carry the essential load for the full 48 hours: **1 783 Ah, about 2.4 tonnes, 4.8 m² of floor** — and there is nowhere in this shelter to put that. If it can, the battery is a **4-hour cabinet, 149 Ah, 204 kg, 0.40 m²**, which fits in bay 5 beside the plant. Part 17 designs the 4-hour cabinet.

And the consequence is recorded rather than resolved. Reopening BV-4 and BV-5 leaves **two DN350 bores open through the post-attack period** — the two bores that fail both EMP criteria (Part 18.4). It sharpens the question of whether bay 8 is inside the EMP boundary at all, without deciding it.

15.7 Mode 3 is limited by the soda lime, and that is the number to remember

CO2 UNSCRUBBED	9 people x 0.020 m ³ /h	= 0.18 m ³ /h	
	0.04 % -> 1.0 % over 184.96 m ³	= 9.9 h	
SODA LIME	40 kg store at 20 kg/day	= 48 h	<- GOVERNS
OXYGEN	2 x 50 L at 150 bar = 15 m ³		
	at 4.5 m ³ /day	= 80 h	

Closed mode covers 48 of the 96 hours of endurance. The other 48 require filtration, which means the shelter is *expected* to come off closed mode and run the trains while the outside is still hostile. That is a design position with an operational consequence, and it is the reason the carbon bed's service life matters as much as the blast rating of the doors.

Nine point nine hours is the number that makes the scrubber non-optional. Against a 96-hour design occupancy, an unscrubbed envelope is not a shelter.

15.8 What the HVAC design cannot close

Item	Why	Class
Total fan static pressure	Five of the eight loss components — louvre, blast valve, pre-filter, HEPA, carbon — are vendor data , and in a CBRN train they dominate the build-up. The derivable part is 161 Pa	[N]
Carbon bed change-out interval	Needs a challenge concentration and vendor breakthrough data	[N]
Dehumidifier duty	DH-1 is confirmed to exist; no latent load and no target relative humidity exist anywhere in the project	[N]
Ambient design temperature	Nowhere in the project, which is why Part 2.11 can state a direction and not a number of hours	[N]
Noise criterion	Nowhere in the project. Terminal selection cannot be closed, and a berthing space for nine is exactly where a noisy diffuser is felt	[N]
Generator heat rejection into bay 8	Not stated	[N]

Quoting a total fan duty without the vendor figures would be a fabricated number, and a fan selected on a fabricated duty is a fan that does not hold the cascade when the filters load up. What the procurement must specify instead is stated exactly: **the clean and dirty pressure drop of every stage; the fan selected on the dirty figure; the hand crank sized on the same duty; and the plenum held at +50 to +100 Pa across the whole range.**

PART 16 — WATER, SEWAGE AND DRAINAGE

Water inside a sealed box is a protective problem before it is a plumbing problem. Every route out is a route in: a drain that leaves the envelope is a blast path, a gas path and an EMP path, and a trap that loses its seal at +100 Pa is an open pipe. That is why the box is **tanked, not drained**, why there is exactly **one** services penetration, and why the sump has three lines of defence of which only two need electricity.

16.1 The four rules that follow from tanking the box

The structure is a **tanked box, not a drained one** (Part 6.4). Four rules follow, and the whole of this Part is their consequence:

1. **Nothing penetrates the roof.** No rainwater outlet, no vent stack, no soil stack through the 2 000 mm cover — that would breach both the radiation mass and the roof membrane. **There is no downpipe on the buried roof and none is added.**
2. **One services envelope crossing.** The service entry plate. The sump rising main uses it; **no second drainage penetration is created.**
3. **Three streams, never combined.** Clean (seepage and condensate) → storm soakaway. Foul (peacetime) → septic tank → soak pit. Decon effluent → 1 000 L tank → **tanker only.**
4. **Drainage is not in the safety path.** No drainage failure may block egress, breach the envelope, or flood an entrance.

16.2 The daily water balance

THE DAILY WATER BALANCE - WHAT ARRIVES, WHERE IT IS HELD, AND WHERE IT IS ALLOWED TO GO

SOURCE	BASIS	RATE	HELD IN	DISCHARGES TO
STRUCTURAL SEEPAGE	0.5 L/m ² /day x 401 m ²	200 L/day	CLEAN SUMP SU-01 3 375 L	STORM SOAKAWAY SK-02
CONDENSATE + WASHDOWN	DH-1 through a 75 deep-seal trap	200 L/day	CLEAN SUMP SU-01 3 375 L	STORM SOAKAWAY SK-02
FOUL, PEACETIME	10 users x 45 lpcd	450 L/day	SEPTIC TANK ST-01 1 125 L	FOUL SOAK PIT SK-01
DECON EFFLUENT	airlock stages 1 and 2	on use	TK-01 1 000 L	TANKER ONLY - OFF SITE
STAIRWELL SURFACE WATER	approach and stairwell	0.10 L/s	SU-02 1 000 L	SK-03 - NOT SIZED [N]

400 L/day INTO A 3 375 L SUMP = 8 DAYS. That is the number that matters: it is longer than the 48-hour closed-mode limit set by the soda lime, by a factor of four.

THE SEEPAGE RATE 0.5 L/m²/day IS A DESIGN ALLOWANCE FOR AN INTACT TANKED STRUCTURE. It is not a measured figure and it is not a leak allowance for a defective one.

Groundwater does not stop because the shelter is sealed. The clean sump pump stays on the essential board through closed mode, and the hand pump covers the battery failing.

Figure 40 The daily water balance: what arrives, where it is held, and where it is allowed to go.

Collected flow inside the envelope:

structural seepage	200 L/day	[C]
condensate and washdown	200 L/day	[C]
TOTAL	400 L/day	= 0.00463 L/s

And the seepage figure's basis, reconstructed because the sheet did not state it:

submerged wall height	(-)2.000 -> (-)6.700	= 4.700 m
external perimeter	2 x (22.0 + 6.2)	= 56.40 m
submerged wall area	56.40 x 4.700	= 265.08 m ²
mat underside	22.0 x 6.2	= 136.40 m ²
	TOTAL	= 401.48 m ²
0.5 L/m ² /day x 401 m ²		= 200 L/day

It is the external envelope standing below the design water table — not the internal wetted area. Stating it matters, because **if the water table is confirmed higher than (–)2.000 the wetted area, the seepage and the sump duty all move together**, and the relationship has to be explicit for anyone re-running it after the piezometer is read.

400 L/day into a 3 375 L sump is eight days of store with no power at all. That is longer than the 48-hour closed-mode limit by a factor of four, and it is the number that makes the drainage design defensible rather than merely present.

16.3 Internal drainage — sized by bore and velocity, not by flow

Item	Adopted	Class
Floor falls	transverse 1:80 in wet areas, 1:100 elsewhere; spine 1:400 east to the sump along Y 3100	[A]
Fall formed in	floor screed — 25 mm minimum at the sump edge rising to 78 mm at the far corner; area-average 52 mm	[A] / [R]
Gravity drains	DN100 minimum at 1:100 minimum	[A]
Capacity, DN100 at 1:100	6.72 L/s at 0.85 m/s — self-cleansing	[R]
Capacity ratio against the design flow	1 450 : 1	[R]
Traps	75 mm deep seal throughout, primed	[A]
Rising main	DN50 , 0.76 m/s at 1.5 L/s; welded or flanged, no push-fit joint inside the envelope	[A]
Clean sump	3.375 m³ , invert (–)7.600, base slab (–)8.000; 2 × 1.5 L/s submersible + hand pump PU-03	[C]

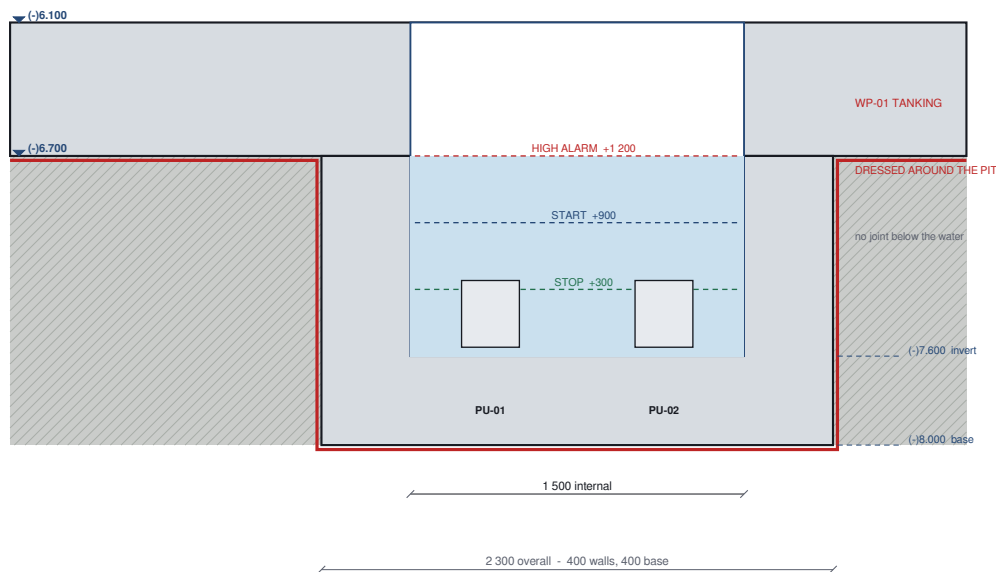
Internal drainage is governed by minimum bore and self-cleansing velocity, not by flow — the capacity ratio is three orders of magnitude. **Stated so that no reviewer looks for a hydraulic sizing calculation the flows do not justify.**

Trap seals against shelter overpressure (Part 2.7.1):

Seal depth	Holds	vs +100 Pa operating	vs +300 Pa test
50 mm	491 Pa	4.9 : 1	1.6 : 1
75 mm (adopted)	736 Pa	7.4 : 1	2.5 : 1
100 mm	981 Pa	9.8 : 1	3.3 : 1

16.4 The clean sump

SU-01 CLEAN SUMP - A PIT IN THE MAT, AND THEREFORE A PENETRATION OF THE TANKING 1 : 34



CAST MONOLITHIC WITH THE MAT IN THE SAME CONTINUOUS POUR. T16 @ 150 each face each way, 4-T20 trimmers each face and side.

Wall and base thickness 300 / 400 was an open item; 400 is held. A construction joint here would be a joint permanently below the water table.

3 375 L OF STORE AGAINST 400 L/day = EIGHT DAYS WITH NO POWER AT ALL.

That store, the hand pump PU-03 and the 48-hour closed-mode limit are the three numbers a drill card needs.

Figure 41 The clean sump SU-01 - a pit cast into the mat, and therefore a penetration of the tanking.

The clean sump is $1\,500 \times 1\,500 \times 1\,500$ internal = 3.375 m^3 , invert $(-)\,7.600$, base slab $(-)\,8.000$, walls 300 and base 400, **cast monolithic with the mat in the same continuous pour**. T16 at 150 each face each way, with 4-T20 trimmers each face and side. The membrane is **dressed around the pit**, not cut and re-lapped at it.

It is a pit in the mat, and therefore a penetration of the tanking. A construction joint here would be a joint permanently below the water table on the one element that cannot be inspected or repaired from outside, which is why it is cast in the same pour as the mat and why the membrane is dressed around rather than through it.

Level above invert	What happens
+300	Stop — the duty pump cuts out
+900	Start — the duty pump cuts in
+1 200	High alarm

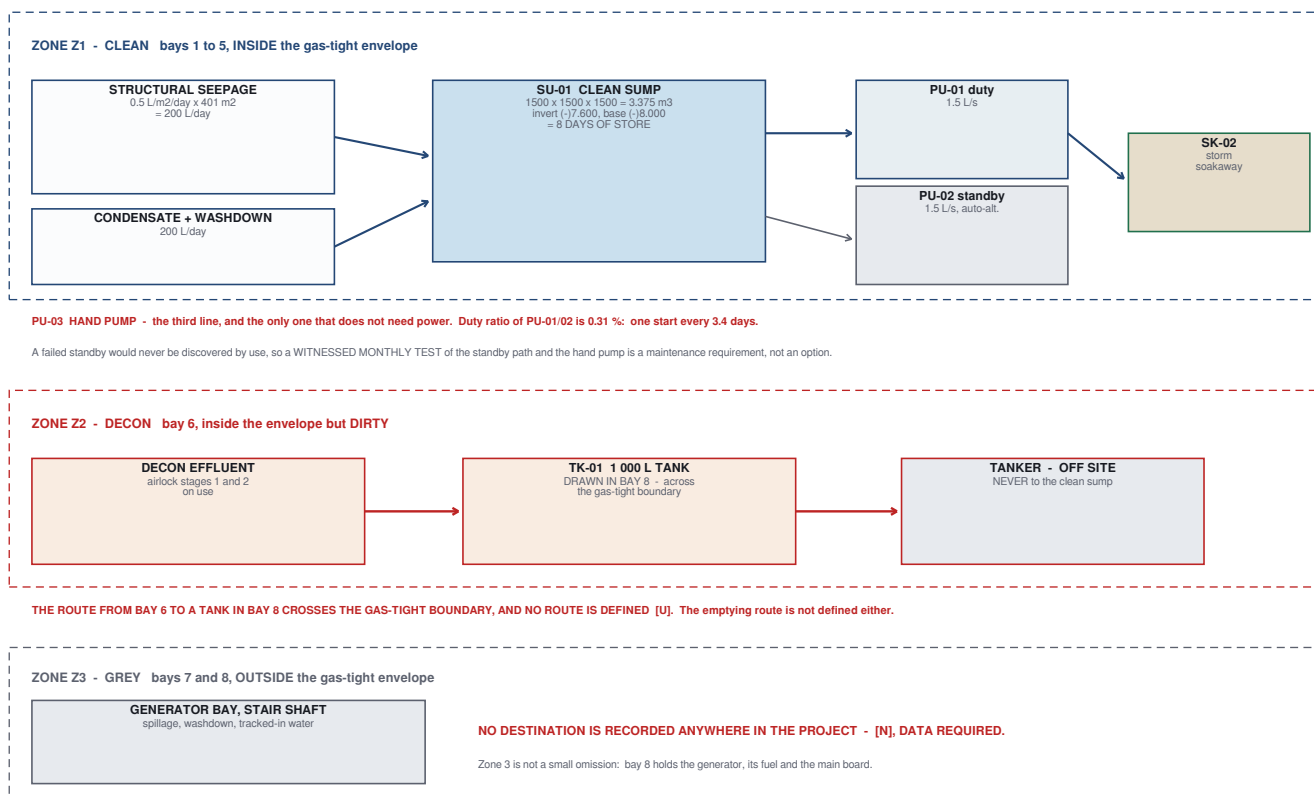
Three lines of defence, and only two need electricity: PU-01 duty and PU-02 standby at 1.5 L/s, auto-alternating, and **hand pump PU-03**, which is independent of power and of the battery. Both filter fans keep their hand crank for the same reason.

The duty ratio is 0.31 % — one start every 3.4 days, 15 minutes per start. Cycling that light is good for the pumps and bad for confidence: **at that duty a failed standby would never be discovered by use**. A witnessed monthly test of the standby path *and* of the hand pump is therefore a maintenance requirement of this design, not an operator's discretion.

16.5 The three hydraulic zones

THE THREE HYDRAULIC ZONES - AND THE ONE THAT HAS NOWHERE TO GO

Water inside a sealed box is a protective problem before it is a plumbing problem: every route out is a route in.



Separately: the stairwell sump SU-02 (1.0 m³) with PU-04 / PU-05 at 2 L/s takes the approach and stairwell surface water to SK-03. 32 h to fill at the worst door-open driving-rain rate.

Figure 42 The three hydraulic zones, and the one that has nowhere to go.

Zone	Extent	Destination	Status
Z1 CLEAN	bays 1–5, inside the gas-tight envelope	clean sump SU-01 in bay 5, then the storm soakaway	COMPLETE [C]
Z2 DECON	bay 6 — inside the envelope but dirty	1 000 L tank TK-01, drawn in bay 8, across the gas-tight boundary	ROUTE UNDEFINED [U]
Z3 GREY	bays 7 and 8, outside the gas-tight envelope	none recorded anywhere	NO DESTINATION [N]

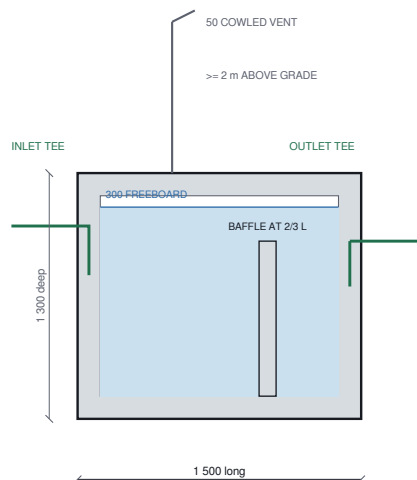
Zone 3 is not a small omission. Bay 8 holds the generator, its fuel system and the main LV board, and bay 7 is the stair shaft down which anything entering the shelter drips. Neither has a recorded drainage destination. **It is carried as a gap rather than solved here, because choosing a destination for a grey zone outside the envelope is a protective decision about what may cross the boundary, not a plumbing one.**

And zone 2 is worse in kind if not in size. The effluent from the decon airlock is the most contaminated water this structure will ever hold, and the tank it goes to is drawn on the far side of the protective boundary from the airlock that produces it. **The route is not defined, and neither is the emptying route.**

16.6 Sewage and foul drainage

FOUL DRAINAGE - SEPTIC TANK AND SOAK PIT TO IS 2470

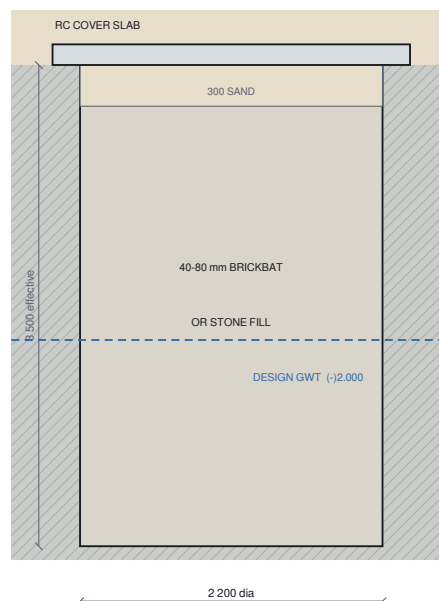
10 users at 45 lpcd = 450 L/day. The tank is sized for the PEACETIME duty: the shelter is not occupied continuously, and the sheltered case belongs to the airlock, not to the lavatory.

ST-01 SEPTIC TANK 1 : 40

$$1.50 \times 0.75 \times 1.00 \text{ liquid} = 1\,125 \text{ L}$$

1 050 L required by IS 2470 (Pt 1) Table 1 - PASS

two compartments · inlet and outlet tees · 300 freeboard

SK-01 / SK-02 SOAK PIT 1 : 55

side area 24.19 m² - the base is assumed blinded by silt

WIDENED FROM 2.000 TO 2.200 DIAMETER, NOT DEEPEMED.

Deepening would drive the pit below the design groundwater table at (-)2.000, where a soak pit cannot soak. Widening also keeps SK-01 and SK-02 one detail, one cover slab and one set of spare materials.

THE 20 L/m²/day ABSORPTION RATE IS ASSUMED [A], AND A PERCOLATION TEST TO IS 2470 (Pt 2) CL. 4 IS MANDATORY.

It, not this calculation, sets the final size. SK-01 is 2.3 % short of its own stated requirement - 21.99 m² against 22.5 m² - and that is recorded, not resized: the test may move the requirement by far more than 2.3 %.

Figure 43 Foul drainage - the septic tank and the soak pit, to IS 2470.

In protective mode the shelter is sealed and uses sealed-cassette chemical toilets — nothing is discharged at all. The septic tank serves **peacetime use only**: design population 10, 45 lpcd = 450 L/day, tank 1.5 × 0.75 × 1.0 m liquid to IS 2470 (Pt 1) Table 1, two compartments, baffle at $\frac{2}{3}$ L, inlet and outlet tees, 50 mm cowled vent ≥ 2 m above grade. Every IS 2470 (Pt 1) check reproduces: **450 L detention + 600 L sludge = 1 050 L required against 1 125 L provided (+7.1 %)**; L/B = 2.0; B = 750; depth 1.00 m; freeboard 300 → overall 1.30 m. **PASS.**

16.7 The operating procedure — peacetime, warning, attack and recovery

The hydraulics do not change between modes. What changes is which routes out of the envelope are allowed to be open, and therefore where each stream is permitted to go. This section states that once, mode by mode, so that it can go on a drill card instead of being reasoned out during an event.

WATER, SEWAGE AND DRAINAGE - WHAT HAPPENS TO EVERY STREAM IN EACH OPERATING MODE

The hydraulics do not change between modes. What changes is which routes out of the envelope are allowed to be open, and therefore where each stream is permitted to go.

	1 PEACETIME unfiltered running	2 PROTECTIVE filtered, +50 Pa	3 CLOSED sealed, 48 h	4 PURGE airlock in use	5 RECOVERY post all-clear
SEEPAGE 200 L/day	sump, pumped	sump, pumped	SUMP, PUMPED	sump, pumped	sump, pumped
CONDENSATE 200 L/day	sump, pumped	sump, pumped	SUMP, PUMPED	sump, pumped	sump, pumped
FOUL 450 L/day	septic, soak pit	septic, soak pit	CASSETTE, SEALED	CASSETTE, SEALED	septic, soak pit
DECON on use	none produced	TK-01, tanker	none produced	TK-01, tanker	TK-01, tanker
STAIRWELL 0.10 L/s	SU-02, soakaway	SU-02, soakaway	outside the envelope	SU-02, soakaway	SU-02, soakaway

IN CLOSED MODE NOTHING LEAVES THE ENVELOPE EXCEPT THROUGH THE RISING MAIN, AND THE RISING MAIN IS THE ONE ROUTE THAT CANNOT BE SHUT.

Groundwater does not stop because the shelter is sealed. Seepage and condensate keep arriving at 400 L/day while every blast valve is closed, so the clean sump pump stays on the essential board through the whole of mode 3 and the hand pump PU-03 is what covers it if the battery fails.

AND THE FOUL STREAM STOPS ENTIRELY, BY DESIGN.

In protective occupation the lavatory uses SEALED-CASSETTE CHEMICAL TOILETS and nothing is discharged at all. The septic tank and the soak pit are PEACETIME plant - which is why they are sized for 10 users at 45 lpcd and not for nine occupants for 96 hours.

THE DECON STREAM IS TANKER-ONLY IN EVERY MODE AND NEVER REACHES GROUND. It is the most contaminated water this structure will ever hold, its 1 000 L tank is drawn across the gas-tight boundary from the airlock that produces it, and NEITHER THE ROUTE TO THE TANK NOR THE ROUTE OUT OF IT IS DEFINED - open item, and the one line of this table that cannot yet be built.

Figure 44 Water, sewage and drainage - what happens to every stream in each of the five operating modes.

16.7.1 What happens to every stream, in every mode

Stream	1 PEACETIME	2 PROTECTIVE	3 CLOSED	4 PURGE	5 RECOVERY
Seepage 200 L/day	sump, pumped out	sump, pumped out	SUMP, PUMPED OUT	sump, pumped out	sump, pumped out
Condensate 200 L/day	sump, pumped out	sump, pumped out	SUMP, PUMPED OUT	sump, pumped out	sump, pumped out
Foul 450 L/day	septic tank → soak pit	SEALED CASSETTE	SEALED CASSETTE	SEALED CASSETTE	septic tank → soak pit
Decon effluent	none produced	TK-01 → tanker only	none produced	TK-01 → tanker only	TK-01 → tanker only
Stairwell surface water	SU-02 → soakaway	SU-02 → soakaway	<i>outside the envelope</i>	SU-02 → soakaway	SU-02 → soakaway

16.7.2 Mode 1 — PEACETIME

The only mode in which anything foul leaves the structure by gravity.

- Lavatory discharges to **ST-01**, the 1 125 L septic tank, and thence to the foul soak pit **SK-01**. Sized for **10 users at 45 lpcd = 450 L/day**, which is the *peacetime* population — shift crews and maintenance — and **not** the nine sheltered occupants.
- The clean sump **SU-01** runs on its ordinary duty cycle: **one start every 3.4 days, fifteen minutes a start, a duty ratio of 0.31 %**.
- Routine, and it is the whole of the maintenance regime that matters:** a **witnessed monthly test** of the standby pump PU-02, of the hand pump PU-03, and of the stairwell standby PU-05. At a 0.31 % duty ratio **a failed standby would never be discovered by use**.
- De-sludge the septic tank on the IS 2470 interval; the chambers IC-01 and IC-02 exist for it.

16.7.3 Mode 2 — PROTECTIVE, and the changeover on warning

The changeover is a short list, and it is short on purpose.

- Stop using the lavatory's gravity discharge. Switch to sealed-cassette chemical toilets.** Nothing foul is discharged for the duration.

2. **Confirm every trap is primed.** A dry trap is an open pipe, and the envelope is about to be held at +50 to +100 Pa. The 75 mm deep seals hold **736 Pa** — 7.4 : 1 on the operating overpressure and 2.5 : 1 on the +300 Pa test — but only while there is water in them.
3. **Confirm the clean sump is at or below its stop level**, so that the store ahead of it is the full 3 375 L.
4. **Confirm PU-01 and PU-02 are on the essential board and that the changeover works.**
5. **Decon effluent begins on first use of the airlock, to TK-01, tanker only, never to ground, in any mode.**

16.7.4 Mode 3 — CLOSED, which is the case the design is really for

Nothing leaves the envelope except through the rising main, and the rising main is the one route that cannot be shut.

- **Groundwater does not stop because the shelter is sealed.** Seepage and condensate keep arriving at **400 L/day** with every blast valve closed.
- **3 375 L of sump against 400 L/day is eight days of store with no power at all** — four times the 48-hour limit that the soda lime puts on closed mode (Part 15.7). **The drainage is not what ends closed mode.**
- **The clean sump pump stays on the essential board through the whole of mode 3**, at 10 per cent duty, drawing 0.033 kW of the 1.283 kW essential load (Part 17.3).
- **If the battery fails, the hand pump PU-03 is the answer**, and it needs no power, no generator and no decision.
- **Foul: sealed cassettes only. Stairwell: outside the envelope, and not a concern until someone has to go out through it.**

The order to work in if the sump is rising and power is lost:

```
1  confirm the duty pump has actually failed -- not the level switch
2  start the standby, PU-02
3  if both are dead, HAND PUMP PU-03
4  record the level and the rate. 400 L/day into 3 375 L is EIGHT DAYS;
    there is time to diagnose, and the right response to a rising sump is
    never to open anything
```

The last line is the point of the whole procedure. Every instinct in ordinary building services says *find a way to let the water out*. Here, **every route out is a route in**, and the store exists precisely so that nobody has to make that trade under pressure.

16.7.5 Mode 4 — PURGE, each entry through the airlock

- The airlock purge is **5 air changes of stage 1 = 64 m³ at 300 m³/h = 12.8 minutes**, which sets the manning rate at **four to five persons an hour** (Part 15.4).
- **Decon effluent from airlock stages 1 and 2 goes to TK-01 on every use.** At 1 000 L the tank is the constraint on how many decon cycles can be run before it must be emptied, and **the emptying route is not defined** — open item.
- **The airlock floor falls 1:30 to segregated gullies** with a 50 mm upstand at W5 and at Blast Door 1, so that decon effluent cannot run into the clean zone. That upstand is a drainage detail doing a protective job.

16.7.6 Mode 5 — RECOVERY, after the all-clear

The order matters, and it is the reverse of the warning sequence with one addition.

1. **Do not restore the foul gravity discharge until the ground outside is assessed.** The septic tank vents to atmosphere and the soak pit discharges to ground; both are outside the protective boundary and neither has been under anyone's control.
2. **Empty TK-01 by tanker before resuming decon operations**, not after.
3. **Check every trap seal before the envelope is de-pressurised**, not after — a seal lost during closed mode is found more easily while there is still a pressure difference to find it with.
4. **Test the standby pumps and the hand pump**, because the monthly regime will have lapsed.
5. **Return the sump to its ordinary duty cycle** and log the volume pumped during the closed period: it is the only direct measurement of seepage this structure will ever produce, and **the 0.5 L/m²/day design allowance has never been verified against anything.**

That last item is worth a line on its own. The design's seepage rate is an allowance for an intact tanked structure, not a measurement. **The volume pumped over a sealed 48-hour period is a free, direct measurement of it**, and nobody will ever be better placed to take it.

16.7.7 What the procedure cannot state

Missing	Consequence
The route from bay 6 to TK-01, which crosses the gas-tight boundary	[U] — the decon line of the table above is the one that cannot yet be built
The emptying route for TK-01	[N] — tanker access, hardstanding and a connection point are all undefined
Any destination for zone 3, bays 7 and 8	[N] — the generator, its fuel and the main board all sit in a bay with no recorded drainage
Cassette type, capacity and the number carried	[N] — which means the foul provision for a 96-hour occupation is stated as a principle and not as a quantity
SK-03 and SK-04 sizes	[N] — both exist on the drawings; neither is sized anywhere

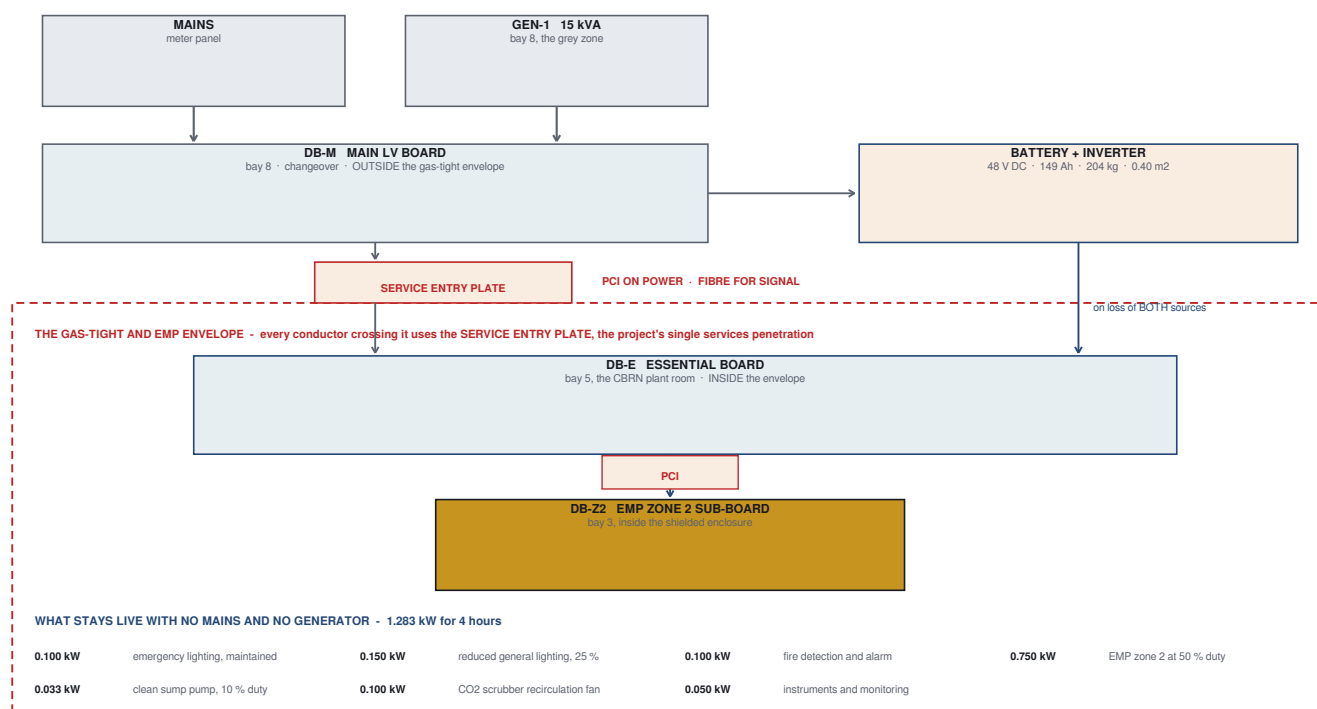
16.8 Findings

Ref	Finding
Soak pit size	$\pi \times 2.0 \times 3.5 = 21.99 \text{ m}^2$ against 22.5 m^2 required is 2.3 % short — arithmetic, not judgement. SK-01 is therefore 2.200 m diameter, depth unchanged at 3.500 m → 24.19 m², +7.5 %. Widened rather than deepened, because deepening drives the pit below the design water table, where it cannot soak at all. The mandatory percolation test still governs the final size, and may move it by far more than 7.5 %
Stairwell catchment	The 0.10 L/s for stairwell surface water is an open-cut figure — 7.2 m^2 of open pit at 50 mm/h — and the approach is now covered with the door at grade, so the catchment with the door shut is zero . 0.10 L/s is retained as a declared conservatism: it changes no pump, pipe or pit, since the stairwell pump is twenty times it, and a door-open driving-rain case is exactly the case a covered stairwell still has to survive
DR-F1	The clean sump cycles once every 3.4 days , 15 min per start, 0.31 % duty ratio . Cycling is fine — but at that duty a failed standby would never be discovered by use. A witnessed monthly test of the standby path and the hand pump is required
DR-F2	A drained screed reduces the finished clear height from the structural 3 200 to between 3 122 (far corner) and 3 175 (sump edge). A client-facing number must say which one it is
DR-F4	A drained floor cannot be built inside the 1.0 kPa mat allowance. At 24 kN/m^3 that buys 42 mm of screed; the adopted grading averages 52 mm = 1.24 kPa , an excess of 0.24 kPa = 25 kN over the 104 m^2 floor — 1.2 % of the mat's own weight, 0.4 % of the uplift, and acting in the FAVOURABLE direction for flotation. Referred to the structural engineer, not assumed
DR-F5	The protective boundary creates three hydraulic zones and two have no drainage destination. Zone 1 (bays 1–5) → clean sump, complete. Zone 2 (bay 6 decon) → its tank, which the sheet draws in bay 8, across the boundary — route undefined. Zone 3 (bays 7–8) → nothing recorded. A protective-design decision
BW-01	Builder's work, requested and not accepted. The one buried drain inside the envelope needs a 300 wide recess in the top of the mat, 150 deep rising to 216, leaving 384 of the 600 mat locally on the line of the peak transverse sagging moment. Structural engineer to accept or refuse; a fallback with no buried drain at all is drawn
DR-D1	Rainfall intensity 50 mm/h is the only intensity in the project. No return period, duration or IDF source

PART 17 — ELECTRICAL AND POWER

A basic package, on purpose. It stops at board level: sources, a load schedule, three boards, the essential / battery system, one single-line diagram. **No circuit schedule, no cable sizing, no luminaire or socket layout, no protection study** — that is a **declared scope boundary**, not a gap. Stating where a package stops is a design act; leaving a reader to discover the edge by looking for something that is not there is not.

ELECTRICAL SINGLE LINE - THREE BOARDS, ONE CABLE ENTRY, AND WHAT STAYS LIVE WITH NOTHING RUNNING



CONNECTED LOAD 6.256 kW = 7.360 kVA AT pf 0.85. GEN-1 AT 15 kVA IS 49.1 % UTILISED - high enough to avoid wet-stacking, low enough to carry growth.

The largest motor is the filter fan at 0.379 kW; even direct-on-line it is about 2.7 kVA, so there is no starting problem. THE PACKAGE STOPS AT BOARD LEVEL - no circuit schedule, no cable sizing, no luminaire layout.

GROUNDWATER DOES NOT STOP BECAUSE THE SHELTER IS SEALED. The hand pump PU-03 and the two hand cranks on the filter fans are what cover the battery failing; no electrical design should obscure them.

Figure 45 Electrical single line - three boards, one cable entry, and what stays live with nothing running.

17.1 The load schedule

Tag	Load	Board	kW	Class
L-01	General lighting, bays 1-8, LED	DB-E	0.520	[A]
L-02	Emergency lighting, maintained, DC	DB-E	0.100	[A]
P-01	Small power, socket outlets	DB-M	1.000	[A]
F-01	Filter train fan, 1 duty of 2	DB-E	0.379	[R]
D-01	Dehumidifier	DB-E	1.000	[A]
U-01	Clean sump pump, 1 duty of 2	DB-E	0.334	[R]
U-02	Stairwell pump, 1 duty of 2	DB-M	0.223	[R]
Z-01	EMP Zone 2 ops / comms	DB-Z2	1.500	[A]
S-01	Fire detection and alarm panel	DB-E	0.100	[A]
B-01	Battery charger / inverter	DB-M	0.800	[A]
G-01	Generator auxiliaries	DB-M	0.300	[A]
	CONNECTED LOAD		6.256 kW	
	at power factor 0.85		7.360 kVA	

Standby units are not counted twice — the standby fan and the two standby pumps never run simultaneously with their duty partners.

GEN-1 rating	= 15.0 kVA	[C]
Connected load	= 7.360 kVA	
UTILISATION	= 49.1 %	
Spare	= 7.64 kVA	
Largest motor, the filter fan 0.379 kW; DOL start	~ 2.7 kVA	

The confirmed 15 kVA is about twice the connected demand, and 49 % sits in the healthy loading band for a diesel set — high enough to avoid wet-stacking, low enough to carry growth. There is no starting problem and the project's own figure needs no change.

17.2 Three boards, and one cable entry

Every conductor crossing the protective envelope uses the service entry plate — the project's single services penetration — with a **pulse-current injection suppressor on power and fibre for signal**, as Part 18 requires. **This package creates no new penetration.**

DB-M main LV in Bay 8 (outside the gas-tight envelope, mains

- generator with changeover); DB-E essential in Bay 5 (inside the envelope, fed from DB-M and

from the battery inverter on loss of both sources); DB-Z2 inside the shielded enclosure in Bay 3, fed from DB-E **through the PCI on the Zone 2 boundary.**

17.3 The essential system, and the battery that carries it

Emergency lighting 0.100 + reduced general lighting 0.150 + fire detection 0.100 + EMP Zone 2 at 50 % 0.750 + clean sump pump intermittent 0.033 + CO₂ scrubber recirculation fan 0.100 + instruments 0.050 = **1.283 kW**, carried for **4 h** on **149 Ah at 48 V** (depth of discharge 0.80, inverter efficiency 0.90).

Groundwater does not stop because the shelter is sealed. The clean sump pump has to stay powered through the closed mode — and **hand pump PU-03 is what covers it if the battery fails.**

And two sources need no electricity at all, both already confirmed: a hand crank on both filter fans, and the hand pump. These are the real last line, the project already has them, and no electrical design should be allowed to obscure them.

17.4 The CO₂ scrubber loop, checked rather than assumed

$Q = \text{production} / (\text{eta} \times C_{\text{target}})$

target CO2	eta 0.50	eta 0.80	eta 0.95	
0.5 %	72.0	45.0	37.9	m3/h
1.0 %	36.0	22.5	18.9	m3/h

ADOPTED 75 m3/h recirculation loop -- the worst cell, because designing to it costs almost nothing.

CHECK a 75 m3/h fan at an assumed 250 Pa across a packed bed draws ~ 11.6 W against the 0.10 kW allowance -- roughly 8 x margin.
NO ELECTRICAL VALUE CHANGES; the allowance is now CHECKED rather than assumed.

Not designed, not invented [N]: the absorber vessel, bed depth, face area, residence time; the **soda lime charge** (17.28 m³ of CO₂ is the duty — converting it to a mass needs vendor absorption capacity); the single-pass efficiency; and **where in Bay 5 the absorber stands**, the tightest bay at 1 560 clear where the filter trains already leave 110 mm at the sides.

17.5 The generator day tank

The fuel is sized from the generator's own duty, and then the design problem turns out to be the two small pipes rather than the tank:

600.6 kWh over 96 h x 0.35 L/kWh [A] = 210.2 L
ADOPTED 210 L usable . 250 L nominal . 275 L bund (110 %)
BAY 8 -- outside the gas-tight envelope, INSIDE the EMP boundary
welded steel, banded, contents gauge, low-level alarm to the panel

THE FILL AND THE VENT ARE THE REAL DESIGN QUESTION, NOT THE TANK.
Both cross the protective boundary, and bay 8 is inside the EMP boundary, so two new bores would each need blast, gas AND EMP treatment -- in the same wall where BV-4/BV-5 already FAIL the EMP criteria.

ADOPTED route both up the EXISTING SH-2 bore and add NO NEW PENETRATION
-- DN25 metallic fill with a lockable cap at the SH-2 head,
DN25 metallic gooseneck vent, both bonded to the shaft earth.

Use what already crosses. Never make a new hole in a protective boundary if an existing one can be shared — the same instinct that keeps the heat rejection on the ventilation air rather than on a new duct.

And the fit has not been checked. Two DN25 lines sharing a 600 × 600 bore with two DN350 blast valves is an arrangement that needs an HVAC and EMP fit check before it is drawn [A]. Fuel type unconfirmed, the 0.35 L/kWh rate [A], no vendor set, no fuel polishing and no tank fire suppression — all [N], all stated.

PART 18 — EMP PROTECTION

EMP is the one hazard in this project that does not care how thick the concrete is. A reinforcement cage is a mesh, a mesh is a low-pass filter, and a low-pass filter has a corner frequency. Everything in this Part follows from that single fact — including the decision to design the shielded enclosure as if the buried box were not there at all.

18.1 Design basis

Item	Value	Class
Performance requirement	80 dB, 10 kHz – 1 GHz, MIL-STD-188-125-1	[C]
Verification standard	IEEE Std 299, full survey	[C]
Incident field strength, waveform, E1/E2/E3 decomposition	NOT HELD, AND NOT NEEDED	[N]
What is protected	Equipment, so the shelter can still function afterwards. Not people	[C]
Shield architecture	Zone 2 designed to the full requirement STANDING ALONE . No credit taken for the concrete box at any frequency	[C]
Reinforcement bar spacing	150 mm, both curtains, both ways — an EMP requirement, stricter than IS 456 Cl. 26.3.3 needs	[C]
Bond strap	flat, ≤ 100 mm long, width : length ≥ 5 : 1	[R]
Earthing target	≤ 5 Ω — a power-safety and lightning number, NOT an EMP one	[C]

A performance requirement in decibels over a band can be met, and verified, from geometry alone. That is why this package could be designed at all without an incident field strength, and it is the single most useful property of the way MIL-STD-188-125-1 is written.

THE THREE EMP ZONES - AND THE ONE THAT CARRIES THE REQUIREMENT

MIL-STD-188-125-1 asks for 80 dB from 10 kHz to 1 GHz. Only one of these three zones delivers it.

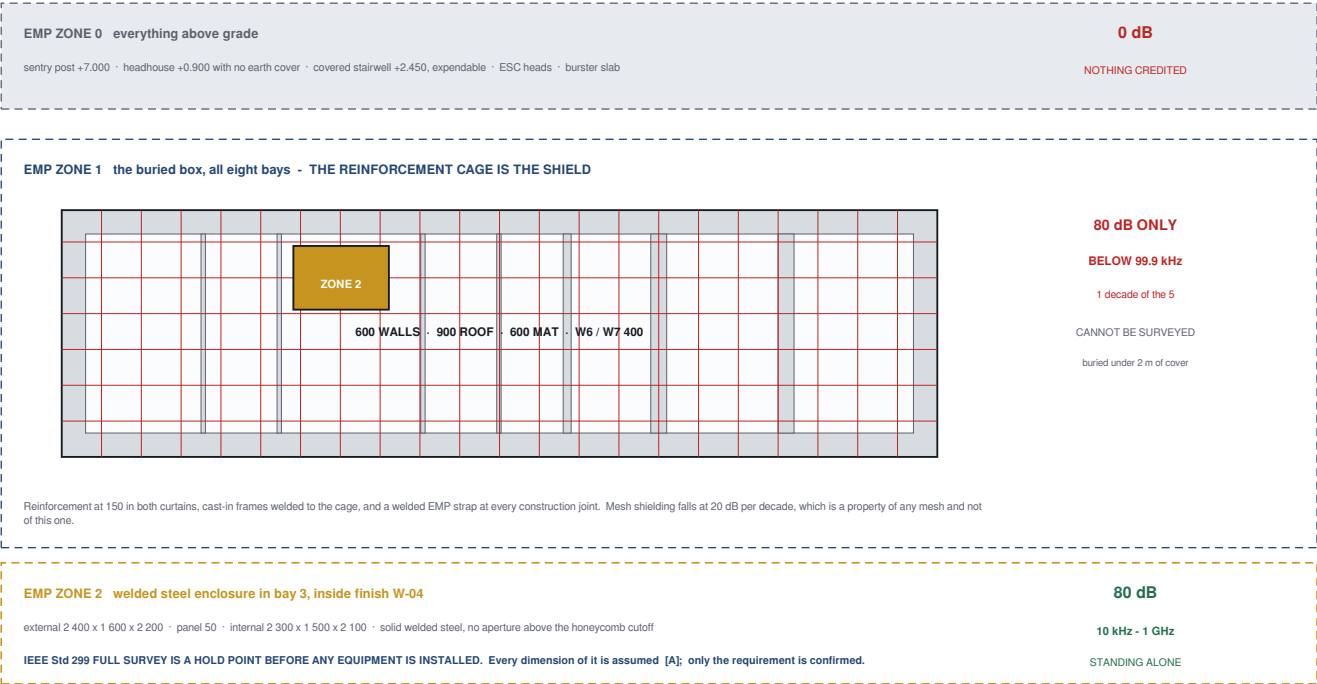


Figure 46 The three EMP zones, and the one that carries the requirement.

18.2 The zone model

Zone	What it is	Performance	Verification
EMP ZONE 0	Everything above grade — sentry post +7.000, headhouse +0.900 with no earth cover , covered stairwell +2.450 (expendable), shaft heads, burster slab	None credited	n/a
EMP ZONE 1	The buried box, all eight bays . The reinforcement cage	99.99 dB at 10 kHz → 0 dB at 999 MHz , pierced by an 8.85 m ² stair void and two 1 400 shafts. 80 dB only below 99.93 kHz	Cannot be surveyed — no accessible exterior under 2 m of cover
EMP ZONE 2	Welded steel enclosure, Bay 3	80 dB, 10 kHz – 1 GHz, STANDING ALONE	IEEE Std 299 full survey — HOLD POINT

The design rule, adopted with the model — the whole package in one line

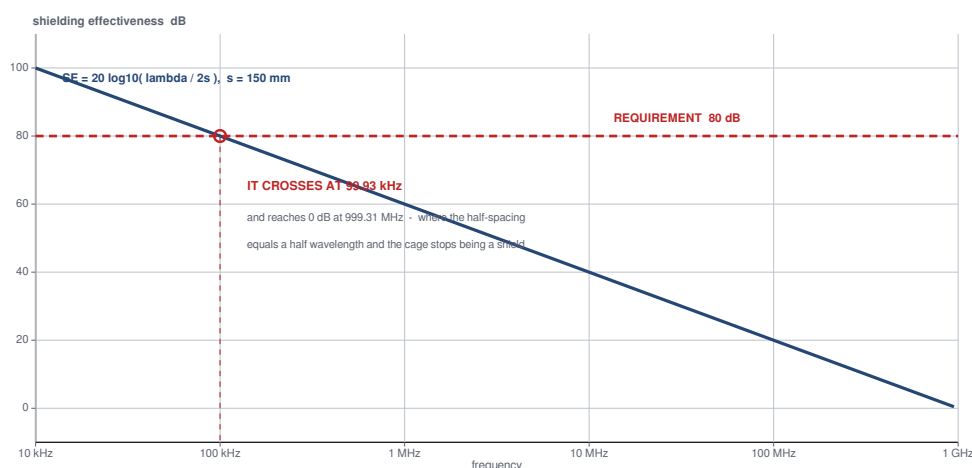
EMP Zone 2 is designed to the full 80 dB standing alone. No attenuation from the concrete box is credited at any frequency.

The cage is then **margin, not design** — which is the only defensible way to use a shield you can never survey, buried under two metres of engineered cover.

18.3 Why the cage cannot be the boundary, in one chart

The shielding effectiveness of a mesh follows $SE = 20 \log_{10}(\lambda / 2s)$, with s the half-spacing. At the project's 150 mm bar spacing that is a straight line on a log-frequency axis falling at **20 dB per decade**, and the requirement it is measured against is flat at 80 dB.

SHIELDING EFFECTIVENESS OF THE REINFORCEMENT CAGE - 20 dB PER DECADE AGAINST A FLAT 80 dB REQUIREMENT



THE CAGE MEETS 80 dB OVER ONE DECADE OF THE FIVE THE STANDARD ASKS FOR. IT IS NOT A SUBSTITUTE FOR ZONE 2.

This is a property of any mesh: a 150 mm cage cannot do better, and closing the grid to reach 1 GHz would need a spacing of about 0.15 mm - a sheet, not a cage.

It also cannot be surveyed. The exterior is under two metres of engineered cover, so the figure above is a calculation and stays one.

Figure 47 Shielding effectiveness of the reinforcement cage against a flat 80 dB requirement.

Frequency	Wavelength	SE dB	Against 80 dB
10 kHz	29 979 m	99.99	+19.99 PASS
30 kHz	9 993 m	90.45	+10.45 PASS
99.93 kHz	3 000 m	80.00	the crossing
100 kHz	2 998 m	79.99	-0.01 FAIL
1 MHz	299.8 m	59.99	-20.01 FAIL
10 MHz	29.98 m	39.99	-40.01 FAIL
100 MHz	2.998 m	19.99	-60.01 FAIL
999.31 MHz	300.2 mm	0.00	the cutoff — half-spacing = half-wavelength
1 GHz	299.8 mm	0.00	-80.00 FAIL

The cage meets 80 dB over one decade of the five the standard asks for. That is not a defect in this cage; it is a property of every cage. Closing the grid until it reached 1 GHz would need a spacing of about **0.15 mm** — a sheet, not a reinforcement cage.

And it cannot be surveyed. The exterior is under two metres of engineered cover, so there is nowhere to stand a transmitter. The figures above are a calculation and they stay one, which is the second reason the enclosure in bay 3 is designed to the full requirement standing alone.

18.4 What the structural design already does right, none of it labelled EMP

1. **The 150 mm bar spacing is a deliberate EMP decision, and it was the right one.** Stricter than IS 456 Cl. 26.3.3 needs; recorded in the master, in the structural package, in eight bar schedules and on the drawings.
2. **The cast-in frames are already specified as EMP bonds** — every blast door frame *"cast in and welded to the cage"*, and the rule is applied even to the **non-blast-rated** headhouse door, because the shield must be electrically continuous.
3. **Every construction joint already carries a welded Cu / galvanised EMP strap.** A construction joint is where a cage loses continuity, and the project already knew it.
4. **There are no movement joints inside the protective envelope, and the reason is recorded:** *a movement joint is a guaranteed blast, gas and EMP discontinuity.*

18.5 Every penetration of the boundary

A bore through a conducting wall is a **waveguide below cutoff**: it attenuates everything below $f_{c} = 1.8412 c / (\pi d)$ for the TE₁₁ mode, at roughly $32 L/d$ decibels beyond it. A bore through a *concrete* wall is not a waveguide at all, and the difference decides five of the ten rows below.

EVERY PENETRATION OF THE PROTECTIVE BOUNDARY - AND WHETHER THE HOLE IS ITS OWN WAVEGUIDE

A bore through a conducting wall attenuates below its cutoff: $f_{c} = 1.8412 c / (\pi d)$ for the TE₁₁ mode, and about $32 L/d$ dB of attenuation beyond it.

TAG	WHAT AND WHERE	BORE	DEPTH	BOUNDED BY	CUTOFF MHz	WBC dB	
BV-1 / BV-2	blast valve, W1, bay 1	100 dia	600	steel	1 757	192	PASS
BV-3	blast valve, W6 at (14998, 4900)	100 dia	400	steel	1 757	128	PASS
BV-4 / BV-5	blast valve, east wall, bay 8	350 dia	600	steel	502	55	FAIL
SEP	service entry plate, north wall	800 wide	600	steel	187	20	FAIL
PD-05	rising main through the SEP	50 dia	600	steel	3 514	384	PASS
ESC 1	escape shaft, bay 1	1 400 dia	3 050	CONCRETE	126	-	FAIL
ESC 2	escape shaft, bay 8	1 400 dia	3 600	CONCRETE	126	-	FAIL
VOID	stair void through the roof, bay 7	3 160 wide	900	CONCRETE	47	-	FAIL

A CONCRETE BORE IS NOT A WAVEGUIDE.

The waveguide-below-cutoff formula needs conducting walls. The two escape shafts and the stair void are concrete, so the cutoff column is what the bore WOULD give if it were metallic; the real figure is the plain aperture value, about 41 dB and 34 dB at 1 MHz.

THE STAIR VOID IS THE FINDING.

It is 3.16 m wide through a 900 slab, it is the entry route, and NO TREATMENT FOR IT EXISTS OR IS PROPOSED ANYWHERE IN THE PROJECT. A bonded conducting hatch works at the escape shafts; nothing equivalent is specified here.

AND THE GENERATOR VALVES REOPEN AFTER THE SHOCK.

BV-4 and BV-5 are ruled to reopen for generator operation, which leaves two DN350 bores open through the post-attack period - the two bores that already fail both criteria. That sharpens the open question of whether bay 8 is inside the EMP boundary at all.

The service entry plate is treated as the shield itself: solid, welded, bonded 360 degrees to its cast-in frame, with each sleeve through it treated one by one.

Figure 48 Every penetration of the protective boundary, and whether the hole is its own waveguide.

Tag	What and where	Bore	Depth	Bounded by	Cutoff MHz	WBC dB	Verdict
BV-1 / BV-2	blast valve, W1, bay 1	100 Ø	600	steel	1 757	192	PASS
BV-3	blast valve, W6 at (14998, 4900)	100 Ø	400	steel	1 757	128	PASS
BV-4 / BV-5	blast valve, east wall, bay 8	350 Ø	600	steel	502	55	FAIL
SEP	service entry plate, north wall	800 wide	600	steel	187	20	FAIL
PD-05	rising main through the plate	50 Ø	600	steel	3 514	384	PASS

Tag	What and where	Bore	Depth	Bounded by	Cutoff MHz	WBC dB	Verdict
ESC 1	escape shaft, bay 1	1 400 Ø	3 050	concrete	126	—	FAIL
ESC 2	escape shaft, bay 8	1 400 Ø	3 600	concrete	126	—	FAIL
VOID	stair void through the roof	3 160 wide	900	concrete	47	—	FAIL

A concrete bore is not a waveguide. The waveguide-below-cutoff formula needs conducting walls. For the two escape shafts and the stair void the cutoff column above is what the bore *would* give if it were metallic; the real figure is the plain aperture value, about **41 dB** and **34 dB** at 1 MHz respectively. **Lining the shaft does not fix it either** — a liner that is not bonded top and bottom is a floating conductor, and one that is bonded only at the head is a quarter-wave antenna at some frequency in the band.

The treatments that work are stated, including for the two that have none. A **bonded conducting hatch** answers each escape shaft, and Part 12 designs the structural half of it. The service entry plate is treated as **the shield itself** — solid, welded, bonded 360° to its cast-in frame, with each sleeve through it treated one at a time. A **honeycomb waveguide panel inboard of the valve** answers BV-4 and BV-5. **Nothing answers the stair void, and nothing is invented for it.**

18.6 EMP Zone 2 — the enclosure

Every dimension in this section is [A]. The project confirms the enclosure is **required** and contains no specification for it.

Host	Bay 3, internal 3 500 × 5 000 × 3 200
External	2 400 × 1 600 × 2 200, at X 5820–8220, Y 3700–5300
Shielded panel	50 mm
Internal clear	2 300 × 1 500 × 2 100 — 3.45 m ² / 7.245 m ³
Shielded envelope area	25.28 m ² — walls 17.60, roof 3.84, floor 3.84
Total seam length	24.80 m
Clearances	300 mm north, south and west; 800 east; 1 000 head — 850 clear above the 150 mm HVAC duct zone

A 300 mm gap is held on every free face so an IEEE Std 299 survey can physically reach every seam. A shielded room you cannot walk around cannot be tested, and **an untested shield is a claim, not a shield.** The enclosure is entirely clear of the Y 2500–3400 circulation route.

The seam length is the risk, not the area. Every one of those **24.8 metres** has to be continuously welded or continuously gasketed, and IEEE Std 299 will find the metre that is not.

The five ways in:

PoE	Treatment	Performance
PoE-1 ACCESS §5.4	RF-gasketed or knife-edge shielded door	≥ 80 dB · type and vendor [N]
PoE-2 VENTILATION §5.5	Honeycomb, 6 mm cell × 25 mm deep	cutoff 29.3 GHz · 133 dB (+53 dB margin)
PoE-3 POWER §5.7.2.1	PCI on every conductor, mounted on the boundary	[N] — no residual quoted, deliberately
PoE-4 SIGNAL §5.7.4.1	Fibre, no metallic strength member or armour	A dielectric is not a penetration
PoE-5 RF §5.7.6	—	Nothing to apply it to — EM-F6

PoE-3 quotes no residual figure on purpose. MIL-STD-188-125-1 specifies pulse-current injection performance **by pulse test**, and the project's register carries the section number only. **Inventing a number would be worse than** [N].

PoE-4 is the bargain of this package. Fibre with no metallic member is **not a penetration at all**. It is the one place this project can buy perfect performance for almost nothing, and it should take it.

18.6.1 The panel is not the shield — the seams are

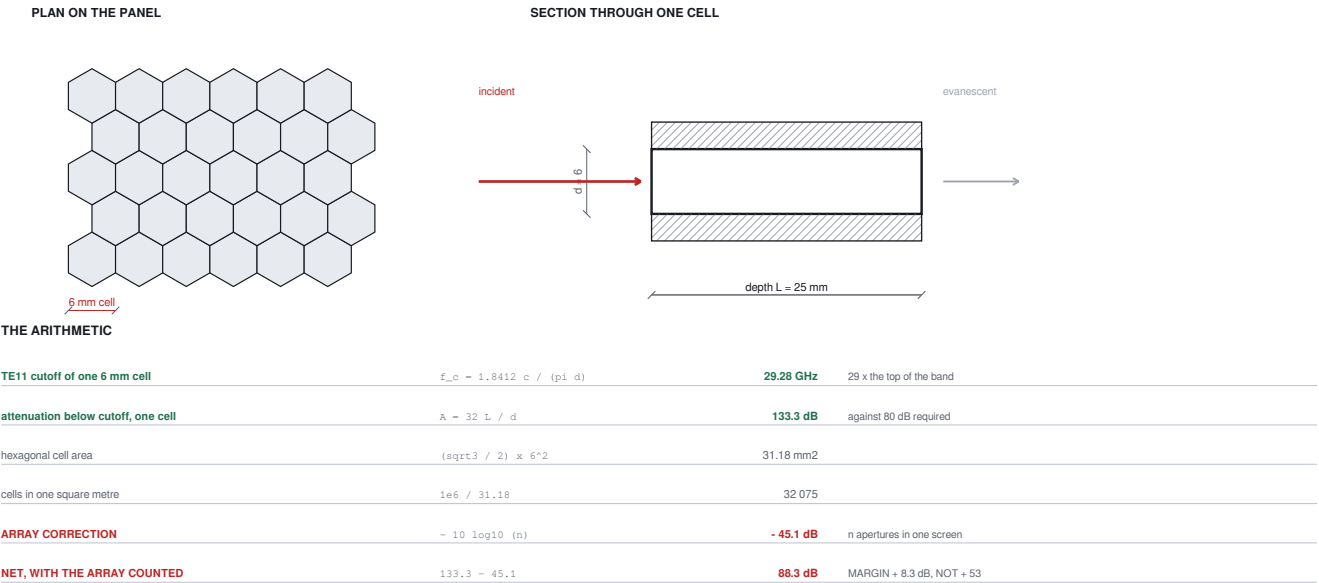
Part 2.8.4 works the arithmetic: for solid low-carbon steel, **about 1 mm gives 80 dB by absorption alone at 10 kHz**, the hardest frequency in the band, and the plane-wave reflection term adds another 97 dB on top of it. A 2 mm skin is a quarter of a thousand decibels.

So no number in this section is about the panel. The enclosure's measured performance will be decided by its **24.8 metres of seam**, its **shielded door**, and its **four points of entry** — and by nothing else. That is the reason the acceptance test is a survey of joints rather than a measurement of steel, and the reason a 300 mm gap is held on every free face so that a surveyor can physically reach every metre of seam.

18.7 The honeycomb waveguide panel, and the margin that is not what it looks like

The enclosure holds powered equipment inside a sealed shelter and must pass air. A ventilation opening in a shield is answered by a **honeycomb waveguide panel** — an array of tubes, each one below cutoff at every frequency in the band.

THE HONEYCOMB WAVEGUIDE PANEL - TENS OF THOUSANDS OF PARALLEL TUBES, EACH BELOW CUTOFF



THE SINGLE-CELL NUMBER IS AN UPPER BOUND, AND THE ARRAY CORRECTION IS WHY VENDOR TEST DATA IS NOT OPTIONAL.
A panel still passes at 88 dB, but a margin of 8 dB is a margin a bad gasket eats. Mount the panel INBOARD of the blast device - the valve takes the pressure, the honeycomb takes the radio frequency - and require a certified attenuation curve for the panel as built, not for one cell.
Two consequences are referred rather than resolved: a honeycomb panel adds pressure drop, and the fan duty was built without one; and the panel's own blast rating is [N].

Figure 49 The honeycomb waveguide panel, and the array correction that turns a 53 dB margin into an 8 dB one.

CELL	6 mm across flats	[A]
DEPTH	25 mm	[A]
TE11 cutoff	$f_{c} = 1.8412 \text{ c} / (\pi \text{ d})$ $= 1.8412 \times 2.998\text{e}8 / (\pi \times 0.006)$ $= 29.28 \text{ GHz}$	-- 29 x the top of the band
Attenuation below cutoff	$A = 32 \text{ L} / \text{d} = 32 \times 25 / 6$	$= 133.3 \text{ dB}$ against 80 required
*** AND THEN THE ARRAY CORRECTION, WHICH THE SINGLE-CELL NUMBER OMITTS ***		
hexagonal cell area	$(\text{sqrt}3 / 2) \times 6^2$	$= 31.18 \text{ mm}^2$
cells in one square metre	$1\text{e}6 / 31.18$	$= 32\ 075$
array term	$- 10 \log_{10} (32\ 075)$	$= - 45.1 \text{ dB}$
NET	$133.3 - 45.1$	$= 88.3 \text{ dB}$
MARGIN OVER 80 dB		$= + 8.3 \text{ dB}$

PR2-F2 — the margin is +8.3 dB, not +53 dB, and the difference matters. 133 dB is the attenuation of one cell. A panel is tens of thousands of cells in one screen, and the classical $-10 \log_{10} (\text{n})$ correction for n identical apertures applies to it. The panel still passes — but eight decibels is a margin a bad gasket eats, where fifty-three is not.
This is recorded, not corrected. The project's own figure of +53 dB is the single-cell value; it is reproduced in this report and the array term is offered alongside it as an observation. What it changes is a procurement requirement, not a design value: require a certified attenuation curve for the panel AS BUILT, not for one cell.

Two consequences are referred rather than resolved. A honeycomb panel adds pressure drop, and the fan duty in Part 15 was built without one — which matters because five of the eight loss components are already vendor data. And the panel's own blast rating is [N]; it is mounted inboard of the blast device so that the valve takes the pressure and the honeycomb takes the radio frequency, but that arrangement is a requirement rather than a tested assembly.

18.8 Bonding and earthing — where the design actually lives

A bond is not a resistor. At EMP frequencies it is an inductor, and its impedance rises linearly with frequency (Part 2.8.7):

Strap	L	X at 1 MHz	X at 10 MHz	X at 100 MHz
600 mm × 25 × 3	512.2 nH	3.22 Ω	32.2 Ω	321.8 Ω
100 mm × 50 × 3 — ADOPTED	38.9 nH	0.24 Ω	2.45 Ω	24.5 Ω

The five bonding rules, and they are the ones that get built wrong:

1. **Every bond $\leq$ 100 mm long.** A 600 mm strap is 322 Ω at 100 MHz — that is not a bond, it is a resistor with a green sleeve on it.
2. **Width-to-length at least 5 : 1.**
3. **Flat strap only** — never a round wire, never a pigtail, never *"loop it round to the nearest stud"*.
4. **Clean bare metal both ends**, protected after making off.
5. **The shield bonds to the structure at ONE place.** A second bond is a loop, and a loop is an antenna.

Earthing, and why $\leq$ 5 Ω is the wrong target to chase here:

Electrode	$\rho = 1\,000\ \Omega\cdot\text{m}$	$\rho = 10\,000\ \Omega\cdot\text{m}$
One 3 m × 16 mm rod	335 Ω	3 349 Ω
Rods needed for 5 Ω, <i>ignoring interaction</i>	67	670
The structure itself — 136.4 m ² , r_{eq} 6.589 m	38 Ω	379 Ω

$\leq$ 5 Ω is a power-safety and lightning requirement from IS 3043 / IEEE 142. It is real and it still applies. It is not an EMP number. An EMP shield works by being **equipotential**, and equipotential is decided by **bonding inductance**, not by earth resistance. The mat and its cage are already a large concrete-encased electrode — an order of magnitude better than a rod, and free, because it is already built. **Bond to the structure; do not chase rods.**

18.9 Findings

Ref	Finding
EM-F1	The stair void is a 2 800 × 3 160 (8.85 m²) aperture that never reaches 80 dB anywhere in the band and is simply open above 47.4 MHz. It does not open into soil — it opens into the headhouse, which is at +0.900 with no earth cover , and thence to grade through a stairwell that is declared expendable. The entry route is an open electromagnetic path from grade to Bay 7 , and Blast Door 1 is the only thing across it — its RF performance is vendor data the project does not have. <i>This is not a criticism of the architecture: a shelter needs a staircase, and the void is where a staircase has to go. It is a statement that the blast boundary and the EMP boundary are not the same surface, and the project has only ever drawn one of them</i>
EM-F2	The cage meets 80 dB over one decade of the five required and gives 0.00 dB at 1 GHz. A genuine low-frequency measure; must not be described as a MIL-STD boundary
EM-F3	"EMP Zone 2" had been named since Rev F with no Zone 1 and no Zone 0 ever defined, and no specification for the enclosure itself. Four revisions, no shield
EM-F4	BV-4 and BV-5 (DN350) are the only blast valves failing both criteria — cutoff 502 MHz, 54.8 dB. Bay 8 is inside the EMP boundary , so the honeycomb waveguide panels inboard of them are required, not conditional — and because the generator runs in closed mode, the panels have to work with the valves open
EM-F5	$\leq$ 5 Ω is not achievable with rods in Deccan basalt (335 Ω per rod at the <i>low</i> resistivity bound) and is not an EMP requirement in any case. The structure is already the better electrode
EM-F6	There is no antenna, mast, feeder or communications design anywhere in this project. §5.7.6 is in the register with nothing to apply it to. An ops room that cannot transmit is an ops room in name only — and an antenna is by definition a deliberate conductor from outside to inside, the hardest EMP penetration there is

18.10 Verification, and what must not be claimed

EMP ZONE 2 — SURVEY IT. Full IEEE Std 299 survey, 10 kHz – 1 GHz, on the completed enclosure with every penetration made off and every panel closed. Acceptance **80 dB**. **This is a HOLD POINT: no equipment is installed before it passes**, because a failed survey means opening seams.

EMP ZONE 1 — DO NOT SURVEY IT, AND DO NOT CLAIM IT. A buried box under 2 m of engineered cover **cannot** be surveyed to IEEE 299 — there is no accessible exterior to put a transmitter on. **Any statement that the box gives 80 dB is unsupportable and should not be made.**

The Zone 2 acceptance survey, and what it consists of:

Standard	IEEE Std 299 , full survey
Band	10 kHz – 1 GHz , covering the low-frequency magnetic, resonance and plane-wave ranges

Condition	Enclosure complete, every penetration made off, every panel closed, the door shut and latched
Coverage	Every seam , the door perimeter , and each of the four points of entry individually
Acceptance	80 dB
Status	HOLD POINT — no equipment is installed before it passes

The hold point is placed there for a reason that is entirely practical: a failed survey means opening seams. Doing that on an empty enclosure is a day's work; doing it on one full of installed equipment is a different job altogether.

What *can* be checked on site, cheaply, at the only moment it can be fixed:

1. **Continuity of the cage across every construction joint — before the pour.** Once it is cast, a missed strap is permanent and invisible.
2. **Continuity of every cast-in frame to the cage — before the pour.** Same reason. This covers both blast door frames, the headhouse door frame, every blast valve sleeve and the service entry plate.
3. **Earth resistance, early** — because in 1 000–10 000 $\Omega\cdot\text{m}$ basalt the answer decides how the earthing is done, and finding out late is expensive.
4. **Bond resistance at every strap after making off**, against a stated value.

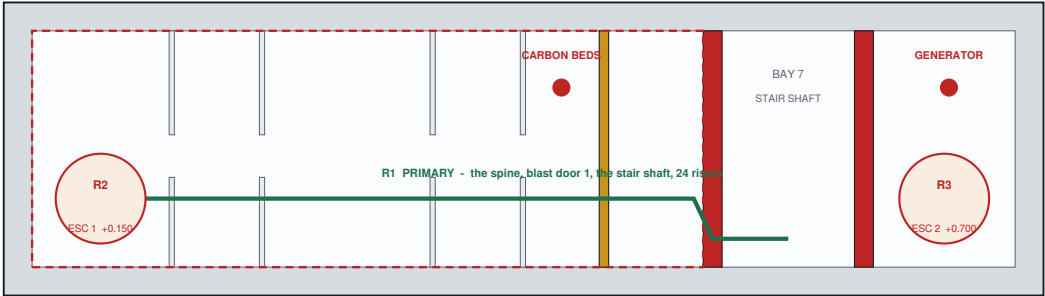
None of these prove shielding effectiveness. All of them catch the mistakes that destroy it — and every one of them has to happen at a moment that passes and does not come back.

Every cage figure in this project is an upper bound. The classical $-10 \log_{10}(n)$ array correction is not applied, the crossings are **tied rather than welded** — and whether they are tied or welded is itself [N] — and no concrete absorption is credited. **A measured cage will be worse than the calculation.**

PART 19 — FIRE AND LIFE SAFETY

The governing fact is Part 2.10: this is a sealed shelter and it cannot be ventilated of smoke.

FIRE AND EGRESS - THREE ROUTES, ONE SMOKE COMPARTMENT 20.8 m LONG, AND A SHELTER THAT CANNOT BE VENTILATED OF SMOKE



WHERE THE FIRE IS DECIDES THE ROUTE

BAY 8 - the generator	R1, or R2 if the spine is smoke-logged BLAST DOOR 2 SHUT. Bay 8 is separately ventilated at 2 600 m³/h. R3 IS INSIDE THE FIRE COMPARTMENT - DO NOT USE IT
BAYS 1 TO 6 - occupied	R1 if the route to bay 7 is clear; otherwise R2 from the west, R1 or R3 from the east bays 1 to 6 are ONE compartment - smoke starting in bay 1 reaches bay 6 unobstructed
BAY 7 - the stair shaft	R2 or R3 the primary route IS the fire. Both blast doors shut
HEADHOUSE OR STAIRWELL	R2 or R3 R1's surface end is blocked; both are outside the boundary and the stairwell is expendable

R1 IS THE ONLY ROUTE THAT DOES NOT REQUIRE CLIMBING A SHAFT, AND THE ONLY ONE USABLE BY AN INJURED PERSON.

THE MOST LIKELY FIRE IN THE SHELTER DENIES ONE OF ITS THREE ESCAPE ROUTES.

ESC 2 is in the same bay as the generator. Both facts are confirmed and neither is wrong alone; the coupling is what had never been stated, and it is why the rule above routes a bay 8 fire away from ESC 2.

A FIRE DURING THE CLOSED MODE CANNOT BE VENTILATED AT ALL, and clearing smoke means opening a blast valve - which breaks the protection the closed mode exists to provide. No rule exists for which hazard takes precedence. That belongs to the client, and it should be made before it is needed rather than during.

Figure 50 Fire and egress - three routes, one smoke compartment 20.8 m long, and the rule that picks a route.

19.1 What the design already does well

1. The largest fire load is already compartmented away from the occupants. The generator is in Bay 8, behind Blast Door 2 in the 400 mm wall W7, and Bay 8 is separated from the occupied bays by Bay 7, the stair shaft. Two barriers and a buffer between the fuel and the people.
2. Bay 8 has its own air path that never touches the gas-tight envelope — 2 600 m³/h through BV-4/BV-5. A generator fire is therefore ventilated, and ventilated away from the occupied zone. The one part of this shelter that can be cleared of smoke is the one part most likely to produce it.
3. The two dead-end bays are exactly the two bays with escape shafts. Bay 1 has ESC 1; Bay 8 has ESC 2. No occupant is ever in a dead end without a way up out of it.
4. Travel distances are short — the longest travel to a route is about 14.6 m, and ESC 1 is 1.5 m away in that same bay.

19.2 Where the fire load is

Bay	Load	Comment
8	Generator, fuel system, exhaust	The largest by far. The set, its fuel system, exhaust and acoustic treatment are not specified and not priced
5	CBRN filter trains	ASZM-TEDA activated carbon. Carbon beds self-heat; a filter fire is a known hazard in protected ventilation
1	Stores	Combustible stores load not scheduled anywhere [N]
3	Ops room, EMP Zone 2 enclosure	Electronics and cabling
4	Berthing, 9 berths	Bedding and personal effects
all	Cabling	Cable type, route, containment and fire-stopping are not specified anywhere [N]

19.3 The evacuation plan

Route	Path	Emerges at
R1 PRIMARY	Any bay → spine → Blast Door 1 → Bay 7 stair shaft → 24R @ 170.8333, 3 flights of 8, 1 200 wide	Headhouse (–)2.000, then 12R up the covered stairwell to the entry door at grade
R2	ESC 1 , Bay 1, 1 400 dia	Head +0.150 — a 6.250 m climb
R3	ESC 2 , Bay 8, 1 400 dia	Head +0.700 — a 6.800 m climb

R1 is the only route that does not require climbing a shaft, and the only one usable by an injured or unconscious person. R2 and R3 are escape shafts, not exits — treat them as the last resort they are.

Decision rule — where the fire is decides the route:

Fire in	Route	Why
Bay 8, generator	R1 , or R2 if the spine is smoke-logged	Blast Door 2 SHUT. Bay 8 is separately ventilated. R3 / ESC 2 is inside the fire compartment — do not use it
Bays 1–6, occupied zone	R1 if the route to Bay 7 is clear; otherwise R2 from the west end, R1 or R3 from the east	Bays 1–6 are one smoke compartment
Bay 7, stair shaft	R2 or R3	The primary route is the fire. Both blast doors shut
Headhouse or covered stairwell	R2 or R3	R1's surface end is blocked; both are outside the boundary and the stairwell is expendable

Immediate actions. Raise the alarm **by voice** — there is no alarm system, and in a 20.8 m shelter with 9 occupants and permanently open 900 mm partition gaps, voice carries. Attack it only while it is small and only if the route behind you is clear. **Shut the blast door between you and the fire — they are the only real barriers in the shelter.** Stop the generator if the fire is in Bay 8 or of unknown origin. Evacuate by the rule above; **do not pass the fire to reach a preferred route.** **Roll call: 9** — the occupancy is a confirmed figure, so the count is unambiguous. **Muster point — NOT DEFINED, because no site plan exists.**

There is no fire team. With 9 occupants, everyone is the fire team. No fire duty, warden role or drill frequency is defined anywhere in the project [N].

19.4 The findings a fire plan here is obliged to raise

Ref	Finding
FS-1	W5 is designated "fire and gas-tight", and a fire separation with an unfilled opening in it is not a fire separation. The compartment it protects contains the activated-carbon filter trains , and escape route R1 has to cross it. The door D-05 is a requirement of this design and is to be designed — Part 12.4
FS-2	Bays 1 to 6 are ONE smoke compartment, 20.8 m long. The four W8 partitions are 110 mm, non-structural, with permanent 900 mm gaps and no doors scheduled. Smoke starting in Bay 1 reaches Bay 6 unobstructed. <i>Not a criticism of the partitions — they were never fire barriers</i> — but it means the only real fire barriers in the shelter are Blast Doors 1 and 2, and W5 once it has its door
FS-3	ESC 2 is in the same bay as the generator. Both facts are confirmed and neither is wrong alone. The coupling had never been stated: the most likely fire in the shelter denies one of its three escape routes — which is exactly why the decision rule routes a Bay 8 fire away from ESC 2
FS-4	A fire during the closed mode cannot be ventilated at all , and clearing smoke means opening a blast valve, which breaks the protection the closed mode exists to provide. No rule exists anywhere in the project for which hazard takes precedence. That belongs to the client, and it should be made before it is needed rather than during
FS-5	Every active fire measure depends on the electrical design. A detection and alarm panel and maintained emergency lighting are on the essential board, on the battery, so the power path exists. What remains is fire engineering — head layout, zoning, detector type, any suppression — and none of it is invented here
FS-6	The six-metre unequipped climb out of each escape shaft. A ladder is designed (Part 10.6), and it still has no fall-arrest, no rest platform, and no answer on the injured person. The three numbers — 6.250 m, 6.800 m, and a 1 400 mm bore — were each recorded separately long before anyone put them in the same picture; it was the escape-plan drawing's vertical profile that made the gap obvious , which is an argument for drawing things

What the plan cannot say, and does not invent: extinguisher type, number and siting; fire ratings of doors and finishes; cable type, containment and fire-stopping; generator fuel type and quantity; muster point and fire service access; drill frequency and warden duties. **Codes beyond NBC 2016 Part 4 and IS 13416 are not in the register, and their numbers are not invented.**

And one trade that must be a conscious decision, not an improvisation. The 1 000 L tank in Bay 1 is the **96-hour potable supply** — 27.8 L/person/day. **It is not firefighting water.** Using it against a fire trades the shelter's endurance for its fire response, and that is the commander's call.

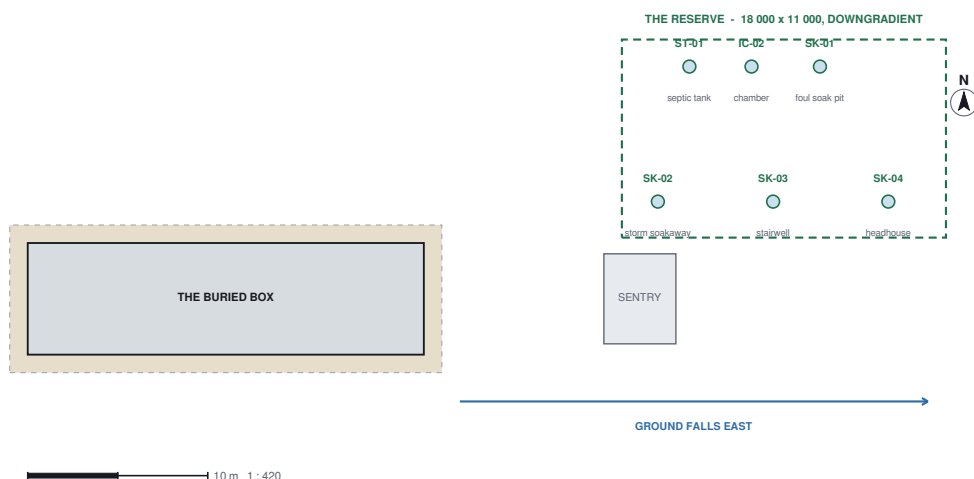
PART 20 — SITE LAYOUT, FINISHES AND CONCEALMENT

20.1 External works — the reserve, and why everything is in it

An **external works reserve** of **18.0 × 10.5 m** at **X 33 000 – 51 000, Y 6 500 – 17 500** — **10.0 m** clear of the main excavation's **east face, downgradient**, and entirely inside the owner's 50 m envelope (worst corner **42.5 m** of 50 m).

Tag	What	Centre (X, Y)	Serves
ST-01	Septic tank 1.5 × 0.75	(36 750, 16 000)	10 users, 450 L/day
SK-01	Foul soak pit 2.2 dia	(44 000, 16 000)	Septic tank effluent — the furthest downgradient of everything
SK-02	Storm soakaway 2.2 dia	(35 000, 8 500)	Clean sump + entry channel. Set 2.10 m from the sentry post against IS 2470's 2.0 m minimum
SK-03	Stairwell soakaway — footprint reserved	(41 400, 8 500)	Stairwell sump. Size recorded nowhere [N]
SK-04	Headhouse soakaway — footprint reserved	(47 800, 8 500)	Headhouse gully. Size recorded nowhere, and its pipe cannot be routed
IC-01 · IC-02	Inspection chambers 600 × 450	(7 500, 9 800) · (40 200, 16 000)	Change of direction; de-sludging

THE DRAINAGE RESERVE - EVERYTHING DOWNGRADIENT, IN ONE PLACE, FOR FOUR STATED REASONS 1 : 420



WHY EVERYTHING IS IN ONE RESERVE, DOWNGRADIENT

- 1 nothing recharges the ground upslope of a box that is flotation-critical at FoS 0.33 in the mat-only stage - and whose side backfill is MORE permeable than the basalt around it
- 2 one percolation-test location, one keep-clear zone, one reserved fallback
- 3 one trench - and where rockhead is 0.9 to 1.5 m, the cost *is* the trench
- 4 concealment - four cover slabs and a 2 m septic vent, grouped 11 to 22 m away, MARK THE DRAINAGE FIELD, NOT THE SHELTER

The layout is anchored to confirmed geometry only, and every offset is relative - so if the perimeter fence turns out to be closer than the reserve's east edge, THE WHOLE RESERVE TRANSLATES AND NOT ONE OFFSET CHANGES.

Figure 51 The drainage reserve, and why everything is in it and downgradient.

Everything in one reserve, on the downgradient side, for four stated reasons:

1. **Nothing recharges the ground upslope or alongside a box that is flotation-critical at FoS 0.33 in the mat-only stage** — and whose side backfill at 95 % MDD is **more permeable than the basalt around it**, so effluent released near it would run *into* the backfill and down the outside of the tanking.
2. **One percolation-test location, one keep-clear zone, one reserved fallback.**
3. **One trench** — three of the runs share a common services trench, and **where rockhead is 0.9–1.5 m the cost *is* the trench.**
4. **Concealment** — four cover slabs and a 2 m septic vent, grouped 11–22 m away, **mark the drainage field, not the shelter.**

The layout is anchored to confirmed geometry only. Nothing is dimensioned from the sentry post, so it survives the sentry position being resolved differently; and **every offset is relative**, so if the perimeter fence turns out to be closer than the reserve's east edge, **the whole reserve translates and not one offset changes**. That is why the fence distance — SG2-V2 — does not block it: **the layout is immune to the answer**.

The IS 2470 offsets — complete for the first time:

foul soak pit to the SEPTIC TANK	>= 5 m	5.40 m	DEMONSTRATED
soak pit to any BUILDING	>= 2 m	10.97 m	DEMONSTRATED, 5.5x over
foul soak pit to any WELL	>= 15 m	--	NOT DEMONSTRABLE

The caveat, and it matters. The 15 m well clearance rests on a statement that there is no well within reach, **not on a survey**. It is recorded as such and **not** as [C] surveyed, and the distinction is deliberate: **if a well is later found within 15 m, the foul soak pit moves**. Foul effluent near a water source is the one clearance nobody should carry on an assumption, and the topographic survey is what converts it from a statement into a fact.

28 of 28 clearance checks pass, and the five pipe runs are fixed except one:

Run	Length	Status
Septic tank → foul soak pit, DN100 at 1:100	5.40 m	FIXED
Service entry plate → storm soakaway, DN50 rising main	27.00 m	FIXED — its vertical leg waits on the entry plate's level, which is [N]
Entry channel → storm soakaway, DN100 at 1:100	32.35 m	FIXED — routed west first, because north would cross the stairwell excavation
Stairwell sump → stairwell soakaway, DN50 rising main	29.60 m	FIXED
Headhouse gully → headhouse soakaway, DN100 at 1:80	[U]	CANNOT BE ROUTED

20.1.1 SG2-F1 — the soak pit's problem is DEPTH, not arithmetic

The soak pit was widened rather than deepened when it was found 2.3 % short (Part 16.7). **The reason that choice was forced, rather than merely preferred, is the finding:**

21–43 % of the pit's required side area is above the design water table, and its only permeable horizon is 0.2–0.5 m thick. A pit driven deeper goes further below (–)2.000, where it cannot soak at all. So the fallback **dispersion field is reserved** rather than the pit being re-sized again — and **the mandatory percolation test governs the final size and form**.

20.1.2 Two more findings that the layout could not resolve

Ref	Finding
SG2-F3	The septic tank is sized for a building that no pipe connects to it. It is sized for <i>"sentry-post shift crews + shelter maintenance"</i> , the drainage package excludes the sentry post from scope , and there is no pipe from the sentry post to the tank anywhere in the project . The layout positions the tank and leaves the connection to the drainage engineer; it does not prejudge it
SG2-F4	The fresh-air intake's plan position. The equipment schedule gave the generator shaft an X range and gave the intake only <i>"west of the box"</i> — no X, no Y, no coordinate . The fresh-air intake of a CBRN shelter is not a minor fitting. RULED (ruling 14, RC4): SH-1 centred (–2 400, 3 100), RECOVERED from the project's own recorded 12.3 m to the entry, to within 28 mm. SH-2's head level is +1.500, the same gooseneck head

And the intake / exhaust separation was settled by geometry, not by meteorology. SH-1 and SH-2 are **25.30 m apart at opposite ends of a 22 m box** — robust whatever the wind does, which is exactly why the site package justified its orientation on access, fall, noise and end-to-end separation and **explicitly not on prevailing wind**.

But the plume question stays open, and it must. A CBRN shelter with no wind direction data **cannot state which way a release drifts**, and 25.30 m of separation says nothing about that. Closing both halves of that question on one argument is exactly the move the open-item register exists to prevent.

20.2 Finishes

No finish specification exists anywhere in this project. The master, the Rev F drawings, the services sheet and the structural package fix concrete grade, cover, waterproofing, crack control and bar spacing — **and nothing about what a surface is finished with.**

So the finishes package does **not** report finishes as project facts. **Every code is a performance requirement derived from something the project does confirm**, and the product that satisfies it is left open. **A schedule that filled them in would be inventing a specification.**

Requirement	Derived from
Decontaminable, coved, joint-free finishes in bays 1–6	The gas-tight CBRN envelope
Wet-area treatment in the lavatory, the plant bay and the airlock	The room uses
Non-slip decontaminable finish in the stair shaft	Those faces are designated a wet / dirty zone — it is why their cover is 30 and not 40
Non-slip external finish in the covered stairwell	Outside the boundary, declared expendable
Hose-down finish and 1:80 fall in the headhouse	Confirmed on the Rev F ground plan
EMP Zone 2 enclosure in the ops room	The finish is subordinate to the shielding
Cast-in fixings only	40 mm cover, 150 mm bar spacing as an EMP requirement, the membrane, the cage
No suspended ceilings or cavities anywhere in the envelope	Duct inspectability, drain inspectability, decontamination
Floor build-up limited	The 1.0 kPa mat allowance, and drainage finding DR-F4
Cast-in stair nosing, no tread build-up	The stair going is FROZEN. An applied nosing or a thick tread finish would change the effective going

Thirteen spaces are scheduled, each with its floor, skirting, wall, ceiling, door, waterproofing, wet-area status and falls. Two entries carry findings rather than finishes:

- **The decon airlock is the dirtiest surface in the shelter** — wet area, falls 1:80 to **segregated** gullies, 50 mm upstand at W5 and at Blast Door 1.
- **The generator bay falls to a gully that has no destination defined** — drainage finding DR-F5.

FN-U1 / D-05 — the finishes schedule recorded the missing W5 door as a data gap; **the fire plan made it a defect**, and it has since been ruled: *design it now*.

20.3 Concealment

The shelter is concealed. The installation is not.

What the design already achieves:

Measure	Value
Topsoil / turf over the whole buried roof	300 mm , 30.867 m ³ , the top layer of the cover — its function is named as <i>concealment</i> , <i>erosion</i> , <i>sheds rain</i>
Turf sourced from the site itself	92.82 m³ stripped and stockpiled for re-use — the re-laid surface is the surface that was there before
Finished grade	crowned, falling 1:50 away — a natural-looking shed, not a flat platform
Berm against the headhouse and covered stairwell	1.5:1 to +0.900
Surplus excavation	1 113.937 m³ re-used on site as engineered fill and crushed rubble
Roof penetrations for drainage	NONE. No roof outlet, no downpipe, no rainwater pipe anywhere on the buried roof
Burster slab	laid to a 1:50 crossfall so infiltration disperses at the berm toe without a pipe

Two of these matter more than the rest. Re-using the site's own stripped turf means the restored surface is **not a different green from its surroundings**, which is what usually gives a buried structure away. And **because the cover is drained without a single pipe**, there is **no manhole, no gully and no outfall anywhere on the roof to break the surface.**

The signature inventory — every element above ground:

Element	Above grade	Note
Sentry post	+7.000 parapet	The tallest thing on the site by 4.5 m
Covered entry stairwell	+2.450 head	Outside the boundary, declared expendable

Element	Above grade	Note
Fresh-air shaft SH-1	+1.500 gooseneck	600 × 600, west of the box
Generator air shaft SH-2	+1.500 gooseneck	600 × 600, east of the box
Headhouse	+0.900 roof top	4 800 × 5 800, no earth cover. The largest above-ground mass
Escape shaft ESC 2	+0.700 head	1 400 dia clear, 1 900 OD collar
Escape shaft ESC 1	+0.150 head	as ESC 2
Soak pit / soakaway covers	at grade	Positions fixed by the site layout
Sentry post spiral stair	external	1 000 R, 250 dia pole

The buried box under 2 m of graded, turfed cover is genuinely hard to see. But a 7 m sentry post 10 m away, a 4.8 × 5.8 m headhouse with no earth cover, a 2.45 m stairwell head and two 1.5 m goosenecks are not concealed by anything, and nothing in the project asks them to be.

This is a statement of fact about the design as recorded, not a criticism of it — the sentry post is a manned observation position and is *meant* to be seen, and the headhouse is the entrance. **But the distinction had never been written down, and a concealment policy that did not say it would be misleading.**

Policy. The turf is a **continuous surface, not a patch** — re-lay the stockpiled site turf; do not import a different species or a different soil colour. **No new spoil heap** — all 1 113.937 m³ of surplus is re-used on site, because **a spoil mound is a permanent signature that outlasts the works.** The berm should **read as ground, not as an engineered line.**

And what the policy cannot say. The concealment bill item — "*camouflage and concealment measures beyond the 300 topsoil / turf layer*" — is tagged [N], quantity *to be verified from final measurement*, note "**no concealment**". **No net, screen, paint scheme, thermal treatment or detection criterion exists anywhere in this project, and none is invented.** Nor can a concealment **layout** be drawn: there is no site plan, so nothing's position on the ground is known except relative to the box.

PART 21 — WORKS MANAGEMENT

A design that cannot be built to its own programme is not finished. This Part is where the structure stops being a set of calculations and becomes a sequence of days, quantities, rates, inspections and decisions. Two things in it are worth reading even by someone who skips the rest: **the critical path runs through the ground, not through the structure, and seventy per cent of the basic cost is protective content and reinforcement, not concrete.**

21.1 What the works management package consists of

Deliverable	Contents
Master programme	MSPDI XML — Microsoft Project's own published schema — plus CSV for any other tool, plus a 7-sheet A3 programme PDF with the critical path marked
Work breakdown structure	279 activities, 18 milestones, 404 logic links , three levels, 14 level-1 packages
Bill of quantities	90 items in 10 sections , whole project
Cost estimate	Priced take-off, five priced parts, seven cost heads
Resource plan	33 resources, 426 assignments
Procurement plan	16 packages , every one tied to a programme activity, with a long-lead register
QA/QC plan	42 ITP items and 16 hold points , each hold point an activity in its own right
Safety and risk register	15 hazard classes, 25 risks
Quantity derivation	About 800 lines of working, with the source of every input named
Consistency audit	65 executed checks, 65 pass

21.2 The bill of quantities

THE BILL OF QUANTITIES - TEN SECTIONS, AND THE QUANTITY THAT DEFINES EACH ONE

■	A	SITE PREPARATION AND EARTHWORKS	1 338 m3 excavation, of which 994 m3 is rock
■	B	MAIN SHELTER - CONCRETE	388.7 m3, all M35 except 17.3 m3 of M15 blinding
■	C	REINFORCEMENT - ALL WORKS	77.33 t ordered, including 5 % wastage
■	D	FORMWORK AND FALSEWORK	1 058 m2, including 3.2 m high falsework to the pressure slab
■	E	ENGINEERED COVER AND OVERBURDEN	205.8 m3 in six layers over 136.4 m2
■	F	WATERPROOFING	532 m2 of continuous tanking
■	G	SENTRY POST - CONCRETE AND FRAME	20.95 m3 M30, 1.97 t of reinforcement
■	H	SENTRY POST - BRICK MASONRY	12.20 m3, 6 400 modular bricks to IS 1077
■	I	ELECTRICAL AND EMP INSTALLATION	EVERY LINE IS [N] - no electrical design package exists
■	J	PRINCIPAL MATERIALS	194.2 t cement, 828.7 t aggregate

THE QUANTITIES ARE TAKEN FROM THE STRUCTURAL CAD PACKAGE, NOT RE-MEASURED FROM THE DRAWINGS.

Six of the eight concrete lines reproduce exactly from the stated geometry; the main staircase and the entry stairwell are composite figures accepted as they stand.

SECTION I IS NOT AN OMISSION - IT IS A DECLARED GAP.

Seven electrical and EMP lines are carried in the bill with no quantity against them, because no design exists to measure. Pricing them would be inventing them.

Figure 52 The bill of quantities - ten sections, and the quantity that defines each one.

§	Section	Items	The quantity that defines it
A	Site preparation and earthworks	9	1 338 m³ of excavation, of which 994 m³ is rock
B	Main shelter — concrete	14	388.7 m³ , all M35 except 17.3 m³ of M15 blinding

§	Section	Items	The quantity that defines it
C	Reinforcement — all works	13	77.33 t ordered , including 5 % wastage
D	Formwork and falsework	7	1 058 m² , including 3.2 m high falsework to the pressure slab
E	Engineered cover and overburden	8	205.8 m³ in six layers over 136.4 m ²
F	Waterproofing	6	532 m² of continuous tanking
G	Sentry post — concrete and frame	8	20.95 m³ M30 , 1.97 t of reinforcement
H	Sentry post — brick masonry and finishes	8	12.20 m³ , about 6 400 modular bricks to IS 1077
I	Electrical and EMP installation	7	every line is [N] — no electrical design package exists to measure
J	Principal materials	3	194.2 t cement, 828.7 t aggregate

21.2.1 The concrete take-off, element by element

Item	Element	m ³	Class
C-01	Mat 600, M35	81.8	[C]
C-02	Pressure slab 900, net of openings , M35	112.0	[C]
C-03	Perimeter walls 600, M35	103.7	[C]
C-04	Walls W6 / W7 400, M35	12.8	[C]
C-05	Wall W5 200, M35	3.20	[C]
C-06	Headhouse walls 400 + roof 500, M35	32.7	[C]
C-07	Main staircase, waist and landings, M35	3.00	[C]
C-08	Entry stairwell — walls, roof, raft, flight, landings, M35	19.9	[C]
C-09	Blinding M15 100 thk under the mat	14.2	[D]
C-10	Blinding M15 100 thk — sump pit, footings F1, stairwell raft	3.05	[D]
C-11	Sump pit SU-01 walls 300 and base 400 below the mat soffit, M35	3.71	[D]
C-12	Escape shaft collars, 250 RC, OD 1900, M35	6.29	[D]
C-13	Local thickenings of the pressure slab 900 → 1200 at openings, M35	3.33	[D]
C-14	Internal partitions W8, 4 No., 110 thk, non-structural	6.21	[D]

Six of the eight structural lines reproduce exactly from the stated geometry. $22.000 \times 6.200 \times 0.600 = 81.84$ for the mat; the roof net of its void and two shafts is 112.03; the perimeter walls are 103.68; W6 and W7 are 12.80; W5 is 3.20; the headhouse is 32.74. **The main staircase and the entry stairwell are composite figures built element by element and are accepted as they stand rather than re-derived.**

21.2.2 The two items that are gaps, not omissions

Section I carries seven electrical and EMP lines with no quantity against any of them, because no electrical design package exists to measure. They are in the bill so that a tenderer sees them and prices the risk, and they carry no quantity so that nobody mistakes an allowance for a measurement. **Pricing them would be inventing them.**

The same applies to the specialist items throughout: **the blast doors, the blast valves, the filter trains, the generator and its fuel system, the EMP Zone 2 enclosure, the escape-shaft hatches and ladders, and all concealment beyond the turf layer.** Each reads *"to be verified from final measurement"*, each is a vendor or specialist item, and **none is priced by guessing.**

21.3 The cost estimate

THE COST ESTIMATE - FIVE PACKAGES OF BASIC COST, AND SEVEN ADDITIONS ON TOP OF THEM

BASIC COST, BY PACKAGE		bar to scale	Rs million	share
I	SURVEY, SITE CLEARANCE AND EXCAVATION		1.91	7.9 %
II	SUBSTRUCTURE AND FOUNDATION WORKS		1.28	5.3 %
III	SUPERSTRUCTURE AND STRUCTURAL RCC		4.15	17.1 %
IV	REINFORCEMENT STEEL AND SHIELDING		6.33	26.1 %
V	CBRN, EMP, CLOSURES AND ANCILLARY SERVICES		10.55	43.5 %

TOTAL BASIC COST	Rs 2 42 20 254	24.22
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ON TOP OF THE BASIC COST - 23.0 % IN SIX LINES

Contingencies	3.0 %	Rs 726 608
Water and electricity provisions	1.0 %	Rs 242 203
Specialist CBRN / EMP and engineering consultant	6.0 %	Rs 1 453 215
Site supervision and quality assurance	2.0 %	Rs 484 405
Statutory clearances and liaisoning	1.0 %	Rs 242 203
Contractor's overhead and profit	10.0 %	Rs 2 422 025

FINAL PROJECT COST**Rs 2 97 90 913****PART V IS 43.5 % OF THE BASIC COST AND PART IV IS 26.1 % - SEVENTY PER CENT OF THIS STRUCTURE IS ITS PROTECTIVE CONTENT AND ITS STEEL, NOT ITS CONCRETE.**

Rates are not verified against any published schedule of rates: no DSR or state SoR reference exists in the project, the CBRN and EMP items are budgetary allowances, and several have no vendor quotation behind them. The owner's own summary reaches Rs 3 00 33 306 from a basic cost of Rs 2 43 36 022, and its cost heads sum Rs 1 00 000 short of the total it states - a recorded conflict, reproduced here and not corrected.

Figure 53 The cost estimate - five packages of basic cost, and the additions on top of them.

Part	Package	Cost ₹	Share
I	Survey, site clearance and excavation	1 912 726	7.9 %
II	Substructure and foundation works	1 283 471	5.3 %
III	Superstructure and structural RCC	4 145 877	17.1 %
IV	Reinforcement steel and shielding	6 333 152	26.1 %
V	CBRN, EMP, closures and ancillary services	10 545 028	43.5 %
	TOTAL BASIC COST	24 220 254	100 %

Cost head	Basis	₹
Contingencies	3.0 %	726 608
Water and electricity provisions	1.0 %	242 203
Specialist CBRN / EMP and engineering consultant	6.0 %	1 453 215
Site supervision and quality assurance	2.0 %	484 405
Statutory clearances and liaisoning	1.0 %	242 203
Contractor's overhead and profit	10.0 %	2 422 025
FINAL PROJECT COST	+ 23.0 %	₹ 29 790 913

Parts IV and V together are 69.6 % of the basic cost. This is not a concrete structure with protective features added; it is a protective installation that happens to be made of concrete. Any value-engineering exercise that starts with the concrete is looking in the wrong place, and any programme risk that threatens the specialist packages threatens two-thirds of the money.

The total is a LOWER BOUND, and it says so. Every unpriced item above is missing *upward*.

And the rates are not verified against any published schedule of rates. No Delhi Schedule of Rates or state schedule reference exists anywhere in the project, the CBRN and EMP items are budgetary allowances, and several have no vendor quotation behind them at all. **The estimate is a budget, not a tender sum, and treating it as the latter would be a category error.**

21.3.1 The owner's summary, and a difference that is reproduced rather than corrected

The project owner's own cost summary reaches ₹ 30 033 306 from a basic cost of ₹ 24 336 022. Applying the same seven percentages to that basic cost gives ₹ 29 933 306 — the heads sum ₹ 1 00 000 short of the total the summary states.

That difference is a recorded conflict in the project, and this report reproduces it rather than correcting it. Nothing downstream is sensitive to ₹ 100 000 on a ₹ 30 million budget, and editing a recorded value away is how a discrepancy stops being visible without stopping being true. The ruling-applied column reconciles exactly, which is how the difference was located in the first place.

21.4 The programme and the critical path

THE CONSTRUCTION PROGRAMME - FOURTEEN WORK PACKAGES OVER 384 CALENDAR DAYS

Monday 2 November 2026 to Saturday 20 November 2027. 326 working days, 279 activities, 18 milestones, and 75 activities at zero total float.

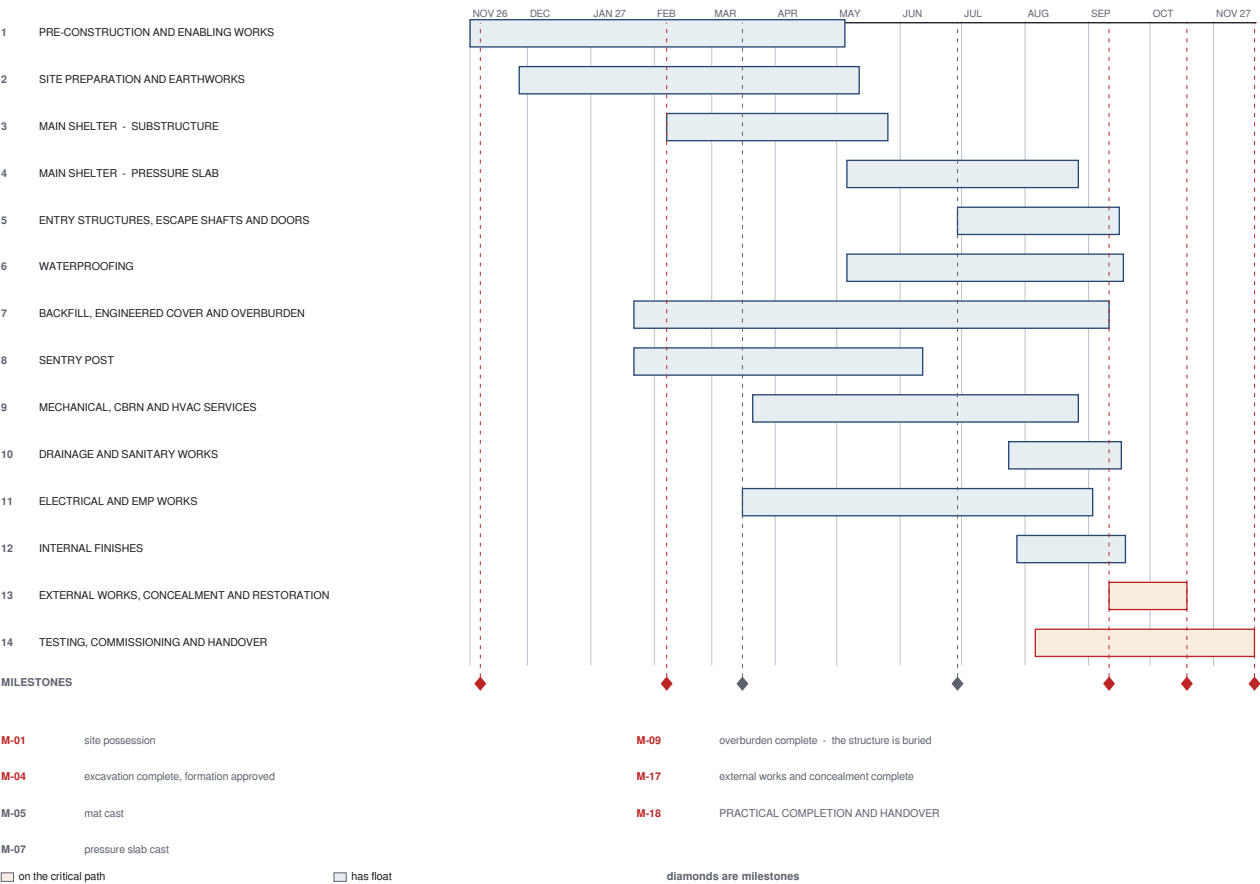


Figure 54 The construction programme - fourteen work packages over 384 calendar days.

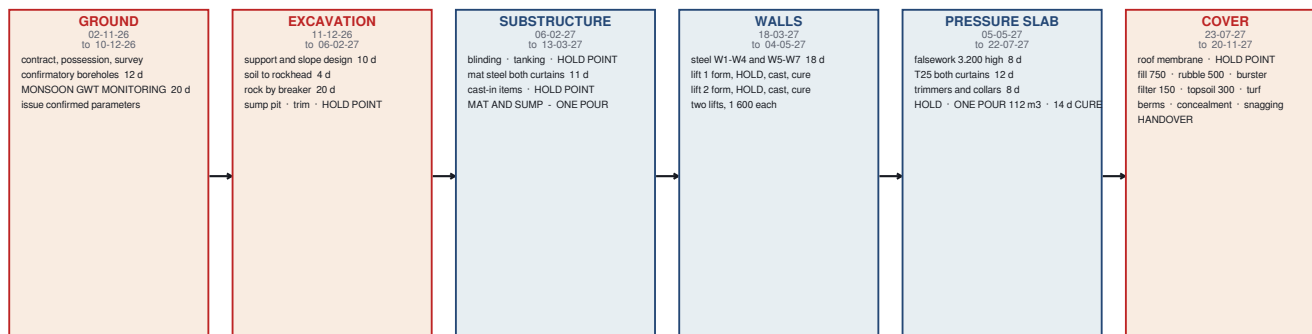
Start	Monday 2 November 2026 [A] , carried from the owner's own schedule
Finish	Saturday 20 November 2027
Duration	326 working days, 384 calendar days
Calendar	Six-day week, Mon-Sat, Sunday non-working, plus five date-certain national holidays
Activities	279, of which 18 are milestones
Critical	75 activities at total float zero

Festival holidays are not in the calendar, and they are not fabricated either. Their dates move year to year, so inventing them would put false precision into a programme. A ten-working-day contingency activity immediately before handover absorbs them

— and that contingency is itself **on the critical path**, which is the honest way to carry it.

21.4.1 Where the critical path actually runs

THE CRITICAL PATH - SEVENTY-FIVE ACTIVITIES, AND THE SIX PLACES IT CAN ACTUALLY BE LOST



NINE HOLD POINTS SIT ON THE CRITICAL PATH. Each one is a day the Engineer must be on site, and each one is a day the contractor cannot buy back.

THE TWENTY DAYS OF MONSOON GROUNDWATER MONITORING ARE THE MOST IMPORTANT TWENTY DAYS IN THE PROGRAMME.

The design groundwater table at (-)2.000 is an assumption. Every uplift number, the whole staged construction sequence and the 1.22 factor of safety at the bare-box stage depend on it. If the monitoring returns a higher table, the answer is found before a single cubic metre of concrete is placed - which is exactly where it should be found.

THE OTHER FIVE PLACES IT CAN BE LOST

- rock excavation, 20 days by breaker - no blasting is permitted, and the quantity carries a declared overrun band
- the two single continuous pours - 81.8 m3 of mat and 112 m3 of pressure slab, neither has a construction joint to fall back on
- the 14-day cure and the props that must stay 14 days after that - three weeks that cannot be compressed
- the roof membrane hold point - once it is covered it cannot be inspected again for the life of the structure
- the ten days of festival and weather contingency at the end, which are already spent in the programme as issued

Figure 55 The critical path condensed to the six places it can actually be lost.

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possession and survey
-> CONFIRMATORY SITE INVESTIGATION          12 d
-> MONSOON GROUNDWATER MONITORING           20 d
-> issue confirmed geotechnical parameters    5 d
-> excavation support and slope design        10 d
-> excavation: soil 4 d, rock 20 d, sump 3 d
-> trim, inspect for red-bole, HOLD POINT
-> blinding, tanking, HOLD POINT
-> mat steel both curtains, HOLD POINT, ONE POUR
-> walls in two lifts, HOLD POINT each
-> falsework, T25 both curtains, trimmers, HOLD POINT
-> PRESSURE SLAB: ONE POUR 112 m3, 14 d CURE, 14 d PROPS
-> roof membrane, HOLD POINT
-> the six cover layers including the burster slab
-> berms, regrading, access, concealment
-> snagging, 10 d contingency, handover
  
```

The design's most important unknown is programmed as work, not as a hope. The twenty days of monsoon groundwater monitoring have **zero float**, while the pressure slab — the governing structural element of the whole project — has **26 days**. That is the correct answer to Part 4.4: the design groundwater table at (-)2.000 is an assumption, every uplift number depends on it, and the flotation factor of safety is **1.22 before the cover goes on**. If the monitoring returns a higher table, the answer arrives **before a single cubic metre of concrete is placed**.

And the monitoring window is not in the monsoon. 12 November to 4 December is post-monsoon in Pune. The activity is on the critical path and is the right activity; **its window is open item SG-V3, and scheduling it is not the same as scheduling it usefully.**

21.4.2 The eighteen milestones

Ref	Milestone	Date	Float
M-01	Site possession and commencement	07-11-26	CRITICAL
M-02	Mobilisation complete	26-11-26	8 d
M-03	All long-lead protective items ordered	23-02-27	230 d
M-04	Excavation complete, formation approved	06-02-27	CRITICAL

Ref	Milestone	Date	Float
M-05	Mat foundation cast	15-03-27	213 d
M-06	Perimeter and protective walls complete	05-05-27	170 d
M-07	Pressure slab cast — the key structural milestone	28-06-27	26 d
M-08	Buried envelope watertight — tanking complete	05-08-27	91 d
M-09	Overburden complete — the structure is buried	10-09-27	CRITICAL
M-10	Sentry post frame complete	20-04-27	182 d
M-11	Sentry post brick masonry complete	28-04-27	175 d
M-12	Sentry post complete	11-06-27	109 d
M-13	CBRN plant installed	26-08-27	23 d
M-14	Drainage installed	16-09-27	13 d
M-15	Electrical installation complete	02-09-27	22 d
M-16	Internal finishes complete	18-09-27	53 d
M-17	External works and concealment complete	18-10-27	CRITICAL
M-18	PRACTICAL COMPLETION AND HANDOVER	20-11-27	CRITICAL

21.4.3 The six places the programme can actually be lost

1. **The twenty days of monsoon groundwater monitoring**, for the reason above.
2. **Rock excavation — 994 m³ by hydraulic breaker, no blasting permitted.** Twenty days at 60 m³/day, and Part 4.7 shows the quantity carries a declared overrun band of **+118 m³, about two extra days**, if rockhead is at the shallow end of the measured range.
3. **The two single continuous pours.** 81.8 m³ of mat and sump in one pour; **112 m³ of pressure slab in one pour.** Neither has a construction joint to fall back on, and a cold joint in the pressure slab of a protective structure is not a defect that can be accepted on paper.
4. **The 14-day cure and the 14-day prop period after it.** Three weeks that IS 456 Table 11 does not allow to be compressed, on the element with the least float in the structural chain.
5. **The roof membrane hold point.** Once the cover is on, the membrane cannot be inspected again for the life of the structure. It is the last chance, and it is 45 minutes of work whose consequence is permanent.
6. **The ten days of festival and weather contingency**, which are already spent in the programme as issued.

21.5 Procurement, and the dependency that starts on day one

Sixteen packages, each tied to a programme activity and each required *before* the activity that installs it. The governing fact of the whole procurement is one sentence:

The blast door cast-in frames must be welded into the W6 and W7 reinforcement cages before those walls are poured. That single dependency puts a vendor lead time onto the critical-path chain from day one, and it is why the blast door enquiry is the **first** procurement activity in the programme.

Ref	Package	Order by	Required on site	Float
P-01	Blast doors 1 and 2	05-12-26	frames at the W6/W7 cages; leaves later	19 d
P-02	CBRN filter trains AHU-1 and AHU-2	11-12-26	mechanical installation	115 d
P-03	Blast valves, 5 No., and sleeves	11-12-26	sleeves cast in / valves later	131 d
P-04	EMP Zone 2 shielded enclosure	17-12-26	electrical and EMP works	148 d
P-05	Generator, 15 kVA	29-01-27	mechanical installation	195 d
P-06	Armoured vision panels, 8 No.	22-02-27	sentry post first storey	173 d
P-07	Submersible pumps and drainage plant	11-01-27	drainage installation	214 d

Nineteen days of float on the item with the longest lead time and the earliest hard dependency. That is the number to watch. Everything else in the procurement has three to seven months of slack; the blast door frames have three weeks.

21.6 Quality: sixteen hold points, placed where an error becomes permanent

A **hold point** in this programme is an activity in its own right: it takes time, it requires the Engineer's attendance, and work does not proceed past it. **Nine of the sixteen sit on the critical path**, which means each one is a day the contractor cannot buy back.

Ref	What is inspected	Why there
H-01	Formation, before anything covers it	The last sight of the founding horizon. Red-bole and vesicular seams are found here or not at all
H-02	Horizontal tanking, before the blinding protection goes on	A membrane under a mat cannot be reached again
H-03	Mat reinforcement, cover and cast-in items	75 mm cover to the bottom curtain, and every sleeve, earth pit and EMP strap
H-04 · H-05	Wall reinforcement, frames and cover — lifts 1 and 2	Including the blast door frames welded to the cage
H-06	Pressure slab pre-pour	The governing element, one continuous pour, T25 at 150 in both curtains
H-07 · H-08	Headhouse walls and roof	The highest utilisation in the project, at 90 %
H-09	Vertical tanking before backfilling — the last chance to see it	The external face of a buried box is visible exactly once
H-10	Roof membrane integrity test before covering	Two metres of cover go on top of it and stay there
H-11 ... H-16	Sentry post founding stratum, footings, first floor, roof, and masonry at both storeys	Including the masonry before it is plastered

Forty-two inspection and test plan items sit behind those hold points, under four inspection types: **HOLD** — work stops until released; **WITNESS** — the Engineer is invited and work may proceed if attendance is waived; **TEST** — a measurement against a stated acceptance criterion; **SURVEILLANCE** — ongoing observation.

Three tests in this project can only be done once, and two of them are pass/fail on the whole installation. The **gas-tightness and overpressure test at +300 Pa**; the **IEEE Std 299 shielding survey on EMP Zone 2**, which is itself a hold point before any equipment is installed; and the **percolation test** that governs the final soak-pit size. A failed shielding survey means opening seams on a completed enclosure.

And the concrete mix is a hold point too. Item Q-06 covers the trial mix to IS 10262 and the integral waterproofing admixture to IS 2645 — the point at which the trial proportions of Part 6.5 are replaced by real ones, **before the first structural pour** rather than after it.

21.7 Resources

Thirty-three resources across 426 assignments. The shape of the histogram is worth one observation rather than a table: the peak is not at the concrete, it is at the **reinforcement**. Seventy-seven tonnes of cut-and-bend steel at 150 mm spacing in both curtains of every element is a bar-fixing job with a concrete job attached to it, and **the programme durations for the steel — 12 days for walls W1-W4, 12 days for the pressure slab curtains, 8 days for the trimmers and collars — are the ones a contractor will challenge first and should not.**

Average steel density is about 191 kg/m³. For a blast-hardened buried box whose bar spacing is set by an **EMP** requirement rather than by strength, and whose sections are thick, that is a plausible figure — and it is a sanity check, not a design check.

21.8 Risk: the five that are design outputs, not contractor problems

The register carries **25 risks in 15 hazard classes**. Five of them are not the contractor's to manage, because they are consequences of the design and are answered inside this report:

Risk	Where it is answered
R-01 Groundwater higher than the assumed (–)2.000	Part 4.4 and Part 9.3.5. Water is nearly two-thirds of the 15.41 kPa/m lateral gradient. The monitoring is on the critical path
R-02 Red-bole or vesicular seams under the mat	Part 9.3.2. A single seam produces the differential-support case that sizes the mat at 84 % . Over-excavate and replace in M15
R-03 Rock quantity exceeds the estimate	Part 4.7 — +118 m³, about two extra days , against a register entry that assumes one
R-12 The 112 m³ pour cannot be completed without a cold joint	Part 21.4.3. Plant standby, a second supply route and a written pour plan — not a decision to be taken at 2 a.m. on the day
R-18 The excavation stands open across a monsoon	Part 4.6 and Part 9.3.5 — the batter, and the formally declared departure from the dewatering mitigation

Two risks in the register are about this design's own record rather than about the works. R-08, that no electrical design package exists below board level, and R-11, that no site plan or ground model exists — which is why the berm volume, the access route, the hardstanding, several drainage runs and the soakaway positions cannot be drawn. **Both are carried as risks because the alternative is to carry them as silence.**

21.9 The rules this project holds its own works management to

- **The project owner's own bill, estimate and schedule GOVERN.** This report's versions are published alongside them, never in place of them.
- **Nothing is retyped.** Source documents are reproduced, not transcribed, so a transcription error cannot enter through the back door.
- **No quantity, rate, date or float is changed by a discipline package.** Every services revision in this project states that it changed none, and each states it separately **so that the claim can be checked rather than believed.**
- **A design change is recorded as a design change.** The sentry post masonry is one (Part 5.9.1). The 8 m³ strip-and-replace under the stairwell raft is a bill item added by a design decision (Part 4.5), **at a rate tagged [A] rather than invented.**

PART 22 — THE ENVIRONMENTAL MANAGEMENT PLAN

A note on the abbreviation, because this project uses it for something else. Everywhere else in this report **EMP** means **electromagnetic pulse** (Part 18). This Part is the **environmental management plan**, and it is called by its full name throughout to keep the two apart.

22.1 What this Part is, and what it is not

NO ENVIRONMENTAL MANAGEMENT PLAN EXISTS IN THIS PROJECT. There is no environmental impact assessment, no consent, no clearance, no monitoring schedule and no statutory correspondence anywhere in the project record. **This Part does not report one; it derives one**, from quantities and methods the project *does* confirm, so that the works have an environmental position instead of no position at all.

Everything derived here is tagged, and nothing statutory is invented. Where a clearance, a consent, a limit value or a monitoring standard would be needed, this Part says which one and leaves it [N]. **A fabricated consent condition would be worse than an admitted gap**, because it would read as though somebody had asked.

The plan is built on the quantities in Part 21 and the methods in Parts 4 and 9, because those are the only environmental facts this project actually holds. **Every impact below is a consequence of a number the design has already fixed.**

22.2 The site's environmental setting, as far as it is known

		Class
Installation	CTW / College of Military Engineering campus, Pune	[C]
Setting	Undeveloped plot, CBRN Live Training Area to the south , perimeter track and nullah line to the east	[C]
Plot boundary, area, levels	NOT DIMENSIONED ANYWHERE	[N]
Surface water	A nullah line east of the plot — the plot falls east toward it	[C] as a relationship, [N] as a distance
Groundwater	Design table (–)2.000, assumed; not encountered in pits about 1.5 m deep	[A]
Surface soil	Black cotton, CH, free swell index 60–65 %	[C]
Rainfall	759.6 mm/yr from the presentation record against 500–600 mm in the soil report — an open conflict	[U]
Monsoon	June to September , 559.7 mm of the annual total in four months	[C]
Ecology, trees, habitat, protected species	Nothing recorded anywhere in the project	[N]
Nearest receptor — dwelling, ward, school, water supply	Nothing recorded	[N]

The nullah is the receptor that matters, and the project cannot say how far away it is. The plot falls east toward it, the drainage reserve is deliberately placed downgradient, and a soak pit discharging foul effluent sits at the far end of that reserve. **Whether the reserve is 40 m or 400 m from the nullah is not recorded, and the answer changes the plan.** Closing it needs the topographic survey — programme activity A1070.

22.3 Environmental aspects and impacts, derived from the project's own quantities

ENVIRONMENTAL ASPECTS - EACH ONE AGAINST THE QUANTITY THIS DESIGN HAS ALREADY FIXED

■	Excavation and spoil	1 338 m ³ , of which 994 m ³ rock	HIGH
■	Rock breaking	hydraulic breaker, NO BLASTING, ~20 days	HIGH
■	Dewatering	continuous, excavation to backfill	HIGH
■	Excavation across a monsoon	559.7 mm onto an open 6.8 m cut	HIGH
■	Decon effluent	1 000 L, TANKER ONLY, never to ground	HIGH
■	Spent CBRN filters	HEPA H14 + ASZM-TEDA carbon, interval [N]	HIGH
■	Concrete	388.7 m ³ + 20.95 m ³ ; 194.2 t cement	MEDIUM
■	Curing water	14 days on 1 058 m ² of formed face	MEDIUM
■	Black cotton spoil	84 to 126 m ³ , destination [N]	MEDIUM
■	Foul effluent, peacetime	450 L/day to septic and soak pit	MEDIUM
■	Fuel	250 L day tank, 275 L bund at 110 %	MEDIUM
■	Generator	15 kVA, 2 600 m ³ /h combustion and cooling air	MEDIUM
■	Spoil RE-USE on site	1 113.937 m ³ re-used, 92.82 m ³ turf	BENEFICIAL
■	Site restoration	site's own turf re-laid, berms 1.5:1	BENEFICIAL

NO ENVIRONMENTAL MANAGEMENT PLAN, ASSESSMENT, CONSENT OR CLEARANCE EXISTS IN THIS PROJECT.

This Part does not report one: it DERIVES one from quantities the project does confirm, so that the works have an environmental position instead of no position at all. Where a consent, a limit value or a monitoring standard would be needed, it says which one and leaves it [N] - a fabricated consent condition would be worse than an admitted gap, because it would read as though somebody had asked.

THE NULLAH IS THE RECEPTOR THAT MATTERS, AND THE PROJECT CANNOT SAY HOW FAR AWAY IT IS. The plot falls east toward it and the drainage reserve is deliberately downgradient. Whether that reserve is 40 m or 400 m from the nullah is not recorded, and the answer changes the plan.

Figure 56 Environmental aspects, each against the quantity this design has already fixed.

Aspect	The quantity the design fixes	Impact	Significance
Excavation and spoil	1 338 m ³ total, of which 994 m ³ rock	Dust, noise, vibration, spoil handling	HIGH
Spoil re-use	1 113.937 m ³ re-used on site; 92.82 m ³ of turf stripped and stockpiled	Avoided haulage, avoided borrow	BENEFICIAL
Rock breaking	Hydraulic breaker, no blasting , ~20 working days	Noise and ground vibration	HIGH
Concrete	388.7 m ³ structural + 20.95 sentry; 194.2 t cement	Embodied carbon, batching water, washout	MEDIUM
Curing water	14 days continuous on 1 058 m² of formed face and 136 m ² of slab	Site water demand in a water-stressed district	MEDIUM
Dewatering	Continuous from excavation to backfill, across a monsoon	Discharge quality, silt, drawdown	HIGH
Excavation across a monsoon	559.7 mm of rain onto an open 6.8 m cut	Silt-laden runoff to the nullah	HIGH
Black cotton spoil	Strip-and-replace 8 m ³ + 76–118 m³ from the batter	A stockpile of very-high-swelling clay with no stated destination	MEDIUM
Foul effluent, peacetime	450 L/day to septic tank and soak pit	Groundwater quality	MEDIUM
Decon effluent	1 000 L tank, tanker only, never to ground	Contaminated liquid waste, off site	HIGH
Fuel	250 L nominal day tank, 275 L bund at 110 %	Spill to ground	MEDIUM
Generator emissions and noise	15 kVA, 2 600 m³/h combustion and cooling air	Air quality, noise	MEDIUM
Spent CBRN filters	HEPA H14 and ASZM-TEDA carbon, change-out interval [N]	Hazardous waste over the life of the structure	HIGH
Site restoration	Turf re-laid from the site's own stockpile; berms graded 1.5:1	Landscape, concealment	BENEFICIAL

22.4 The controls, tied to the activity that needs them

22.4.1 Excavation, dust and spoil

- **No blasting.** Rock is removed by hydraulic breaker — already the method in the programme, and the single largest environmental decision in the works. It removes the blast-vibration, fly-rock and airblast questions entirely, at a cost of about 20 working days.
- **Damp down the rock face and the haul route** during breaking. No water quantity is stated; the curing demand already drives the site water supply and dust suppression adds to it [A].
- **Batter the soil cap 1:1 minimum** (Part 4.6). This is a slope-stability measure first and an erosion measure second, and it does both.
- **All surplus is re-used on site as engineered fill and crushed rubble.** No off-site spoil disposal is programmed, and **no new spoil heap is permitted** — a mound is a permanent signature that outlasts the works, which is a concealment requirement and an environmental one at the same time.
- **The black cotton spoil is the exception, and it has no stated destination.** The strip-and-replace and the batter together produce 84–126 m³ of very-high-swelling clay that **must not** go back under any structure and **should not** be re-laid as the concealment topsoil (SG-V7). **Its destination is [N] and is a decision for the works, not a detail.**

22.4.2 Water, silt and dewatering

- **Silt control at the point of discharge is the single most important control on this site**, because the plot falls east toward a nullah and the excavation is open across a monsoon. Settlement before discharge is required; **the pond size, the discharge standard and the consent are all [N]**.
- **Dewatering runs continuously from the start of excavation until backfill and cover are complete** — a structural requirement first (Part 9.3.5), with an environmental consequence attached to it. **The programme departs from it in writing**, which means the *hydraulic* risk and the *discharge* risk are both live across the 2027 monsoon.
- **No dewatering discharge quality standard exists in the project [N]**. What can be stated is the source: groundwater from basalt at 6.8 m, plus surface water from an open cut in black cotton clay. **The second is the silt problem; the first is not.**
- **Three streams stay separate** (Part 16.1). Clean to the storm soakaway; foul to the septic tank and soak pit; **decon effluent to a tanker and off site, never to ground, under any circumstances.**
- **The percolation test is an environmental control as well as a hydraulic one.** A soak pit in basalt that does not percolate is a foul discharge looking for a path, and the path in fractured basalt is the joint system.

22.4.3 Noise and vibration

- **The dominant source is the hydraulic breaker**, over about 20 working days, on an operational military installation whose working hours are already constrained (R-25).
- **No noise limit, no receptor distance and no permitted-hours condition exists anywhere in this project [N]**. What the design can state is the source, the duration and the direction: the works sit next to a **CBRN Live Training Area**, which is itself a noise source, and the nearest sensitive receptor is unrecorded.
- **Ground vibration from breaking is the control that protects the works as well as the neighbours** — the same fractured basalt that transmits vibration is the founding horizon of the structure, and the trial-pit record already describes the broken-rock layer as *"having lot of cracks, fissures, voids and discontinuities."*

22.4.4 Materials, waste and carbon

- **194.2 t of cement is the largest single embodied-carbon item in the works**, and Part 6.5 fixes 400 kg/m³ for the M35 for durability reasons that are not negotiable — **very severe exposure permanently below the water table**. A blended cement would reduce the carbon and is compatible with the durability case in principle; **no blended cement is specified in this project, and IS 269 OPC 43 is what the bill procures [C]**.
- **Formwork is 1 058 m² and is not a single-use item.** Re-use rates are [A].
- **Reinforcement is ordered at 77.33 t including 5 % wastage** — the wastage is declared, which is the first step to reducing it.
- **Concrete washout is a controlled waste, not a ground discharge.** No washout facility is in the bill [N].

22.4.5 The operational-phase item nobody has costed

Spent CBRN filters are hazardous waste, they are produced for the whole life of the structure, and this project holds no change-out interval, no disposal route and no cost for them.

A HEPA H14 that has been in service in a protected ventilation train is contaminated with whatever it removed, and an ASZM-TEDA carbon bed is contaminated with whatever it adsorbed. The change-out interval needs a challenge concentration and vendor breakthrough data, **both [N]** (Part 15.8). The disposal route needs a licensed facility and a transport chain, [N].

This is stated here because it is the one environmental liability of this design that outlives the works by decades, and because a structure whose ventilation is its reason for existing cannot be handed over without an answer to what happens to its filters.

22.5 Monitoring

What	When	Against what	Status
Groundwater level	Standpipe piezometer, through a full monsoon	The design assumption (-)2.000	Programmed — but the window is post-monsoon (SG-V3)
Dewatering discharge — silt	Continuous during discharge	No standard exists	[N]
Dust at the boundary	During rock breaking	No standard, no boundary	[N]
Noise at the nearest receptor	During rock breaking	No limit, no receptor recorded	[N]
Ground vibration	During rock breaking	No limit recorded	[N]
Backfill compaction	95 % MDD, IS 2720 Part 28	Confirmed, and it is in the programme	[C]
Percolation, before the soak pits are built	Once	IS 2470 Part 2 Cl. 4 — mandatory	[C] , and it governs
Concrete: trial mix, cubes, slump, cover	Continuous	IS 456, IS 10262 — hold point Q-06	[C]

Two of these are in the programme and are real; five have no standard to measure against. That is the honest state of environmental monitoring on this project, and the fix is a single document — a consent or a clearance condition — that does not exist.

22.6 What the environmental management plan cannot supply

Missing	Why it matters
Environmental clearance / consent to establish and operate	Whether one is required at all for defence works on an operational installation is a statutory question this project does not answer [N]
Baseline air, noise, water and ecology survey	There is no baseline to compare any monitoring against
Plot boundary, levels, receptor distances, the distance to the nullah	Every offset in this Part is a relationship, not a dimension
Tree survey and vegetation record	92.82 m ³ of turf is stripped; what else is on the plot is unrecorded
Discharge standards for dewatering, washout and foul effluent	Nothing can be measured against nothing
Spent-filter disposal route and cost	The operational liability above
Construction environmental management plan as a contract document	This Part is a design-stage derivation and is not a substitute for one

Read this Part as a statement of position, not as a compliance document. It says what the works will do to the ground, the water, the air and the neighbours, in terms of quantities the design has already fixed — and it says, item by item, where a consent authority would have to supply the number that turns a control into a compliance obligation.

PART 23 — WHAT THIS DESIGN DOES NOT DEMONSTRATE

This part must be reproduced in every future revision. It is the boundary of the design's validity.

23.1 What IS demonstrated

Demonstrated to IS 456, with IS 13920 for the sentry post frame: ultimate flexural capacity; diagonal-tension shear; minimum and maximum steel; bar spacing; cover; development and anchorage; deflection; crack-width provisions; biaxial column interaction; punching and one-way shear in the footing; and ductile detailing to IS 13920.

23.2 What is NOT demonstrated, and NOT claimed

1. **Support rotation and ductility ratio for the blast case.** $\theta \leq 2^\circ$ corresponds to $\delta \leq 87 \text{ mm} = \text{span}/57$ and requires a **non-linear SDOF check** (IS 4991 Fig. 6 / Biggs) or explicit non-linear FE with strain-rate-dependent materials. **Phase 3.**
2. **Shock-wave propagation down the entry shaft** and the resulting door loading — CFD or shock tube. **Specialist.**
3. **Transient soil-structure interaction** — participating mass, arching, interface pressure ratios.
4. **Transient soil-structure THERMAL behaviour** — the sibling of (3), and the missing piece of RC4-V1.
5. **Blast door, valve and hatch performance** — **vendor-tested, not a civil deliverable.**
6. **Global flotation cannot be shown by an elastic-mat model.** The springs take tension, so in the model the mat never lifts. It is a hand check on real dimensions (Part 9.3.5).
7. **The berm's actual pressure-transmission factor** — removed from the critical path by designing the headhouse walls to the upper bound.
8. **Any analysis result whatsoever.** STAAD.Pro has not been run in this project. No moment, displacement, shear or reaction exists for the underground box or for the three mesh models, and **the mesh convergence conclusion is outstanding.**
9. **The negative phase and rebound as a separate load case** — allowed for physically at the hatches, not analysed.
10. **EMP Zone 1 shielding effectiveness.** It is a calculation and it will stay a calculation; a buried box cannot be surveyed to IEEE 299.

A conventional static STAAD.Pro analysis gives the correct DEMAND. It is not, and must never be presented as, proof of blast resistance.

IS 4991:1968 Cl. 1.1 expressly EXCLUDES nuclear explosions from its scope. It is used here for its loading rules and dynamic material strengths only, as a documented conservative extrapolation. **Every structural capacity is computed to IS 456.**

PART 24 — ASSUMPTIONS AND OPEN ITEMS

24.1 How to read this part

Three registers, and the distinction between them is the point:

Register	What it holds
24.2 Assumptions	Values the design uses that have no test and no client confirmation . Seven are closed; the seven that remain are exactly the site investigation
24.3 Open items	Questions that need information the project does not contain . Eighteen are open . They are not inconsistencies — each is a single position the project holds, with nothing contradicting it
Appendix D	Every decision, conflict and revision behind the two registers above , kept out of the design narrative and held as traceability

THE OPEN REGISTER - EIGHTEEN OPEN ITEMS AND SEVEN ASSUMPTIONS, CLASSIFIED SO THE COUNT CAN BE CHECKED

OPEN ITEMS

THE FOUR A REVIEWER SHOULD SEE FIRST

open, narrowed by a ruling

open, annotated but NOT narrowed

open, untouched

open, new

9 + 2 + 6 + 1

of which the four to read first

an investigation that reached about 1.5 m against a formation at (-)6.800 · two 1 400 mm holes in the protective boundary whose bonding is undesigned · a CBRN shelter that cannot state which way a release drifts · a sealed box with a 4 kW heat surplus and an unmodelled rejection path

ASSUMPTIONS

closed by judgement

STILL OPEN

EVERY ONE OF THE SEVEN THAT REMAIN IS A PHYSICAL TEST ON THIS PLOT · a borehole, a piezometer, a plate load test, a percolation test, a packer test. Not one of them is an argument anybody can win at a desk.

Figure 57 The open register - eighteen open items and seven assumptions, classified so the count can be checked by reading.

24.2 Assumptions — seven closed, seven open

CLOSED (7) — every one by judgement, and every one the conservative or immaterial choice:

#	Assumption	Ruling
A6	$K_a = 1.0$ saturated used; dry berm ≈ 0.5 not relied on	ACCEPTED as the conservative choice — the full 383 kPa is taken rather than crediting unverifiable berm attenuation
A9	Sentry infill not separated → $R = 3.0$	KEPT . $R = 5.0$ gives A_n 0.060 against 0.100, so 73.18 kN is $\approx 1.67 \times$ a separated SMRF's demand . Separating would need an SMRF this project does not claim, and would undo the tie detail
A10	Wind $k_1 = 1.08$, 100-year life	ACCEPTED — a 100-year life on a load path seismic beats 2.4 : 1
A11	$b_{eff} = 2.5$ m for the HW3 line load	ACCEPTED ON THE MARGIN — 41 % on the conservative one-way bound leaves more than half the capacity spare
A12	Trapezoid factor 0.7946 used for both BM and FEM	ACCEPTED ON THE MARGIN . The factor itself is exact; the simplification is worth ≈ 2 %, inside B2's 89 %
A13	Sentry frame has no member releases	CLOSED ON THE EVIDENCE, and it needed no ruling . The register asked for the .std; the file had been in the workspace all along and nobody re-read it . MEMBER RELEASE appears nowhere in it
A14	Poisson's ratio 0.20	ACCEPTED as the standard value for concrete

STILL OPEN (7) — and every one needs a physical test on this plot:

A1	rockhead 1.5–2.0 m	boreholes
A2	DESIGN GWT (-)2.000	piezometer, read through a FULL monsoon

A3	SBC 3240 kPa	plate load / core testing
A4	ks 100 000 - 500 000 kN/m ³	plate load test -- BOTH bounds
A5	K0 0.50, gamma 20/21	site investigation
A7	soak-pit absorption 20 L/m ² /day	PERCOLATION TEST, IS 2470 Pt 2 Cl. 4, MANDATORY
A8	structural seepage 0.5 L/m ² /day	packer permeability tests

That is the sentence worth carrying into a viva. The assumption register is no longer a mixed bag of judgement calls and missing data. **Everything a competent engineer could decide has been decided. What is left is precisely the site investigation the project has never had —** and the reason is Part 4.2.2: the only investigation available is **off-site** and **reached about 1.5 m against a formation at (–)6.800.**

A4 deserves one extra line. The upper-bound model **now exists** and is verified by diff against the reference model. **It is still [ASSUMED] at both bounds, and neither has been run.**

24.3 Open items — eighteen

24.3.1 The four a reviewer should look at first

Ref	Item
SG–V2	The investigation reached about 1.5 m against a formation at (–)6.800. 5.3 m of unlogged ground under the whole structure — including the red-hole case that sizes the mat
EM–V6	Two 1 400 mm holes in the protective boundary whose bonding is still undesigned. The structural half is now designed (Part 10.5); the EMP half is not
SG2–V4	A CBRN shelter that cannot state which way a release drifts. The intake/exhaust half is closed on geometry; the plume half stays open, and must
RC4–V1	A sealed box with a 4 kW heat surplus and an unmodelled rejection path (Part 2.11)

24.3.2 Open, narrowed by a ruling (9)

Ref	Where it stands now
U8	Parapet confirmed at 300 × 150 and exact; one dimension — the roof projection — is recorded nowhere. 4.162 kN/m continues to be used as given; no member force changes
WM–V9	Excavation face ruled — batter the soil cap, vertical in rock. The slope-stability assessment is still outstanding
FS–V7	Ladder designed (Part 10.6). No fall-arrest, no rest platform, and the injured-person question
EM–V6	Structural half designed, EMP half open
EL–V2	Day tank sized at 250 L nominal in Bay 8, fill and vent up the existing shaft. Fuel type, rate and vendor open; the shared-bore fit is unchecked
EL–V7	Duty computed — 0.18 m ³ /h of CO ₂ , a 75 m ³ /h loop, 8 × margin on the fan allowance. Absorber and soda-lime charge open
SG–V6	Stairwell raft in black cotton soil — specified and measured (8 m ³). Rate [A]
SG2–V4	Intake/exhaust closed at 25.30 m separation; PLUME half open
RC4–V1	Narrowed from "no rejection path exists" to "the rejection path is the ground through the structure, and it has not been modelled"

24.3.3 Open, annotated but NOT narrowed (2) — and this pair is instructive

Ref	The problem
EM–V2	Every EMP Zone 2 dimension is [A], pending an equipment schedule
EL–V3	There is no equipment schedule for EMP Zone 2 — only a 1.50 kW allowance

These two are each holding the other's placeholder, and it is worth naming. The enclosure is **sized to fit a bay**; the 1.50 kW load is **chosen to give the enclosure a basis.** **Neither is evidence.** They corroborate each other and neither corroborates anything. **What breaks the loop is an operational equipment list — a client input, not a calculation.**

24.3.4 Open, untouched (6)

Ref	Item
WM-V 6	Sentry seismic weight after the masonry change. The direction is certain and favourable — brick is lighter, so W falls, V_b falls, and every member is over-designed. Nothing is unsafe and nothing is inconsistent. What is missing is a STAAD re-run to put a number on the margin — confirmation, not risk
EM-V 4	No communications design of any kind exists. No antenna, mast or feeder, so MIL-STD §5.7.6 has nothing to apply to
EM-V 5	No pipe material is specified for any run in the project. Metallic → bond it 360° to the entry plate and its exterior becomes shield. Plastic → the bore is an aperture <i>and</i> the water column is a conductor, needing a metallic spool piece nobody has specified
SG-V 1	The geotechnical data is off-site. No investigation has ever been made on the project plot
SG-V 2	Depth of investigation — the governing item
SG-V 7	The concealment turf may be an expansive clay. If the site's own topsoil is the CH horizon, it cracks in the dry season and pumps fines into the granular filter, which is the one thing the filter exists to stop. The load is unaffected

24.3.5 Open, new (1)

Ref	Item
DR-A2-V 1	The clean sump has no cover, and Bay 5 cannot be crossed without one (Part 14.7). Needs the cover itself: type, depth, imposed load duty , frame and rebate, fixing, and how it is lifted to withdraw the pumps

24.3.6 A declared scope boundary — not a gap

Ref	What does not exist	Where it was declared
EL-V 5	No circuit, cable, luminaire or socket schedules. The electrical package stops at board level	The package's own opening paragraph. It closes when a detailed electrical design is commissioned, not by any work in this project

This row was moved out of the open-item register on purpose, so that a **declared decision is never counted as an outstanding problem** — which is what was happening.

PART 25 — VERIFICATION PERFORMED FOR THIS REPORT

25.1 What was checked, and how

This report reproduces several hundred numbers from the project record. **Every one of them was independently recomputed from its own inputs before it was printed here**, by a script that is part of this package:

```
Project Report/Scripts/report_verify.py
-> Project Report/Calculations/REPORT_VERIFICATION_OUTPUT.txt

python3 "Project Report/Scripts/report_verify.py"
```

Each check takes the **inputs** a value is derived from, does the arithmetic **independently** — the formulas are written out again in the script, not copied — and compares the result with the value the project prints.

What a PASS means, stated precisely. *The printed number follows from the printed inputs. It is not a design check and it is not an analysis.* It does not re-derive the engineering judgement behind a value, it does not run STAAD, and it cannot tell you whether an assumption is true. What it does catch is the class of defect this project has already been bitten by twice: a number that no longer agrees with its own derivation.

25.2 The result

CHECKS EXECUTED	559
PASS	553
KNOWN / RECORDED difference	6
UNEXPLAINED FAIL	0

Coverage, by section of the script:

§	Subject	What it recomputes
1	Blast loading and structural dynamics	DLF and its limit case, the design pressure, the reflection coefficient, 50 psi in kPa, E_c , the participating mass, the flexural stiffness, the roof natural frequency and period, and t_d/T at both ends of the duration band
2	The engineered cover and the roof total	All six layer loads, their sum, the declared allowance , the total thickness, the crossfall drop, COMB 103, the static ULS case and the blast : static ratio
3	Soil, water and flotation	γ' , the lateral gradient and the water share, the pressures at soffit and floor, the uplift intensity and total at both box lengths, all four flotation factors of safety , the COMB 102 check at both lengths, the net pressure at formation, every bearing utilisation, and the whole rock-excavation quantity table
4–8	Element design, box and headhouse	Every d , $M_{u,lim}$, A_{st} provided, M_u , utilisation, x_u/d , shear force, τ_v , p_t , V_{us} , A_{sv}/s_v and link spacing for W1–W4, W6/W7, the mat, the roof, the openings, the cantilever pad, the headhouse roof and walls, and HW3 — plus the shaft fill time that produced Finding F1
9	Stairs	Both pitch angles and their cosines, every load term, effective spans, moments, areas, capacities, utilisations, the total-rise closure, and both opening-corner angles
10	Escape shaft head hatches	The leaf force, the hole area, the plate moment and shear, the flat-plate thickness and its 505 kg mass , both climb heights and both rung pitches
11	Seismic and wind	Both A_h values, both base shears, the $R = 3.0$ conservatism factor , the model-versus-hand weight gap and its two explanations to the kilonewton , all three period estimates, the full wind chain, and the portal distribution
12	Sentry post members	Effective spans and the two-way ratio, slab moments and capacities, both beams' governing combinations, capacities, capacity-design shears and earthquake shares , the 5-T16 bar-fit check that forced 3-T20 , the column's slenderness, steel, P_{uz} , biaxial sum and confinement, the footing in service and at ULS, punching, anchorage, and the whole lintel design including the 60° arching bound
13	Development, laps and cover	L_d/ϕ for both grades in tension and compression, every tabulated L_d , and the lap policy margin
14	CBRN ventilation	All eleven figures the HVAC package reproduces , plus the air-change rates, the scrubber matrix at six duty points and the recirculation fan power
15	Thermal balance	The load that is dissipated inside the envelope, the occupant gain, the 96-hour energy, all three bounds — adiabatic, concrete and film — the 13.2 K rise, the cooling duty, and the ventilation rejection arithmetic
16	EMP	Cage shielding effectiveness at five frequencies, the mesh cutoff, the highest frequency meeting 80 dB , the wavelength coincidence at 1 GHz, four TE11 cutoffs, six waveguide attenuations, the stair void, the honeycomb, both bond-strap inductances and reactances , the earth rod, the structure electrode, and the Zone 2 enclosure's area, seam length and volume

§	Subject	What it recomputes
17	Drainage and external works	The wetted envelope and the seepage it produces, the sump storage, all three trap-seal depths , the soak pit before and after widening, the septic tank checks, and every catchment
18	Electrical	The connected load and its kVA, generator utilisation and spare, the essential load, both battery cases and the factor of twelve between them , and the fuel quantity
19	Quantities and cost	Every bar mass from its unit mass and length, both totals, the concrete take-off, the steel density, and both cost summaries head by head
20	Register arithmetic	The open-item counts, the assumption counts and the drawing counts add up
21	Concrete mix design	Both target mean strengths and which branch of IS 10262 Cl. 5.2 governs ; the water content at each slump and after the admixture reduction; the cement from the water/cement ratio; the resulting free w/c against the IS 456 cap and the cement against both the minimum and the shrinkage maximum ; the coarse aggregate fraction and its congestion correction; the absolute-volume closure on 1.000 m³ ; both aggregate masses, both fresh densities and both mixes by mass
22	Programme and cost	The calendar span and the Sundays in it, the six-day working count, the activity total, the basic cost re-summed from its five priced parts , the share of each, and the final project cost re-derived from the seven percentage heads
23	EMP shielding physics	Skin depth in steel at three decades and the thickness that alone gives 80 dB ; the absorption of a 2 mm skin and the plane-wave reflection term; the honeycomb TE11 cutoff, its 32 L/d attenuation, the cell area, the number of cells in a square metre and the array correction on them ; and the hatch leaf force and flat-plate moment at $v = 0.30$ for steel

25.3 The six differences found — reported, never corrected

Master rule M.11 and the project operating guide both forbid editing a recorded value away. Every difference below is an observation offered to the master, not a change. **None of them moves a dimension, load, thickness, bar, quantity, rate, date or float.**

#	Where	Computed	Printed	Assessment
1	Underground box wall shear stress — $525 \times 10^3 / (600 \times 0.8 \times 21600)$	0.0506 N/mm²	0.063 N/mm ²	Not previously recorded. Both values are negligible by two orders of magnitude; the conclusion " <i>IS 13920 Cl. 10.4 boundary elements NOT triggered</i> " is unaffected and no adopted value depends on it
2	Sentry footing F1, ULS eccentricity — $M_u/P_u = 43.9 / 276.8$	0.159 m	0.128 m	Not previously recorded. The footing is governed by IS 456 Cl. 26.5.2.1 minimum steel (720 against 136 mm ² /m required) and its depth by starter anchorage , so no adopted bar, spacing or dimension moves
3	Sentry footing F1, $q_{u,max}$ — the printed expression $276.8/2.25 \times (1 + 6 \times 0.128/1.5)$	186.0 kPa	190.0 kPa	Same item as 2. One-way shear at 0.012 and punching at 0.065 against $\tau_c = 1.369$ are unaffected on either figure
6	The honeycomb waveguide panel's margin — the single-cell 133.3 dB against the same panel with its ~32 000 cells counted, $-10 \log_{10}(n)$	+8.3 dB	+53 dB	The panel still PASSES. What changes is a procurement requirement — require a certified attenuation curve for the panel as built , not for one cell — not a design value (Part 18.7)
5	The programme's working-day count — a raw Mon–Sat count of the 384-day span, less the five date-certain holidays	325 days	326 days	ONE DAY , and it is a property of the programme's own activity calendar rather than of any design value. No date, duration or float moves on it
4	The project owner's cost summary — the seven heads against the printed final	₹ 2 99 33 306	₹ 3 00 33 306	ALREADY RECORDED — conflict R-14 , exactly ₹ 1 00 000 apart, " <i>reported, not corrected</i> ". The verification pass reproduced a known finding , which is the outcome you want from a check like this. The revised estimate reconciles to the rupee and has no such difference

Five observations about that result are worth making.

The two footing items are one item. An eccentricity and the bearing pressure computed from it are the same arithmetic seen twice, and both land on an element whose steel is set by a code minimum and whose depth is set by anchorage. **That is precisely why it survived four revisions without being caught: nothing downstream of it is sensitive to it.**

The wall shear item is the same shape. A figure quoted to show that something is negligible is a figure nobody re-checks, because the conclusion is right either way.

And the fourth was already in the register. Reproducing a recorded finding independently is the best evidence that the check is working — and reproducing it **from the owner's own published numbers** is what makes it safe to leave alone.

The fifth is a single day, and it is worth keeping for exactly that reason. A programme that states 384 calendar days, a six-day week and five holidays should close on 325 working days, and it states 326. Nothing in this design depends on it, no float changes, and it is far more likely to be a property of how a scheduling tool counts a finish milestone than an error — **but the whole point of running the check is that a one-day difference is reported rather than rounded into agreement.**

And the sixth is the only one of the six that changes what somebody has to do. 133 dB is the attenuation of **one** honeycomb cell; a panel is tens of thousands of cells in one screen, and the classical array correction applies to it. The margin is **+8.3 dB, not +53** — still a pass, but eight decibels is a margin a bad gasket eats, where fifty-three is not. **It changes no design value and no dimension. It changes a procurement clause,** and it is the clearest example in this report of why the checks are run on things that already look settled.

25.4 What this report could not verify, and did not attempt

Not verified	Why
Any analysis result	STAAD.Pro is not available. No result exists to check
The mesh convergence conclusion	Same reason. The three models are built and validated; none has been run
Every [ASSUMED] value	An assumption is not an arithmetic error. k_s, the design water table, the safe bearing capacity, K_o, the soak-pit absorption rate, the seepage rate and the rockhead band are all tests, not sums
Vendor performance	Blast doors, blast valves, filter trains, the shielded door, the PCI, the honeycomb, the escape hatches
Anything tagged [N]	There is nothing to check. The projection dimension, the ambient design temperature, the opening heights, the pipe materials, the entry plate's level and size, the sump cover, the fan's five vendor loss components
The drawings themselves	Checked by the drawing QA tool, not by this report — Part 14.3

25.5 Consistency checks made against the project record

Beyond the arithmetic, the following were checked by reading:

- Every reinforcement arrangement in Part 13 matches the calculation in Parts 9 to 11.** No bar size, spacing or count appears in the register that does not appear in a calculation.
- Every evidence tag is carried unchanged.** No [ASSUMED], [UNRESOLVED] or [NOT AVAILABLE] item is converted, downgraded or deleted anywhere in this report.
- The open-item and assumption counts in Part 24 agree with their own tables** — $9 + 2 + 6 + 1 = 18$ open, $18 + 16 + 1 = 35$ rows, $7 + 7 = 14$ assumptions.
- 3a. Every one of the forty-four figures is bounds-checked,** and every one carries a caption in the register that numbers it. The gate is **zero primitives outside a frame** and **zero strings past the text measure**, and it is enforced by code rather than by eye — Appendix C.
 - The drawing counts in Part 14 sum to 80,** and the sheet sizes 75 A1 + 4 A4 + 1 A0 sum to 80.
 - The frozen staircase geometry is reproduced exactly** — 24 risers, 170.8333 mm, 280 mm tread, 3 flights × 8, total rise 4 100 mm — **and nothing in this report changes any of it.**
 - Every code clause cited here appears in the project's own register.** No clause was added, and codes held by title only are named as such.

APPENDIX A — NOTATION

Symbol	Meaning
p_{so}	Peak incident static overpressure of the blast wave
p_r	Reflected overpressure on a surface normal to the wave
q	Dynamic (kinetic) pressure of the air flow behind the front
t_d	Positive phase duration
T, f_1	Fundamental natural period and frequency
μ	Ductility ratio — maximum displacement / yield displacement
DLF	Dynamic load factor, $\mu/(\mu - 0.5)$
K_a	Soil transmission factor for a buried element
K_0	Coefficient of earth pressure at rest, $1 - \sin \phi$
$\gamma, \gamma_{sat}, \gamma', \gamma_w$	Bulk, saturated, submerged and water unit weights
k_s	Modulus of subgrade reaction (kN/m^3)
f_{ck}, f_y	Characteristic concrete and steel strengths
$f_{ck,dyn}, f_{y,dyn}$	The same, increased 25 % for the blast case only
E_c	Modulus of elasticity of concrete, $5000\sqrt{f_{ck}}$
d, d'	Effective depth; cover to the centroid of compression steel
$A_{st}, A_{sv}/s_v$	Tension steel area; shear link area per unit length
$M_p, M_u, M_{u,lim}$	Plastic mechanism moment; design capacity; limiting singly-reinforced capacity
$x_u, x_{u,max}$	Neutral axis depth; its limiting value, 0.46 d for Fe500
$\tau_v, \tau_c, \tau_{c,max}$	Nominal shear stress; permissible concrete shear stress; its maximum
p_t	Tension steel percentage
V_{us}	Shear to be carried by links
L_d, τ_{bd}	Development length; design bond stress
$A_h, Z, I, R, S_a/g$	Seismic design coefficient and its four inputs
V_b, W	Seismic base shear; seismic weight
P_{uz}, α_n	Column pure-axial capacity; biaxial interaction exponent
A_{sh}, l_o	Special confining hoop area; confining length
SE	Shielding effectiveness, dB
WBC	Waveguide below cutoff
T16 @ 150 EF EW	16 mm bars at 150 mm centres, Each Face, Each Way
4L	Four-legged links

Evidence classes — [C] confirmed · [R] reconstructed · [A] assumed · [U] unresolved · [N] not available.

APPENDIX B — WHERE EACH CODE CLAUSE IS USED

Clause	Subject	Used in
IS 456 Cl. 6.2.3.1	$E_c = 5000\sqrt{f_{ck}}$	Part 6.1
IS 456 Cl. 13.4	Construction joints	Part 6.4
IS 456 Cl. 22.2(a)	Effective span	Parts 10.1, 11.2, 11.7
IS 456 Cl. 23.2.1 + Fig. 4	Deflection, modification factor	Parts 10.1, 10.2, 11.2
IS 456 Cl. 24.4, 24.5 + Fig. 7	Two-way slabs; load to beams	Parts 7.7, 10.4, 11.2
IS 456 Cl. 25.1.2	Short column	Part 11.5
IS 456 Cl. 25.4	Minimum eccentricity	Part 11.5
IS 456 Cl. 26.2.1, 26.2.1.1, 26.2.5.1	Development length, laps	Part 6.3
IS 456 Cl. 26.3.3	Bar spacing	Parts 6.2, 11.2 — the project's 150 mm cap is stricter
IS 456 Cl. 26.4.2, .1, .2 + Table 16	Cover	Part 6.2
IS 456 Cl. 26.5.1.1, .2, .5, .6	Beam steel and links	Parts 9.1–9.4, 10.3, 11.3, 11.7
IS 456 Cl. 26.5.2.1, .2	Slab steel, maximum bar diameter	Parts 9.4, 10.1–10.3, 11.2, 11.6
IS 456 Cl. 26.5.3.1, .2	Column steel and ties	Part 11.5
IS 456 Cl. 31.6, .1, .3.1	Punching shear	Parts 9.3 (declared N/A), 11.6
IS 456 Cl. 32.2, 32.5(a)(b)(c)	Walls — minimum steel, two curtains	Parts 9.1, 10.2, 10.4
IS 456 Cl. 33.1(b), 33.2	Stairs — effective span, step load	Parts 10.1, 10.2
IS 456 Cl. 34.2.4.1	Footing one-way shear	Part 11.6
IS 456 Cl. 36.4.2	Partial safety factors for materials	Part 6.1
IS 456 Cl. 38.1, (c), (f)	Limit state, stress block, $x_{u,max}$	Parts 9–11 throughout
IS 456 Cl. 39.1, 39.3, 39.6	Columns, biaxial bending	Part 11.5
IS 456 Cl. 40.1, 40.2.1.1, 40.2.3, 40.4(a)	Shear	Parts 9.1–9.4, 10.1–10.4
IS 456 Tables 3, 5	Exposure, mix requirements	Part 6.1
IS 456 Tables 19, 20	$\tau_c, \tau_{c,max}$	Parts 9–11 throughout
IS 456 Tables 26, 27	Two-way slab coefficients	Parts 5.3, 10.3, 11.2
IS 456 Table 28	Effective length of columns	Part 11.5
IS 456 Annex D, D-1.1 to D-1.8	Two-way slab detailing, corner torsion	Part 11.2
IS 456 Annex G-1.1(b), (c)	$M_u, M_{u,lim}$	Parts 9–11 throughout
IS 875 (Pt 1) Table 1	Unit weights	Parts 6.1, 7.2
IS 875 (Pt 2)	Imposed loads	Parts 7.2, 7.7, 10.1, 10.2
IS 875 (Pt 3) Cl. 7.2 etc.	Wind	Part 7.8.3
IS 1893 (Pt 1) Cl. 6.3.2.2	Directional combination	Part 11.5
IS 1893 (Pt 1) Cl. 6.4.2	A_h	Part 7.8
IS 1893 (Pt 1) Cl. 7.6.2, (c)	Fundamental period	Part 7.8.2
IS 1893 (Pt 1) Cl. 7.6.3	Base shear and its distribution	Parts 7.8, 11.1
IS 1893 (Pt 1) Cl. 7.11.1	Storey drift	Part 11.5
IS 13920 Cl. 6.1	Beam section limits	Part 11.3
IS 13920 Cl. 6.2.1(b), .2, .3, .4	Beam steel limits	Part 11.3
IS 13920 Cl. 6.3.3, 6.3.4 , 6.3.5.1, .2	Capacity-design shear; $\tau_c = 0$	Parts 11.3, 11.4
IS 13920 Cl. 7.2.1	Strong column – weak beam	Part 11.5
IS 13920 Cl. 7.4, 8.1, 8.2	Column hoops, special confining reinforcement	Part 11.5
IS 13920 Cl. 10.4	Boundary elements — checked, not triggered	Part 7.8.1
IS 3370 (Pts 1, 2)	Crack width 0.2 mm; 0.35 % surface-zone steel	Parts 6.1, 9.1
IS 1904 Table 1	Presumptive bearing capacity	Parts 4.3, 11.6
IS 12070 Cl. 6, Table 2	Settlement on rock; broken bedrock	Parts 4.3, 9.3

Clause	Subject	Used in
IS 1498, IS 2720 Pts IV, VIII, X, XIII, XL, 28	Soil classification and testing	Parts 4.2, 4.5
IS 2470 (Pt 1) Cl. 6.2–6.9, Table 1	Septic tank	Part 16.6
IS 2470 (Pt 2) Cl. 4, 5	Percolation test — mandatory ; dispersion	Parts 14.2, 15.1, 18.2
IS 4991 Cl. 1.1	EXCLUDES NUCLEAR — quoted in every deliverable	Parts 3.1, 17.2
IS 4991 Cl. 6.2.1	Reflected pressure, recessing of blast valves	Parts 2.7.3, 14.1.4
IS 4991 Cl. 7.2 + Table 3	Buried roof and walls, K_a	Parts 2.1.4, 7.1, 7.5
IS 4991 Cl. 7.4	Drag on bermed faces	Part 7.5 (superseded basis)
IS 4991 Cl. 10.3.1	Dynamic f_{ck} and f_y	Parts 2.2.5, 7.1
IS 4991 Cl. 10.3.1.1	No dynamic increase on shear; +25 % bond, refused	Parts 2.2.5, 6.3, 9.3
IS 4991 Cl. 10.3.3	$DLF = \mu/(\mu - 0.5)$	Parts 2.2.3, 7.1
IS 4991 Cl. 11.1, 11.2	No wind/EQ with blast; no live load on the roof	Parts 7.4, 7.9
IS 4991 Fig. 6	SDOF support rotation — Phase 3	Parts 2.2.3, 17.2
SP 34 Cl. 5.5	Opening-corner detailing	Part 10.2.1
NBC 2016 Part 4	Stair geometry, 1 100 guarding, means of escape	Parts 9.5.2, 10.1, 14.5
UFC 3-340-02 §4-27, §4-30	Opening trimmers; direct shear	Part 9.4.4
MIL-STD-188-125-1 §5.4–§5.7.6	EMP — access, WBC, PCI, fibre, RF	Parts 2.8, 14.4
IEEE Std 299	Shielding effectiveness survey — HOLD POINT	Part 18.7
IEEE 142 / IS 3043	Earthing, $\leq 5 \Omega$	Parts 2.8.5, 14.3
FEMA 453	0.25 cfm/ft ² ventilation rate	Part 15.1
EN 1822	HEPA H14 at MPPS	Part 2.7.2
IS 1077	Sentry post brick masonry	Parts 5.9.1, 11.7
IS 732, IS 694, IS 1554 (Pt 1)	Electrical installation and cables	Part 17
IS 13416	Construction-phase fire safety	Part 19
BS 8666	Bar shape codes	Part 13.6

APPENDIX C — REPRODUCING THIS REPORT AND THE PROJECT

C.1 This report

```
Project Report/
Documentation/MASTER_PROJECT_REPORT.md    the report SOURCE -- edit this
Scripts/report_render.py                  Markdown -> PDF typesetter
Scripts/report_figures.py                 the 44 drawn figures
Scripts/report_verify.py                   independent recomputation
Calculations/REPORT_VERIFICATION_OUTPUT.txt its output
MASTER_PROJECT_REPORT.pdf                 the deliverable

python3 "Project Report/Scripts/report_verify.py"    run the checks
python3 "Project Report/Scripts/report_render.py"    rebuild the PDF
```

Never edit the PDF. Edit the Markdown and re-run the renderer. The renderer adds no content of its own beyond the cover, the running head and foot, the automatically paginated contents list and the list of figures.

The figures are code, not images. `report_figures.py` draws all forty-four from the `GEOM`, `LEV` and `COVER` constants at the top of the file, in **project coordinates** — so changing a dimension there changes every figure that uses it, and no figure can drift from the geometry it is drawn from. Three directives place them in the source:

```
<!-- FIG: fig_name -->    a numbered, captioned figure, in document order
<!-- LOF -->              the list of figures
<!-- PAGEBREAK -->       a forced page break
```

Every figure is bounds-checked before it is published. `report_figures.check_all()` walks each drawing and reports any line, rectangle, circle, polygon or **string** that falls outside its own frame or runs past the text measure — because reportlab does not clip, so anything drawn outside a figure silently bleeds onto whatever follows it. **The gate is zero.**

The figures are NOT the issued drawings. They carry no title block, no revision box and no bar mark, they are drawn at reading scale rather than at a plotting scale, and they must never be used for setting out or for fabrication. They are also **not** reconstructions of the seven absent S-series sheets (Part 14.6). The issued package is the 80 DXF of Part 14.

C.2 The project

To rebuild	Do this
A discipline drawing package	<code>cd "<Package>/Scripts" && python3 <prefix>_build_all.py</code> . Edit the generator, never the DXF
The drawing index and QA report	Re-run <code>qa_report_data.py</code> then <code>make_index.py</code> . A new package must first register its DXF/ prefix in DISCIPLINE and its name in ORDER
The works management package	<code>wm_build_all.py</code> regenerates everything from <code>wm_data.py</code> + <code>wm_content.py</code>
The S-series structural sheets	Not possible. The toolchain is not in the workspace (Part 14.6) and must not be fabricated
Any STAAD result	Not possible. STAAD.Pro is not available in this environment

The ten Rev F architectural DXFs are source, not build artefacts, and may be edited directly. Everything else generated in this project is a build artefact: **edit the generator and re-run.**

C.3 The rule that makes regeneration safe

Every discipline package in this project has verified that it **regenerates byte-identically**. That property is worth not breaking casually — it is what makes it possible to apply a change to a generator and be certain that nothing else moved.

APPENDIX D — TRACEABILITY: DECISIONS, CONFLICTS AND REVISION HISTORY

Parts 1 to 25 state the design as it stands, in the present tense, without narrating how it got there. This appendix is where the history lives, so that nothing is lost by keeping it out of the design.

The full record, with the reasoning behind every change, is in the master's Part H. This is an index to it, and where the two differ the master governs.

D.1 Revision history in brief

Rev	Date	What it did
A–C	—	Architectural layout development. Rev C lengthened the box to give ESC 2 its 750 mm clearance
—	—	Cover reduced 4.0 m → 2.0 m. Saved 2 m of rock excavation, 2 m of shaft, one stair flight and 2 m of headroom
D	—	Design report issued — basis of design, loads, materials, geotechnics. Current for loads and materials; superseded for the entrance and the headhouse roof
E–F	—	The approach was replaced: a dog-leg open ramp became a covered entry stairwell entered at grade. The open cut had no gravity outfall, a 2 m pit collects dense CBRN agents, and lidding at grade gave zero headroom
F	—	The ten architectural drawings issued. Current for all geometry
M1	3 Sep 2026	W6/W7 200 → 400; box 21 600 → 22 000. Approved and implemented. Finding F1
MS1	3 Sep 2026	Three mesh-density models built and validated
BIM-P1 / P1B	3 Sep 2026	Revit scripts authored — not executed; Revit is not available
SC1	4 Sep 2026	30 reinforcement drawings , 8 bar bending schedules, 70.46 t of steel
DR1 · HV1 · FN1	5 Sep 2026	Drainage, HVAC and the schedule of finishes
WM1	7 Sep 2026	Works management; and SP-B1, the one instructed design change
QA1	9 Sep 2026	65 DXF inspected and corrected in place
BS1 · SP-B2	10 Sep 2026	Burster slab laid to falls; sentry post lintels and wall ties
WM2 · WM3	10 Sep 2026	The owner's own bill, estimate and schedule adopted , then revised alongside
RC1	10 Sep 2026	Every inconsistency that could be ruled on the evidence, ruled
CAM1/2 · FS1/2	10 Sep 2026	Concealment policy and fire plan — neither existed before — then relocated and drawn
EM1	10 Sep 2026	The project's first EMP design
EL1	11 Sep 2026	The project's first electrical design , deliberately basic

Rev	Date	What it did
RC2	11 Sep 2026	Two owner rulings. Neither changed a number
SG1 · SG2	11 Sep 2026	The project's first site and geotechnical section , and its first site layout
QA2 · MS2 · RC3	11 Sep 2026	Drawing re-scan at 80 sheets; the .std input-file corrections ; project-wide reconciliation
DR-A1	11 Sep 2026	Drainage items drawn on the Rev F architectural sheets
RC4	11 Sep 2026	Sixteen owner rulings, item by item
RC5 · RC6 · RC7 · RC8	12 Sep 2026	Sixteen more rulings; K.2 falls from fourteen assumptions to seven
RC9	12 Sep 2026	A count this project got wrong, corrected — and made countable
RC10	12 Sep 2026	What the rulings left stale in the packages — six artefacts, register raised, not yet applied
DR-A2	12 Sep 2026	The sump pit head and a key plan; DR-A2-V1 opened
PR1	13 Sep 2026	The master project report, first issue — 19 parts. No design value changed; 498 checks executed; four differences reported and none corrected
PR2	13 Sep 2026	This report . Re-voiced as a single as-built statement of the design; 25 parts and 44 drawn figures ; the soil report, the M35 and M30 mix designs, the openings and closures, the five services parts, the works management expansion and the environmental management plan added. No design value changed

D.2 The conflicts and the rulings that closed them

D.2.1 The conflicts closed on the evidence

#	Conflict	Resolution
C1	Drawings Rev F, report Rev D — the report describes a dog-leg open ramp; Rev F is a covered entry stairwell	DXF governs . The precedent for everything after it
C2	The report designs the headhouse roof for 3.0 m span + 1.0 m cover; Rev F is 4.0 m + no cover	Recomputed . T20 @ 150 survives at 90 %; shear links new and mandatory
C3	The report calls the stair-shaft walls 600 thk; the Rev F plan shows 200	DXF governs → triggers Finding F1 → Modification M1
C4	Blast door leaf: plan 1200, section 900	1200 × 2100 adopted — agrees with the report's own 1 564 kN leaf force
C5	Plan dimension "7300 covered stairwell external" against 6 800 in the plotted geometry	6 800 adopted
C6	The report checks the mat in flexure only	One-way shear checked → the T12 @ 250 × 250 link grid
C7 / C9	The report uses W = 505 kN for the sentry post; the hand check gives 592.7 kN; STAAD gives 731.80 kN	The STAAD value governs , and Part 7.8.2 explains the gap to the kilonewton
C8	The report adopts a sentry post height of 6 850 mm; the elevation shows roof +6.700, parapet +7.000	Drawn levels adopted ; h = 6.250 m from the base at +0.450
C10	The report designs the headhouse walls for 113 kPa drag only	Raised to the full 383 kPa envelope (Part 7.5)
C11–C15	Five defects in the entry stairwell and underground .std files — wrong concrete grade, missing roof imposed load, missing waterproofing load, wrong fill unit weight and berm level, and four mat corner joints with no vertical restraint	All five resolved , each by bringing the file to the master rather than the master to the file

#	Conflict	Resolution
C16	The roof/platform junction — one clause said the stairwell roof becomes the 500 headhouse roof over the platform	RULED AT 250. It was the only place in the project saying 500 (Part 5.8)
C17	Cover build-up: 40.65 stated, 39.15 summed	RULED. 40.65 held, and the table made self-consistent (Part 7.3)
C18	Sump pit base: the register says 400, a sheet note says 300	RULED AT 400. $(-)8.000 - (-)7.600 = 0.400$ is arithmetic
C19	Soak pit 21.99 m ² against its own stated 22.50 required	RULED. Widened to 2.200 m dia → 24.19 m², +7.5 %
C20	A superseded Rev E catchment on an issued sheet	RULED. Rev F governs; the figure is retained as a declared conservatism and both are labelled
C21	Filter duty: the master said 2 × 250 m ³ /h, the sheet said 300 in nine places	RULED AT 300 m³/h, and the master was corrected. 250 fails the sheet's own FEMA criterion, so it was not the cautious option but a demonstrably inadequate one
ERR-1	A yield-line coefficient computed with the simply-supported constant instead of the fixed-four-edges one — 324.4 against 162.2	Corrected, and recorded as an error rather than quietly fixed. No reinforcement changed, because the one-way value was the one adopted (Part 10.3)

D.2.2 The owner's rulings, revision by revision

Revision	Rulings	Net effect on the open register
RC2 (11 Sep)	2 — the three-zone EMP model adopted ; the generator may run in closed mode	Neither changed a number
RC4 (11 Sep)	16, item by item — 13 actioned, 3 deliberately left open	33 → 25
RC5 (12 Sep)	4 — heat rejection, the hatches, the day tank, and one left open — plus A13 closed on the evidence	25, three narrowed, none closed
RC6 (12 Sep)	The excavation face, and six assumptions closed	K.2 14 → 7
RC7 (12 Sep)	4 — the scrubber duty, the mains, four source defects, the well	25 → 19
RC8 (12 Sep)	3 — the fence, the EM/EL circularity named, one reclassified	19 → 17
RC9 (12 Sep)	None. It corrected a count	17, verifiable by reading
DR-A2 (12 Sep)	None. It drew two things and found one	17 → 18

Three of those deserve to be read as a set, because they are about the project's own discipline rather than about the design.

RC9 **corrected a count this project got wrong.** Revisions RC4 to RC8 ruled faithfully and recorded every ruling — **but thirteen rows of the register itself were never annotated**, so a reader counting the table got 34 while the narratives claimed 21, then 15, then 13. **All three counts were wrong**, and the first was wrong arithmetically on its own terms. The fix was **not to restate the number but to make the register countable**: every row is now classified, and the totals can be verified by reading rather than by trusting a sentence.

RC10 **found the same failure mode one level down.** RC4 to RC9 ruled on thirty-two items and **did not touch the discipline packages**, so six artefacts still say something the master has settled — the EMP penetration register still calls the honeycomb panels conditional, the HVAC schedule still locates the intake as "*west of the box*", the escape-route schedule still records two climbs with no means of climbing them. **Nothing was silently edited**: all six are build artefacts regenerated from their packages' scripts, and applying the register is a package-by-package job with a regeneration check that **has not been done**.

The lesson is one this project already wrote down: a change that stops at "reinforcement" and never reaches "drawings" is not finished. RC4 to RC9 stopped at the master. The failure mode appeared twice in two days — once inside the master, once between the master and its packages — **and both times the fix was to make the thing countable rather than to trust a narrative.**

Master project report, revision PR2 · 13 September 2026 · FOR REVIEW — NOT FOR CONSTRUCTION. Authority: master/MASTER_PROJECT_STATE.md. Where this report and the master disagree, the master governs. No design value, evidence tag, quantity, rate, date or float is changed by this report, no .std file was touched, STAAD.Pro was not run, and the main staircase is unchanged — 24 risers, 170.8333 mm riser, 280 mm tread, 3 flights × 8, total rise 4 100 mm.